Supplementary material: Complex energy learning from finite magnetic-rod observations

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The proofs concern the full 64-coordinate constant-mass mechanical model and its three-coordinate conservative restriction defined in the article. A separate full-coordinate neural certificate retains the exact source operators and all 64 transverse coordinates of the same saved weights; its cubic-corrected twist is distinct from the target-family twist. The selected network has two complex split-tanh heads and 918 real parameters. Its checkpoint SHA-256 prefix is `f51bcea12c2f`. All numerical candidate conclusions below refer to that checkpoint when identified as calibrated. Earlier time-step and fitting diagnostics are explicitly distinguished. Certificates for other architectures and momentum-dependent historical models are not inherited.

The force/coefficient interface uses exact decimal scales $\ell = 1/1000$ m and $E_s = 3/500000000000$ J. Network feature scales are their exact saved binary values and need not equal these decimal numbers. A numerical recorded-state IVP, a compiled simulation trajectory and a physical measurement are different objects; their distinct error requirements remain stated in each result.

S1 Complex canonical coordinates and symplectic discretization of the saved full-coordinate energy

S1.1 The fixed mechanical and network object

Let $n = 64$, let $M = M_0 + M_r \succ 0$ be the exact sum of the saved source matrices, and let $K_0 \succ 0$ be the saved elastic matrix. All saved floating-point entries and network parameters are interpreted as exact real numbers. With $b = e_{64}$, the conservative Hamiltonian is

$$H_N(x,p) = \tfrac{1}{2}p^T M^{-1} p + U_N(x), \qquad U_N(x) = E_0(x) + \kappa[1-\cos(b^T x)] + F_N(x), \quad E_0(x) = \tfrac{1}{2}x^T K_0 x. \tag{S1}$$

The energy correction is precisely the saved configuration graph

$$\begin{aligned} F_N(x) &= c_E E_0(x) + E_{\mathrm{net}}\left[N(s(x)) - N(0) - \sum_{j=1}^{3}\gamma_j s_j(x)\right], \\ s(x) &= \left((Bx)_1/Q_1, (Bx)_2/Q_2, (Bx)_3/Q_3, E_0(x)/E_{\mathrm{net}}\right), \qquad \gamma_j = \partial_j N(0). \end{aligned} \tag{S2}$$

Here $B = (V^T M V)^{-1} V^T M$, where V is the full-column-rank preparation basis supplied by the source record. Its saved rational entries, the source matrices, and all operations defining B are interpreted exactly; a separately rounded runtime projection buffer is not substituted. Here N is the real readout of the saved $4 \to 12 \to 12 \to 1$ complex split-tanh energy head. Together with its dissipation head and the two real skips the model has 918 real parameters; the present conservative Hamiltonian does not depend on the dissipation head. The checkpoint is `f51bcea12c2f2640c0ea33c3493a691d8ab240578018a6d44a9f68904a1a60a9`. There is no learned

momentum dependence and no configuration-dependent mass. The conservative subsystem sets the dissipation and magnetic input force to zero while retaining the magnetic potential in (S1).

For $q = Bx$ and $E = E_0(x)$ write the right-hand side of (S2) as $F(q, E)$. Differentiation of both features gives

$$\nabla_x F_N = B^T F_q + F_E K_0 x, \qquad \nabla U_N = (1 + F_E)K_0 x + \kappa \sin(b^T x)b + B^T F_q. \tag{S3}$$

In particular,

$$(F_q)_j = \frac{E_{\text{net}}}{Q_j}[\partial_j N(s) - \gamma_j], \qquad F_E = c_E + \partial_4 N(s). \tag{S4}$$

The fourth feature is not removed by the affine correction. The Hamilton equations are consequently

$$\dot{x} = M^{-1}p, \qquad \dot{p} = -\nabla U_N(x), \qquad (H_N)_{pp} = M^{-1}, \quad (H_N)_{xp} = 0. \tag{S5}$$

These formulas hold on the full 128-dimensional transverse phase space. They use neither invariance of the three-coordinate trial space nor the scalar target-family prior.

S1.2 Dimensionless canonical variables and the complex equation

The configuration entries contain both displacements and rotations. To avoid adding a physical configuration to a physical momentum, choose a fixed invertible configuration scale L_s, an energy scale $\mathcal{E} > 0$ and an action scale $\mathcal{A} > 0$. Define dimensionless variables by

$$x = L_s Q, \qquad p = \mathcal{A}L_s^{-T}P, \qquad t = (\mathcal{A}/\mathcal{E})\tau, \qquad h(Q, P) = \mathcal{E}^{-1}H_N(L_s Q, \mathcal{A}L_s^{-T}P). \tag{S6}$$

The physical one-form and two-form pull back to

$$p^T dx = \mathcal{A}P^T dQ, \qquad \sum_j dx_j \wedge dp_j = \mathcal{A}\sum_j dQ_j \wedge dP_j. \tag{S7}$$

Thus the common action factor is removed together with the energy/time normalization; the configuration/momentum transform alone is conformally symplectic with constant factor $\mathcal{A}$. Direct use of the chain rule, rather than a change of units alone, gives

$$\begin{aligned} Q' = \partial_P h = A_s P, \qquad & P' = -\partial_Q h = -g(Q), \\ h(Q, P) = \tfrac{1}{2}P^T A_s P + \mathcal{U}(Q), \qquad & A_s = \frac{\mathcal{A}^2}{\mathcal{E}}L_s^{-1}M^{-1}L_s^{-T} \succ 0, \\ \mathcal{U}(Q) = U_N(L_s Q)/\mathcal{E}, \qquad & g(Q) = L_s^T \nabla U_N(L_s Q)/\mathcal{E}. \end{aligned} \tag{S8}$$

A prime means differentiation with respect to τ; the matrix A_s is not the action scale $\mathcal{A}$.

Now define complex canonical coordinates and their inverse by

$$z = (Q + iP)/\sqrt{2}, \quad \bar{z} = (Q - iP)/\sqrt{2}, \qquad Q = (z + \bar{z})/\sqrt{2}, \quad P = (z - \bar{z})/(i\sqrt{2}), \tag{S9}$$

and put $\mathfrak{h}(z, \bar{z}) = h(Q(z, \bar{z}), P(z, \bar{z}))$. The Wirtinger chain rules are

$$\partial_z \mathfrak{h} = (h_Q - ih_P)/\sqrt{2}, \qquad \partial_{\bar{z}} \mathfrak{h} = (h_Q + ih_P)/\sqrt{2}. \tag{S10}$$

Consequently the same real system is

$$z' = -i\partial_{\bar{z}} \mathfrak{h}, \qquad \bar{z}' = i\partial_z \mathfrak{h}, \qquad \omega = \sum_j dQ_j \wedge dP_j = i\sum_j dz_j \wedge d\bar{z}_j. \tag{S11}$$

For example, for real differentiable observables the corresponding Poisson bracket is

$$\{f,k\} = -i\sum_j \left[(\partial_{z_j} f)(\partial_{\bar{z}_j} k) - (\partial_{\bar{z}_j} f)(\partial_{z_j} k)\right], \qquad \{z_j, \bar{z}_k\} = -i\delta_{jk}. \tag{S12}$$

This is a complex coordinate representation of a real scalar Hamiltonian. It does not require that $\mathfrak{h}$ be holomorphic in z alone, or identify a hidden complex neuron with a canonical state.

The analytic extension used for this saved network is equally precise. Each complex affine layer is replaced by its real block matrix $\left(\begin{smallmatrix} W_R & -W_I \\ W_I & W_R \end{smallmatrix}\right)$, and the split activation acts componentwise by real tanh. Complexifying these real variables gives a holomorphic continuation wherever its tanh denominators are nonzero. If Z, W are independent variables, the canonical extension is

$$\mathfrak{h}^{\mathbb{C}}(Z,W) = h^{\mathbb{C}}\left((Z+W)/\sqrt{2}, (Z-W)/(i\sqrt{2})\right). \tag{S13}$$

The physical slice has $W = \bar{Z}$; no conjugation is performed during holomorphic differentiation in the independent variables. The existing source-bound analytic-domain certificate concerns the same realified potential. No literal-complex-tanh replacement is made here.

S1.3 Deriving the Störmer–Verlet difference scheme

Because (S8) is separable, its kinetic and potential subsystems have exact flows

$$D_s(Q,P) = (Q + sA_sP, P), \qquad K_s(Q,P) = (Q, P - sg(Q)). \tag{S14}$$

Their symmetric composition $\Phi_\delta = K_{\delta/2} \circ D_\delta \circ K_{\delta/2}$ gives

$$\begin{aligned} P_{n+1/2} &= P_n - \tfrac{\delta}{2} g(Q_n), \\ Q_{n+1} &= Q_n + \delta A_s P_{n+1/2}, \\ P_{n+1} &= P_{n+1/2} - \tfrac{\delta}{2} g(Q_{n+1}). \end{aligned} \tag{S15}$$

This is the classical Störmer–Verlet construction (Hairer 2010, Lecture 2, Theorem 2 and Section 5, Example 3). It is explicit apart from application of the fixed inverse mass; it does not solve a nonlinear implicit network equation. In physical time with step $\Delta t = (\mathcal{A}/\mathcal{E})\delta$, its identical form is

$$\begin{aligned} p_{n+1/2} &= p_n - \tfrac{\Delta t}{2} \nabla U_N(x_n), \\ x_{n+1} &= x_n + \Delta t M^{-1} p_{n+1/2}, \\ p_{n+1} &= p_{n+1/2} - \tfrac{\Delta t}{2} \nabla U_N(x_{n+1}). \end{aligned} \tag{S16}$$

Both endpoint forces must include (S3). Eliminating momenta gives the associated position difference equation

$$M(x_{n+1} - 2x_n + x_{n-1}) = -(\Delta t)^2 \nabla U_N(x_n). \tag{S17}$$

For the complex version of each substep, recover Q, P from (S9) and apply

$$z \xmapsto{K_s} z - isg(Q)/\sqrt{2}, \qquad z \xmapsto{D_s} z + sA_sP/\sqrt{2}, \tag{S18}$$

with the conjugate updates for $\bar{z}$. Intermediate values of Q, P are updated after each substep. This is the same difference map in complex coordinates, rather than a new discretization obtained by inserting complex states into a real-output network routine.

Exact symplecticity and reversibility. Put $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$. Since A_s and $G(Q) = Dg(Q)$ are symmetric,

$$DK_s = \begin{pmatrix} I & 0 \\ -sG(Q) & I \end{pmatrix}, \qquad DD_s = \begin{pmatrix} I & sA_s \\ 0 & I \end{pmatrix}, \qquad (DK_s)^T J DK_s = (DD_s)^T J DD_s = J. \tag{S19}$$

The chain rule therefore gives $(D\Phi_\delta)^T J D\Phi_\delta = J$. The physical map preserves $\sum dx_j \wedge dp_j$ by (S7); the complex map preserves $i\sum dz_j \wedge d\bar{z}_j$. Also $K_s^{-1} = K_{-s}$ and $D_s^{-1} = D_{-s}$, whence $\Phi_{-\delta} = \Phi_\delta^{-1}$. These identities concern exact mathematical evaluation of the map. They do not assert exact conservation of the nonlinear energy.

The physical linear stability threshold. Let $H_2 = U_N''(0) \succ 0$ and choose a mass-normalized basis S with $S^T M S = I$ and $S^T H_2 S = \mathrm{diag}(\omega_1^2, \ldots, \omega_n^2)$. The transformation $x = S\xi$, $p = MS\pi$ is canonical, since $S^{-T} = MS$. Each physical-time mode of the linearized Verlet map is

$$\begin{pmatrix} \xi_{j,n+1} \\ \pi_{j,n+1} \end{pmatrix} = \begin{pmatrix} 1 - r_j^2/2 & \Delta t \\ -\Delta t \omega_j^2 (1 - r_j^2/4) & 1 - r_j^2/2 \end{pmatrix} \begin{pmatrix} \xi_{j,n} \\ \pi_{j,n} \end{pmatrix}, \qquad r_j = \Delta t \omega_j. \tag{S20}$$

Its determinant is one and its trace is $2 - r_j^2$. For $0 < r_j < 2$ it preserves the positive quadratic form $\pi_j^2 + \omega_j^2(1 - r_j^2/4)\xi_j^2$, so its powers are bounded. At $r_j = 2$ the matrix is $\left(\begin{smallmatrix} -1 & \Delta t \\ 0 & -1 \end{smallmatrix}\right)$, whose powers are generally unbounded. For $r_j > 2$ one eigenvalue has modulus greater than one. Thus, for a positive step, the full linearized map is power bounded exactly when

$$0 < \Delta t < 2/\omega_{\max}. \tag{S21}$$

The existing exact-source candidate frequency enclosures give

$$\omega_{\max} \le \frac{2441081236299847251}{17592186044416}\ \mathrm{rad\,s^{-1}}. \tag{S22}$$

Their exact rational arithmetic therefore supplies the sufficient linear condition

$$0 < \Delta t < \frac{35184372088832}{2441081236299847251}\ \mathrm{s} \simeq 1.44134376052818 \times 10^{-5}\ \mathrm{s}. \tag{S23}$$

This reuses a received spectrum rather than running an integration experiment. It is a condition for the equilibrium linearization of the conservative saved-source model, not a nonlinear safety guarantee, a control hold duration, an error-tolerance choice, or a discrete KAM certificate. The finite-time floating-point integrations in article Section 3.2 are a separate numerical check; this spectral threshold uses no integration output.

S1.4 A full local and global second-order estimate

The following estimates make explicit the domain condition that is otherwise hidden in order notation. Work in the dimensionless coordinates and choose a convex region containing the configurations of the exact solutions and all compared kick/drift stages. Assume $\mathcal{U} \in C^3$ there and, in a fixed compatible vector norm,

$$\|A_s\| \le a, \quad \|g\| \le G_0, \quad \|Dg\| \le G_1, \quad \|D^2 g\| \le G_2, \quad \|A_s P(\tau)\| \le V \tag{S24}$$

The bounds on $g, Dg, D^2 g$ hold uniformly throughout the stated convex configuration region; only the bound on $A_s P(\tau)$ is restricted to the relevant exact arcs. These are assumptions for this estimate, not numerical constants supplied by a finite training loss. Let $0 < \delta \le \delta_0$ and use the phase norm $\|(Q,P)\|_\times = \max(\|Q\|, \|P\|)$.

Starting at the same (Q,P), write $Q_e(s), P_e(s)$ for the exact solution. Two integrations of the equations give

$$Q_e(\delta) = Q + \delta A_s P - \int_0^\delta (\delta - s) A_s g(Q_e(s))\, ds. \tag{S25}$$

The Verlet endpoint $Q_v = Q + \delta A_s P - \delta^2 A_s g(Q)/2$ then satisfies

$$\|Q_v - Q_e(\delta)\| \le a G_1 V \int_0^\delta (\delta - s) s\, ds = C_Q \delta^3, \qquad C_Q = a G_1 V/6. \tag{S26}$$

For $f(s) = g(Q_e(s))$,

$$f''(s) = D^2 g(Q_e(s))[A_s P_e(s), A_s P_e(s)] - Dg(Q_e(s)) A_s g(Q_e(s)), \qquad \|f''\| \le W := G_2 V^2 + G_1 a G_0. \tag{S27}$$

The integral remainder of the trapezoidal rule is bounded by $W\delta^3/12$. Replacing its exact endpoint by Q_v adds at most $\delta G_1\|Q_v - Q_e(\delta)\|/2$. Consequently

$$\|P_v - P_e(\delta)\| \le C_P\delta^3, \qquad C_P = \frac{G_2V^2 + G_1aG_0 + \delta_0 aG_1^2V}{12}, \qquad \|\Phi_\delta - \varphi_\delta\|_\times \le C\delta^3, \quad C = \max(C_Q, C_P). \tag{S28}$$

This proves local order three without assigning an unspecified constant to the network.

On compared stage segments, the two shear Lipschitz constants give

$$\mathrm{Lip}(\Phi_\delta) \le (1 + \delta G_1/2)^2(1 + \delta a) \le e^{\Lambda\delta}, \qquad \Lambda = G_1 + a > 0. \tag{S29}$$

For $T = N\delta$ and matching initial conditions, iteration of $e_{n+1} \le e^{\Lambda\delta}e_n + C\delta^3$ yields

$$e_N \le C\delta^3\frac{e^{\Lambda T} - 1}{e^{\Lambda\delta} - 1} \le C\delta^2\frac{e^{\Lambda T} - 1}{\Lambda}. \tag{S30}$$

Thus the method has global order two on a fixed time interval while the stated exact and numerical stages remain in the prescribed region. A buffer-domain bootstrap may establish this containment once the right-hand side is smaller than the buffer width; containment is not assumed to follow from symplecticity alone.

If an implementation makes a per-step error at most η_n relative to the mathematical Verlet map, the same argument instead gives

$$e_N \le e^{\Lambda T}e_0 + C\delta^2\frac{e^{\Lambda T} - 1}{\Lambda} + \sum_{n=0}^{N-1} e^{\Lambda(N-1-n)\delta}\eta_n. \tag{S31}$$

Network evaluation and inverse-mass solve errors belong to η_n; they are not covered by the exact shear identity. No numerical value of these implementation errors, a selected integration step, or the domain-dependent error constants is supplied here.

S1.5 Distinguishing integration, learning and KAM statements

For a target with the same normalized kinetic energy let $g_*(Q)$ be its conservative force, and suppose on the common solution region that $\|g - g_*\| \le \varepsilon$ and that one of the two complete Hamiltonian vector fields has Lipschitz constant L_*. The integral difference equation gives, for identical initial states,

$$\|\varphi_\tau^N - \varphi_\tau^*\|_\times \le \varepsilon\frac{e^{L_*\tau} - 1}{L_*}, \tag{S32}$$

with τ on the right when $L_* = 0$. Combining this estimate with (S30) separates a δ^2 integration term from a learning term that does not vanish when the step is reduced. Finite node losses do not by themselves imply the required uniform force bound. The existing coefficient allowances, source priors and candidate analytic certificates must be used with their own stated domains and quantifiers to obtain any such bound.

The continuous local elliptic KAM certificate concerns the flow of (S1). Symplecticity and second-order convergence of Φ_δ do not turn it into a KAM certificate for the discrete map: map nonresonances may also involve $\delta k \cdot \omega - 2\pi m$, and a discrete KAM application needs its own hypotheses and bounds. No finite numerical torus radius or all-time trajectory error is deduced from (S30).

Finally, let $D_0 = \alpha M_0 \succ 0$ with $\alpha > 0$, let $a_N(x) \ge 0$ be the saved configuration-dependent squared-readout dissipation factor, and let $g_{\mathrm{mag}}(x) = \mu\cos(b^Tx)b$ be the configuration input port. The full port model has the additional force $-a_N(x)D_0M^{-1}p + g_{\mathrm{mag}}(x)u$. When u is independent of momentum its phase divergence contains $-a_N(x)\,\mathrm{tr}(D_0M^{-1})$, which is negative wherever $a_N > 0$; such a flow cannot preserve the canonical phase volume there. The conservative shear proof is therefore not a symplecticity claim for the damped, forced or feedback-controlled model. Its established power and sampled-work safety bounds supply the separate control argument. Nor does changing canonical coordinates prove that the complex network representation outperforms a real network.

S2 Source-conditioned target regularity and the legacy-model boundary

The native source files distinguish the generated benchmark from an unknown physical device. The actual configured and solved models used for all eleven development configurations retain an Euler–Bernoulli beam with isotropic linear elasticity, spatially constant Young's modulus, fixed cross-section and geometric nonlinearity disabled. The only added weak contribution is rotational inertia. The specified point load is the magnetic moment already represented by the known cosine potential. These are properties of the saved source models, independently of the fitted neural weights.

For the ideal fixed-mesh semidiscrete model specified by these settings, the existing full-FE energy derivation gives

$$U_E(x) = \tfrac{1}{2}x^T K_E x + \kappa(1 - \cos(b^T x)), \qquad K_E = (E/E_{\mathrm{nom}})K_0.$$

The source modulus is $E_{\mathrm{nom}} = 5 \times 10^6$ Pa and the generated target modulus is $E = 5.3 \times 10^6$ Pa. Consequently the material correction relative to the nominal simulator is exactly

$$W_*(x) = \beta E_0(x), \qquad \beta = \frac{3}{50}, \qquad E_0(x) = \tfrac{1}{2}x^T K_0 x.$$

Here $E_0(x)$ denotes the nominal elastic energy. This source-conditioned identity applies to the ideal model specified by the saved settings; it is not inferred from the finite trajectory observations and does not certify exact arithmetic inside a compiled solver.

On the same restriction $x = V\ell z$ and normalization E_s, put $w_*(z) = W_*(V\ell z)/E_s$. Then

$$w_*(z) = \frac{\beta\ell^2}{2E_s} z^T K_{03} z, \quad b_1(w_*) \le \frac{|\beta|\ell^2 R}{E_s} \max_i \sum_j |(K_{03})_{ij}|, \quad b_2(w_*) \le \frac{|\beta|\ell^2}{E_s} \max_{i,j} |(K_{03})_{ij}|,$$

and $b_3(w_*) = b_4(w_*) = b_5(w_*) = 0$. The source-conditioned material fifth-order remainder therefore vanishes. The known magnetic term has not been set to zero: it is retained exactly and cancels only in the material learning error. For the specified three-coordinate conservative restriction the quadratic Hessian is

$$K_* = (53/50)K_{03} + \kappa d d^T.$$

The exact source ratio $3/50$ lies inside the previously received scalar compatibility interval, so the existing uniform positive spectrum and finite-order-divisor certificate covers this ideal simulator member. This is a source-conditioned simulator conclusion, not a completed learning-to-KAM transfer for an unknown device.

Historical geometrically nonlinear models have different kinematics, coordinate charts and discretizations. They supply no interchangeable derivative prior for this fixed-mesh target without an independent coordinate transport and remainder estimate.

Thus the source files justify the displayed regularity bounds for the ideal generated benchmark. They do not identify an unknown material's analytic tail or quadratic spectrum, resolve the legacy nonlinear inertia mapping, establish complex-network superiority, or prove physical-target KAM. For an unknown physical target these remain independent constitutive and measurement premises.

S3 Learning a three-dimensional conservative correction

The learning variable can be the configuration rather than the complete canonical state when the kinetic energy is known and the unknown term is a configuration potential. This reduction concerns the domain on which an error function must be learned. It does not reduce the number of degrees of freedom of the Hamiltonian system.

S3.1 The restricted physical model and its learning dimension

Let A and K_3 be fixed symmetric positive definite three-by-three mass and elastic matrices, respectively. Let $d \in \mathbb{R}^3$ and $\kappa > 0$ be the fixed magnetic port and bias coefficient. Consider the real conservative Hamiltonians

$$\begin{aligned} H_\theta(q,p) &= \tfrac{1}{2}p^{\mathsf{T}}A^{-1}p + U_0(q) + W_\theta(q), \\ H_*(q,p) &= \tfrac{1}{2}p^{\mathsf{T}}A^{-1}p + U_0(q) + W_*(q), \\ U_0(q) &= \tfrac{1}{2}q^{\mathsf{T}}K_3 q + \kappa[1-\cos(d^{\mathsf{T}}q)]. \end{aligned} \tag{S33}$$

Here $q, p \in \mathbb{R}^3$, the mass is not fitted, and damping and time-dependent input are excluded from this conservative subsystem. The target W_* and a candidate W_θ are functions of q alone.

The energy-feature candidate has this structure on its trial subspace. More precisely, let M, K_0 be the full mass and elastic matrices and let V be a fixed full-column-rank trial matrix. Set $A = V^{\mathsf{T}}MV$, $K_3 = V^{\mathsf{T}}K_0V$, and $B = A^{-1}V^{\mathsf{T}}M$, so that $BV = I_3$. Its full potential correction is $F_\theta(Bx, x^{\mathsf{T}}K_0x/2)$, and its restriction is

$$W_\theta(q) = F_\theta(q, E_0(q)), \qquad E_0(q) = \tfrac{1}{2}q^{\mathsf{T}}K_3q. \tag{S34}$$

The four feature entries $(q, E_0(q))$ have only three independent real variables on this surface. For example, the centered scalar readout is

$$\begin{aligned} s(q) &= (q_1/Q_1, q_2/Q_2, q_3/Q_3, E_0(q)/E_*), \\ W_\theta(q) &= c_{\mathrm{E}}E_0(q) + E_*\left[N_\theta(s(q)) - N_\theta(0) - \sum_{j=1}^{3}\frac{q_j}{Q_j}\partial_j N_\theta(0)\right]. \end{aligned} \tag{S35}$$

The positive Q_j, E_* are the frozen exported feature scales; N_θ is the real scalar readout of the real or realified complex network. Equation (S35) implies $W_\theta(0) = 0$ and $\nabla W_\theta(0) = 0$ because $\nabla E_0(0) = 0$. It does not impose zero quadratic, cubic, or quartic corrections. The coefficient c_{E} occurs linearly.

For arbitrary full configurations, $E_0(x)$ is not determined by Bx. In that case

$$V^{\mathsf{T}}\nabla_x F_\theta(Bx, E_0(x)) = F_{\theta,q} + F_{\theta,E}V^{\mathsf{T}}K_0x,$$

whereas the restricted force is $\nabla_q W_\theta = F_{\theta,q} + F_{\theta,E}K_3q$ at $(q, E_0(q))$. Thus unrestricted full-field observations require the separate preparation, transfer, and conservative-label error estimates. The reduction below does not assert autonomous closure of a projected full trajectory.

Fix $\ell > 0$, $E_s > 0$, and $R > 0$, and use the normalized configuration cube and potentials

$$\Omega = [-R, R]^3, \qquad q = \ell z, \qquad w_\theta(z) = W_\theta(\ell z)/E_s, \quad w_*(z) = W_*(\ell z)/E_s. \tag{S36}$$

In the physical interface $\ell = 10^{-3}$ and $E_s = 6 \times 10^{-12}$ J. The normalization E_s need not be identified with the exact exported binary value E_* in (S35). Write $f_\theta = \nabla_z w_\theta$, $f_* = \nabla_z w_*$ and $g_\theta = f_\theta - f_*$. A normalized conservative-force label means a label for f_*, including its sign and the factor ℓ/E_s. For a vector field g, write $\|g\|_{\infty,\Omega} = \max_j \sup_{z\in\Omega}|g_j(z)|$.

Proposition S1. Exact reduction of the error domain. *Let Π be any probability law for (Z, P) on $\Omega \times \mathcal{P}$, with configuration marginal ν. For $1 \le p < \infty$, the physical canonical vector-field error obeys*

$$\begin{aligned} X_{H_\theta} - X_{H_*} &= (0, -(E_s/\ell)g_\theta(z)), \\ \mathbb{E}_\Pi\|X_{H_\theta}(\ell Z, P) - X_{H_*}(\ell Z, P)\|_p^p &= (E_s/\ell)^p\mathbb{E}_\nu\|g_\theta(Z)\|_p^p. \end{aligned} \tag{S37}$$

Moreover, a uniform potential-error bound on Ω is the same bound, multiplied by E_s, on every momentum fiber above that cube. No positive density in the momentum variables is needed for these identities.

Proof. The common kinetic and reference potential terms cancel in $H_\theta - H_*$. Its physical configuration gradient is $(E_s/\ell)g_\theta$, and its momentum gradient is zero. Applying the canonical matrix $J = \left(\begin{smallmatrix} 0 & I_3 \\ -I_3 & 0 \end{smallmatrix}\right)$ gives the first identity. The integrand is independent of P, so integration with respect to the joint law gives its integral against the configuration marginal. The potential statement follows from the same cancellation and the definition of $w_\theta - w_*$.

S3.2 A configuration-space version of the statistical energy bound

The statistical component used here is the uniform bounded-loss generalization inequality in Chen, Matsubara and Yaguchi, *KAM Theory Meets Statistical Learning Theory: Hamiltonian Neural Networks with Non-zero Training Loss*, AAAI 2022, Lemma 2. Their Poincare–Sobolev step is on the complete phase domain. Proposition S1 permits that step to be performed on Ω for the present model.

Let Θ be a hypothesis class fixed before sampling. Suppose $Z_1, \dots, Z_N$ are iid with law ν whose Lebesgue density satisfies $\rho \geq m > 0$ on Ω. For a fixed finite $p \geq 1$, suppose

$$0 \leq \|g_\theta(z)\|_p^p \leq \Lambda_p \quad (z \in \Omega,\ \theta \in \Theta).$$

Let $\overline{\mathfrak{R}}_{p,N}$ be a valid upper bound for the empirical Rademacher complexity of this true loss class, with the normalization of the cited lemma. If observed labels Y_i satisfy $\|Y_i - f_*(Z_i)\|_p \leq \eta$ for every i, set

$$\begin{aligned} \widehat{L}_{p,N}(\theta) &= N^{-1} \sum_i \|f_\theta(Z_i) - Y_i\|_p^p, \\ A_{p,N}(\theta, \delta) &= \left(\widehat{L}_{p,N}(\theta)^{1/p} + \eta\right)^p + 2\overline{\mathfrak{R}}_{p,N} + 3\Lambda_p \sqrt{\frac{2\log(4/\delta)}{N}}. \end{aligned} \tag{S38}$$

With probability at least $1 - \delta$, simultaneously for all θ,

$$\mathbb{E}_\nu \|g_\theta\|_p^p \leq A_{p,N}(\theta, \delta). \tag{S39}$$

Indeed, apply the cited lemma to the iid pairs $(Z_i, f_*(Z_i))$. The triangle inequality for the finite-sample L^p norm bounds the unobserved true empirical loss by the first term of (S38). This argument allows arbitrary label errors satisfying the stated deterministic cap; it does not require those errors to be iid. The class bounds and the complexity bound must still be justified for the whole fixed class.

Theorem S2. Three-dimensional energy reception. *Assume $p > 3$, $w_*, w_\theta \in W^{1,p}(\Omega)$, and the preceding statistical hypotheses. Define*

$$e_\theta = w_\theta - w_* - \frac{1}{|\Omega|} \int_\Omega (w_\theta - w_*)\, dz.$$

On the event (S39), uniformly in θ,

$$\|e_\theta\|_{C^0(\Omega)} \leq C_{p,R} \left(\frac{A_{p,N}(\theta, \delta)}{m|\Omega|}\right)^{1/p}, \qquad C_{p,R} = 2R\, 3^{1-1/p} \frac{p}{p-3}. \tag{S40}$$

The physical error is E_s times this quantity on every momentum fiber. If $W_\theta(0) = W_*(0) = 0$, the absolute error in this fixed physical energy gauge is at most twice the right-hand side of (S40), multiplied by E_s.

Proof. First take a continuously differentiable scalar u. Convexity of the cube and the segment identity imply, for $z \in \Omega$,

$$|u(z) - \overline{u}| \leq \operatorname{diam}_{p'}(\Omega) \int_0^1 \frac{1}{|\Omega|} \int_\Omega \|\nabla u(z + t(y - z))\|_p \, dy\, dt, \qquad p' = \frac{p}{p-1}.$$

Holder's inequality for the inner probability integral and the change of variables $x = z + t(y - z)$ bound that inner integral by $t^{-3/p}(|\Omega|^{-1} \int_\Omega \|\nabla u\|_p^p)^{1/p}$. Here $z + t(\Omega - z) \subset \Omega$ and the Jacobian is t^3. The integral of $t^{-3/p}$ is $p/(p-3)$, and $\operatorname{diam}_{p'}(\Omega) = 2R3^{1-1/p}$. This proves the asserted deterministic inequality for smooth u. Smooth approximation in $W^{1,p}$ on a cube extends it to the continuous representative: apply the same inequality to differences of smooth approximants

after mean centering to obtain uniform convergence, and identify the limit with their L^p limit. Finally, the density and risk bounds give $\int_\Omega \|\nabla e_\theta\|_p^p \, dz \le A_{p,N}/m$. Proposition S1 gives the momentum-fiber statement. For the anchored gauge, $w_\theta(z) - w_*(z) = e_\theta(z) - e_\theta(0)$; the triangle inequality gives the additional factor two.

Thus $p = 4$ is sufficient for this first-order energy estimate; the corresponding six-dimensional argument requires $p > 6$. The pth root in (S40) cannot be omitted for a pth-power loss. Ordinary mean-square training has $p = 2$ and does not itself meet $p > 3$. If an independent uniform force bound $\|g_\theta\|_\infty \le B_0$ is available, then for $p \ge 2$, $\mathbb{E}\|g_\theta\|_p^p \le B_0^{p-2}\mathbb{E}\|g_\theta\|_2^2$; this is an additional boundedness argument, not equivalence of Sobolev embeddings in different dimensions.

S3.3 A mean-square route with a quantitative force regularity prior

The following deterministic result can use any valid population-risk bound, including (S39) with $p = 2$. It avoids replacing mean-square training by an unexecuted fourth-power objective, at the cost of an independent force Lipschitz bound.

Theorem S3. A boundary-valid three-dimensional cone estimate. *Let $g : \Omega \to \mathbb{R}^3$ be continuous and let every component be L-Lipschitz with respect to Euclidean distance. Suppose ν has density at least $m > 0$ and $\mathbb{E}_\nu\|g\|_2^2 \le a$. For every $0 < h \le R$,*

$$\|g\|_{\infty,\Omega} \le Lh + \sqrt{\frac{6a}{m\pi h^3}}. \tag{S41}$$

Consequently define the sufficient uniform-force allowance

$$\varepsilon_3(a; L, m, R) = \inf_{0<h\le R} \left(Lh + \sqrt{\frac{6a}{m\pi h^3}} \right). \tag{S42}$$

When $a = 0$ the allowance is zero. If $a, L > 0$, its minimizing radius is

$$h = \min\left\{ R, \left(\frac{3}{2L}\sqrt{\frac{6a}{m\pi}} \right)^{2/5} \right\}.$$

In the interior regime the allowance is proportional to $a^{1/5}$.

Proof. Fix $z \in \Omega$ and a component j, and put $b = |g_j(z)|$. For each coordinate choose the direction toward the more distant face of the cube. The corresponding octant of the radius-h ball centered at z is contained in Ω for $h \le R$, including when z is on a face, edge, or corner. Its volume is $\pi h^3/6$. Throughout this octant $|g_j| \ge (b - Lh)_+$, so

$$a \ge m\frac{\pi h^3}{6}(b - Lh)_+^2.$$

If $b \le Lh$, (S41) already holds; otherwise solve the last inequality for b. Take the supremum in z, j and then the infimum in h. If $a = 0$, continuity and a positive density imply $g = 0$ everywhere. For $a, L > 0$, differentiation of the displayed one-variable objective gives the stated minimizer. If $L = 0$, the objective is minimized at $h = R$; a sharper bound in this constant-field case is $\|g\|_{\infty,\Omega} \le \sqrt{a}$.

For a d-dimensional cube the same proof uses the inward orthant volume $c_d h^d$, where $c_d = |B_1^d|/2^d$, and gives $\inf_h\{Lh + \sqrt{a/(mc_d h^d)}\}$. Its interior risk exponent is $1/(d+2)$: $1/5$ for three configuration variables and $1/8$ for six phase variables. This improves the interpolation exponent. It does not remove the need for iid configuration sampling, a density lower bound, class-wide bounds, or a small observed training loss.

If $g_\theta = \nabla(w_\theta - w_*)$ has the allowance ε, integration on line segments gives

$$\sup_\Omega |w_\theta - w_* - w_\theta(0) + w_*(0)| \le 3R\varepsilon. \tag{S43}$$

For the mean-zero energy gauge, the sufficient bound is instead $6R\varepsilon$. Both follow by integrating the componentwise gradient bound along a segment in the convex cube, and, for the latter, averaging the resulting difference between two points.

S3.4 Reception of the Hessian and the cubic and quartic jets

Small force error does not alone control the derivatives needed by the normal form. The next result lists the additional orders separately. For a multi-index α, $|\alpha|$ denotes its total degree, $\alpha! = \prod_i \alpha_i!$, and D^α denotes the corresponding ordinary real partial derivative.

Theorem S4. Low-order derivative reception. *Suppose $g = \nabla(w_\theta - w_*)$ and $\|g\|_{\infty,\Omega} \le \varepsilon$. For $k \in \{1,2,3\}$, assume $g \in C^{k+1}(\Omega)$ and that every component derivative of total order $k+1$ has absolute value at most B_{k+1} on Ω. For $|\alpha| = k$, every component satisfies*

$$|D^\alpha g_j(0)| \le J_k(\varepsilon, B_{k+1}, R) := \inf_{0<h\le R/k} \left(\frac{2^k \varepsilon}{h^k} + \frac{k B_{k+1} h}{2} \right). \tag{S44}$$

When $\varepsilon, B_{k+1} > 0$, take

$$h = \min\{R/k, (2^{k+1}\varepsilon/B_{k+1})^{1/(k+1)}\}.$$

In the interior regime the bound equals $(k+1)B_{k+1}^{k/(k+1)}\varepsilon^{1/(k+1)}$.

Let $c_{\theta,\beta}$ and $c_{*,\beta}$ be ordinary Taylor coefficients of the normalized potentials, so their definition includes division by $\beta!$. For $d = 2,3,4$ and $|\beta| = d$,

$$|c_{\theta,\beta} - c_{*,\beta}| \le \frac{J_{d-1}(\varepsilon, B_d, R)}{\beta!}. \tag{S45}$$

In particular, the normalized Hessian operator error is at most $3J_1$.

Proof. For a coordinate unit vector e_i define $\Delta_{h,i}u(z) = u(z + he_i) - u(z)$ and $\Delta_h^\alpha = \prod_i \Delta_{h,i}^{\alpha_i}$. Choose $i_1, \dots, i_k$ with α_i occurrences of coordinate i. Repeated use of the fundamental theorem of calculus gives

$$h^{-k}\Delta_h^\alpha g_j(0) = \int_{[0,1]^k} D^\alpha g_j \left(h \sum_{r=1}^k t_r e_{i_r} \right) dt.$$

All evaluation points and their segments from zero lie in Ω when $h \le R/k$. The derivative bounds imply that the integrand differs from $D^\alpha g_j(0)$ by at most $B_{k+1} h \sum_r t_r$. Its integral is at most $kB_{k+1}h/2$. On the other hand, the binomial coefficients in Δ_h^α have absolute sum 2^k, so $|\Delta_h^\alpha g_j(0)| \le 2^k\varepsilon$. Combining the two estimates proves (S44). Direct minimization gives the stated radius and interior value. Zero bounds follow by taking limits.

For $|\beta| = d$, choose j with $\beta_j > 0$ and set $\alpha = \beta - e_j$. Then $D^\beta(w_\theta - w_*) = D^\alpha g_j$. Divide by $\beta!$ to obtain (S45). Every Hessian entry is at most J_1; its maximum row sum and hence its symmetric operator norm are at most $3J_1$.

The three required priors are bounds on $D^3(w_\theta - w_*)$, $D^4(w_\theta - w_*)$, and $D^5(w_\theta - w_*)$, respectively. A fifth-derivative bound alone is not substituted for the third- and fourth-derivative bounds. A certified sum of the candidate and target bounds suffices for each order. The force Lipschitz constant in Theorem S3 instead concerns $D^2(w_\theta - w_*)$ and is a separate, lower-order prior. For physical derivatives the degree-d allowance is multiplied by $E_s\ell^{-d}$; for coefficients of $W(\ell z)$ it is multiplied only by E_s.

Here is an explicit spectral consequence. Suppose the physical target Hessian $K_* = D_q^2(U_0 + W_*)(0)$ is positive definite. Set

$$\zeta = 3E_s\ell^{-2}J_1, \qquad \upsilon = \|A^{-1}\|_2\zeta. \tag{S46}$$

Then $\zeta < \lambda_{\min}(K_*)$ implies $K_\theta > 0$. If $\omega_{*,1} \le \omega_{*,2} \le \omega_{*,3}$ are the positive target frequencies and $\upsilon < \omega_{*,1}^2$, the corresponding ordered candidate frequencies satisfy

$$|\omega_{\theta,j} - \omega_{*,j}| \le \frac{\upsilon}{\sqrt{\omega_{*,1}^2 - \upsilon} + \omega_{*,1}} =: \Delta_\omega. \tag{S47}$$

Indeed, the mass-normalized symmetric Hessian perturbation has norm at most υ. The variational eigenvalue bounds therefore give $|\omega_{\theta,j}^2 - \omega_{*,j}^2| \le \upsilon$; factor the difference of squares to obtain (S47). Thus a positive target divisor margin

$$\Delta_* := \min_{k\in\mathbb{Z}^3,\, 1\le|k|_1\le 4} |k\cdot\omega_*|$$

remains positive for the candidate if $4\Delta_\omega < \Delta_*$. This finite-order condition supports the degree-four normal form; it is not an all-integer Diophantine assertion.

The six quadratic, ten cubic, and fifteen quartic coefficient intervals in (S45), with their physical scales, can be passed to a spectrum and Birkhoff normal-form verifier. The verifier must include the cubic homological correction to the quartic term and prove a nonzero twist determinant uniformly over the admitted intervals. A center-point determinant is insufficient for the whole family. If these conditions and the separate analytic hypotheses of the local elliptic KAM criterion hold for a selected candidate with an equilibrium at the origin, that criterion applies to this same candidate. The learning estimates supply accuracy and jet uncertainty; they do not replace any of those KAM hypotheses.

S3.5 What the dimension reduction establishes and what it leaves open

The real estimates above do not establish a quantitative pole-free complex neighborhood. A realified split-tanh network is continued in the complexified physical coordinates, with independent continued real and imaginary channels. Evaluating a real-part operation at complex arguments is not that continuation. A quantitative outer-domain bound, target regularity when needed, and the same selected weights are required for analytic interpolation or an analytic KAM certificate. Existing strip-interpolation and local normal-form results may then be used with their own stated hypotheses.

Two examples separate the necessary assumptions. First, on the fixed cube take $u_a(z) = \exp(-|z|_2^2/a^2)$ and subtract its mean. These are real-analytic functions. Their force mean-square integral is at most $3\pi^{3/2}a/2^{3/2}$, while their mean-zero sup norm does not tend to zero: the mean is at most $\pi^{3/2}a^3/|\Omega|$ and $u_a(0) = 1$. Thus three dimensions do not make $W^{1,2}$ embed in C^0, even within real-analytic functions without uniform regularity bounds. Second,

$$v_k(z) = k^{-2}[1 - \cos(kz_1)]$$

has uniformly vanishing force, a force Lipschitz constant at most one, and $\partial_{11}v_k(0) = 1$ for every k. Its fourth derivative at zero has magnitude k^2. Therefore neither a small force nor the cone estimate alone gives small Hessian and quartic errors. The separately stated derivative priors or a higher-order jet receiver are substantive.

The present statements implement the configuration-potential part of the proposed learning-to-KAM route. They do not assert that an optimizer attains a small population risk. In particular, repeated velocities at one configuration, two finite-difference step diagnostics at the same configuration, or correlated trajectory windows do not provide iid configuration coverage with a positive three-dimensional density. A force-label cap received at one prepared point is not a population-risk estimate. Actual selected weights must have their own training record and all the required class, label, regularity, analytic, and normal-form bounds. The arguments apply to real and realified complex potentials alike and do not prove a complex-network advantage or a safety statement for a driven dissipative full-state system.

S4 Real regularity bounds for the current energy-feature network

This statement concerns the selected 918-real-parameter architecture, with two complex $4 \to 12 \to 12 \to 1$ branches. Only the energy branch enters the following conservative bounds. The checkpoint identity is `f51bcea12c2f2640c0ea33c3493a691d8ab240578018a6d44a9f68904a1a60a9`. The result does not reuse the earlier 736-parameter phase-space class.

Let $w_N(z) = W_N(\ell z)/E_s$, with $\ell = 1/1000$ and $E_s = 3/500000000000$ J. The exported feature-energy scale E_* retains its exact saved binary value, separately from E_s. For a scalar function f, let $b_k(f)$ be an upper bound for the absolute value of every ordered mixed partial derivative of order k on the cube $|z|_\infty \le R$.

Proposition S5. A derivative majorant for the actual graph. *Let x_k bound every mixed derivative of order k of a scalar preactivation. Define the exponential partial Bell polynomials by*

$$B_{0,0} = 1, \qquad B_{k,j}(x) = \sum_{i=1}^{k-j+1} \binom{k-1}{i-1} x_i B_{k-i,j-1}(x),$$

with the other zero-index cases zero. Put $t_1 = 1, t_2 = 2, t_3 = 4, t_4 = 16, t_5 = 16$. Then

$$b_k(\tanh \circ f) \le \sum_{j=1}^{k} t_j B_{k,j}(x), \qquad 1 \le k \le 5.$$

For an affine layer, the corresponding derivative bounds are the absolute-weighted sums of its input bounds. These two operations, applied to the realification of the complex layers and to the quadratic energy feature, bound $D^2 w_N, D^3 w_N, D^4 w_N, D^5 w_N$ on the entire cube. The subtraction of the three first-feature derivatives at the origin changes only the first derivative and the constant term.

Proof. Write $t = \tanh x$, so $|t| \le 1$ for real x. The first five derivative polynomials factor as

$$1 - t^2, \quad -2t(1 - t^2), \quad -2(1 - t^2)(1 - 3t^2), \quad 8t(1 - t^2)(2 - 3t^2), \quad 8(1 - t^2)(2 - 15t^2 + 15t^4).$$

The stated bounds follow from $0 \le t^2 \le 1$; for the last expression, $2 - 15u + 15u^2 \in [-7/4, 2]$ when $u \in [0, 1]$. The multivariate chain rule is the sum over partitions of the k ordered differentiation positions. Grouping partitions by their block sizes gives the displayed Bell recurrence. Absolute values and the triangle inequality give the affine operation.

For $e(z) = \ell^2 z^{\mathsf{T}} K_3 z/2$, the feature bounds used are

$$b_1(e) \le R\ell^2 \max_i \sum_j |(K_3)_{ij}|, \qquad b_2(e) \le \ell^2 \max_{i,j} |(K_3)_{ij}|, \qquad b_k(e) = 0 \ (k > 2).$$

The first three features have first derivative at most ℓ/Q_j and no higher derivatives. Divide the fourth feature by the exact E_*, propagate through the two realified hidden layers and the scalar readout, multiply by E_*/E_s, and add $|c_{\mathrm{E}}| b_k(e)/E_s$. This is precisely the graph used for the selected potential. Its constant and linear subtractions do not affect orders two through five. All reported arithmetic uses rational numbers. Absolute stiffness entries are rounded upward to dyadics before the recurrence, and final bounds are rounded upward once more. Every operation in the majorant is monotone in these nonnegative inputs, so both roundings preserve the inequalities.

The resulting conservative upward decimal bounds for the same selected weights are

R	$b_2(w_N)$	$b_3(w_N)$	$b_4(w_N)$	$b_5(w_N)$
$1/2$	0.324211	0.0108748	0.0376192	0.154884
$1/64$	0.322470	0.00368385	0.0136982	0.0339880

These are bounds on normalized derivatives, not ordinary Taylor coefficients or measured errors. For an independently justified target bound $b_k(w_*)$, the sum $b_k(w_N) + b_k(w_*)$ bounds the error derivative of order k. Thus the candidate side of the separate third-, fourth-, and fifth-derivative assumptions in a force-to-jet reception theorem can be received without substituting one derivative order for another. No target bound follows from the candidate graph.

Two containing classes have also been bounded by the same recurrence: first, frozen hidden layers with realified readout ℓ^1 norm at most one and $|c_{\mathrm{E}}| \le 1$; second, the complete current graph with every trainable real parameter component at most two in absolute value and $|c_{\mathrm{E}}| \le 1$. The selected weights belong to both classes. These containing classes were defined after the preceding training; they do not retroactively provide a same-data selection event. Fresh independent validation can instead treat the selected output as fixed. Neither class construction supplies the unknown target's regularity, the sampling density, a population risk bound, a pole-free complex neighborhood for every member, or physical-target KAM.

S5 Finite-trajectory conservative-force reception

S5.1 Removing finite-window bias without assuming exact observations

The preceding finite label is an average along a trajectory. It is not an instantaneous force measurement. This distinction persists even when the momentum increment and the magnetic integral are computed exactly. Let $s_\sigma(t)$ be the dimensionless projected sum of the unknown conservative force and dissipative force on the physical trajectory with sign label $\sigma \in \{+,-\}$. More precisely, for the full native model let

$$s_\sigma(t) = \frac{\ell}{E_s} V^T \{\nabla(E_{\mathrm{el},0} + \mathcal{W}_*)(x_\sigma(t)) + D_*(x_\sigma(t))v_\sigma(t)\}.$$

Here $E_{\mathrm{el},0}(x) = x^T K_0 x/2$ is the nominal elastic potential and $\mathcal{W}_*$ is its unknown full-coordinate correction; its reduced restriction is $W_*(q) = \mathcal{W}_*(Vq)$. The symbol E_s remains the constant energy scale. Put

$$\begin{aligned} P_\sigma &= (\ell/E_s)V^T M v_\sigma, \\ m_\sigma &= (\ell/E_s)V^T b\{\kappa \sin(b^T x_\sigma) - \mu u_\sigma \cos(b^T x_\sigma)\}, \\ f_* &= (\ell/E_s)V^T \nabla \mathcal{W}_*(x_{\mathrm{prep}}). \end{aligned}$$

The sign convention is $\dot{P}_\sigma + s_\sigma + m_\sigma = 0$; the input u_σ is the applied transverse magnetic field. Thus an exact observed-momentum expression is

$$Y(T) = -\frac{1}{2}\sum_{\sigma=+,-}\left\{\frac{P_\sigma(T) - P_\sigma(0)}{T} + \frac{1}{T}\int_0^T m_\sigma(t)\,dt\right\} - f_0.$$

The magnetic quantities B_m, ϵ_m below refer to this scaled vector m_σ, and ϵ_p refers to P_σ. The prepared full-coordinate position may include saved rounding; the additional transfer to an ideal reduced design node is stated below. The known magnetic force is accounted for separately. Define $S = (s_+ + s_-)/2$ and $f_0 = \ell V^T K_0 x_{\mathrm{prep}}/E_s$. With exact momentum increments and exact magnetic integrals, the paired label is

$$Y(T) = \frac{1}{T}\int_0^T S(t)\,dt - f_0.$$

This identity uses the physical equations. Exact initial-position and opposite-velocity symmetry would remove the initial dissipative term; the following result instead retains an explicit initial allowance.

Proposition S6. A finite-window force receiver. *Suppose $S \in C^3([0,4T])$, $T > 0$, and $\sup_{[0,4T]} \|S'''\|_\infty \le B_3$. Put*

$$\mathcal{R}_T Y = \tfrac{8}{3}Y(T) - 2Y(2T) + \tfrac{1}{3}Y(4T).$$

If f_* is the desired conservative correction force at the prepared configuration and $\|S(0) - f_0 - f_*\|_\infty \le I_0$, then

$$\|\mathcal{R}_T Y - f_*\|_\infty \le I_0 + \tfrac{1}{3}B_3 T^3. \tag{FW1}$$

For a receiver based on saved endpoint momenta and composite trapezoids, assume additionally that: each scaled projected endpoint momentum has error at most ϵ_p; each sampled projected magnetic force has error at most ϵ_m; and the exact projected magnetic force on each signed trajectory has second derivative bounded by B_m. Use exact times, with endpoints $T, 2T, 4T$ and maximum panel width h. Then the observed receiver satisfies

$$\|\widehat{\mathcal{R}_T Y} - f_*\|_\infty \le I_0 + \frac{B_3 T^3}{3} + \frac{5B_m h^2}{12} + 5\epsilon_m + \frac{11\epsilon_p}{2T}. \tag{FW2}$$

All bounds concern the same physical model, signed trajectories and normalization. None of $I_0, B_3, B_m, \epsilon_p, \epsilon_m$ is inferred solely from a small observed weak residual or a coarse–fine difference.

Proof. The three weights sum to one. Their first and second moments at $(T, 2T, 4T)$ vanish. Thus constants are preserved and the linear and quadratic temporal terms are removed. Taylor's formula with integral remainder gives

$$\mathcal{R}_T Y - (S(0) - f_0) = \int_0^{4T} K_T(u) S'''(u)\, du,$$

where, writing $v = u/T$, $K_T(u) = T^2 k(v)$ and

$$k(v) = \begin{cases} v^2(12 - 7v)/24, & 0 \le v \le 1, \\ \{(4 - v)^3 - 12(2 - v)^3\}/72, & 1 \le v \le 2, \\ (4 - v)^3/72, & 2 \le v \le 4. \end{cases}$$

This kernel is nonnegative: on $[1, 2]$ use $4 - v \ge 3(2 - v) \ge 0$; the other pieces have the stated sign directly. Exact integration gives $\int_0^4 k(v)\, dv = 1/3$. This proves (FW1), including the initial allowance.

The absolute sum of the averaging weights is 5. A composite trapezoidal average has error at most $B_m h^2/12$, and the sampled-force error contributes at most ϵ_m. Averaging the two signs retains these common bounds. For either sign, the momentum combination has endpoint coefficients, in the order $0, T, 2T, 4T$,

$$\frac{1}{T}\left(-\tfrac{7}{4},\ \tfrac{8}{3},\ -1,\ \tfrac{1}{12}\right).$$

Their absolute sum is $11/(2T)$. These facts give (FW2).

If the two signs have different exact saved clocks, each receiver must be formed on its own clock. Exact Lagrange weights are obtained from the three actual durations; the corresponding kernel and observation constants must then be received for those durations. Changing the negative-velocity label does not change the direction of time.

Interpretation of the saved-data diagnostics. The predeclared diagnostic evaluated all windows of $1, 2, 4, 8, 12, 16, 24, 32, 64, 128, 256$ panels, all specified first-order pairs, and all specified second-order triples. No model was refitted and no best window was selected. At direction 3 on the finer record, the old 12-panel label differs from the declared material force by about 2.15835×10^{-4} in its largest component. The decomposition contains an elastic time-average drift of about 2.16072×10^{-4}, an even dissipative contribution of about 3.32821×10^{-7}, and a negative averaged weak-residual contribution of magnitude about 5.69840×10^{-7}. These are signed vector terms; their maxima must not be added as if their signs and maximizing coordinates agreed.

The fixed $1, 2, 4$-panel formula reduces the observed discrepancy from the declared material force to about 2.32485×10^{-8} at direction 2 and 1.76758×10^{-7} at direction 3 on the finer records. These are evaluator diagnostics. In contrast, the first-order $12, 24$-panel formula makes the direction-3 discrepancy about 4.28863×10^{-4}, illustrating why a temporal-bias model matters. The old 12-panel momentum slope has endpoint-noise factor $1/(6T)$; the new formula has factor $11/(2T)$, which is 33 times larger. Reduced bias therefore comes with increased sensitivity to endpoint errors. The saved-data comparison supplies neither the continuous observation allowances nor the derivative bounds in (FW2).

The refined receiver uses the recorded observations and known native operators. Constitutive evaluator coefficients enter only the subsequent comparison. This calculation does not identify a general unknown quartic, satisfy iid sampling assumptions, or establish a KAM transfer. Its role is to expose and remove a specific finite-window bias before further configuration measurements are interpreted.

A fourth window and the regularity required by the jet receiver. The C^5 potential regularity used in the fourth-order jet receiver below also motivates one further temporal cancellation, provided the dissipative law and input have the corresponding regularity along the trajectories. For $S \in C^m([0, a_m T])$ define $A(t) = \int_0^1 S(ut)\, du$, including $A(0) = S(0)$. Then $\|A^{(m)}\|_\infty \le \mathcal{B}_m/(m + 1)$ when $\|S^{(m)}\|_\infty \le \mathcal{B}_m$. Polynomial interpolation at the positive nodes $a_1 T, \dots, a_m T$, evaluated at zero, consequently has error at most

$$\frac{\mathcal{B}_m T^m \prod_i a_i}{(m + 1)!}.$$

Indeed the scalar interpolation remainder applies on the convex hull of zero and all interpolation nodes, component by component. No statistical independence or measured derivative is used in this formula.

For $(a_1, a_2, a_3, a_4) = (1, 2, 4, 8)$ the weights are

$$(64/21, -8/3, 2/3, -1/21),$$

and the continuous temporal allowance is $8B_4T^4/15$. Under the endpoint, magnetic and initial allowances of Proposition S6, the four-window version obeys

$$\|\widehat{\mathcal{R}_T^{(4)}Y} - f_*\|_\infty \le I_0 + \frac{8B_4T^4}{15} + \frac{15B_mh^2}{28} + \frac{45\epsilon_m}{7} + \frac{45\epsilon_p}{7T}.$$

Here B_4 bounds the fourth time derivative of the paired nonmagnetic force, while B_m retains its meaning as a magnetic second-derivative bound. The magnetic weight norm is $45/7$. The combined momentum coefficients at $0, T, 2T, 4T, 8T$ are

$$T^{-1}(-15/8, 64/21, -4/3, 1/6, -1/168),$$

whose absolute sum is also $45/(7T)$. Relative to the three-window formula, endpoint-noise amplification increases by the factor $90/77$. This modest additional cost does not by itself guarantee a better result: the fourth temporal derivative and all observation allowances must still be justified. No observed temporal fit supplies those bounds.

The predeclared four-window diagnostic did not improve the fixed three-window result on either new configuration at either time step. For example, at direction 3 on the fine record its maximum declared-family discrepancy is approximately 1.77537×10^{-7}, compared with 1.76758×10^{-7} for three windows in the same floating diagnostic. These are evaluations of saved finite labels, not continuous-error bounds. The interval reception in Proposition S7 uses its own exact native arithmetic. The negative comparison motivates a separate time-step diagnostic rather than further increasing the extrapolation order.

Proposition S7. Certified arithmetic for a fixed finite-window force label. *Use the same exact native full matrices M, K_0, V, tip vector b, and $d = V^Tb$. For each signed observation record $\sigma \in \{+, -\}$, let $\tau_{\sigma,i}$ be its own saved clock and set*

$$p_{\sigma,i} = V^TMv_{\sigma,i}, \qquad m_{\sigma,i} = \kappa \sin(b^Tx_{\sigma,i}) - \mu B_{y,\sigma,i} \cos(b^Tx_{\sigma,i}).$$

The vectors $x_{\sigma,i}, v_{\sigma,i}$ are the observed full 64-coordinate vectors, assembled directly from the recorded w, θ, w_t, θ_t fields. At the fixed panel counts $n \in \{1, 2, 4\}$, define $T_{\sigma,n} = \tau_{\sigma,n} - \tau_{\sigma,0}$ and

$$A_{\sigma,n} = \frac{p_{\sigma,n} - p_{\sigma,0}}{T_{\sigma,n}} + \frac{d}{T_{\sigma,n}} \sum_{i=0}^{n-1} \frac{\tau_{\sigma,i+1} - \tau_{\sigma,i}}{2}(m_{\sigma,i} + m_{\sigma,i+1}).$$

All twelve saved clocks start exactly at zero and satisfy, as exact rational identities, $T_{\sigma,2} = 2T_{\sigma,1}$ and $T_{\sigma,4} = 4T_{\sigma,1}$. Their actual-clock Lagrange weights at zero are therefore $(8/3, -2, 1/3)$; their sum is one, their first two time moments vanish, and their absolute sum is five.

For each of the six prescribed signed pairs, let x_g^{prep} be that configuration's own prepared rounded full-coordinate position. With the fixed normalization $S = (10^{-3})/(6\,10^{-12}) = 500000000/3$, the stored-data label is

$$\ell_g = -\frac{S}{2} \sum_{\sigma\in\{+,-\}} \left(\frac{8}{3}A_{\sigma,1} - 2A_{\sigma,2} + \frac{1}{3}A_{\sigma,4}\right) - SV^TK_0x_g^{\mathrm{prep}}.$$

Exact rational native matrix arithmetic and Arb arithmetic at 256 bits enclose all six labels, using every signed trace's actual saved times in both the slope and trapezoid. No clock, prepared position, or sign is substituted from another configuration.

Only after the completed label process and its output hash are recorded, the evaluator reads the declared stiffness coefficient. The saved readbacks give $a_E = E_{\mathrm{mod}}/(5\,10^6) = 53/50$ in these cases. Against the declared initial formula

$$f_g^{\mathrm{decl}} = (a_E - 1)SV^TK_0x_g^{\mathrm{prep}},$$

the normalized maximum component errors obey the strict bounds

configuration	*coarse diagnostic*	*fine diagnostic*
original shifted	$9.469\,10^{-8}$	$2.239\,10^{-8}$
direction 2	$9.842\,10^{-8}$	$2.325\,10^{-8}$
direction 3	$7.478\,10^{-7}$	$1.768\,10^{-7}$

Each table entry is an exact rational upper bound on $\|\ell_g - f_g^{\mathrm{decl}}\|_\infty$. All six pairs use the same previously fixed panel counts; no pair or window is selected by error.

Proof. The receiver binds all twelve observation files, the prepared full-coordinate positions, and the native geometry by source hashes. It verifies the saved time and node identities and the native full field/DOF mapping. Binary64 observations and prepared positions are interpreted as exact saved numbers; the native mass is the exact sum $M_0 + M_r$. For every signed clock, the rational Lagrange calculation uses the three actual durations and verifies the constant, linear, and quadratic moment identities before computing the interval label. The label subprocess completes with exit code zero and writes an immutable hashed result before the evaluator subprocess is allowed to parse E_{mod}. The second process evaluates the declared formula and certifies the six displayed rational relaxations.

This result certifies arithmetic on stored finite observations and a subsequent comparison with a declared initial constitutive formula. The vanishing time moments are algebraic identities; no bound on a continuous-time extrapolation remainder, quadrature defect, native solver error, observation error, or unknown physical force follows from them. The records remain development and paired discretization diagnostics. No existing label, dataset purpose, or data split is changed; no training archive is exported, and no new training or native solve is performed.

The initial allowance and its recorded-state part. Use all 64 free transverse displacement/rotation coordinates in their fixed native ordering, and write $E_{\mathrm{el},0}(x) = x^T K_0 x/2$, $\lambda_F = \ell/E_s$ and $d = V^T b$. For the physical signed trajectories define the scaled full-state quantities

$$\begin{aligned} P_\sigma(t) &= \lambda_F V^T M v_\sigma(t), \\ m_\sigma(t) &= \lambda_F d\{\kappa \sin(b^T x_\sigma(t)) - \mu B_{y,\sigma}(t)\cos(b^T x_\sigma(t))\}, \\ g(x) &= \lambda_F V^T \nabla(E_{\mathrm{el},0} + \mathcal{W}_*)(x), \qquad R(x) = \lambda_F V^T D_*(x), \\ s_\sigma(t) &= g(x_\sigma(t)) + R(x_\sigma(t))v_\sigma(t). \end{aligned}$$

The full equation gives $\dot P_\sigma = -s_\sigma - m_\sigma$. Thus the momentum and magnetic terms use precisely the normalization of Proposition S6. At the prepared rounded position x_{prep}, put

$$f_0 = \lambda_F V^T K_0 x_{\mathrm{prep}}, \qquad f_* = g(x_{\mathrm{prep}}) - f_0.$$

Let $\widehat{x}_\sigma, \widehat{v}_\sigma$ denote the actual saved binary64 first-row full-coordinate values, rather than the intended initial conditions. Define

$$\widehat{\delta}_0 = \tfrac{1}{2}\sum_{\sigma=\pm}\{g(\widehat{x}_\sigma) + R(\widehat{x}_\sigma)\widehat{v}_\sigma\} - g(x_{\mathrm{prep}}), \qquad I_0^{\mathrm{rec}} = \|\widehat{\delta}_0\|_\infty.$$

This quantity alone does not bound the physical initial allowance. For example, suppose the physical initial states lie within $\delta_{x,\sigma}, \delta_{v,\sigma}$ of their saved values in the fixed coordinate norm. If g, R have Lipschitz bounds L_g, L_R on the relevant position neighborhoods and $\|R\|_\infty \le \overline{R}$ there, then a sufficient allowance is

$$I_0 = I_0^{\mathrm{rec}} + U_0, \qquad U_0 = \tfrac{1}{2}\sum_{\sigma=\pm}\{L_g\delta_{x,\sigma} + L_R\delta_{x,\sigma}\|\widehat{v}_\sigma\|_\infty + \overline{R}\delta_{v,\sigma}\}.$$

Indeed, insert and subtract the recorded-state forces and use $R(x)v - R(\widehat{x})\widehat{v} = R(x)(v-\widehat{v}) + (R(x) - R(\widehat{x}))\widehat{v}$. This proves $\|S(0) - f_0 - f_*\|_\infty \le I_0$ without imposing exact position or opposite-velocity symmetry.

Exact reception for the declared linear family. For the saved evaluator family,

$$\nabla(E_{\mathrm{el},0}+\mathcal{W}_*)(x)=a_E K_0 x, \qquad D_*=a_C D_0, \qquad D_0=\alpha_0 M_0,$$

and both damping matrices are constant. The existing generators fix the mesh and linear geometry, use the mass-proportional coefficient `alpha_mass`, and set the stiffness-proportional coefficient to zero. The rotary patch changes inertia only. The frozen native matrix construction independently defines $D_0=\alpha_0 M_0$ before runtime rounding. Here the readbacks are $a_E=53/50$ and $a_C=5754906970707699/5004266931050173$. Put $G=\lambda_F a_E V^T K_0$ and $C=\lambda_F a_C V^T D_0$. The recorded defect is the exact rational vector

$$\widehat{\delta}_0=G\left(\frac{\widehat{x}_+ +\widehat{x}_-}{2}-x_{\mathrm{prep}}\right)+C\frac{\widehat{v}_+ +\widehat{v}_-}{2}.$$

If componentwise physical initial-error radii $\rho_{x,\sigma},\rho_{v,\sigma}\in\mathbb{R}^{64}_{\geq 0}$ are supplied, a sharper directly computable sufficient term is

$$U_0=\left\|\frac{|G|}{2}(\rho_{x,+}+\rho_{x,-})+\frac{|C|}{2}(\rho_{v,+}+\rho_{v,-})\right\|_\infty .$$

These radii have not been certified for the current physical observations.

Exact native rational arithmetic on all twelve saved numerical initial states gives the following strict rational upper bounds. Every sign and time-step record is received separately, with no projection or replacement by x_{prep}.

	I_0^{rec}		δ_{prep}
configuration	coarse	fine	
original shifted	$9.871\,10^{-15}$	$2.861\,10^{-15}$	$1.606\,10^{-18}$
direction 2	$2.782\,10^{-14}$	$1.390\,10^{-14}$	$1.907\,10^{-18}$
direction 3	$1.125\,10^{-13}$	$5.730\,10^{-14}$	$3.931\,10^{-18}$

The last column is a distinct preparation-transfer allowance. For each configuration, the frozen exact modal preparation $q=\ell z$ gives $x_{\mathrm{ideal}}=V\ell z$, whereas the stored label uses its own rounded x_{prep}. For the declared correction force,

$$\delta_{\mathrm{prep}}=\|\lambda_F V^T(a_E-1)K_0(x_{\mathrm{prep}}-V\ell z)\|_\infty .$$

Both time-step records share this preparation allowance. Adding it transfers the force-error budget from the rounded configuration to the ideal jet point; a point representation of an interval-certified label also requires its certified arithmetic radius. Arb comparison shows that each present fixed three-window discrepancy is strictly greater than its corresponding I_0^{rec} upper bound. Consequently the recorded initial pairing defect cannot by itself explain those discrepancies. This comparison does not bound U_0 or assign the remaining discrepancy to a particular source.

Endpoint momentum accuracy required by the force budget. For the FW2 majorant of Proposition S6 to fit a force tolerance η, its nonnegative allowances must satisfy

$$\epsilon_p\leq\frac{2T}{11}\left(\eta-I_0^{\mathrm{rec}}-U_0-\frac{B_3T^3}{3}-\frac{5B_m h^2}{12}-5\epsilon_m\right),$$

with a nonnegative bracket. This is a requirement on that certified error budget, not an assertion that cancellations in the actual error are impossible. For every current record the exact first saved duration and coefficient are

$$T=\frac{5902958103587057}{75557863725914323419136}\ \mathrm{s},$$
$$\frac{2T}{11}=\frac{5902958103587057}{41556825049252877805248}\ \mathrm{s}\simeq 1.420454545\,10^{-8}\ \mathrm{s}.$$

After subtracting the recorded I_0^{rec} alone, a force tolerance 10^{-7} therefore leaves an optimistic scaled endpoint-momentum ceiling of approximately $1.42045\,10^{-15}$; the uninstantiated terms can only reduce it. No continuous solver or observation accuracy at that scale has been certified by this calculation.

Derivative formulas without an asserted derivative bound. For the declared full equation set $A = a_E K_0$, $D = a_C D_0$, $\theta = b^T x$, $q_\theta = b^T v$, $r_\theta = b^T a$, and $m = \kappa \sin\theta - \mu B_y \cos\theta$. Then

$$
\begin{aligned}
a &= -M^{-1}(Ax + Dv + bm),\\
j &= -M^{-1}(Av + Da + b\dot{m}), \qquad k = -M^{-1}(Aa + Dj + b\ddot{m}),\\
\dot{m} &= (\kappa\cos\theta + \mu B_y \sin\theta) q_\theta - \mu \dot{B}_y \cos\theta,\\
\ddot{m} &= (\kappa\cos\theta + \mu B_y \sin\theta) r_\theta + (-\kappa\sin\theta + \mu B_y \cos\theta) q_\theta^2\\
&\quad + 2\mu \dot{B}_y \sin\theta\, q_\theta - \mu \ddot{B}_y \cos\theta,\\
s_\sigma''' &= \lambda_F V^T (A j_\sigma + D k_\sigma).
\end{aligned}
$$

Here $a = x''$, $j = x'''$, and $k = x''''$. A bound on these formulas over certified trajectory neighborhoods would supply B_3 and, from $\lambda_F d\ddot{m}$, the magnetic second-derivative allowance. No such neighborhood bounds are supplied by the saved initial-state arithmetic or by the coarse–fine records.

S5.2 A fixed-window time-refinement diagnostic

To test the time-discretization sensitivity of the stored-data receiver in S7, two additional native COMSOL solves were declared before execution. They retain the third preparation $q = (0, 0, 1/2000)\,\mathrm{m}$ and the prescribed velocities $\dot{q} = \pm 100 q$. The source model, material parameters ($E = 5.3 \times 10^6\,\mathrm{Pa}$ and mass-damping coefficient $0.6389230223455499\,\mathrm{s}^{-1}$), drive, spatial mesh, initial-field expressions, generalized-α settings and tolerances were retained. Only the internal step was reduced from 3.90625×10^{-8} to $1.953125 \times 10^{-8}\,\mathrm{s}$. Both signs retain the same 257 requested output times over $20\,\mu\mathrm{s}$, with output spacing $7.8125 \times 10^{-8}\,\mathrm{s}$.

For every sign and internal step, the fixed receiver uses output indices $(1, 2, 4)$. Its own recorded increasing clock is interpreted exactly; the three actual durations satisfy the dyadic relations and yield the predeclared weights $(8/3, -2, 1/3)$. No window, weight or model parameter was selected from these results. An independent 256-bit Arb receiver reconstructed the projected momentum and magnetic quadrature from the saved displacement, rotation, velocity and drive observations. Material coefficients enter only the subsequent evaluator comparison.

Let $\widehat{f}_h$ denote the paired conservative correction label, and let $f_* = (\ell/E_s)(a_E - 1)K_{03} q$ be the correction in the declared material family, with $a_E = 53/50$. This diagnostic uses the ideal full-coordinate reference $x_{\mathrm{prep}} = Vq$; the preceding fixed stored receiver uses the prescribed rounded full-coordinate position. All three time steps are re-received under the same ideal convention and therefore are not bitwise the earlier rounded-reference quantity. An exact rational calculation bounds the corresponding change in the normalized total reference force by 6.944×10^{-17}. The finite-data arithmetic enclosures give the following upper bounds, rounded upward:

internal step h (s)	$\|\widehat{f}_h - f_*\|_\infty$
7.8125×10^{-8}	7.477611×10^{-7}
3.90625×10^{-8}	1.767437×10^{-7}
1.953125×10^{-8}	3.398021×10^{-8}

The maximum component of the difference between consecutive labels decreases from approximately 5.710174×10^{-7} to 1.427635×10^{-7}, a ratio of approximately 4.000. The finest discrepancy is approximately 5.20 times smaller than that at the previous fine step. Thus the previous observed discrepancy is not a demonstrated numerical floor.

The prescribed initial fields are fixed, but their realized initial states are retained rather than assumed exact. At the three steps, the signed-pair position differences are approximately 1.5625×10^{-14}, 7.8125×10^{-15} and 3.9062×10^{-15} in the native coordinate convention; the velocity antisymmetry defects are approximately 8.1400×10^{-8}, 4.0700×10^{-8} and 2.0350×10^{-8}. These changes can contribute to the observed discrete sensitivity. Both new native solves and the arithmetic receiver terminated with actual exit code zero; a separate 512-bit direct endpoint calculation encloses its results inside the

256-bit reception intervals. This experiment supplies a three-step diagnostic for one declared material and one preparation. It does not instantiate the continuous bounds in S6, separate all error sources, identify unknown constitutive physics, or certify a KAM transfer. No training or checkpoint update was performed.

S5.3 A posteriori enclosures for the recorded-state native IVPs

The preceding time-step comparisons do not certify the error of a continuous trajectory. To supply such a bound for a specified mathematical model, consider

$$M\ddot{x} + D\dot{x} + Kx + b\,g(b^T x, t) = 0, \qquad g(\theta, t) = \kappa \sin\theta - \mu B(t)\cos\theta,$$

where $M = M_0 + M_r$, $K = (53/50)K_0$, $D = \alpha_* M_0$ and $b = e_{64}$. The mass and stiffness entries are interpreted as the exact rational values of the saved finite-element matrices. The drive is the exact decimal law $B(t) = 10^{-4}\cos(11.41692154656479\,t)$. The coefficient α_* is the saved binary value of the material readback. These conventions specify a mathematical IVP; they do not identify the solver's compiled transcendental evaluations with exact arithmetic.

For each sign, let Q be the piecewise cubic Hermite curve through the saved full-coordinate positions and velocities, with $\dot{Q}$ its derivative. The comparison solution starts at that sign's own saved numerical state. Define

$$\begin{aligned} R_Q &= M\ddot{Q} + D\dot{Q} + KQ + b\,g(b^T Q, t), \\ r^2 &= (x-Q)^T K (x-Q) + (\dot{x}-\dot{Q})^T M (\dot{x}-\dot{Q}), \\ c_M^2 &= b^T M^{-1} b, \qquad c_K^2 = b^T K^{-1} b. \end{aligned}$$

Lemma S8. Residual-to-state enclosure. *Suppose M, K are symmetric positive definite, D is symmetric positive semidefinite, and $|B(t)| \le A$. On a time cell, assume $\sqrt{R_Q^T M^{-1} R_Q} \le \bar{\rho}$. Put $L_g = \kappa + \mu A$ and $a = L_g c_M c_K/2$. If $r(t_i) \le r_i$, then throughout that cell*

$$r(t_i + \tau) \le e^{a\tau} r_i + \bar{\rho}\frac{e^{a\tau} - 1}{a}, \qquad 0 \le \tau \le h_i. \tag{CT1}$$

The initial value is $r_0 = 0$ for the recorded-state IVP specified above.

Proof. Write $e = x - Q$. The error equation gives, almost everywhere,

$$r\dot{r} = -\dot{e}^T D\dot{e} - (b^T\dot{e})\{g(b^T x, t) - g(b^T Q, t)\} - \dot{e}^T R_Q.$$

The global force bound $|g(\theta, t) - g(\eta, t)| \le L_g|\theta - \eta|$ and the two energy-dual inequalities yield

$$\begin{aligned} |b^T e| \le c_K\|e\|_K, \qquad |b^T\dot{e}| \le c_M\|\dot{e}\|_M, \\ \|e\|_K\|\dot{e}\|_M \le r^2/2, \qquad |\dot{e}^T R_Q| \le r\bar{\rho}. \end{aligned}$$

Thus $\dot{r} \le ar + \bar{\rho}$. A regularized norm establishes the comparison also at $r = 0$. Since Q and $\dot{Q}$ are continuous across the saved nodes, integrating this inequality gives (CT1) without a jump contribution at cell boundaries.

Whole-cell residual reception. Use the degree-nine rational polynomial obtained by replacing the magnetic term in R_Q by

$$b\left\{\kappa(\theta - \theta^3/6) - \mu A(1 - \omega_B^2 t^2/2)(1 - \theta^2/2)\right\}, \qquad \theta = b^T Q.$$

Its Bernstein coefficient vectors β_k on each cell give the uniform mass-dual bound $\max_k \sqrt{\beta_k^T M^{-1}\beta_k}$, by convexity of that norm. If $|\theta| \le \Theta$ and $0 \le \omega_B t \le 1$, a valid additional mass-dual remainder is

$$c_M\left\{\frac{\kappa\Theta^5}{120} + \frac{\mu A}{24}\left(\Theta^4 + (\omega_B t_{i+1})^4\right)\right\}. \tag{CT2}$$

Indeed, the sine remainder is at most $\Theta^5/120$; the product of the two cosine approximations has error at most $(\Theta^4 + (\omega_B t)^4)/24$, since $|1 - \omega_B^2 t^2/2| \le 1$. All polynomial operations and matrix identities are rational. Interval square roots and exponentials supply the outward bounds in (CT1).

For the two half-fine direction-3 records, this reception verifies all eight cells of the fixed four-panel prefix. Exact positive elimination pivots establish definiteness of M, K, M_0. The final radii are bounded by 2.589070×10^{-13} and 2.688128×10^{-13} in the norm of Lemma S8. The growth constant satisfies $a < 479.200675\,\mathrm{s}^{-1}$. These bounds hold for the continuous mathematical IVPs, throughout each cell, rather than only at sampled residual values.

Limits of a generic endpoint estimate. At a saved node the scaled momentum error is at most

$$C_P r(t_i), \qquad C_P = (\ell/E_s) \max_j \sqrt{(V^T M V)_{jj}}.$$

Substitution into the fixed receiver gives a paired momentum allowance less than 0.035010, which exceeds the intended 10^{-7} force tolerance. This is a valid but insufficient upper bound, rather than a lower bound on the actual data error. A direct projected-force reception below retains the spatial and temporal cancellations discarded by this generic estimate.

The certificate concerns solutions starting at the saved numerical states. It supplies neither an error bound for physical measurements, nor a verification of the compiled COMSOL equation or its dense output. A physical initial-state discrepancy requires an additional nonzero initial allowance in (CT1). No learned potential jet or physical KAM transfer follows from the trajectory enclosure alone.

S6 A continuous reference bound from energy contraction

The reference IVP in Supplementary Section S5.3 can be bounded over the full interval $0 \le t \le T_f = 3/5$ without an exponential factor containing the largest elastic frequency. Retain its exact saved source operators, formed in the rational domain before further rounding, and write

$$\begin{aligned} M\ddot{x} + D\dot{x} + Kx + \kappa b \sin\theta = b f(t)\cos\theta, \qquad \theta = b^T x, \\ K = \tfrac{53}{50} K_0, \qquad D = \alpha_* M_0, \qquad b = e_{64}, \\ f(t) = f_a \cos(\nu t), \qquad f_a = 10^{-4}\mu, \qquad \nu = \frac{1141692154656479}{10^{14}}, \\ \alpha_* = \frac{5754906970707699}{9007199254740992}. \end{aligned} \tag{S48}$$

The values of μ and κ are the same exact source constants as in the previous IVP certificates. The frequency is the exact decimal drive coefficient. The material readback α_* is interpreted as its exact binary rational, rather than the nominal damping coefficient. These conventions specify a mathematical source IVP; a comparison with a compiled simulation or physical measurement requires its own equation and observation error interface.

S6.1 A full-time angle bound with the nonlinear terms retained

Set $c_K^2 = b^T K^{-1} b$ and define

$$\mathcal{E}(x, v) = \tfrac{1}{2} v^T M v + \tfrac{1}{2} x^T K x + \kappa(1 - \cos b^T x).$$

Since $\kappa > 0$, one has $|\theta| \le \sqrt{2} c_K \sqrt{\mathcal{E}}$ and

$$\frac{d\mathcal{E}}{dt} = -v^T D v + f(t) \frac{d}{dt} \sin\theta. \tag{S49}$$

Let $\mathrm{TV}\, f$ denote total variation on $[0, T_f]$, and put

$$\begin{aligned} B_E &= \mathcal{E}(x_0, v_0) + |f(0)\sin\theta_0|, \\ a_E &= \sqrt{2} c_K \left(\|f\|_\infty + \mathrm{TV}\, f \right), \\ \Theta_E &= \frac{\sqrt{2} c_K}{2} \left(a_E + \sqrt{a_E^2 + 4B_E} \right). \end{aligned} \tag{S50}$$

Lemma S9. Global energy angle bound. *The solution of* (S48) *exists throughout* $[0, T_f]$ *and satisfies* $|b^T x(t)| \le \Theta_E$ *there. For the present drive,*

$$\mathrm{TV}\, f = f_a\{5 - \cos(\nu T_f)\}. \tag{S51}$$

Proof. The finite-dimensional right-hand side is globally Lipschitz in the state on this finite interval, giving existence and uniqueness. Integrating (S49) by parts and using $D \succeq 0$ gives

$$\mathcal{E}(t) \le B_E + |f(t)\sin\theta(t)| + \int_0^t |f'(s)\sin\theta(s)|\, ds.$$

For $\mathcal{Q} = \max_{0\le t\le T_f}\sqrt{\mathcal{E}(t)}$, $|\sin\theta| \le |\theta| \le \sqrt{2}c_K\mathcal{Q}$ implies $\mathcal{Q}^2 \le B_E + a_E\mathcal{Q}$. The nonnegative root bound yields (S50). It uses the complete sine and cosine source, with no prior small-angle assumption. Finally $2\pi < \nu T_f < 3\pi$. The variation from 0 through the two interior extrema is $4f_a$, and the remaining variation is $f_a(1 - \cos(\nu T_f))$, proving (S51).

S6.2 The energy contraction and two-component residual bound

Let $K_l = K + \kappa bb^T$, $\mathsf{P} = \mathrm{diag}(K_l, M)$, and use the real state $\zeta = (x, v)$. The complete linear source has matrix

$$\mathsf{A} = \begin{pmatrix} 0 & I \\ -M^{-1}K_l & -M^{-1}D \end{pmatrix}.$$

Direct multiplication gives

$$\mathsf{A}^T\mathsf{P} + \mathsf{P}\mathsf{A} = \mathrm{diag}(0, -2D) \preceq 0, \qquad \|e^{t\mathsf{A}}\zeta\|_{\mathsf{P}} \le \|\zeta\|_{\mathsf{P}} \quad (t \ge 0). \tag{S52}$$

No commutation of the mass, stiffness, and damping matrices is assumed.

For any differentiable reference pair $(\widehat{x}, \widehat{v})$, define both residuals

$$\begin{aligned} d_x &= \dot{\widehat{x}} - \widehat{v}, \\ d_f &= M\dot{\widehat{v}} + D\widehat{v} + K_l\widehat{x} - bf(t), \\ \delta(t) &= \sqrt{d_x^T K_l d_x + d_f^T M^{-1} d_f}. \end{aligned} \tag{S53}$$

In particular $\dot{\widehat{x}} = \widehat{v}$ is not assumed. Set $c_M^2 = b^T M^{-1} b$ and, for any established bound $|\theta(t)| \le R$, define $r_R = \kappa R^3/6 + f_a R^2/2$.

Proposition S10. Continuous nonlinear-source reference tube. *The exact solution of* (S48) *satisfies*

$$\|\zeta(t) - \widehat{\zeta}(t)\|_{\mathsf{P}} \le \|\zeta_0 - \widehat{\zeta}_0\|_{\mathsf{P}} + \int_0^t \delta(s)\, ds + tc_M r_R, \qquad 0 \le t \le T_f. \tag{S54}$$

The same absolute bound controls the K_0/M *mechanical norm.*

Proof. Write the true equation as

$$M\ddot{x} + D\dot{x} + K_l x = b\{f(t) + r(t,\theta)\}, \qquad r = \kappa(\theta - \sin\theta) + f(t)(\cos\theta - 1).$$

The global real Taylor inequalities give $|r| \le r_R$ along the already bounded true trajectory. The associated first-order forcing has P norm $c_M|r|$. Variation of constants and (S52) bound its contribution by $tc_M r_R$. The reference's two defects have first-order norm δ, giving the other terms in (S54) by the same argument. Thus no nonlinear stability exponent is needed for this forcing-along-the-true-solution estimate. Finally $K_l \succeq K_0$, so the P norm dominates the K_0/M norm.

S6.3 An explicit finite-exponential reference

The reference used here is the mathematical function

$$\begin{aligned}\widehat{x}(t) &= \mathrm{Re}\left\{\sum_{j=1}^{128}\xi_j e^{\lambda_j t} + \xi_p e^{i\widehat{\nu}t}\right\},\\ \widehat{v}(t) &= \mathrm{Re}\left\{\sum_{j=1}^{128}\upsilon_j e^{\lambda_j t} + \upsilon_p e^{i\widehat{\nu}t}\right\},\\ \widehat{\nu} &= \frac{3213577726426711}{281474976710656}.\end{aligned} \tag{S55}$$

Each saved real and imaginary amplitude and each pole is interpreted as an exact binary rational. The amplitude vectors already contain the homogeneous coefficients. A rounded spectral decomposition is used only to propose this function: no exact eigenvector or spectral identity is inferred from it. In particular, pre-multiplying and rounding a basis and coefficient vector defines these saved amplitudes; it is not identified with the exact product of their separately saved factors.

The homogeneous complex defect vectors are

$$r_{x,j} = \lambda_j\xi_j - \upsilon_j, \qquad r_{f,j} = M\lambda_j\upsilon_j + D\upsilon_j + K_l\xi_j.$$

For a pair of complex vectors define

$$\mathcal{D}(r_x, r_f)^2 = \sum_{\sigma\in\{\mathrm{Re},\mathrm{Im}\}}\{\sigma(r_x)^T K_l\sigma(r_x) + \sigma(r_f)^T M^{-1}\sigma(r_f)\}.$$

This bounds the real defect norm for every common complex phase by Cauchy–Schwarz. All $a_j = \mathrm{Re}\,\lambda_j$ are strictly negative, so the homogeneous integrated defect is bounded by

$$\varepsilon_h = \sum_{j=1}^{128}\mathcal{D}(r_{x,j}, r_{f,j})\frac{e^{a_j T_f} - 1}{a_j}. \tag{S56}$$

The particular vectors have residuals $r_{x,p} = i\widehat{\nu}\xi_p - \upsilon_p$ and $r_{f,p} = i\widehat{\nu}M\upsilon_p + D\upsilon_p + K_l\xi_p - bf_a$, giving $\varepsilon_p = T_f\mathcal{D}(r_{x,p}, r_{f,p})$. This uses a forcing at $\widehat{\nu}$, whereas the source uses ν. The difference contributes the additional bound

$$\varepsilon_\nu = c_M f_a|\widehat{\nu} - \nu|\,T_f^2/2, \tag{S57}$$

by $|\cos(\widehat{\nu}t) - \cos(\nu t)| \le |\widehat{\nu} - \nu|t$. Together with the explicitly computed initial defect, these quantities define

$$\varepsilon_l = \|\zeta_0 - \widehat{\zeta}(0)\|_{\mathsf{P}} + \varepsilon_h + \varepsilon_p + \varepsilon_\nu.$$

It is a full continuous-time error bound to the exact linear source.

S6.4 Angle slabs and two noncircular refinements

Let $\chi^2 = b^T K_l^{-1} b$. An already established full-time error $\varepsilon^{(r)}$ gives $|\theta(t)| \le |b^T\widehat{x}(t)| + \chi\varepsilon^{(r)}$. To preserve cancellation in the slowly varying angle, divide $[0, T_f]$ into $N = 3000$ slabs of width $h = 1/5000$ and midpoints m_k. Let J_s contain the 36

poles with $|\lambda_j| < 10000$ and let J_f contain the remaining 92. Define

$$\begin{aligned} L_\theta &= |b^T\xi_p|\,|\widehat{\nu}| + \sum_{j\in J_s} |b^T\xi_j|\,|\lambda_j|, \\ A_f &= \sum_{j\in J_f} |b^T\xi_j|, \\ \Theta_k &= \left| \mathrm{Re}\left\{ b^T\xi_p e^{i\widehat{\nu}m_k} + \sum_{j\in J_s} b^T\xi_j e^{\lambda_j m_k} \right\} \right| + \tfrac{h}{2}L_\theta + A_f. \end{aligned} \tag{S58}$$

Since $a_j < 0$, the slow derivative is bounded by L_θ for all $t \geq 0$, and the omitted midpoint tail is bounded by A_f. Thus $|b^T\widehat{x}(t)| \leq \Theta_k$ on the entire kth slab. All 128 terms remain in the reference; the split is used only for this scalar enclosure.

Start with the proved energy bound $R = 33/10000$ and $\varepsilon^{(0)} = \varepsilon_l + T_f c_M r_R$. For $r = 0, 1$ set

$$\begin{aligned} R_k^{(r)} &= \Theta_k + \chi\varepsilon^{(r)}, \\ \varepsilon^{(r+1)} &= \varepsilon_l + c_M \sum_{k=1}^{N} h\left\{ \frac{\kappa}{6}(R_k^{(r)})^3 + \frac{f_a}{2}(R_k^{(r)})^2 \right\}. \end{aligned} \tag{S59}$$

If $\varepsilon^{(r)}$ is valid on the whole interval, the dual angle bound makes every $R_k^{(r)}$ valid there. Substitution into the nonlinear remainder proof of Proposition S10 gives $\varepsilon^{(r+1)}$ on the whole interval as well. A partial-time integral is at most the full sum of these nonnegative terms. Induction therefore justifies both refinements. No smaller tube is assumed to prove itself, and no small root of a fixed-point inequality is used as an a priori error bound.

S6.5 Source constants and the full-time numerical bound

For the actual saved initial state common to the coarse and fine positive-velocity records, use the fixed normalization

$$n_0 = \sqrt{x_0^T K_0 x_0 + v_0^T M v_0} \simeq 2.029128417959 \times 10^{-6}. \tag{S60}$$

Both the numerator norm and this initial state match the K_0/M mechanical error criterion. The denominator is not replaced by a time-dependent norm or by the K_l/M norm. Exact-source arithmetic and outward scalar enclosures give

$$\begin{aligned} c_M &< 4.134198337 \times 10^{6}, & \chi &< 369.020558, \\ \mathrm{TV}\, f &< 8.312933059 \times 10^{-9}, & \Theta_E &< 0.003292257522, \\ T_f c_M r_R &< 3.592709307 \times 10^{-8}, & \varepsilon_l/n_0 &< 5.740601207 \times 10^{-7}. \end{aligned} \tag{S61}$$

The separate linear defect terms satisfy

$$\begin{aligned} \varepsilon_h &< 5.408628234 \times 10^{-13}, & \varepsilon_p &< 6.239762808 \times 10^{-13}, \\ \varepsilon_\nu &< 2.839908971 \times 10^{-19}, & \|\zeta_0 - \widehat{\zeta}(0)\|_{\mathsf{P}} &< 2.316288243 \times 10^{-18}. \end{aligned}$$

The slab calculation gives $L_\theta < 0.465092260$ and $A_f < 4.381231894 \times 10^{-5}$. It bounds the reference angle by $\max_k \Theta_k < 0.000755947809$. The coarse bound and successive refinements satisfy

$$\begin{aligned} \varepsilon^{(0)}/n_0 &< 0.017706251410, \\ \varepsilon^{(1)}/n_0 &< 0.000236201704, \\ \varepsilon^{(2)}/n_0 &< 0.000222887810. \end{aligned} \tag{S62}$$

The corresponding final absolute radius is below $4.522679884 \times 10^{-10}$. The prescribed initial state was also bounded separately; its final normalized radius is below 0.000222888884. It is not silently substituted for the actual saved state. The source and residual bounds use exact rational matrix entries and interval scalar evaluation; an independent rational reconstruction and higher-precision enclosure confirm the stated inequalities.

The last line of (S62) certifies all continuous times for the exact function (S55). If a numerical evaluator returns a sampled pair $\widetilde{\zeta}(t_k)$ with a separately proved bound $\|\widetilde{\zeta}(t_k) - \widehat{\zeta}(t_k)\|_{K_0,M} \le \varepsilon_{\mathrm{eval}}$, its source-error allowance is $\varepsilon^{(2)} + \varepsilon_{\mathrm{eval}}$ before normalization. Any coordinate transformation used in scoring needs its own consistent transfer allowance. No such sampling or chart identity is assumed by the continuous-source certificate itself.

S6.6 Conditional force-derivative bounds from a trajectory tube

Consider the declared native full-coordinate system

$$M\ddot{x} + Kx + D\dot{x} + b\,g(b^T x, t) = 0, \qquad g(\theta, t) = \kappa \sin\theta - \mu B(t)\cos\theta,$$

where $x \in \mathbb{R}^{64}$, $b = e_{64}$, $M = M_0 + M_r$, $K = (53/50)K_0$ and $D = \alpha_* M_0$. The saved material readback gives the exact binary coefficient $\alpha_* = 5754906970707699/9007199254740992$. In particular its ratio to the saved nominal damping coefficient is $5754906970707699/5004266931050173$, rather than an assumed exact $23/20$. The specified drive is

$$B(t) = 10^{-4}\cos(11.41692154656479\,t).$$

Here the displayed decimal drive defines an exact continuous law; the helper also accepts explicit rational overrides if a different literal convention is adopted. This convention does not assert exact evaluation of that law by the numerical solver.

Put $v = \dot{x}$, $a = \ddot{x}$, $j = x^{(3)}$ and $k = x^{(4)}$. Differentiating the smooth declared equation gives

$$\begin{aligned} a &= -M^{-1}(Kx + Dv + bg),\\ j &= -M^{-1}(Kv + Da + b\dot{g}),\\ k &= -M^{-1}(Ka + Dj + b\ddot{g}). \end{aligned}$$

Let $\omega = b^T v$ and $\alpha = b^T a$. Then

$$\begin{aligned} \dot{g} &= (\kappa\cos\theta + \mu B\sin\theta)\omega - \mu B'\cos\theta,\\ \ddot{g} &= (\kappa\cos\theta + \mu B\sin\theta)\alpha + (-\kappa\sin\theta + \mu B\cos\theta)\omega^2\\ &\quad + 2\mu B'\sin\theta\,\omega - \mu B''\cos\theta. \end{aligned}$$

Consequently the normalized projected restoring-and-dissipative force $s = (\ell/E_s)V^T(Kx + Dv)$ and the separately treated magnetic force $m = (\ell/E_s)V^T b\,g$ satisfy

$$s^{(3)} = (\ell/E_s)V^T(Kj + Dk), \qquad m'' = (\ell/E_s)V^T b\,\ddot{g}.$$

Subtracting a constant nominal force from s does not change its third derivative.

Projection before interval bounds. Bounding all coordinates of j and k separately can discard useful spatial cancellation. For an arbitrary fixed rational λ, define

$$z = \begin{pmatrix} x \\ v \end{pmatrix}, \quad A_\lambda = \begin{pmatrix} 0 & I \\ -M^{-1}(K + \lambda bb^T) & -M^{-1}D \end{pmatrix}, \quad u = \begin{pmatrix} 0 \\ -M^{-1}b \end{pmatrix}, \quad r = g - \lambda\theta.$$

Then $z' = A_\lambda z + ur$ and, for every constant row c,

$$cz^{(n)} = cA_\lambda^n z + \sum_{q=0}^{n-1} cA_\lambda^{n-1-q} u\, r^{(q)}, \qquad 1 \le n \le 3. \tag{DB1}$$

The helper first forms these rational rows exactly. Its default $\lambda = \kappa$ puts the linear magnetic stiffness in A_λ. With $r_\theta = \kappa\cos\theta + \mu B\sin\theta - \lambda$ and $r_{\theta\theta} = -\kappa\sin\theta + \mu B\cos\theta$,

$$r' = r_\theta\omega - \mu B'\cos\theta, \qquad r'' = r_\theta\alpha + r_{\theta\theta}\omega^2 + 2\mu B'\sin\theta\,\omega - \mu B''\cos\theta.$$

The magnetic derivative is recovered as $g'' = \lambda\alpha + r''$. The quantity α itself follows from (DB1) with $c = (0, b^T)$ and $n = 1$; hence this procedure is not circular. For the three rows $c_i = (\ell/E_s)(V^T K, V^T D)_i$, (DB1) with $n = 3$ directly encloses $s_i^{(3)}$. Optional coordinate rows provide a, j, k when needed. Near zero, the evaluation uses $\cos\theta - 1 = -2\sin^2(\theta/2)$ and the degree-seven polynomial for $\sin\theta - \theta$, with rigorous remainder $|\theta|^9/9!$, to retain the linear cancellation.

Required tube and its exact dual bound. On each time segment, the caller must independently certify

$$(x - \widetilde{x})^T K(x - \widetilde{x}) + (v - \widetilde{v})^T M(v - \widetilde{v}) \le R^2. \tag{DB2}$$

The radius R is a norm radius, not an energy. If an energy error $\frac{1}{2}(\delta x^T K \delta x + \delta v^T M \delta v) \le E$ is supplied, one may use $R = \sqrt{2E}$. Exact positive LDL pivots verify that M and K are positive definite. For a row $c = (c_x, c_v)$, define

$$\rho(c)^2 = c_x K^{-1} c_x^T + c_v M^{-1} c_v^T.$$

The Cauchy–Schwarz inequality in this energy metric gives

$$|c(z - \widetilde{z})| \le R\rho(c). \tag{DB3}$$

For a polynomial center $\widetilde{z}(u), 0 \le u \le 1$, the scalar polynomial $c\widetilde{z}(u)$ is formed before bounding its range. Its exact Bernstein coefficients enclose that range; (DB3) supplies the additional tube interval. Interval sine and cosine, the exact drive derivatives and (DB1) then produce componentwise bounds $B_{3,i}$ for $s_i^{(3)}$ and $B_{m,i}$ for m_i''. For the paired force $S = (s_+ + s_-)/2$, the bound is $(B_{3,i,+} + B_{3,i,-})/2$. Bounds over several segments are obtained by their maxima. A common magnetic bound for both signs is obtained by taking the maximum over signs, components and segments.

The method supplies the derivative inputs required by S6 conditional on (DB2). A cubic Hermite curve through the saved observations does not establish (DB2), and neither its interpolation residual nor a coarse–fine difference alone certifies the initial error or a continuous trajectory tube. No numerical tube radius is certified in this component.

S6.7 Reception of the continuous derivative bounds

The completed energy-tube calculation encloses the two declared native full-coordinate IVPs on the first four saved output panels. Each IVP starts from its own saved initial state. Its uniform cell radii use the metric $\delta x^T K \delta x + \delta v^T M \delta v$ with $K = (53/50)K_0$ and $M = M_0 + M_r$. The received all-prefix radius of each cell is applied to the entire cell; a terminal radius is not mistaken for an endpoint-only bound.

Applying the frozen procedure in S6.6 to these verified tubes gives the following componentwise bounds for the paired third derivative, rounded upward:

$$(B_{3,1}, B_{3,2}, B_{3,3}) \le (2.993942 \times 10^{10},\ 4.270068 \times 10^{10},\ 6.523766 \times 10^{12}).$$

For each component, the maximum is first taken over all four cells of each sign and the two signed bounds are then averaged. A common bound on the normalized projected magnetic second derivative, over all cells, signs and components, is

$$B_m \le 2.604809 \times 10^7.$$

The actual signed clocks are received separately and their dyadic relations are verified. For their fixed first-panel duration $T \simeq 7.8125 \times 10^{-8}$ s, the derivative contributions to S6 satisfy

$$\frac{B_3 T^3}{3} \le 1.036925 \times 10^{-9}, \qquad \frac{5B_m h^2}{12} \le 6.624371 \times 10^{-8},$$

where h is the largest actual saved panel width. Their sum is at most 6.728064×10^{-8}.

The tube producer and derivative receiver both terminated with actual exit code zero. This receives B_3 and B_m for the stated mathematical IVPs; it does not certify physical measurement error or the compiled or interpolated COMSOL trajectory. Moreover, the generic energy-to-momentum observation allowance remains approximately 0.03501, so the generic endpoint

estimate alone does not establish a force tolerance of 10^{-7}. The derivative calculation neither sets the initial pairing allowance to zero nor claims that the full instantaneous-force error bound has been closed.

S6.8 Receiving a fixed label through projected force averages

Proposition S11. Reception through a projected force average. *Use all* 64 *free transverse displacement/rotation coordinates in the fixed native ordering. Let* M, K *be constant symmetric positive definite matrices, let* D *be constant symmetric positive semidefinite, let* V_j *denote column* j *of* V*, and let* $\lambda_F = \ell/E_s > 0$*. For the declared evaluator family these matrices are*

$$M = M_0 + M_r, \qquad K = a_E K_0, \qquad D = a_C D_0 = \alpha_{\mathrm{mass}} M_0,$$

with the native sum and the declared coefficients interpreted exactly. For each sign $\sigma \in \{+, -\}$*, let* x_σ *be the solution of the declared recorded-state initial-value problem*

$$M x_\sigma'' + D x_\sigma' + K x_\sigma + b\, m_\sigma(b^T x_\sigma, t) = 0, \qquad m_\sigma(\theta, t) = \kappa \sin\theta - \mu B_{y,\sigma}(t) \cos\theta.$$

Its initial position and velocity are, by definition, that sign's own saved first-row values. This defines a mathematical initial-value problem and does not identify the physical initial state with a saved measurement.

Let Q_σ *be the piecewise cubic Hermite interpolant of that sign's saved positions and velocities, with* Q_σ' *interpolating the saved velocities. Use each sign's actual saved clock, and assume that the received exact clock identities are*

$$t_{\sigma,0} = 0, \quad t_{\sigma,1} = T, \quad t_{\sigma,2} = 2T, \quad t_{\sigma,4} = 4T, \qquad T > 0.$$

The interior time $t_{\sigma,3}$ *need only lie strictly between* $2T$ *and* $4T$*; it need not equal* $3T$*. Write* $I_{\sigma,i} = [t_{\sigma,i}, t_{\sigma,i+1}]$ *and* $h_{\sigma,i} = t_{\sigma,i+1} - t_{\sigma,i}$ *for* $0 \le i \le 3$*. Define*

$$e_{x,\sigma} = x_\sigma - Q_\sigma, \qquad e_{v,\sigma} = x_\sigma' - Q_\sigma', \qquad r_\sigma^2 = e_{x,\sigma}^T K e_{x,\sigma} + e_{v,\sigma}^T M e_{v,\sigma}.$$

In particular, $e_{x,\sigma}(0) = e_{v,\sigma}(0) = 0$*. Suppose certified nonnegative numbers* $R_{\sigma,i}, J_{\sigma,i}$ *satisfy*

$$r_\sigma(t) \le R_{\sigma,i}, \qquad \int_0^t r_\sigma(s)\, ds \le J_{\sigma,i} \quad (t \in I_{\sigma,i}).$$

A sufficient choice from whole-cell energy bounds alone is $J_{\sigma,i} = \sum_{q=0}^{i} h_{\sigma,q} R_{\sigma,q}$*.*

Define the averaging kernel and the normalized nonmagnetic force by

$$\mathcal{W}_T(t) = \begin{cases} 7/4, & 0 \le t < T, \\ -11/12, & T \le t < 2T, \\ 1/12, & 2T \le t \le 4T, \end{cases} \qquad \mathfrak{s}(z, w) = \lambda_F V^T (Kz + Dw).$$

Thus $k = (7/4, -11/12, 1/12, 1/12)$ *gives the value of* $\mathcal{W}_T$ *on the four saved cells, regardless of the location of* $t_{\sigma,3}$*. Set*

$$C_{Q,\sigma} = \frac{1}{T} \sum_{i=0}^{3} k_i \int_{I_{\sigma,i}} \mathfrak{s}(Q_\sigma, Q_\sigma')\, dt,$$
$$C_{\mathrm{true},\sigma} = \frac{1}{T} \sum_{i=0}^{3} k_i \int_{I_{\sigma,i}} \mathfrak{s}(x_\sigma, x_\sigma')\, dt,$$
$$C_Q = \tfrac{1}{2}(C_{Q,+} + C_{Q,-}), \qquad C_{\mathrm{true}} = \tfrac{1}{2}(C_{\mathrm{true},+} + C_{\mathrm{true},-}).$$

The integrands defining C_Q *are polynomials. For the received rational native matrices, normalization, clocks and saved values,* C_Q *is therefore an exact rational polynomial integral.*

Let $\widehat{F}$ *be the already fixed observation-only raw force receiver formed from projected momentum increments and magnetic trapezoids with averaging weights* $(8/3, -2, 1/3)$*. Its label process is completed and hashed before declared evaluator*

coefficients are read. Let x_{prep} be a fixed reference preparation, either its own rounded position or the exact ideal reference Vq. Use the same reference for both the nominal subtraction and the declared target, and put

$$f_0 = \lambda_F V^T K_0 x_{\mathrm{prep}}, \qquad \widehat{Y} = \widehat{F} - f_0, \qquad \Delta_Q = \widehat{Y} + f_0 - C_Q = \widehat{F} - C_Q.$$

Changing the reference requires only the exact known nominal adjustment to $\widehat{Y}$; it does not change the fixed raw label $\widehat{F}$. The same fixed raw receiver must be enclosed when receiving Δ_Q. In particular, if directed Arb arithmetic produces the interval $\mathbf{\Delta}_{Q,j}$, use $d_{Q,j} = \sup\{|z| : z \in \mathbf{\Delta}_{Q,j}\}$ as the upper bound on $|\Delta_{Q,j}|$. A point representation of an interval-certified label must retain its certified arithmetic radius in this enclosure.

For each projected component define

$$\begin{aligned} A_{D,j} &= (V_j^T D M^{-1} D V_j)^{1/2}, \\ A_{KM,j} &= (V_j^T K M^{-1} K V_j)^{1/2}, \\ L_j &= (V_j^T K V_j + V_j^T D M^{-1} D V_j)^{1/2}, \\ B_{\sigma,i,j} &= \min\{L_j R_{\sigma,i},\ A_{D,j} R_{\sigma,i} + A_{KM,j} J_{\sigma,i}\}, \\ \Gamma_j &= \frac{\lambda_F}{2T} \sum_{\sigma\in\{+,-\}} \sum_{i=0}^{3} |k_i| h_{\sigma,i} B_{\sigma,i,j}. \end{aligned}$$

Then $|(C_Q - C_{\mathrm{true}})_j| \le \Gamma_j$.

Suppose, in addition, that

$$S(t) = \tfrac{1}{2} \sum_{\sigma\in\{+,-\}} \mathfrak{s}(x_\sigma(t), x'_\sigma(t)) \quad \textit{satisfies} \quad S \in C^3([0,4T]), \qquad \sup_{[0,4T]} \|S'''\|_\infty \le B_3.$$

The fixed saved-data receiver obeys the componentwise bound

$$\left|\widehat{Y}_j - (S_j(0) - f_{0,j})\right| \le |\Delta_{Q,j}| + \Gamma_j + \frac{B_3 T^3}{3} \le d_{Q,j} + \Gamma_j + \frac{B_3 T^3}{3}.$$

For the declared conservative correction at the chosen reference, define

$$f_*^{\mathrm{prep}} = \lambda_F V^T (K - K_0) x_{\mathrm{prep}}, \qquad \delta_0^{\mathrm{rec}} = S(0) - \lambda_F V^T K x_{\mathrm{prep}}, \qquad I_0^{\mathrm{rec}} = \|\delta_0^{\mathrm{rec}}\|_\infty.$$

Since the recorded-state initial values are the saved numbers, δ_0^{rec} is exactly computable for this declared linear family and its supplied exact reference. In particular,

$$\begin{aligned} |\widehat{Y}_j - f_{*,j}^{\mathrm{prep}}| &\le d_{Q,j} + \Gamma_j + \frac{B_3 T^3}{3} + |\delta_{0,j}^{\mathrm{rec}}|, \\ \|\widehat{Y} - f_*^{\mathrm{prep}}\|_\infty &\le \max_j (d_{Q,j} + \Gamma_j) + \frac{B_3 T^3}{3} + I_0^{\mathrm{rec}}. \end{aligned}$$

Proof. The dual norm of the linear functional $(e_x, e_v) \mapsto V_j^T(Ke_x + De_v)$ in the product energy norm gives

$$|V_j^T(Ke_x + De_v)| \le \left(V_j^T K V_j + V_j^T D M^{-1} D V_j\right)^{1/2} \left(e_x^T K e_x + e_v^T M e_v\right)^{1/2} = L_j r.$$

Also $e_x' = e_v$ and $e_x(0) = 0$, so Cauchy–Schwarz in the mass norm yields

$$\begin{aligned} |V_j^T K e_x(t)| &= \left| \int_0^t V_j^T K e_v(s)\, ds \right| \le A_{KM,j} \int_0^t r(s)\, ds, \\ |V_j^T D e_v(t)| &\le A_{D,j} r(t). \end{aligned}$$

Both estimates apply to the same error. On a whole cell their minimum is therefore the stated $B_{\sigma,i,j}$. Integrating against the absolute value of the kernel, then taking the half-sum of the two signs, proves $|(C_Q - C_{\mathrm{true}})_j| \le \Gamma_j$.

For every continuous vector function s, direct collection of the three averaging intervals gives

$$\frac{8}{3}\frac{1}{T}\int_0^T s(t)\,dt - 2\frac{1}{2T}\int_0^{2T} s(t)\,dt + \frac{1}{3}\frac{1}{4T}\int_0^{4T} s(t)\,dt = \frac{1}{T}\int_0^{4T} \mathcal{W}_T(t)s(t)\,dt.$$

Consequently C_{true} is the three-window average of S. The Peano-kernel bound of Proposition S6 therefore gives $\|C_{\mathrm{true}} - S(0)\|_\infty \le B_3 T^3/3$. Now use the exact decomposition

$$\widehat{Y} - (S(0) - f_0) = \Delta_Q + (C_Q - C_{\mathrm{true}}) + (C_{\mathrm{true}} - S(0)).$$

The componentwise triangle inequality proves the reception bound. Finally, $S(0) - f_0 = f_*^{\mathrm{prep}} + \delta_0^{\mathrm{rec}}$ gives both prepared-target bounds. The stated interval reception includes the finite arithmetic radius in $d_{Q,j}$ and hence preserves these inequalities.

This organization receives the fixed momentum-and-trapezoid label directly against a continuous nonmagnetic force average. Its Δ_Q already includes the discrepancy of that complete saved-data receiver from the Hermite force average. Separate endpoint-momentum, sampled-magnetic-force and magnetic-quadrature allowances are therefore not added again to this bound. A valid B_3 certificate for the same declared trajectories is still required.

The recorded initial defect is distinct from an unknown physical initial-state allowance. For example, if the same declared force law applies and independently certified componentwise physical initial radii are $\rho_{x,\sigma}, \rho_{v,\sigma}$, a sufficient additional term is

$$U_0 = \left\| \frac{\lambda_F |V^T K|}{2}(\rho_{x,+} + \rho_{x,-}) + \frac{\lambda_F |V^T D|}{2}(\rho_{v,+} + \rho_{v,-}) \right\|_\infty, \qquad I_0 \le I_0^{\mathrm{rec}} + U_0.$$

Here absolute values of matrices are componentwise, and I_0 denotes the physical initial force defect. No such radii, and hence no value of U_0, are established by this proposition. In particular, $e_x(0) = e_v(0) = 0$ for a recorded-state mathematical initial-value problem does not establish $U_0 = 0$ for a physical observation. A claim about a physical trajectory additionally requires initial-state, input and equation/model-discrepancy bounds that apply to that same trajectory; the declared recorded-state tubes alone do not supply them.

The scope is postlabel certification for the declared recorded-state initial-value problems using evaluator coefficients. The statement does not identify a general unknown constitutive force, certify a native compiled equation or its dense output, or establish a complex-network advantage. It changes neither the fixed saved raw label nor any data purpose or training split.

S6.9 Received continuous force bound at the direction-3 configuration

The fixed half-fine direction-3 receiver is now received by Proposition S11. The raw force label was already fixed by the saved observations before this certification. An independent 512-bit recomputation overlaps each of its earlier interval enclosures. The declared material coefficients enter only the subsequent certificate. They are not supplied as labels during that observation-only construction.

The reference preparation in this reception is the ideal point $\bar{x} = V(0, 0, 1/2000)^T$. Accordingly the same nominal force is subtracted there, and the recorded initial allowance includes the whole difference of the two saved initial states from this reference. The two mathematical solutions start at their respective saved first rows; no physical measurement is declared exact.

The components of the force-error budget are listed below. All displayed values are rounded upwards. The quantity $d_{Q,j}$ bounds the complete fixed raw-label minus Hermite nonmagnetic-average discrepancy; it is not a separately fitted residual. The tube term Γ_j uses the minimum of the two rigorously valid projected estimates in the proposition. Totals are rounded upward from the exact component sums; separately rounded displayed entries need not sum to the displayed total.

j	$d_{Q,j}$	Γ_j	$B_{3,j}T^3/3$	$\lvert\delta_{0,j}^{\mathrm{rec}}\rvert$	Total
1	2.999876×10^{-9}	1.502520×10^{-9}	4.758743×10^{-12}	9.950403×10^{-15}	4.507164×10^{-9}
2	4.277506×10^{-9}	6.079664×10^{-10}	6.787091×10^{-12}	1.418104×10^{-14}	4.892274×10^{-9}
3	3.434153×10^{-8}	9.209175×10^{-10}	1.036925×10^{-9}	2.889657×10^{-14}	3.629940×10^{-8}

The largest resulting normalized force-error bound is 3.629940×10^{-8}. Thus this one fixed receiver satisfies the declared 10^{-7} force tolerance for the two recorded-state mathematical IVPs.

This conclusion is stronger than a comparison of saved values with an algebraic material formula: it includes a whole-cell continuous trajectory enclosure and a third-derivative temporal-bias allowance. It remains a postlabel certificate for the specified material family. It supplies no physical measurement/model-discrepancy bound. The eight additional jet-design positions are received separately in Section S6.10. This two-IVP result does not validate a newly trained checkpoint, identify an arbitrary unknown quartic potential, or transfer the candidate-only KAM result to the physical target. The generic endpoint-based 0.035010 upper bound is retained as a valid negative result; the direct projected reception proves a different, sharper inequality for the same fixed data.

S6.10 Actual observations at all eleven design configurations

The eight additional positions of the exact fourth-order design were implemented in native COMSOL. Each position uses $q = \ell z^{(i)}$ and the two prescribed velocities $\dot{q} = \pm 25 q\,\mathrm{s}^{-1}$. The existing three positions retain their own earlier prescribed velocities and their finest available saved records. A fixed internal step of 1.953125×10^{-8} s is used for the sixteen new solves; each exports 257 full-field samples over $20\,\mu$s with the same source-decimal cosine drive and fixed shifted material. Exact preparation arithmetic bounds every new prescribed initial energy by 5.322879×10^{-12} J. This is a feasibility bound for the prescribed initial values, not a claim that their numerical realization has zero error. All sixteen native processes completed successfully, and the source MPH remained unchanged.

The observation-only label process receives each sign's own saved clock, positions, velocities and magnetic field. It uses the same fixed three windows and weights at every node, subtracting the nominal force at the exact ideal design point. The declared material coefficients do not enter the force labels. The generator material is public in its protocol, so this is not a hidden-target or blind test. Each label is then independently received against its own continuous recorded-state IVPs by Proposition S11. The resulting maximum normalized force allowances, rounded upwards, are

Node	Configuration	Force allowance
0	original	2.274829×10^{-8}
1	direction2	2.363062×10^{-8}
2	direction3	3.629940×10^{-8}
3	jet_node03	4.582830×10^{-8}
4	jet_node04	4.582371×10^{-8}
5	jet_node05	2.530279×10^{-8}
6	jet_node06	3.630746×10^{-8}
7	jet_node07	3.629510×10^{-8}
8	jet_node08	4.582309×10^{-8}
9	jet_node09	1.576921×10^{-8}
10	jet_node10	3.631862×10^{-8}

All eleven selected records meet the fixed 10^{-7} tolerance. The older direction-3 step-1 and step-2 bounds remain approximately 7.531×10^{-7} and 1.797×10^{-7} and do not meet that tolerance. They are retained rather than removed by a posteriori window selection. The three existing finest records were selected by the smallest saved internal step before calculating these bounds. The new eight groups have only their single predeclared step.

All eleven groups are development/training configurations. Neither the 257 time samples nor the two signs at a position are declared independent configuration samples. The smaller $25q$ velocity also reduces damping excitation relative to the earlier $100q$ directions; no improvement in damping identifiability is inferred solely from the added position count.

S7 From prepared forces to a low-order potential jet

Set $\ell = 10^{-3}\,\mathrm{m}$, $E_s = 6\,10^{-12}\,\mathrm{J}$, and $w(z) = W(\ell z)/E_s$. Here W is the conservative correction: the known nominal elastic and prescribed magnetic forces are removed with the same normalization. Thus observations approximate the positive gradient $\nabla_z w$, rather than the negative mechanical force. A prepared point means the restriction $x = V\ell z$ of the full native state; departures from this prescribed state belong to the label error.

Proposition S12. Three-point symmetric Hessian reception. *Suppose explicitly that* $w(z) = z^T Hz/2$ *with a constant symmetric matrix* H*. Let*

$$z^{(0)} = (a,b,c)^T = \left(\frac{1}{2}, \frac{1256189}{200000000}, \frac{26951}{12500000}\right)^T, \quad z^{(1)} = (0, \tfrac{1}{2}, 0)^T, \quad z^{(2)} = (0, 0, \tfrac{1}{2})^T.$$

For $h = (H_{11}, H_{22}, H_{33}, H_{12}, H_{13}, H_{23})^T$*, stack the three blocks*

$$A(z) = \begin{pmatrix} z_1 & 0 & 0 & z_2 & z_3 & 0 \\ 0 & z_2 & 0 & z_1 & 0 & z_3 \\ 0 & 0 & z_3 & 0 & z_1 & z_2 \end{pmatrix}$$

into $A \in \mathbb{Q}^{9\times 6}$*. This matrix has rank six. For* $L = (A^T A)^{-1} A^T$*, any finite labels* $y = Ah + e$ *with separately received component bounds* $|e_m| \le \eta_m$ *satisfy*

$$|(Ly - h)_i| \le \sum_{m=1}^{9} |L_{im}| \eta_m.$$

The Hessian in physical q *coordinates is* $E_s \ell^{-2} H$*.*

Proof. The third and second axis points determine the off-diagonal entries, H_{22} and H_{33}; then $a \neq 0$ determines H_{11} from the first component at $z^{(0)}$. Thus A has full column rank and $LA = I_6$. Apply L to $y = Ah + e$ and use the triangle inequality. The certificate also verifies these identities exactly using all nine scalar observations.

The numerical application uses the existing fine-step signed labels at these three configurations. Each label is obtained from its actual finite observation window; it is not replaced by the force at the initial point. Its arithmetic radius is added to its individual received force error before applying L. The available numerical errors were evaluated against the declared quadratic material family. They are therefore not general measurement-error certificates for an arbitrary unknown physical law. Proposition S12 allows arbitrary symmetric H only conditional on independently valid errors for that H. Without the quadratic assumption, third- and higher-order contributions cannot be silently absorbed into the Hessian.

Proposition S13. An eleven-point receiver for the complete fourth jet. *Assume* $w \in C^5$ *on a neighborhood of* $[-1/2, 1/2]^3$*, with* $\nabla w(0) = 0$*. Fix the additive gauge* $w(0) = 0$*, and write*

$$w(z) = \sum_{2 \le |\alpha| \le 4} c_\alpha z^\alpha + R_5(z).$$

Append the following eight positions to the preceding three:

$$\begin{array}{cccc} (\frac{1}{2}, -\frac{1}{4}, -\frac{1}{2}), & (-\frac{1}{2}, 0, -\frac{1}{2}), & (-\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}), & (0, -\frac{1}{4}, \frac{1}{2}), \\ (\frac{1}{2}, \frac{1}{4}, \frac{1}{2}), & (-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}), & (-\frac{1}{2}, \frac{1}{4}, \frac{1}{4}), & (-\frac{1}{2}, -\frac{1}{4}, \frac{1}{2}). \end{array}$$

For the 31 *ordinary monomials of degrees two through four, define*

$$G_{(i,j),\alpha} = \alpha_j (z^{(i)})^{\alpha - e_j}, \qquad J = (G^T G)^{-1} G^T.$$

The term is zero when $\alpha_j = 0$*. Exact rational arithmetic gives* $\operatorname{rank} G = 31$ *and* $JG = I_{31}$*. Eleven configurations attain the necessary lower bound* $3m \ge 31$*.*

If $y_{ij} = \partial_j w(z^{(i)}) + e_{ij}$*,* $|e_{ij}| \le \eta_{ij}$*, and* $|\partial_j R_5(z^{(i)})| \le \tau_{ij}$*, then*

$$|(Jy - c)_\alpha| \le \sum_{i,j} |J_{\alpha,(i,j)}| (\eta_{ij} + \tau_{ij}). \tag{S63}$$

For a common componentwise fifth-derivative bound $|D^\nu w| \le M_5$ *for all* $|\nu| = 5$ *on the cube, one may use*

$$\tau_{ij} = \frac{M_5}{24} \|z^{(i)}\|_1^4.$$

Proof. The exact 33×31 matrix and its 31×33 left inverse are stored in the certificate, which verifies $JG = I_{31}$ without a floating rank threshold. The gradient Taylor expansion gives $y = Gc + t + e$. Multiplication by J proves (S63). For each $\partial_j w$, apply the third-degree Taylor formula with fourth-order remainder along the segment from 0 to z. The multinomial identity gives $\sum_{|\nu|=4} |z|^\nu/\nu! = \|z\|_1^4/24$. This derives a gradient-tail bound directly; it does not differentiate an unquantified scalar remainder.

The maximal exact left-inverse row sums, rounded upward here, are 39.124439, 52.294639, and 125.879240 within degrees two, three, and four. They multiply a uniform scalar force-error cap at this fixed half-unit design. The old symmetric 33-configuration design has the smaller constants $25890/6589$, 4, and $104/9$ on the same cube, while also receiving three potentially nonzero linear coefficients. The new design minimizes configuration count under the explicit zero-gradient assumption, at the cost of greater deterministic noise amplification. Using the three existing points plus all eight equal-radius corners would not suffice: that matrix has rank 30, with null polynomial $z_1^4 - z_1^2/2$.

At the preceding three-configuration stage, only the first three positions had observations. That stage did not supply the other eight labels or their error caps. The actual reception below now supplies them for the declared quadratic family; a general unknown target still requires its own fifth-derivative bound. The certificate gives each coefficient's separate contribution from the existing nine scalar error bounds, from future errors, and from M_5. A new label procedure, including finite-window extrapolation, may supply different bounds at each position without changing this exact left inverse.

For consistency with a subsequent jet-to-normal-form calculation, $D^\alpha w(0) = \alpha! c_\alpha$: in particular, $H_{ii} = 2c_{2e_i}$ and $H_{ij} = c_{e_i+e_j}$ for $i \neq j$. The physical derivative is multiplied by $E_s \ell^{-|\alpha|}$. For a fixed candidate, its known signed finite-design tail alias can be subtracted as in the existing force-to-jet receiver; this never removes an unknown target tail or an observation error. The resulting interval jet must still satisfy the spectral, canonical-normalization, Birkhoff nondegeneracy and quantitative analytic KAM conditions uniformly. Neither full design rank nor this three-position quadratic reception is itself a KAM conclusion or an identification of a full 64-coordinate unknown potential. The construction supplies the explicit local-jet interface required between prepared-force learning and those later gates.

As an initial-energy feasibility witness only, $q = \ell z$ and $\dot q = \pm 25 q\,\mathrm{s}^{-1}$ at the proposed positions give exact full-field preparations $x = Vq$, $v = V\dot q$. Under the already declared material interval, their initial energy is below $0.90(6\,10^{-12}\ \mathrm{J})$. At this earlier preparation stage, the velocity was only a feasibility witness: relative to $\pm 100q$, its initial linear damping-force signal is four times smaller and its kinetic energy is sixteen times smaller. A subsequent experimental choice must also meet a separate dissipation-excitation and observation error budget. No COMSOL solve or realized initial-state guarantee is included in this preparation calculation.

S8 Full-coordinate observability of the four-feature material jet

This section concerns an additional, explicitly stated target class. A general unknown potential on $\mathbb{R}^{64}$ need not factor through the four inputs used by the network. We suppose here that its material correction has the local form

$$W(x) = E_s \Phi(u(x), e(x)), \quad u(x) = Bx/\ell, \quad e(x) = E_0(x)/E_s, \quad E_0(x) = \tfrac{1}{2} x^T K_0 x, \tag{S64}$$

where $\ell = 1/1000$ m and $E_s = 3/500000000000$ J are exact decimal normalizations, $BV = I_3$, $K_3 = V^T K_0 V$, and K_0 is positive definite. The local function Φ is real analytic, has no constant or linear u terms, and is otherwise unrestricted. These are target assumptions, not conclusions from the finite observations. The selected network has this factorization by construction, with

$$\begin{aligned} \Phi_\theta(u,e) = c_E e + \frac{E_{\mathrm{net}}}{E_s}\Big[& N_\theta(Du, \lambda e) - N_\theta(0) \\ & - D_{s_q} N_\theta(0) Du \Big], \\ D = \mathrm{diag}(\ell/Q_j), \qquad & \lambda = E_s/E_{\mathrm{net}}. \end{aligned} \tag{S65}$$

Here Q_j, E_{net} keep their saved binary values interpreted as exact numbers. No equality of these binary scales with ℓ, E_s is asserted. The invertible constant rescaling does not change any dimensions below. The original 918-coordinate checkpoint

and the newly refitted 914-coordinate comparison models share this feature geometry; no KAM certificate for one set of weights is transferred to another.

Assign weight one to each u_j and weight two to e. Denote by $\mathcal{P}_{2:4}$ the ordinary polynomial space spanned by $u^\alpha e^b$ with $2 \le |\alpha| + 2b \le 4$. It has

$$\dim \mathcal{P}_{2:4} = (6+1) + (10+3) + (15+6+1) = 42. \tag{S66}$$

These are coefficients of the material correction, before restoring the known elastic and magnetic potentials. Weighted degree is physical degree after substitution in (S64). Let

$$a(u) = \frac{\ell^2}{2E_s} u^T K_3 u, \qquad A(u,e) = e - a(u). \tag{S67}$$

On the trial space $x = V\ell u$, one has $e = a(u)$.

Proposition S14. The eleven unobserved feature-jet coordinates. *The restriction map*

$$\mathcal{R} : \mathcal{P}_{2:4} \longrightarrow \mathbb{R}[u_1, u_2, u_3]_{2:4}, \qquad \mathcal{R}P = P(u, a(u)),$$

is surjective of rank 31. *Its kernel is exactly*

$$\ker \mathcal{R} = A\,\mathcal{P}_{\le 2}, \qquad \mathcal{P}_{\le 2} = \operatorname{span}\{1, u_i, u_i u_j \ (i \le j), e\}, \tag{S68}$$

and has dimension 11. *In particular, complete knowledge of the restricted material jet does not identify the full-coordinate jet even inside this four-feature target class.*

Proof. The monomials with $b = 0$ give all 31 degree-two through degree-four polynomials, proving surjectivity. Polynomial division in the variable e by the monic polynomial $e - a(u)$ gives $P = AQ + P(u, a(u))$. Since a is weighted homogeneous of degree two, the quotient has weighted degree at most two; conversely every displayed AQ belongs to $\mathcal{P}_{2:4}$ and restricts to zero. Multiplication by the nonzero polynomial A is injective, and the quotient space has $1 + 3 + 6 + 1 = 11$ basis elements.

The existing eleven-node energetic design receives projected forces $V^T \nabla W(V\ell u_i)$ and the 31 coefficients obtained from those same force labels. Its exact 33-by-31 gradient design has a retained left inverse. Consequently it receives $\mathcal{R}P$, rather than all 42 coordinates of P. Every AQ has identically zero restricted potential and restricted gradient, not merely a zero value at these eleven points. This observation concerns the point/jet energy objective; it does not say that arbitrary AQ is invisible to all full-field trajectories or to the separate weak-residual loss.

Proposition S15. What complete forces at the old nodes could add. *Put $q_i = \ell u_i$ and*

$$n_i = (K_0 V - B^T K_3) q_i. \tag{S69}$$

Suppose the existing projected gradient design has rank 31. *Let $J = \{i : n_i \neq 0\}$, and let r be the rank of the evaluation matrix of all ten ordinary quadratic polynomials in u at the nodes in J. Then the design that evaluates the complete* 64*-coordinate gradients of $E_s P$ at the same nodes has rank exactly $31 + r$. If $r = 10$, its one-dimensional kernel is*

$$\operatorname{span}\{A^2\}. \tag{S70}$$

Proof. Vanishing of all complete forces implies vanishing of the projected forces, hence $P = AQ$ by the rank-31 assumption. Write $Q = c_0 + c^T u + u^T C u + c_e e$, with an ordinary quadratic convention for the six independent entries. At a trial-space node, $A = 0$ and

$$\nabla_x (E_s A Q) = Q(u_i, a(u_i)) n_i.$$

The scalar polynomials $Q(u, a(u))$ fill the entire ten-dimensional quadratic polynomial space; the map from the eleven coordinates of Q to that space has kernel $\operatorname{span}\{A\}$. The normal constraints therefore add exactly r independent rows. When $r = 10$, Q is a multiple of A, proving the last assertion.

The polynomial $A(x)^2$ has zero value and complete gradient on the whole trial space, while its full-coordinate physical fourth jet is nonzero. This is an observability statement. A nonzero quartic term does not by itself prove either singular or nonsingular Birkhoff twist; its effect on KAM nondegeneracy needs a separate normal-form calculation. Nor does exact algebraic rank imply usable reconstruction at the available noise level. The current strict force allowances cover the three projected channels, not 64 certified force components. Existing full-field position and velocity arrays are therefore a possible input to a new force receiver, not already received complete-force labels.

Proposition S16. Eleven normal labels, or one after full-force reuse. *There is a nonzero direction w satisfying $Bw = 0$ and $V^T K_0 w = 0$. Put $k_w = w^T K_0 w > 0$. On*

$$x = V\ell u + tw, \qquad e = a(u) + \delta, \qquad \delta = t^2 k_w/(2E_s) > 0, \tag{S71}$$

the normalized directional energy-gradient label is

$$\mathcal{L}_w W(u,t) := \frac{w^T \nabla W(x)}{t k_w} = \partial_e \Phi(u, a(u) + \delta). \tag{S72}$$

For an exactly weighted-degree-four target P, subtract any known restricted representative and write its remaining part as AQ. Take the ten quadratic interpolation nodes

$$0, \quad \pm h e_i \ (i = 1,2,3), \quad h(e_i + e_j) \ (i < j), \qquad h \neq 0,$$

at one positive offset δ_1, and one additional label at $u = 0$, $\delta_2 > 0$, $\delta_2 \neq \delta_1$. These eleven labels identify all eleven coefficients of Q. The absolute determinant of their design matrix, with the ordinary quadratic basis, is

$$16|h|^{15}|\delta_2 - \delta_1| > 0. \tag{S73}$$

Eleven scalar linear labels are necessary when only the restricted 31 coefficients have been received. If instead the old complete-force design has rank 41, one label (S72) at $u = 0$ and any $\delta > 0$ completes its rank to 42.

Proof. The stacked constraints on w have at most six independent rows in dimension 64, so their common kernel has dimension at least 58. Positive definiteness gives $k_w > 0$. Expansion of E_0 gives (S71). Along w, the coordinate u is constant and $D_w e = t k_w/E_s$, proving (S72). For $Q = c_0 + c^T u + u^T C u + c_e e$,

$$\partial_e (AQ)\big|_{e=a(u)+\delta} = c_0 + c^T u + u^T C u + c_e\{a(u) + 2\delta\}.$$

At fixed δ_1 the first ten rows receive an ordinary quadratic. The ten-node quadratic evaluation determinant has absolute value $8|h|^{15}$: the three signed-axis pairs give $2|h|^3$ each and the three mixed nodes give $|h|^2$ each. Subtracting the $u = 0$ first row from the last row isolates $2(\delta_2 - \delta_1)c_e$; the known $a(u)$ column is a combination of the quadratic columns. This proves (S73). The dimension lower bound is elementary rank–nullity. In the rank-41 case the remaining term is cA^2, whose normalized normal derivative at $u = 0$ is $2c\delta$.

This direction also proves that the 42 feature polynomials give 42 distinct physical jets: substitution $e = a(u) + k_w t^2/(2E_s)$ is injective, by its nonzero highest power of t^2. All native matrices are rational source objects, so a rational w can be constructed. Rational nonzero t_1, t_2 with different squares give positive rational offsets and an exact rational interpolation design; taking a square root is unnecessary for its construction.

Finite labels require a separately bounded remainder. For an analytic target write $\Phi = P + R$, where the physical terms of R start at degree five. At nonzero interpolation nodes, normal observations receive $\partial_e P + \partial_e R$, not the jet derivative alone. If received measurement/force-reconstruction errors are η_i and separate valid tail allowances are $\rho_i \geq |\partial_e R(u_i, a(u_i) + \delta_i)|$, then for the eleven-label design matrix D_N the coefficient error satisfies

$$|\widehat{q} - q| \leq |D_N^{-1}|(\eta + \rho).$$

Uncertainty of an already received restricted representative contributes its own additional propagated label allowance. Equivalently one can receive all 42 rows together with their joint error matrix. Analyticity alone supplies no numerical ρ_i. The prior $R = 0$ is a genuine finite-degree target restriction, and must not be inferred from the network architecture or from a restricted $M_5 = 0$ statement.

For example, set $s = t\sqrt{k_w/(2E_s)}$ and consider the dimensionless lifted remainder $\widetilde{R}(u, s) = R(u, a(u) + s^2)$. If every ordered fifth partial of the lifted target is bounded by M_5 on the relevant convex segments, its Taylor remainder obeys

$$|\partial_e R(u, a(u) + s^2)| \le \frac{M_5(\|u\|_1 + |s|)^4}{48|s|}, \qquad s \neq 0.$$

This follows by Taylor's formula for the derivative in s, followed by division by $2s$. This is an additional target bound on the lifted four-dimensional domain, not a candidate-only derivative bound or a bound on the three-dimensional restriction.

Finally, the stipulated scalar family $W_* = \beta E_0$ corresponds to $\Phi_* = \beta e$. Its global form fixes the otherwise unobserved directions by assumption; current loss-to-parameter and family KAM certificates are valid precisely with that source prior. Relaxing it requires the additional observable rows and tail/error allowances above, then an independently received full-coordinate spectral and cubic-corrected twist criterion. This section proves neither that the new labels are already available nor that their eventual errors meet such a KAM tolerance.

Exact-source instance for the current full-coordinate model. For the retained native rational source, the elastic matrix has 64 positive exact LDL pivots, $BV = I_3$, and $V^T K_0 V = K_3$ exactly. The retained eleven nodes and their exact gradient left inverse receive the rank-31 restriction in Proposition S14. An exact-source calculation gives

$$\begin{aligned} \operatorname{rank}(K_0 V - B^T K_3) &= 3, \\ n_i \neq 0 \quad (i = 1, \dots, 11), & \\ \operatorname{rank}\big[u_i^\alpha\big]_{i=1,\dots,11,\ |\alpha|\le 2} &= 10. \end{aligned} \tag{S74}$$

Consequently the 704 complete-force rows at those same nodes would have rational rank 41, with kernel $\operatorname{span}\{A^2\}$. One additional normalized normal-gradient row at $u = 0$ and a positive off-trial energy offset raises the rank to 42. These statements describe the design, not an already received set of complete-force observations.

The rank calculation does not use a floating-point singular-value threshold. All source rationals are reduced modulo each of the two primes $p_1 = 1000000007$ and $p_2 = 1000000009$; every used denominator is nonzero in each field. The retained 41-by-41 full-force minors have residues 518511633 and 46316120, respectively. After the new normal row, the retained 42-by-42 minors have residues 541265412 and 980655374. A nonzero modular minor proves that the corresponding rational minor is nonzero. The exact nonzero vector A^2 annihilated by all 704 old rows supplies the matching upper bound 41; the augmented matrix has only 42 columns. The exact restriction and alias identities, the active quadratic evaluation rank 10, and the normal-geometry rank 3 were also checked.

For a constructive normal design, exact rational row reduction of $[B; V^T K_0]$ gives rank 6, so its common kernel has dimension 58. A retained rational w satisfies both zero constraints and $k_w = w^T K_0 w > 0$. Taking $h = 1/4$, $t_1 = 2^{-16}$, and $t_2 = 2t_1$ gives

$$\delta_1 = \frac{k_w t_1^2}{2E_s} > 0, \qquad \delta_2 = 4\delta_1, \qquad |\det D_N| = 16h^{15}(\delta_2 - \delta_1) > 0.$$

The exact direction, all interpolation rows and their inverse, the selected modular minors, and the exact full-force design are retained as reproducibility artifacts. Physical admissibility of these new configurations and useful force-reconstruction precision remain to be received. No new COMSOL solution, force measurement, or training run is part of this calculation.

Thus reuse of complete forces at the old configurations, if they can be received with valid errors, could replace ten of the eleven new scalar labels in Proposition S16. The existing certified allowances still cover only the three projected force channels. Neither the rank result nor the retained full-field position/velocity histories supplies the missing complete-force error bounds or the analytic target-tail allowances. Membership of an unknown material in the four-feature class remains a separate model assumption, and no unknown-target KAM conclusion follows from these rank certificates alone.

S9 Normal-force bounds from the full-field observations

Use the exact source normalizations $\ell = 1/1000$ and $E_s = 3/500000000000$, and the four-feature notation $u = Bx/\ell$, $e = x^T K_0 x/(2E_s)$, $a(u) = \ell^2 u^T K_3 u/(2E_s)$, and $A = e - a(u)$. For a fixed old node $q_i = \ell u_i$, define

$$x_i = Vq_i, \quad n_i = (K_0 V - B^T K_3)q_i, \quad d_i = M^{-1} n_i, \quad c_i = d_i^T n_i > 0, \quad r_i = d_i/c_i. \tag{S75}$$

The exact source has $n_i \neq 0$ at all eleven retained nodes. Since $B = (V^T M V)^{-1} V^T M$ and $BV = I$, one has $V^T n_i = 0$, $Bd_i = 0$, and $r_i^T K_0 x_i = 1$. Here r_i is a normalized force functional in the stored SI coordinate convention; it is not an ordinary SI displacement direction to be added to the state. The scalar label $r_i^T \nabla W$ is dimensionless. These are identities for the exact source matrices, not identities asserted for every rounded implementation buffer.

Proposition S17. A normal force component. *Suppose a material correction has the polynomial decomposition*

$$W(x) = E_s\{R(u) + A(u,e)Q(u,e)\},$$

where R is any restricted representative. Then

$$r_i^T \nabla W(x_i) = Q(u_i, a(u_i)). \tag{S76}$$

In particular, the representative contributes exactly zero. Writing $Q = S(u) + cA$ identifies an ordinary quadratic S with ten coordinates; the coefficient of A^2 remains unobservable on the trial space.

Proof. At x_i, $A = 0$ and $E_s \nabla_x A = n_i$, whereas $\nabla_x R = B^T \nabla_u R/\ell$. The identities $Br_i = 0$ and $r_i^T n_i = 1$ give the result. The substitution $e = a(u)$ maps the eleven-dimensional Q space onto the ten-dimensional ordinary quadratic space with kernel $\mathrm{span}\{A\}$.

For an arbitrary fixed direction r, consider the declared mathematical initial-value problem

$$M\ddot{x} + D\dot{x} + Kx + b\, g(b^T x, t) = 0, \quad K = \tfrac{53}{50} K_0, \quad D = \alpha M_0, \tag{S77}$$

with the same exact source-decimal magnetic drive and coefficients as the retained full-coordinate Hermite certificates. Each exact solution starts at its own saved first state. This is the declared simulation reference; a tube for this equation is not a certificate for a general unknown constitutive law or for the compiled COMSOL equation.

Let T be the first saved time step and keep the already fixed Richardson windows $(T, 2T, 4T)$. With endpoint weights $(-7/4, 8/3, -1, 1/12)$ at nodes $(0, 1, 2, 4)$ and cell weights $w = (7/4, -11/12, 1/12, 1/12)$, form the observation-only signed label

$$L_r^\sigma = -\frac{1}{T} \sum_{j \in \{0,1,2,4\}} a_j r^T M v^\sigma(t_j) - \sum_{k=0}^{3} \frac{w_k h_k}{2T}\, r^T b\{g_{\mathrm{saved}}(t_k) + g_{\mathrm{saved}}(t_{k+1})\}. \tag{S78}$$

Only saved positions, velocities, times and magnetic inputs, and known nominal source matrices enter this label. The pair label for the material component is

$$y_i = \tfrac{1}{2}(L_{r_i}^+ + L_{r_i}^-) - r_i^T K_0 x_i = \tfrac{1}{2}(L_{r_i}^+ + L_{r_i}^-) - 1.$$

The label uses no material correction coefficient. No old projected-force allowance is reused.

Let Q^σ denote the retained cubic Hermite interpolant and let R_k^σ be its previously certified all-prefix energy-tube radius:

$$\|(x - Q, \dot{x} - \dot{Q})\|_{K,M} \leq R_k^\sigma.$$

This full-coordinate tube is independent of the output projector. Define

$$L_r^2 = r^T K r + r^T D M^{-1} D r, \qquad A_{D,r}^2 = r^T D M^{-1} D r, \tag{S79}$$

$$A_{KM,r}^2 = r^T K M^{-1} K r, \qquad J_k^\sigma = \sum_{j \le k} h_j R_j^\sigma. \tag{S80}$$

The newly calculated directional allowance is

$$\Gamma_r^\sigma = \sum_k \frac{|w_k| h_k}{T} \min\{L_r R_k^\sigma,\, A_{D,r} R_k^\sigma + A_{KM,r} J_k^\sigma\}. \tag{S81}$$

Both bounds follow by Cauchy–Schwarz in the certified energy metric; the second also uses $x(0) - Q(0) = 0$ and integration of the velocity error. All gains in this expression are recomputed for r_i.

Let C_r^σ be the exact weighted integral of $r^T(KQ^\sigma + D\dot{Q}^\sigma)$. Let $B_{3,r}^\sigma$ bound the third time derivative of $r^T(Kx + D\dot{x})$ on the retained tubes. The existing generic full-coordinate derivative recurrence is evaluated with the new row $[r^T K, r^T D]$; its old three projected outputs are not used. Average signed quantities with an overbar and put

$$I_{0,r} = \left| r^T \{K(\bar{x}_0 - x_i) + D\bar{v}_0\} \right|. \tag{S82}$$

The fixed receiver then gives

$$|y_i - \tfrac{3}{50}| \le \eta_i := |\bar{L}_r - \bar{C}_r| + \bar{\Gamma}_r + \bar{B}_{3,r} T^3/3 + I_{0,r} \tag{S83}$$

for the declared reference equation. Magnetic sample and quadrature discrepancies are already included in the raw-minus-Hermite term. The exact initial discrepancy is retained even when the saved signed states nearly cancel; it is not set to zero by the intended preparation. Arithmetic enclosure radii are included when a saved midpoint is used.

For the eleven ordinary quadratic evaluations, retain a deterministic exact left inverse J_N of the eleven-by-ten evaluation matrix. For a genuinely degree-four member of the feature target class, the coefficient enclosure is

$$|\widehat{s} - s| \le |J_N| \eta.$$

If the target is analytic rather than exactly degree four, independently certified normal tail allowances must be added before this propagation. The reference-equation certificate above does not provide such unknown target bounds. A finite bound may also be too large to identify small nonlinear coefficients; such rows remain in the result without changing the projector, time window, or retained configurations.

For another target equation, the observation-only label formula remains well defined, but its error certificate must be replaced by a tube for that equation, valid directional force/tube sensitivity and third-time- derivative bounds, and an initial-state allowance for that target. These quantities are not inherited from the declared linear material reference merely because the observation files are the same.

Bounds at all eleven configurations. The normal-force construction uses both velocity signs and the fixed $(1, 2, 4)$ windows at all eleven prescribed nodes. Directional gains, third-time-derivative bounds and initial-state allowances are computed for each r_i using the full-coordinate Hermite tubes of the declared reference IVPs. The original three projected-force error bounds are not reused for these new directions.

Node 0 is a particularly weak normal channel: $c_0 \simeq 2.063\,10^{-18}$. Its endpoint-momentum and sampled-magnetic contributions are approximately -633648.03 and 633653.23 before the nominal subtraction of one. These large additive values explain the cancellation in this normalization; they are not separate noise bounds. The certified error decomposition at that node is $|\bar{L} - \bar{C}| \simeq 4.15706$, $\bar{\Gamma} \simeq 0.0238312$, $\bar{B}_3 T^3/3 \simeq 0.0310829$, and $I_0 \simeq 4.13\,10^{-7}$. Thus the raw-minus-Hermite term dominates this certificate. The bound alone does not establish an actual-error lower bound. Separately, comparison with the declared reference value gives a posterior label discrepancy of approximately 4.14617; this comparison was not used to choose a window or a projector.

Bounds for all ten quadratic coordinates. The original fixed first-lexicographic left inverse is retained. It gives maximum coefficient radii by ordinary degree $0, 1, 2$ of approximately 1.77923, 17.7908, and 56.9309. The predetermined refinement

Node	Normal material label y_i	Certified error bound η_i
0	4.206169974	4.211977
1	0.059962646	$3.788056\,10^{-5}$
2	0.059999578	$4.428778\,10^{-7}$
3	0.059999468	$5.584729\,10^{-7}$
4	0.059999465	$5.613862\,10^{-7}$
5	0.059999416	$6.126460\,10^{-7}$
6	0.059999579	$4.416293\,10^{-7}$
7	0.059999575	$4.459154\,10^{-7}$
8	0.059999468	$5.581601\,10^{-7}$
9	0.059999634	$3.839009\,10^{-7}$
10	0.059999581	$4.398915\,10^{-7}$

Table S1. Dimensionless normal-force labels on the existing records. Labels are rounded displays; error bounds are rounded upward from the certified exact rational bounds and include arithmetic midpoint radii. All rows are retained. Their error certificate concerns the declared reference IVPs, whose material normal value is $3/50$.

rule is to enumerate every ten-node subset of the eleven-node evaluation matrix, skip only exact singular matrices, propagate each valid coefficient interval, and intersect all resulting intervals. All eleven subsets are exactly nonsingular in this instance. Each interval is a necessary consequence of the same simultaneous label constraints, so their intersection remains an enclosure. No preferred subset is selected by its error, and all eleven observations remain in the construction.

For the ordinary quadratic coefficients of $S(u) = Q(u, a(u))$, the maximum intersection half-widths satisfy

$$r_S^{(0)} < 1.3700\,10^{-4}, \qquad r_S^{(1)} < 1.5880\,10^{-4}, \qquad r_S^{(2)} < 9.7100\,10^{-4}.$$

These are radii about the certified, generally biased coefficient centres. They are not bounds on the coefficients' absolute values. For example, the degree-two coefficient intervals together give the distinct absolute bound $1.8780\,10^{-3}$ in this normalization. The constant interval contains $3/50$, and every nonconstant interval contains zero, as required for the declared linear material reference.

This calculation turns the old full-field records into additional directional force information for that reference equation and an explicit ten-coordinate coefficient interface. The A^2 coordinate remains unobserved on every old trial configuration. An unknown four-feature analytic target additionally needs target-valid trajectory and force-error bounds and a third-order remainder allowance for its normal-force function S (or an equivalent lifted-potential tail bound). The vanishing remainder of the declared scalar material family is a source prior, not a conclusion for an unknown target. No interval here certifies a full-coordinate Birkhoff-twist tolerance or transfers a KAM result to an unknown material.

S10 Trajectory information about the last feature-jet coordinate

Keep the exact source definitions $u = Bx/\ell$, $e = x^T K_0 x/(2E_s)$, $a(u) = \ell^2 u^T K_3 u/(2E_s)$, and $A = e - a(u)$. At each of the eleven old ideal initial configurations, retain the fixed normal functional r_i constructed without trajectory outcomes, with $Br_i = 0$ and $r_i^T K_0 V q_i = 1$. It is not adjusted at later times. Write the weighted-four material polynomial as

$$P(u, e) = R(u) + A(u, e)S(u) + c_A A(u, e)^2,$$

where R has the 31 restricted coefficients and S is an ordinary quadratic with ten coefficients.

Proposition S18. Cancellation along a full-coordinate trajectory. *For every full-coordinate state x, not only the initial trial plane,*

$$r_i^T \nabla_x (E_s P) = J_i(x)\{S(u(x)) + 2c_A A(x)\}, \qquad J_i(x) = r_i^T K_0 x. \tag{S84}$$

Proof. $D_{r_i} u = 0$ and $E_s D_{r_i} A = r_i^T K_0 x$. Apply the product rule. Both $D_{r_i} R$ and $D_{r_i} S$ vanish.

Thus uncertainties in the 31 coefficients of R do not contaminate this normal observable. On a trajectory that leaves the trial plane, the remaining coefficient has sensitivity $2A(x)J_i(x)$. Its pointwise nonzero value, or a nonzero numerical matrix rank, does not establish a coefficient bound at the available error level.

For a prescribed horizon $T_n = t_n - t_0$, define $\mathcal{W}_n f = T_n^{-1} \int_{t_0}^{t_n} f(t)\,dt$. The original three horizons have $n = j, 2j, 4j$. Keep both the three individual averages and their prescribed combination

$$\mathcal{R}_j = \tfrac{8}{3}\mathcal{W}_j - 2\mathcal{W}_{2j} + \tfrac{1}{3}\mathcal{W}_{4j}.$$

All signed records and all predetermined nodes are retained. For a full-coordinate equation with linear damping αM_0, the weak observation has the linear form

$$Y_{i,n} = \sum_{|\gamma|\le 2} G_{i,n,\gamma} s_\gamma + H_{i,n} c_A + D_{i,n}\alpha, \tag{S85}$$

where, using the same average or fixed combination on each term,

$$G_{i,n,\gamma} = \mathcal{W}_n[J_i(x)u(x)^\gamma], \quad H_{i,n} = 2\mathcal{W}_n[A(x)J_i(x)], \quad D_{i,n} = \mathcal{W}_n[r_i^T M_0 \dot{x}].$$

Here Y consists of the negative momentum difference, the known magnetic integral and the known nominal elastic integral. In a reference-only branch with a stipulated known $D = \alpha M_0$, subtract the last term. If α is unknown, it remains a twelfth column. Pairing initial velocities does not remove its integral on an entire trajectory. State-dependent learned damping is a different target class and is not represented by this scalar nuisance column.

All polynomial columns can be integrated on a cubic Hermite centre: J and u have time degree three, A has degree six, and Ju^γ and $2AJ$ have degree at most nine. Saved binary64 observations are interpreted exactly before any rounding. An outward interval evaluation of these integrals certifies arithmetic for the saved Hermite centre, not closeness to a physical trajectory.

What a valid trajectory tube would add. Suppose on a cell a target-valid tube gives $\|x - Q\|_K \le R$ and $\|\dot{x} - \dot{Q}\|_M \le R$. The nominal operator $C = K_0 - B^T K_3 B$ gives $A(x) = x^T Cx/(2E_s)$. With centre bounds $U_k \ge |u_k(Q)|$, $J_Q^{\max} \ge |J(Q)|$ and $A_Q^{\max} \ge |A(Q)|$, put

$$\epsilon_{u_k} = R\|(B_k/\ell)K^{-1/2}\|, \quad \epsilon_J = R\|r^T K_0 K^{-1/2}\|,$$
$$\epsilon_A = \frac{R\sup\|K^{-1/2}CQ\| + \frac{1}{2}R^2\|K^{-1/2}CK^{-1/2}\|}{E_s}.$$

These yield the following cellwise errors:

$$|\Delta(Ju^\gamma)| \le \epsilon_J \prod_k (U_k + \epsilon_{u_k})^{\gamma_k} + J_Q^{\max} \sum_k \gamma_k \epsilon_{u_k} (U_k + \epsilon_{u_k})^{\gamma_k - 1} \prod_{l \ne k} (U_l + \epsilon_{u_l})^{\gamma_l},$$
$$|\Delta(2AJ)| \le 2\{\epsilon_A J_Q^{\max} + \epsilon_J (A_Q^{\max} + \epsilon_A)\}.$$

Zero-exponent derivative terms are omitted. Integrate with absolute values of the fixed window weights. Endpoint momentum errors, magnetic integration/sample errors, nominal elastic errors and, when appropriate, damping errors must be certified as well. A three-component force allowance cannot substitute for these new functional errors.

For one known-damping row, let H_Q be the centre sensitivity, ϵ_H its tube/arithmetic error, and let E include observation error and all propagated S-coefficient and column uncertainties. A finite coefficient bound requires a separated denominator:

$$|H_Q| > \epsilon_H, \qquad |c_A| \le \frac{|Y_Q - G_Q \widehat{s}| + E}{|H_Q| - \epsilon_H}.$$

The sign or numerical rank of H_Q alone is insufficient. For an unknown scalar damping coefficient, the augmented columns must also separate c_A from α at the certified precision.

The old verified tubes cover only the first four saved cells of each selected record and only the declared reference equation. Longer predetermined windows may be examined for saved-data geometry, but do not inherit those tube/error values. Likewise an analytic four-feature target needs a separately bounded normal-force remainder beyond $S + 2c_A A$; a

general unknown 64-dimensional potential need not have this feature factorization. None of these target assumptions follows from the existing twenty-two simulation records.

Finite intervals on the first four cells. Only the $j = 1$ windows have the certified first-four-cell tube and functional-error bounds. Among the fixed rows, 34 known-damping rows have a separated interval denominator. Intersecting every such valid interval gives

$$-12.872332490981 \le c_A \le 12.872689597309. \tag{S86}$$

When a single unknown constant α is retained, all 946 row pairs are examined, of which 355 have a separated interval determinant. Their simultaneous necessary intervals give

$$-16.009760737420 \le c_A \le 16.009085357190. \tag{S87}$$

The corresponding constant-damping interval is $-798.726283550263 \le \alpha \le 800.418421268464$ in s^{-1}. These outer decimal enclosures are displays of exact rational interval endpoints. They are not confidence intervals. Every valid row or row pair enters the intersection; an inseparable row is retained without dividing by an interval containing zero. Independent interval reconstruction gives the same exact intersections.

The intervals are finite for the declared reference trajectory problem but remain far wider than the A^2 twist tolerance in Supplementary Section S17. The longer fixed windows $j > 1$ supply saved-Hermite geometry only; no target-valid trajectory or force-error bound is assigned to them. Thus later off-plane motion supplies algebraic sensitivity to A^2 without establishing its smallness or an unknown material transfer.

S10.1 Composition of observation, preparation, and jet allowances

The finite-window receiver approximates a force at the prepared rounded full state x_i^{prep}, whereas the polynomial design uses the exact point $\bar{x}_i = V\ell z^{(i)}$. This distinction must be retained. Let

$$F_c(x) = (\ell/E_s)V^T\nabla\mathcal{W}_*(x), \qquad w(z) = \mathcal{W}_*(V\ell z)/E_s.$$

Then $\nabla_z w(z^{(i)}) = F_c(\bar{x}_i)$. Suppose a separate preparation allowance satisfies

$$\|F_c(x_i^{\mathrm{prep}}) - F_c(\bar{x}_i)\|_\infty \le d_i.$$

For example, $d_i = L_{c,i}\|x_i^{\mathrm{prep}} - \bar{x}_i\|_\infty$ suffices if F_c has the corresponding Lipschitz constant on the joining segment. This allowance concerns the correction force: the nominal force is already subtracted at the actual prepared position in the finite label.

Corollary S19. A conditional physical-observation to jet interface. *For every design position, suppose the hypotheses and the common-clock form of Proposition S6 hold, with valid constants for that position. Let r_i enclose the arithmetic error of the reported label midpoint, and define*

$$b_i = I_{0,i} + \frac{B_{3,i}T_i^3}{3} + \frac{5B_{m,i}h_i^2}{12} + 5\epsilon_{m,i} + \frac{11\epsilon_{p,i}}{2T_i} + d_i + r_i.$$

Then each reported component $\hat{y}_{ij}$ approximates $\partial_j w(z^{(i)})$ with absolute error at most b_i. Under the fifth-derivative and equilibrium assumptions of Proposition S13, the same exact left inverse gives

$$|(J\hat{y} - c)_\alpha| \le \sum_{i,j} |J_{\alpha,(i,j)}| \left(b_i + \frac{M_5}{24}\|z^{(i)}\|_1^4\right).$$

For independently computed candidate coefficients $c_{\theta,\alpha}$, a sufficient candidate-to-target allowance is this right-hand side plus $|c_{\theta,\alpha} - (J\hat{y})_\alpha|$. If the two signs use different clocks, their individual clock-dependent constants must replace the common-clock terms before this composition.

Proof. Apply the finite-window inequality to the prepared-state target, then the preparation and arithmetic triangle inequalities. This gives the componentwise force allowance b_i at the exact design node. Substitute it into (S63); the final statement is another scalar triangle inequality. No statistical or optimizer property is required for these deterministic implications.

The stored-data interval radii certify r_i. Comparisons against a declared constitutive formula do not by themselves certify $\epsilon_{p,i}, \epsilon_{m,i}, B_{3,i}, B_{m,i}$ or M_5 for an unknown physical target. A recorded-state initial-defect calculation also does not supply a true-state measurement bound. All eleven design positions and their stated error and tail allowances are required for this receiver. Their actual reception for the declared quadratic family is supplied below; a general unknown physical target still requires its own tail and measurement bounds. Once admitted, these ordinary coefficient intervals still have to pass the separate spectral, Birkhoff, analytic-domain and KAM gates; they do not inherit the candidate-only KAM conclusion automatically. The KAM verifier receives the total potential jet after restoring the known nominal elastic and prescribed magnetic coefficients, with the physical factors $E_s \ell^{-|\alpha|}$ applied consistently.

For the declared linear material family, Proposition S11 provides an alternative to the separate endpoint and magnetic budgets. At a reference design node where that proposition has been instantiated, the force error may instead be received componentwise as

$$b_{ij} = d_{Q,ij} + \Gamma_{ij} + \frac{B_{3,ij} T_i^3}{3} + |\delta^{\mathrm{rec}}_{0,ij}|,$$

provided the nominal subtraction and target use the same reference node. Any additional reported-midpoint radius must also be retained. The componentwise b_{ij} replaces b_i in the left-inverse sum above; the fifth-derivative tail and the other design-node requirements remain. The preceding numerical example received only the direction-3 pair. The next subsection receives every required node separately. No physical measurement allowance is supplied by either reception.

S10.2 Instantiated eleven-node coefficient intervals and their limits

The actual label midpoints $\hat{y} \in \mathbb{R}^{33}$ and their componentwise continuous-force and arithmetic allowances η now instantiate the exact left inverse in Proposition S13. For the declared simulation family $\mathcal{W}_*(x) = \beta x^T K_0 x/2$, the reduced correction is quadratic, its origin gradient vanishes and $M_5 = 0$. These are explicit constitutive-family assumptions, not properties inferred from eleven point observations. In that family,

$$c_* = J\hat{y} + e_c, \qquad |(e_c)_\alpha| \le r_\alpha := \sum_{i,j} |J_{\alpha,(i,j)}| \eta_{ij}.$$

Every one of the 31 received intervals contains the corresponding independently evaluated declared coefficient. Their maximum radii within degrees two, three and four are bounded respectively by

$$4.282817 \times 10^{-7}, \qquad 8.141379 \times 10^{-7}, \qquad 1.808799 \times 10^{-6}.$$

The known magnetic quartic has maximum normalized ordinary coefficient approximately 5.363314×10^{-9}. After restoring those prescribed coefficients, all fifteen total quartic coefficient intervals contain zero. This is an information and sign-ambiguity diagnostic; it is not itself a calculation of the Birkhoff twist determinant. In particular, finite design rank and small force error do not establish the uniform twist and analytic conditions for an unknown physical target. For a general C^5 target, the independent fifth-derivative tail in the preceding receiver must also be restored.

S11 A coherent finite-data ambiguity in the quartic material jet

The following result concerns identifiability from the received finite supervision interface. It does not replace the stipulated quadratic constitutive family used in the target-family theorem. The actual interface contains eleven configurations, thirty-three force components, and thirty-one ordinary potential coefficients.

Write $q = \ell z$, with $\ell = 10^{-3}$ and $E_s = 6 \times 10^{-12}$ J. Let z_i be the eleven exact design nodes, $y \in \mathbb{R}^{33}$ the saved binary64 correction-force centers interpreted as exact rationals, and $\eta \in (0, \infty)^{33}$ their original component-specific received allowances. Let G be the sampled gradient matrix for all ordinary monomials of degrees two through four. Its saved rational

left inverse J satisfies $JG = I_{31}$. The received coefficient center and radius are, without an independence assumption,

$$c^{\mathrm{obs}} = Jy, \qquad r^{\mathrm{obs}} = |J|\eta.$$

The prescribed magnetic potential remains $\kappa(1-\cos(d^T q))$ with the exact native positive κ and native port vector d. The material correction alone is denoted by W.

Proposition S20. One common admissible error vector with vanishing quartic twist. *For the received data, there is a parameter $\widetilde{\beta} \in [0, 0.1]$ such that the entire material correction*

$$\widetilde{W}(q) = \frac{\widetilde{\beta}}{2} q^T K_{03} q + \frac{\kappa}{24}(d^T q)^4 \tag{S88}$$

has zero origin gradient and identically zero fifth material derivative, and satisfies all thirty-three original force intervals and all thirty-one original coefficient intervals simultaneously. In particular, the coefficient inequalities arise from the same error vector as the force inequalities.

The conservative Hamiltonian

$$\widetilde{H}(q,p) = \tfrac{1}{2} p^T M_3^{-1} p + \tfrac{1}{2} q^T K_{03} q + \widetilde{W}(q) + \kappa(1-\cos(d^T q))$$

has a positive definite quadratic part and satisfies $\widetilde{H}_3 = \widetilde{H}_4 = 0$ exactly. Consequently its degree-four Birkhoff twist is the zero matrix. Thus the finite force/jet intervals, even together with the stated zero-gradient and $M_5 = 0$ material-tail conditions, do not uniformly imply the nondegenerate degree-four local KAM criterion in a material class admitting quartic corrections.

Proof. All calculations in this proof use exact rational arithmetic. Let $a \in \mathbb{Q}^{31}$ be the ordinary coefficient vector of $\ell^2 z^T K_{03} z/(2E_s)$. For $|\alpha| = 4$, put

$$b_\alpha = \frac{\kappa \ell^4}{E_s \alpha!} d^\alpha,$$

and set all other entries of b to zero. The ordinary coefficients and sampled gradients of (S88) are $c(\beta) = \beta a + b$ and $f(\beta) = G(\beta a + b)$, respectively. For each force component form

$$|\beta(Ga)_m + (Gb)_m - y_m| \le \eta_m.$$

For each coefficient also form

$$|\beta a_\alpha + b_\alpha - (Jy)_\alpha| \le (|J|\eta)_\alpha.$$

When the multiplier of β is nonzero, each inequality gives an exact closed rational interval; when it vanishes, the residual inequality is checked directly. Intersecting all sixty-four constraints and $[0, 1/10]$ gives a nonempty interval. Its exact endpoints are retained in the machine-readable result. A decimal enclosure, wider than those exact endpoints, is

$$[\beta_-, \beta_+] \subset [0.05999999779279021,\ 0.05999999877671872].$$

Choose its exact rational midpoint as $\widetilde{\beta}$. Direct substitution verifies

$$\max_m \frac{|f_m(\widetilde{\beta}) - y_m|}{\eta_m} < 0.982670319892135, \qquad \max_\alpha \frac{|c_\alpha(\widetilde{\beta}) - (Jy)_\alpha|}{(|J|\eta)_\alpha} < 0.236855585657115.$$

For the same vector $e = f(\widetilde{\beta}) - y$, exact multiplication verifies $c(\widetilde{\beta}) - Jy = Je$, using $JG = I$. The coefficient constraints therefore preserve all cross-component correlations. They are not selected independently from their marginal intervals.

The polynomial $\widetilde{W}$ has no linear term and degree at most four, so $\nabla \widetilde{W}(0) = 0$ and $D^5 \widetilde{W} \equiv 0$. The exact Sylvester principal minors verify $K_{03} \succ 0$, and $1 + \widetilde{\beta} > 0$. Thus

$$\widetilde{H}_2 = \tfrac{1}{2} p^T M_3^{-1} p + \tfrac{1}{2} q^T [(1+\widetilde{\beta}) K_{03} + \kappa d d^T] q$$

is positive definite. Finally the Taylor expansion of the retained cosine gives

$$\widetilde{H} = \widetilde{H}_2 + \frac{\kappa}{720}(d^T q)^6 + O(\|q\|^8).$$

Both cubic and quartic homogeneous polynomials vanish before any linear canonical transformation. The cubic homological correction and the quartic resonant projection are therefore zero. The degree-four Birkhoff twist is exactly zero.

This is a coherent ambiguity of the finite deterministic supervision interface. It is not a claim that the alternative material reproduces all saved COMSOL time histories, is the actual physical material, or is realizable by the fixed network architecture. It does not contradict the target-family KAM theorem under the stronger hypothesis $W_* = \beta E_0$, which excludes the quartic term in (S88). A vanishing degree-four twist also does not imply absence of invariant tori: higher-order nondegeneracy may be relevant. The result identifies the particular criterion that cannot be inferred uniformly from this interface.

The executable certificate performs 1,289 exact checks. The protocol, source hashes, full common interval, coherent error vector, and actual successful child-process exit are retained together. No network update or new native solve is used in this construction.

S12 Exact finite-node ambiguity without a target-tail prior

The preceding quartic ambiguity uses the original nonzero force allowances and retains a zero fifth material derivative. A separate construction explains why improving finite-node precision alone does not replace an independent target-tail assumption in an unrestricted analytic class. Let $a_1, \dots, a_m \in \mathbb{R}^3 \setminus \{0\}$ be a finite set of configuration nodes and set

$$\Phi(z) = \prod_{i=1}^{m} \frac{\|z - a_i\|_2^2 \|z + a_i\|_2^2}{\|a_i\|_2^4}, \qquad \Delta w(z) = \frac{\kappa \ell^4}{24 E_s}(d^T z)^4 \Phi(z).$$

The squared norms in this expression are real polynomials and continue to complex arguments as sums of squares, without complex conjugation.

Proposition S21. An even analytic correction invisible to all retained forces. *The polynomial Δw is even and nonnegative on real configurations. Its gradient vanishes exactly at every a_i and $-a_i$. Its homogeneous terms of degrees zero through three vanish, and its fourth homogeneous term is $\kappa \ell^4 (d^T z)^4/(24E_s)$. Thus replacing the material correction $\beta q^T K_{03} q/2$ by*

$$W_{\mathrm{alt}}(q) = \tfrac{\beta}{2} q^T K_{03} q + E_s \Delta w(q/\ell)$$

preserves all these sampled forces and the positive equilibrium Hessian while cancelling the prescribed magnetic quartic. Its cubic and quartic total Hamiltonian terms, and therefore its degree-four Birkhoff twist, vanish. The alternative material remains real entire and even.

Proof. Each factor is an even polynomial, equals one at the origin, and has zero gradient there. At its own signed node, its value and gradient are zero because a squared Euclidean distance and its gradient vanish there. The product rule proves the node identities, including in the presence of other factors. Consequently $\Phi(z) = 1 + O(\|z\|^2)$, and

$$E_s \Delta w(q/\ell) = \tfrac{\kappa}{24}(d^T q)^4 + O(\|q\|^6).$$

This cancels the fourth term in $\kappa(1 - \cos(d^T q))$. There is no cubic term, so the cubic homological correction also vanishes. The positive quadratic part is unchanged. Every factor is nonnegative on real arguments; hence the material addition is nonnegative there. Polynomiality gives the claimed entire continuation.

For the eleven actual nonzero rational nodes, exact arithmetic verifies all twenty-two signed-node identities and the fifteen ordinary quartic coefficient cancellations. The resulting polynomial has degree 48. Independent univariate restrictions also exhibit nonzero fifth derivatives inside the configuration cube. Thus this example has $M_5 \neq 0$ on that cube and is explicitly outside the stipulated zero-tail material family. Its sampled correction forces agree exactly with those for $\beta = 3/50$, whose

compatibility with the saved force allowances is checked separately. Its higher-order tail aliases the fourth coefficient under the finite receiver: if b is its fourth coefficient vector and h its sampled higher-order force tail, then $Jh = -b$. The receiver's independent target-tail term cannot be omitted even if the retained forces were measured exactly.

This construction concerns the information in finite pointwise forces. It does not assert that the alternative material generates the saved transient records or belongs to the fixed network class. Neither this example nor a zero degree-four twist proves absence of invariant tori; they delimit inference by the particular nondegenerate quartic criterion.

S13 A finite received-jet loss for the unchanged complex model

Let $\mathcal{A} = \{\alpha \in \mathbb{N}_0^3 : 2 \le |\alpha| \le 4\}$, so $|\mathcal{A}| = 6 + 10 + 15 = 31$. All coefficients below are ordinary monomial coefficients; $D^\alpha w(0) = \alpha!\, c_\alpha$. Write $q = \ell z$, $\ell = 1/1000$, and $E_s = 3/500000000000$. The conservative correction of the retained complex model is

$$W_\theta(q) = c_E E_0(q) + E_{\mathrm{net}} \left[N_\theta(q/Q_*, E_0(q)/E_{\mathrm{net}}) - N_\theta(0) - \sum_{j=1}^{3} \gamma_{\theta,j} q_j / Q_{*,j} \right],$$

where $E_0(q) = \frac{1}{2} q^T K_{03} q$ and $\gamma_{\theta,j} = \partial_j N_\theta(0)$. Here N_θ is the real readout of the existing split-tanh complex network, and E_{net} and Q_* are the saved checkpoint scales. Define $c_{\theta,\alpha}$ by the degree-four Taylor polynomial of $w_\theta(z) = W_\theta(\ell z)/E_s$. The prescribed magnetic potential is kept in the full Hamiltonian, but is not added to these correction coefficients or to the correction labels.

Let $\widehat{c} = Jy$ and $r \ge 0$ be the already received exact centers and radii from the fixed eleven-node force design. Assume that its received deterministic statement is

$$|c_{*,\alpha} - \widehat{c}_\alpha| \le r_\alpha, \qquad \alpha \in \mathcal{A}. \tag{JG1}$$

For the current receipt, the zero target fifth-derivative tail and $\nabla W_*(0) = 0$ follow from the explicit declared family $W_* = \beta E_0$. They are family priors, not conclusions learned from the eleven positions. A generic unknown target instead needs its own centering and fifth-derivative hypotheses in (JG1).

Fix $s_\alpha = \max\{r_\alpha, 10^{-8}\} > 0$ before any optimizer update, and define the original-center mathematical objective

$$L_{\mathrm{jet}}(\theta) = \frac{1}{31} \sum_{\alpha \in \mathcal{A}} \left(\frac{c_{\theta,\alpha} - \widehat{c}_\alpha}{s_\alpha} \right)^2. \tag{JG2}$$

The training inputs in (JG2) are the received centers and radii. No known-zero cubic or quartic coefficient is substituted as a label.

Proposition S22. Received-jet fitting bound. *For every parameter vector of the same model, (JG1)–(JG2) imply*

$$|c_{\theta,\alpha} - c_{*,\alpha}| \le |c_{\theta,\alpha} - \widehat{c}_\alpha| + r_\alpha \le s_\alpha \sqrt{31 L_{\mathrm{jet}}(\theta)} + r_\alpha \quad (\alpha \in \mathcal{A}). \tag{JG3}$$

If a reported joint score is $L_{\mathrm{joint}} = \lambda_{\mathrm{jet}} L_{\mathrm{jet}} + L_{\mathrm{other}}$ with fixed $\lambda_{\mathrm{jet}} > 0$ and $L_{\mathrm{other}} \ge 0$, then L_{jet} in the last expression may be replaced by $L_{\mathrm{joint}}/\lambda_{\mathrm{jet}}$.

Proof. The first inequality is the triangle inequality and (JG1). Each nonnegative summand in $31 L_{\mathrm{jet}}$ is at most their sum, which gives the second inequality after taking square roots. Finally $L_{\mathrm{jet}} \le L_{\mathrm{joint}}/\lambda_{\mathrm{jet}}$. No descent, convexity, optimizer convergence or independent sampling assumption is needed. A point-force loss by itself cannot replace L_{jet} in this argument.

The component residual in the first bound of (JG3) is normally tighter than the bound derived from a single aggregate loss. In a final source-bound check, let $[a_\alpha, b_\alpha]$ enclose the exact-real coefficient of the actual saved weights. A rigorous computable bound is

$$|c_{\theta,\alpha} - c_{*,\alpha}| \le \Delta_\alpha + r_\alpha, \qquad \Delta_\alpha = \max\{|a_\alpha - \widehat{c}_\alpha|, |b_\alpha - \widehat{c}_\alpha|\}. \tag{JG4}$$

For fixed exact positive scales, the rational expression $\overline{L} = (1/31)\sum_\alpha(\Delta_\alpha/s_\alpha)^2$ is an upper bound for (JG2). Thus it also validates the aggregate version of (JG3). Directly receiving the candidate coefficients in (JG4) requires no candidate fifth-order remainder. Candidate remainder and analytic-domain bounds may still be needed for later claims about functions or Hamiltonian transport beyond the finite jet.

The Torch polynomial implementation is optimizer arithmetic. Its float64 value is not itself a certified upper bound for (JG2). If it uses rounded centers $\widetilde{c}_\alpha$ and fixed rounded positive scales $\widetilde{s}_\alpha$, the corresponding exact-real loss yields instead

$$|c_{\theta,\alpha} - c_{*,\alpha}| \le \widetilde{s}_\alpha\sqrt{31\widetilde{L}_{\mathrm{jet}}} + r_\alpha + |\widetilde{c}_\alpha - \widehat{c}_\alpha|.$$

This statement still presupposes the exact-real candidate coefficient; final Arb reception supplies that link despite optimizer rounding of operators, ℓ, E_s and network calculations. Recording the actual normalization and rounding centers prevents an unreported change of loss.

At present the received maximum fourth-order coefficient radius is 1.808799×10^{-6} as an upward decimal bound. Even zero loss therefore does not guarantee that every received quartic error bound is below 10^{-6}. The jet is a deterministic transformation of the same eleven force records, not an independent additional dataset. Reducing (JG2) retains the full complex energy and dissipation heads; it neither replaces the architecture by a scalar prior nor supplies a physical-target spectral, twist or KAM certificate.

S13.1 Precision-weighted calibration of the final complex readout

For fixed hidden layers, let $h_\psi(s) \in \mathbb{C}^{12}$ be the last hidden representation. Its real scalar energy readout is $N_{a,b,\psi}(s) = \mathrm{Re}[(a+ib)^T h_\psi(s) + b_0]$. The centering operation in W_θ is linear in this readout: it removes its constant and its three linear configuration terms, while retaining the response through the quadratic energy feature. Consequently, with the saved feature scales and fixed hidden layers,

$$W_\theta(q) = \sum_{k=0}^{24} \xi_k \Phi_{\psi,k}(q), \qquad \xi = (c_E, a_1, \dots, a_{12}, b_1, \dots, b_{12}).$$

The output bias cancels under centering and is held fixed as a gauge. The $\Phi_{\psi,k}$ include the nominal energy skip and the centered real and imaginary hidden readouts, with their actual signs.

Proposition S23. A finite linear fitting block. *For fixed ψ, the 33 normalized force components at the eleven design nodes and the 31 normalized correction coefficients form a linear map $A_\psi\xi \in \mathbb{R}^{64}$. If $\hat{y}$ denotes the force midpoints, $\tilde{c}$ the saved jet centers, and $\eta_i > 0$ and $\tilde{s}_\alpha > 0$ the fixed received scales, the numerical block objective is the weighted least-squares expression*

$$\left\| D^{-1}\left(A_\psi\xi - \begin{pmatrix}\hat{y}\\ \tilde{c}\end{pmatrix}\right)\right\|_2^2, \qquad D = \mathrm{diag}(\eta_1, \dots, \eta_{33}, \tilde{s}_1, \dots, \tilde{s}_{31}). \tag{PW1}$$

This is a block of the unchanged complex model. It changes neither its function graph nor the number of stored real parameters.

Proof. The centered potential is linear in ξ. Differentiation at a fixed configuration and extraction of a Taylor coefficient are linear operations, so stacking their values gives $A_\psi\xi$. Positive diagonal scaling gives (PW1). No independence of the 64 rows is asserted: the jet rows are deterministic transformations of the same 33 force observations.

The implemented step uses float64 automatic differentiation to form A_ψ and a single least-squares solve, without a ridge penalty. Its numerical rank and condition estimate are diagnostics, not an exact-rank or exact-optimality certificate. The method was selected after the disclosed earlier fitting results; it is not a blind test. The step fixes the already trained hidden layers and the complete dissipation head. It performs one readout-block solve and zero gradient updates; the preceding full-layer training remains part of its documented history.

The force scales in (PW1) are observation-error allowances. The preceding gradient stage instead normalized point-force residuals by the observed signal RMS while normalizing coefficients by their much smaller received errors. The scores of

these two objectives are therefore not directly comparable. Each term of (PW1) is the square of a residual divided by its scale, namely $(r/s)^2$.

The proof of fit is independent of the numerical solver. At the ideal exact design node, let an outward enclosure $[f_i^-, f_i^+]$ contain the force component defined by the saved weights and exact native operators. Set $\delta_i^f = \max\{|f_i^- - \hat{y}_i|, |f_i^+ - \hat{y}_i|\}$. If $|f_{*,i} - \hat{y}_i| \le \eta_i$, then

$$|f_{\theta,i} - f_{*,i}| \le \delta_i^f + \eta_i.$$

The corresponding coefficient statement is (JG4), with the appropriate saved-center rounding allowance. In particular, lying inside a data interval does not mean equality with its unknown target. Neither finite-node fit nor a small coefficient loss alone implies an analytic norm bound, a physical-target twist condition, or a complex-network advantage.

S14 A local KAM criterion for the actual conservative network

Consider the conservative restriction of the full-observation model,

$$H_N(q,p) = \tfrac{1}{2} p^{\mathsf{T}} A^{-1} p + U_N(q), \qquad A = A^{\mathsf{T}} > 0. \tag{S89}$$

Here $q, p \in \mathbb{R}^3$, and U_N includes the elastic, prescribed magnetic, and learned potential terms. The centering implemented in the network gives $\nabla U_N(0) = 0$. Dissipation and the time-dependent transverse magnetic input are set to zero in (S89). The statement below concerns this restriction as its own Hamiltonian system. It does not assume that the restriction is an invariant subsystem of the full finite-element dynamics.

Proposition S24. Local criterion based on the selected weights. *Suppose the real-coordinate continuation of U_N is holomorphic near $q = 0$, and suppose the following conditions hold for the same weights:*

1. *$K_N = D^2 U_N(0)$ is positive definite;*
2. *the positive generalized frequencies of (K_N, A) obey $k \cdot \omega_N \neq 0$ for all integer k with $0 < |k|_1 \le 4$;*
3. *the degree-four Birkhoff normal form is*

$$h_N(I) = \omega_N \cdot I + \tfrac{1}{2} I^{\mathsf{T}} T_N I, \qquad \det T_N \neq 0. \tag{S90}$$

Then every neighborhood of $(q,p) = (0,0)$ contains a positive-measure family of invariant Lagrangian three-dimensional tori of H_N with Diophantine frequencies. Neither a specified nonzero torus radius nor a specified frequency is asserted.

Proof. Positivity gives a real linear canonical change of coordinates to $H_2 = \sum_{j=1}^3 \omega_{N,j}(X_j^2 + Y_j^2)/2$. Let $a_j = (X_j + iY_j)/\sqrt{2}$, so that $\{a_j, \bar{a}_k\} = -i\delta_{jk}$ and $I_j = a_j \bar{a}_j$. For a homogeneous monomial $a^\alpha \bar{a}^\beta$,

$$\{H_2, a^\alpha \bar{a}^\beta\} = i\, \omega_N \cdot (\alpha - \beta) a^\alpha \bar{a}^\beta.$$

The second condition solves every cubic homological equation and all quartic equations except the action monomials $\alpha = \beta$. For $H_3 = \sum h_{\alpha\beta} a^\alpha \bar{a}^\beta$, choose

$$\chi_3 = \sum_{|\alpha|+|\beta|=3} \frac{i h_{\alpha\beta}}{\omega_N \cdot (\alpha - \beta)} a^\alpha \bar{a}^\beta.$$

Thus $\{H_2, \chi_3\} = -H_3$. Under the corresponding time-one canonical pullback the effective quartic term is

$$H_{4,\mathrm{eff}} = H_4 + \tfrac{1}{2}\{H_3, \chi_3\}.$$

Its action projection gives (S90); a quartic generating function removes its other monomials. These finite polynomial flows are analytic and locally invertible for sufficiently small (X, Y). Consequently

$$H_N \circ \Psi = h_N(I) + R_5(X,Y), \qquad R_5 = O(\|(X,Y)\|^5). \tag{S91}$$

All normal-form transformations and the remainder in this equation refer to H_N, including its learned cubic terms.

Fix action and time units before the following estimates. More explicitly, for fixed $S_0, \omega_0 > 0$, replace (X, Y) by $(X, Y)/\sqrt{S_0}$, H by $H/(S_0\omega_0)$, and time t by $\omega_0 t$. The resulting equations are canonical, and the strict conditions above are invariant under this fixed rescaling. Use the same symbols for these dimensionless variables. Fix a balanced action box, for example $1 < J_j < 2$, and put $I = \rho^2 J$. This substitution is used to describe shrinking domains; Hamilton's equations and the following KAM application still use the physical canonical actions I. On a slightly larger complex action box and a fixed angle strip $|\operatorname{Im}\theta| < s$, a branch of $\sqrt{I_j}$ is analytic, and

$$X_j = \sqrt{2I_j}\cos\theta_j, \qquad Y_j = -\sqrt{2I_j}\sin\theta_j$$

has $\{\theta_j, I_k\} = \delta_{jk}$ and size $O(\rho)$. Analyticity of (S91) therefore supplies constants $C, M, c_h > 0$, independent of sufficiently small ρ, such that the perturbation has bound $F_\rho = C\rho^5$, the action Hessian of h_N has norm at most M, and the frequency parameterization is analytic on a complex neighborhood of width $h_\rho = c_h\rho^2$. The last assertion follows directly from the affine inverse $I = T_N^{-1}(\omega - \omega_N)$ and $\det T_N \neq 0$. Choose c_h sufficiently small, using $\|T_N^{-1}\|$ and the distance of the balanced box from $I_j = 0$, so that all complex centers $I_0(\omega)$ together with their action offsets $|\delta I| < r_\rho$ remain in the larger complex action box. This ensures that the square-root branches, fixed angle strip, and bound $C\rho^5$ are valid uniformly for every frequency parameter in that neighborhood.

Apply Theorems A and B of Pöschel (2001), using $\tau > 2$, $\nu = \tau + 1$, fixed $s \le 1$, and

$$\alpha_\rho = a\rho^{2+\beta}, \qquad 0 < \beta < \tfrac{1}{2}, \qquad r_\rho = \sqrt{C/M}\,\rho^{5/2}, \qquad a > 0. \tag{S92}$$

In the frequency-parameter formulation the local perturbation includes both the Taylor quadratic part of h_N and R_5, with bound $Mr_\rho^2 + F_\rho = 2C\rho^5$. The theorem's sufficient smallness inequality is met if

$$C\rho^{1-2\beta} \le \frac{\gamma^2 a^2 s^{2\nu}}{4M}, \tag{S93}$$

where $\gamma > 0$ depends only on dimension and τ. Indeed (S93) is exactly $F_\rho \le \gamma^2\alpha_\rho^2 s^{2\nu}/(4M)$. The remaining requirements $\alpha_\rho s^\nu \le h_\rho$, $r_\rho = o(\rho^2)$, and $r_\rho, s, h_\rho \le 1$ also hold after decreasing ρ. Here $1 - 2\beta > 0$ is essential.

For completeness the Diophantine parameter set is nonempty with positive measure on these shrinking domains. The frequency image is an affine image of a box of side ρ^2. A resonance slab for k has thickness at most $C_k\alpha_\rho/|k|^{\tau+1}$ in frequency space, with a uniform geometric factor depending on T_N. Summing the cross-sectional estimates over integer $k \neq 0$ converges when $\tau > 2$. Including the boundary layer, the excluded fraction is at most a constant times $\alpha_\rho/\rho^2 = a\rho^\beta$, which tends to zero. The Lipschitz estimates in Theorem B preserve positive measure for the torus family. Transforming back by the canonical analytic changes proves the claim in every neighborhood of the equilibrium.

Consequences and limits for learning. This criterion permits changes in the learned quadratic, cubic, and quartic terms. It requires their recomputation from the actual selected network, including the cubic contribution to the quartic normal form. The finite nonresonance checks concern the linear frequencies only through order four; the Diophantine frequencies of the resulting tori are selected from the nearby nonlinear frequency image. No all-integer Diophantine property of ω_N is inferred from a finite search.

The three strict conditions are open conditions on the degree-four jet. A rigorous force-to-jet receiver can therefore supply a sufficient learning criterion: an enclosure of the admissible degree-four jets must pass positivity, the finite frequency gaps, and the twist test uniformly. An independent analytic continuation hypothesis is still needed for the remainder. Such a receiver need not bound a large quadratic change by the much smaller reference magnetic quartic; the quadratic change is included in the new spectrum.

An interval test on the actual network is a posterior criterion for that model. Its success alone does not prove that the physical target has the same low-order jet, that an optimizer will find such weights, or that a finite-data loss guarantees its conditions. A jet penalty based on the prescribed magnetic law is a stated physical prior when the observations cannot resolve those coefficients. It must not be reported as experimental identification of an unknown nonlinearity.

For a specified finite action range the constants in (S93), the normal-form domain, and the chosen Diophantine parameter set must be bounded on that range. The local constant C bounds the remainder after the cubic and quartic canonical

transformations; a bound for the original potential's Taylor remainder cannot be substituted for it without controlling these transformations. The local existence conclusion does not by itself certify such a range, identify an individual trajectory as a torus, or transfer a torus to the full damped, driven finite-element model. These limitations also prevent interpreting this result as a control-safety theorem.

S14.1 Received coefficient-guided fit and calibrated checkpoint

Starting from the initial eleven-configuration fit, 200 complete outer cycles update the energy branch by one strong-Wolfe LBFGS iteration and the dissipation branch by one Adam step with the conservative contribution detached. There are 432 energy closure evaluations and 200 dissipation Adam updates. Every complex weight layer changes in both its real and imaginary coordinates. The selected cycle is 200. The coefficient loss drops from approximately 50743.76 to 4.399638, but the maximum point-force residual rises to 2.434500×10^{-4}. Independent coefficient evaluation bounds the largest declared-family quartic discrepancy by 3.342145×10^{-6}. This stage improves the initial high-order error but does not satisfy all received intervals.

The precision-weighted step (PW1) is then applied to those saved hidden layers. It performs one 25-coordinate linear solve with no ridge penalty. Only c_E and the final complex energy weight change; all other saved coordinates remain bitwise identical to the warm start. The full two-branch graph retains its 918 real parameters. The selected skip is approximately 0.06002256805 and the Euclidean norm of its 24 real neural readout coordinates is approximately 0.0017053353, compared with 0.29265564 before calibration. This change is a final-readout calibration of a previously trained network, not a new full-layer gradient run.

Arb evaluation at 512 bits receives this exact saved graph with exact native operators at the ideal rational nodes. All 33 force residual intervals lie inside their individual received allowances, and all 31 coefficient intervals lie inside the original data intervals. The learner's once-rounded coordinates are separately recorded and are not used as substitutes for the ideal-node check. The maximum finite-node residual has upward bound 1.797797×10^{-8}; the maximum residual-to-allowance ratio is below 0.790301. The corresponding maximum candidate-to-target force bound, using the triangle inequality and the admitted observation error, is 5.023184×10^{-8}. These statements concern the eleven configurations and the explicitly admitted target family.

The strict exact-real upper bound for the actual saved-center coefficient loss is

$$\overline{L}_{\mathrm{jet}} < 1.082009 \times 10^{-2}.$$

Its computation uses rational endpoint bounds on an independent outward expansion, not the float64 training loss. The original data centers and the exported training centers are kept distinct, with the latter's exact rounding allowance included. The maximum normalized coefficient-error bounds are

Degree	Data-based triangle bound	Declared-family comparison
2	4.616863×10^{-7}	3.635946×10^{-8}
3	8.174276×10^{-7}	1.768735×10^{-8}
4	2.066906×10^{-6}	3.908709×10^{-8}

The middle column follows from the data intervals and (JG4). The final column additionally evaluates the explicitly declared quadratic simulation target and is an evaluator comparison, not a training label or a guarantee for an unknown constitutive law. The quartic discrepancy has thus decreased from more than 6.345877×10^{-4} for the initial point-force fit to less than 3.908709×10^{-8} after the disclosed coefficient-guided and final-readout stages.

The calibrated candidate's three-coordinate conservative restriction is pole-free on $|z_j| \le 1/64$ and has frequencies approximately

$$(11.73745102,\ 72.26615415,\ 326.09695871)\ \mathrm{rad\,s^{-1}}.$$

Its cubic-corrected quartic Birkhoff twist has two positive and one negative eigenvalues and determinant near -5.271108×10^{19}, strictly excluding zero. Its positive Hessian and finite-order nonresonance satisfy the qualitative local criterion of Proposition S24 for the same saved conservative restriction. No specified finite torus radius or physical-target KAM transfer follows from this candidate-specific calculation.

Finite data also admit a scalar correction. For the comparison subfamily $W(q) = \zeta E_0(q)$, force and coefficient constraints are linear inequalities in an unknown ζ. Exact rational intersection of all 33 force and 31 coefficient intervals is nonempty. Its endpoints are approximately 0.0599999981693 and 0.0600000004402. An exact rational midpoint witness satisfies all 64 inequalities. No material value is assigned to ζ in this calculation. The zero cubic and quartic terms here define this comparison subfamily; they are not extra labels supplied to the complex network. Consequently the received finite data cannot require a nonlinear correction or establish a complex-network advantage. This limitation persists after the substantial improvement in the calibrated network's coefficient fit. All stages use the same development data, without an independent validation or probability claim.

S15 An exact-source KAM certificate for the full-coordinate learned Hamiltonian

This section treats the finite conservative Hamiltonian defined by the saved network weights and the exact interpretation of their mechanical sources. Let $x \in \mathbb{R}^{64}$ collect the 32 native transverse displacement and rotation pairs, and let $p \in \mathbb{R}^{64}$ be the canonical momenta. The source entries of M_0, M_r, K_0 are interpreted as their exact saved binary values, whereas V is the exact rational basis retained in the source operator record. The sum and inverse in

$$M = M_0 + M_r, \qquad P = \operatorname{diag}(Q_1, Q_2, Q_3)^{-1}(V^{\mathsf{T}} M V)^{-1} V^{\mathsf{T}} M \tag{S94}$$

are exact mathematical operations. The positive feature scales Q_a and energy scale E are the exact binary values in the same saved checkpoint; they are not redefined by rounded decimal expressions. Let $N(l, e)$ denote the real energy readout of the realification of that fixed complex network. With $b = e_{64}$, define

$$\begin{aligned} \mathcal{H}_N(x,p) &= \frac{1}{2} p^{\mathsf{T}} M^{-1} p + U_N(x), \\ U_N(x) &= (1 + c_E) E_0(x) + \kappa[1 - \cos(b^{\mathsf{T}} x)] \\ &\quad + E\left[N\left(Px, \frac{E_0(x)}{E} \right) - N(0,0) - N_l(0,0) \cdot Px \right], \qquad E_0(x) = \frac{1}{2} x^{\mathsf{T}} K_0 x. \end{aligned} \tag{S95}$$

Thus all 64 transverse configuration coordinates remain in the Hamiltonian; no invariance of $\operatorname{range}(V)$ is assumed. This is the exact-source mathematical graph with the same saved weights as the trained model. Numerical evaluation uses rounded matrix buffers and floating-point operations, so bitwise equivalence with every runtime operation is not part of this statement. The reproducibility manifest identifies the saved weights, source arrays, and interval certificates used below.

The actual quadratic spectrum and analytic domain. All derivatives of N in this section are evaluated at $(l, e) = (0, 0)$. The centering in (S95) gives $\nabla U_N(0) = 0$, while the energy feature retains its quadratic contribution. In particular,

$$H := U_N''(0) = f K_0 + E P^{\mathsf{T}} N_{ll} P + \kappa b b^{\mathsf{T}}, \qquad f = 1 + c_E + N_e > 0. \tag{S96}$$

The source-bound spectral certificate establishes $M > 0$, $H > 0$, and 64 simple generalized eigenvalues in disjoint positive intervals. Taking their positive square roots gives the actual neural frequencies ω_i and the strict finite-order bound

$$|k \cdot \omega| \geq \frac{545883796425}{281474976710656} > 0, \qquad k \in \mathbb{Z}^{64}, \quad 0 < |k|_1 \leq 4. \tag{S97}$$

These frequencies include the learned Hessian correction in (S96).

Realification makes the network analytic in the real configuration coordinates. Complexifying those real coordinates, the analytic-domain certificate excludes every hidden $\tanh$ pole throughout

$$|w_j| < 2^{-40}\,\mathrm{m}, \qquad |\theta_j| < 2^{-40}\,\mathrm{rad}, \qquad j = 1, \ldots, 32.$$

For example, the first- and second-layer imaginary variations from their real centers are bounded by 6.597×10^{-8} and 5.018×10^{-8}, respectively, both strictly below $\pi/2$. The remaining affine operations, centering polynomial, mechanical

polynomial, and cosine introduce no poles. The kinetic term is entire in p. This polydisc supplies an analytic neighborhood; its component radius is not a certified torus-amplitude bound.

Lemma S25. Enclosures of the actual modal projections. *Let S_0 be a square real proposal matrix, with its saved binary entries interpreted exactly, and put*

$$G = S_0^{\mathsf{T}} M S_0, \qquad A = S_0^{\mathsf{T}} H S_0, \qquad g = \|G - I\|_\infty < 1.$$

For each column define

$$\mu_i = \frac{A_{ii}}{G_{ii}}, \qquad d_i = (A - \mu_i G)e_i, \qquad \epsilon_i = \frac{\|d_i\|_2}{\sqrt{(1-g)G_{ii}}}.$$

The interval $[\mu_i - \epsilon_i, \mu_i + \epsilon_i]$ contains a generalized eigenvalue. If it excludes every certified spectral interval except the one for mode i, intersecting the two intervals yields a refined enclosure J_i of that eigenvalue. If

$$\gamma_i = \operatorname{dist}\left(\mu_i, \bigcup_{j \neq i} J_j\right) > 0,$$

the sign of an actual M-normalized eigenvector s_i can be chosen so that

$$\left\| s_i - \frac{S_0 e_i}{\sqrt{G_{ii}}} \right\|_M \leq \frac{\sqrt{2}\epsilon_i}{\gamma_i}. \tag{S98}$$

For every real row h^{T} this also gives

$$\left| h^{\mathsf{T}} s_i - \frac{h^{\mathsf{T}} S_0 e_i}{\sqrt{G_{ii}}} \right| \leq \frac{\|h^{\mathsf{T}} S_0\|_2}{\sqrt{1-g}} \frac{\sqrt{2}\epsilon_i}{\gamma_i}. \tag{S99}$$

Proof. Symmetry gives $\|G - I\|_2 \leq g$ and therefore $G \geq (1-g)I > 0$. In particular, S_0 is invertible. Set $u_i = S_0 e_i / \sqrt{G_{ii}}$ and $r_i = (H - \mu_i M)u_i$. Then

$$r_i^{\mathsf{T}} M^{-1} r_i = \frac{d_i^{\mathsf{T}} G^{-1} d_i}{G_{ii}} \leq \frac{\|d_i\|_2^2}{(1-g)G_{ii}}.$$

Expansion in an M-orthonormal eigenbasis bounds the distance of μ_i from the spectrum by ϵ_i. The interval exclusion test identifies the enclosed eigenvalue because the previously certified spectrum is complete and simple. In this eigenbasis the sum of the squared coefficients of u_i outside mode i is at most ϵ_i^2/γ_i^2. Choose the sign of s_i to make its coefficient nonnegative. The inequality $1 - \sqrt{1-t} \leq t$ for $0 \leq t \leq 1$ then proves (S98). Finally,

$$h^{\mathsf{T}} M^{-1} h = (h^{\mathsf{T}} S_0) G^{-1} (S_0^{\mathsf{T}} h) \leq \frac{\|h^{\mathsf{T}} S_0\|_2^2}{1-g},$$

and Cauchy–Schwarz in the mass metric proves (S99).

The cubic-corrected twist in the same actual chart. Let $S = (s_1, \ldots, s_{64})$, $\Omega = \operatorname{diag}(\omega_i)$, and $L = S\Omega^{-1/2}$. Since $S^{\mathsf{T}} M S = I$ and $S^{\mathsf{T}} H S = \Omega^2$, the change

$$x = LX, \qquad p = ML\Omega Y = MS\Omega^{1/2} Y \tag{S100}$$

is canonical and gives $\mathcal{H}_2 = \sum_i \omega_i (X_i^2 + Y_i^2)/2$. Apply Lemma S25 to the three rows of P and to b^{T}, and divide by the positive enclosures of $\sqrt{\omega_i}$. This encloses the actual projections

$$Z_{ai} = (PL_i)_a, \qquad t_i = b^{\mathsf{T}} L_i.$$

The nominal stiffness in this chart follows from the actual Hessian identity:

$$R_{ij} := L_i^{\mathsf{T}} K_0 L_j = \frac{\delta_{ij}\omega_i - E\sum_{a,b=1}^{3} Z_{ai} N_{ab} Z_{bj} - \kappa t_i t_j}{f}. \tag{S101}$$

Thus only these four projected rows are needed to propagate the full modal uncertainty. Define

$$v_i = \sum_{a=1}^{3} N_{ae} Z_{ai}, \qquad Q_{ij} = \sum_{a,b=1}^{3} N_{abe} Z_{ai} Z_{bj}.$$

Differentiating (S95) gives the actual cubic derivative tensor and the fourth-derivative pairs:

$$\begin{aligned} C_{ijk} &= v_i R_{jk} + v_j R_{ik} + v_k R_{ij} + E \sum_{a,b,c=1}^{3} N_{abc} Z_{ai} Z_{bj} Z_{ck}, \\ D_{iijj} &= \frac{N_{ee}}{E}(R_{ii}R_{jj} + 2R_{ij}^2) + Q_{ii}R_{jj} + R_{ii}Q_{jj} + 4Q_{ij}R_{ij} \\ &\quad + E \sum_{a,b,c,d=1}^{3} N_{abcd} Z_{ai} Z_{bi} Z_{cj} Z_{dj} - \kappa t_i^2 t_j^2. \end{aligned} \tag{S102}$$

Here C, D are derivative tensors, not ordinary polynomial coefficients. Consequently the direct quartic action Hessian has diagonal entries $D_{iiii}/8$ and off-diagonal entries $D_{iijj}/4$.

To include the cubic correction, put $a_i = (X_i + iY_i)/\sqrt{2}$ and $I_i = a_i \bar{a}_i$. If $\mathcal{H}_3 = \sum h_{\alpha\beta} a^\alpha \bar{a}^\beta$, the finite-order bound (S97) permits the choice

$$\chi_3 = \sum_{|\alpha|+|\beta|=3} \frac{i h_{\alpha\beta}}{\omega \cdot (\alpha - \beta)} a^\alpha \bar{a}^\beta, \qquad \{\mathcal{H}_2, \chi_3\} = -\mathcal{H}_3.$$

The twist T is defined, in this same chart, by

$$\frac{1}{2} I^{\mathsf{T}} T I = \Pi_{\text{action}} \left[\mathcal{H}_4 + \frac{1}{2}\{\mathcal{H}_3, \chi_3\} \right]. \tag{S103}$$

The action projection retains all repeated-index terms and their cross contributions. In particular, the actual network is not treated as an even potential with vanishing cubic jet.

Proposition S26. Certified nondegeneracy of the full-coordinate twist. *For the saved learned Hamiltonian* (S95)*, the outward calculation encloses the actual T in* (S103)*. For $\mathcal{T} = \Omega^{-1} T \Omega^{-1}$ and the saved exact-binary preconditioner B_0, it proves*

$$\|I - B_0 \mathcal{T}\|_\infty < \frac{2353}{10^9} < 1. \tag{S104}$$

Hence T is nonsingular.

Proof. The four-feature source jet is enclosed using outward ball arithmetic at 384-bit precision. The modal residual test succeeds for all 64 modes. Its certified Gram defect is below 1.248×10^{-14}, its mass-normalized residuals are below 1.347×10^{-5}, and the eigenvector distances in (S98) are below 7.472×10^{-10}. The resulting projection enclosures and the identities (S101)–(S102) enclose the modal cubic tensor and all quartic pairs. The finite cubic homological sum in (S103), including its repeated-index terms, is evaluated with outward arithmetic and strictly nonzero divisor enclosures. This yields an enclosure of the full actual $\mathcal{T}$. The preconditioner is only a proposal: the final calculation encloses every entry of $I - B_0\mathcal{T}$ and bounds every absolute row sum to prove (S104). The Neumann criterion makes $B_0\mathcal{T}$ invertible, so $\mathcal{T}$ and then T are invertible. No definiteness of T is needed.

Theorem S27. Local KAM tori of the fixed full-coordinate learned Hamiltonian. *Every open neighborhood of $(x, p) = 0$ contains a family of real-analytic invariant Lagrangian 64-tori of* (S95) *whose union has positive 128-dimensional Lebesgue measure. The original Hamiltonian flow on each torus is conjugate to a Diophantine rigid rotation.*

Proof. The analytic domain, positive quadratic spectrum, and finite-order nonresonance established above permit real-analytic canonical normalization through degree four. With the actual cubic correction retained, the result is

$$\mathcal{H}_N \circ \Phi = \omega \cdot I + \frac{1}{2} I^{\mathsf{T}} T I + R_5(X, Y), \qquad I_i = \frac{X_i^2 + Y_i^2}{2}, \qquad R_5 = O(\|(X, Y)\|^5). \tag{S105}$$

The finite canonical maps are analytic and invertible near the equilibrium, and Proposition S26 establishes the required invertibility of T.

Fix action and time units for the following scaling argument. Choose a bounded positive action domain $\overline{B} \Subset (0,\infty)^{64}$ with fixed analytic buffers and a fixed complex angle strip. The square roots in $X_i = \sqrt{2I_i}\cos\vartheta_i$, $Y_i = -\sqrt{2I_i}\sin\vartheta_i$ then have uniform analytic branches. For $I = aJ$ with $a > 0$, set

$$\mathcal{H}_a(J,\vartheta) = a^{-2}(\mathcal{H}_N \circ \Phi)(aJ,\vartheta) = \frac{\omega}{a}\cdot J + \frac{1}{2}J^{\mathsf{T}}TJ + f_a(J,\vartheta), \qquad \|f_a\| = O(a^{1/2}). \tag{S106}$$

The scaling of actions is conformally symplectic; the Hamiltonian $\mathcal{H}_a$ uses the constant time change $\sigma = at$. It preserves invariant sets and the Lagrangian property.

The integrable frequency map $J \mapsto \omega/a + TJ$ has the fixed invertible derivative T. Its image is a translation of TB. Hence the analytic widths, Hessian bounds, inverse Hessian bounds, and geometric resonance-strip estimates can be chosen independently of a on an interior action domain. Apply the parameterized Theorems A–B of Pöschel (2001) with $\tau = 64 > 63$ and Diophantine constant $\gamma_a = a^{1/8}$. After balancing the local action radius against the perturbation, $r_a = O(a^{1/4})$, their sufficient smallness condition is met because

$$\frac{\|f_a\|}{\gamma_a^2} = O(a^{1/4}) \longrightarrow 0.$$

More explicitly, the perturbation including the local integrable quadratic Taylor term is $O(a^{1/2})$, whereas $\gamma_a r_a = O(a^{3/8})$. Thus the corresponding smallness ratio is $O(a^{1/8}) \to 0$, and the condition comparing γ_a with the fixed complex frequency width also holds. The excluded fraction of the translated frequency domain tends to zero with γ_a; the Lipschitz parameter estimates in Theorem B preserve a positive-measure torus family. No all-integer Diophantine condition on the equilibrium vector ω is required.

Returning to original time multiplies every torus frequency by $a > 0$, which preserves its Diophantine property with a correspondingly smaller positive constant. The action scaling multiplies volume by $a^{64} > 0$, and the canonical maps preserve volume. The resulting torus union therefore has positive original phase-space measure. Its Cartesian size is $O(\sqrt{a})$, so choosing a sufficiently small places it in any prescribed open neighborhood of the equilibrium. This proves the assertion using the general fifth-order remainder; an even-potential sixth-order remainder is unnecessary.

The conclusion concerns the finite 64-coordinate transverse conservative mathematical graph with the same saved network weights. It does not include the additional axial coordinates, damping or forcing, the continuous PDE, or an unknown physical material. It supplies neither a computable nonzero torus radius nor Lyapunov stability of every nearby trajectory. It also does not establish finite-data identification, transfer to a physical target, or superiority of complex over real network representations.

S16 A full-coordinate robustness criterion at the saved neural model

The base Hamiltonian in this section is the same verified 918-parameter exact-source conservative graph, with all 64 configuration coordinates. Its equilibrium Hessian, canonical chart, finite-order divisors, and cubic-corrected twist are verified inputs. This differs from the earlier three-coordinate material-family calculation. No tolerance from that calculation is used here.

Write the verified chart as $x = LX$ and $H_2 = \frac{1}{2}\sum_{j=1}^{64}\omega_j(X_j^2 + Y_j^2)$. With $a_j = (X_j + iY_j)/\sqrt{2}$, one has $\{a_j, \bar{a}_k\} = -i\delta_{jk}$ and $I_j = a_j\bar{a}_j$. Let $\|P\|_1$ be the sum of absolute ordinary coefficients of a homogeneous polynomial in $(a,\bar{a})$. Let $d_3 > 0$ be a verified lower bound on all cubic homological divisors, $h_3 \geq \|H_3\|_1$, and $t \geq \|T^{-1}\|_2$ for the verified full 64×64 Birkhoff twist. In particular H_3 is not assumed zero.

Consider a real analytic conservative perturbation $\Delta U = \Delta U_3 + \Delta U_4 + O(|x|^5)$ with no terms of degree at most two. Thus the equilibrium Hessian and the canonical chart remain exactly the verified ones. Suppose its canonical polynomials satisfy $\|\Delta H_3\|_1 \leq \epsilon_3$ and $\|\Delta H_4\|_1 \leq \epsilon_4$.

Proposition S28. A criterion retaining the nonzero neural cubic. *If*

$$t\left[2\epsilon_4+\frac{18h_3\epsilon_3+9\epsilon_3^2}{d_3}\right]<1, \tag{S107}$$

then the perturbed full-coordinate quartic twist is nonsingular. Together with real analyticity and the inherited fixed Hessian and order-four nonresonance, this gives the same qualitative local elliptic KAM conclusion: every open neighborhood of the equilibrium contains a positive-measure union of invariant Lagrangian 64*-tori.*

Proof. Use the same pullback convention as in the verified base certificate: $\{H_2,\chi_3\}=-H_3$ and $\{H_2,\Delta\chi_3\}=-\Delta H_3$. The change in the effective quartic is

$$\Delta H_4^{\mathrm{eff}}=\Delta H_4+\tfrac{1}{2}\big(\{H_3,\Delta\chi_3\}+\{\Delta H_3,\chi_3\}+\{\Delta H_3,\Delta\chi_3\}\big).$$

For homogeneous degrees m,n, the canonical complex bracket satisfies $\|\{P,Q\}\|_1\le mn\|P\|_1\|Q\|_1$. The divisor bound gives $\|\chi_3\|_1\le h_3/d_3$ and $\|\Delta\chi_3\|_1\le\epsilon_3/d_3$. Balanced quartic projection does not increase the coefficient one-norm. Extraction of its action Hessian has Frobenius norm at most twice that one-norm. Consequently

$$\|\Delta T\|_2\le 2\epsilon_4+(18h_3\epsilon_3+9\epsilon_3^2)/d_3.$$

Condition (S107) makes $\|T^{-1}\Delta T\|_2<1$, so $T+\Delta T$ is invertible. No term of degree at least five changes the Hessian, the finite-order divisors or this twist. Apply the already verified finite-dimensional local elliptic KAM construction. The resulting neighborhood may depend on the higher-order analytic remainder; no common numerical radius is claimed.

The 13 *cubic and* 22 *quartic feature directions.* Use the exact source normalizations $\ell=10^{-3}$, $E_s=6\times10^{-12}$ and

$$u=Bx/\ell,\quad e_s=x^TK_0x/(2E_s),\quad a(u)=\ell^2u^TK_3u/(2E_s),\quad A=e_s-a(u).$$

The saved neural fourth input is $e_{\mathrm{net}}=x^TK_0x/(2E_{\mathrm{net}})=(E_s/E_{\mathrm{net}})e_s$, where E_{net} is the fixed saved binary64 neural energy scale. The homogeneous cubic feature directions are the ten monomials in u of degree three and the three monomials Au_i. The homogeneous quartic directions are the fifteen monomials in u of degree four, the six monomials Au_iu_j with $i\le j$, and A^2. There are seven separate quadratic directions, namely six u_iu_j and A; uncertainty in those directions is excluded by the present fixed-Hessian hypothesis and requires an additional spectral analysis.

Let $U_i\ge\|u_i(LX)\|_1$ and $C_A\ge\|A(LX)\|_1$ in the same canonical complex convention. Set

$$h_k(U)=\sum_{|\alpha|=k}\prod_i U_i^{\alpha_i},\quad C_3=E_s\{h_3(U)+C_Ah_1(U)\},\quad C_4=E_s\{h_4(U)+C_Ah_2(U)+C_A^2\}.$$

Specifically write $\Delta U_3=E_s[\delta R_3+A\delta S_1]$ and $\Delta U_4=E_s[\delta R_4+A\delta S_2+\delta c_AA^2]$. Dimensionless bounds η_3,η_4 on the ordinary coefficients inside these respective brackets imply $\epsilon_3\le C_3\eta_3$ and $\epsilon_4\le C_4\eta_4$. These constants are computed anew from the verified 64-mode chart; they are not derivatives or three-mode constants.

How the constants are enclosed. The saved modal intervals give $Z=PL$ and $v=b^TL$. All derivatives below are taken at the origin in the saved four-feature jet $N(s_1,s_2,s_3,e_{\mathrm{net}})$, with $s=Px$. Write $f=1+c_E+N_4(0)$, where $N_4=\partial N/\partial e_{\mathrm{net}}$, and $N_{ss}=(N_{ij})_{i,j\le3}$, where $N_{ij}=\partial^2N/(\partial s_i\partial s_j)$. The exact-source Hessian identity gives

$$R=L^TK_0L=\{\mathrm{diag}(\omega)-E_{\mathrm{net}}Z^TN_{ss}Z-\kappa v^Tv\}/f.$$

Thus C_A is bounded by $E_s^{-1}\sum_{i,j}|(R-\ell^2U_X^TK_3U_X)_{ij}|$, where $(U_X)_{ij}=Q_iZ_{ij}/\ell$; and $\|u_i(LX)\|_1=\sqrt{2}\sum_j|(U_X)_{ij}|\le U_i$. The coefficient one-norm of the neural H_3 is bounded by substituting these verified modal intervals into the physically weighted-degree-three terms of the saved four-feature jet. Magnetic cosine contributes no cubic term at the origin.

Let $\Omega = \mathrm{diag}(\omega_1, \dots, \omega_{64})$ and $N_T = \Omega^{-1} T \Omega^{-1}$. A fixed approximate inverse J with $\|JN_T - \mathrm{Id}\|_\infty \le d < 1$ gives

$$\|T^{-1}\|_2 \le \|T^{-1}\|_\infty \le \frac{\|J\|_\infty}{(1-d)\omega_{\min}^2}.$$

This uses symmetry of the exact inverse T^{-1}. All source binary parameters, the exact rational K_3, modal intervals and arithmetic bounds retain their original identities.

A simultaneous cubic and quartic sufficient neighborhood. For the exact-source 918-parameter graph with checkpoint prefix `f51bcea12c2f`, outward computation on its verified 64-mode chart gives $C_3 < 5.539762 \times 10^{13}$, $C_4 < 1.326507 \times 10^{24}$, $h_3 < 1.060037 \times 10^{6}$, $t < 4.060190 \times 10^{-5}$ and $d_3 > 1.696884$. The exact rational bounds, rather than these rounded displays, yield

$$(\eta_3, \eta_4) = (2^{-57}, 2^{-69}), \qquad t\left[2C_4\eta_4 + \frac{18h_3C_3\eta_3 + 9C_3^2\eta_3^2}{d_3}\right] < 0.357977 < 1. \tag{S108}$$

This is one sufficient neighborhood with an exactly fixed quadratic jet. It includes changes to the nonzero cubic and its homological correction. Supplementary Section S17 gives a direct quartic bound under $\Delta H_3 = 0$; Section S18 combines that response with cubic errors while fixing only H_2.

The seven quadratic directions, a general unknown feature factorization, target-valid observation errors and analytic remainders are separate hypotheses. Neither criterion places the present unknown target inside the stated coefficient neighborhood. An exceeded sufficient bound does not prove absence of invariant tori. No numerical torus radius or dissipative conclusion is assigned.

S17 A direct quartic response with the quadratic and cubic terms fixed

The reference is the exact-source conservative Hamiltonian H_N^{64} of the article, with the same 918 real parameters, checkpoint prefix `f51bcea12c2f`, exact nominal mass and elasticity operators, and complete 64-mode canonical chart. The fourth neural input is E_0/E_{net}; coefficients below use the distinct exact source scale $E_s = 6 \times 10^{-12}$ J. Let $x = LX$, $I_i = (X_i^2 + Y_i^2)/2$, and $\Omega = \mathrm{diag}(\omega_1, \dots, \omega_{64})$. The full reference cubic H_3 and its homological correction are retained. Both H_2 and H_3 are held exactly fixed in this section.

Use $u = Bx/\ell$, $\ell = 10^{-3}$ m, and $A = x^T(K_0 - B^T K_3 B)x/(2E_s)$. Write

$$u_r = v_r^T X, \qquad A = \frac{X^T C_\perp X}{2E_s}, \qquad C_\perp = L^T(K_0 - B^T K_3 B)L.$$

The 22 quartic directions P_j are the fifteen $E_s u^\gamma$ with $|\gamma| = 4$, the six $E_s A u_r u_s$ with $r \le s$, and $E_s A^2$. Define their action-Hessian responses T_j by $\langle P_j \rangle = I^T T_j I/2$, where brackets denote independent angle averages in the fixed canonical chart.

Lemma S29. Exact action-Hessian responses. *For $P = E_s \prod_{m=1}^4 (v_m^T X)$, the response is*

$$(T_P)_{ii} = 3E_s \prod_{m=1}^{4} v_{m,i}, \tag{S109}$$

$$(T_P)_{ik} = E_s \sum_{\substack{S \subset \{1,2,3,4\} \\ |S|=2}} \prod_{m \in S} v_{m,i} \prod_{m \notin S} v_{m,k}, \quad i \ne k. \tag{S110}$$

Repeated factors occupy distinct positions in the six-term sum. For $P = E_s A u_r u_s$, the response is

$$(T_P)_{ii} = \tfrac{3}{2}(C_\perp)_{ii} v_{r,i} v_{s,i}, \tag{S111}$$

$$(T_P)_{ik} = \tfrac{1}{2}\{(C_\perp)_{ii} v_{r,k} v_{s,k} + (C_\perp)_{kk} v_{r,i} v_{s,i}\} + (C_\perp)_{ik}(v_{r,i} v_{s,k} + v_{r,k} v_{s,i}), \quad i \ne k. \tag{S112}$$

For $P = E_s A^2$, the response is

$$(T_P)_{ii} = \frac{3(C_\perp)_{ii}^2}{4E_s}, \qquad (T_P)_{ik} = \frac{(C_\perp)_{ii}(C_\perp)_{kk} + 2(C_\perp)_{ik}^2}{2E_s} \ (i \neq k). \tag{S113}$$

Proof. Since $X_i = \sqrt{2I_i}\cos\theta_i$, $\langle X_i^4 \rangle = 3I_i^2/2$ and $\langle X_i^2 X_k^2 \rangle = I_i I_k$ for $i \neq k$. Every other quartic monomial has a zero angle average. A coefficient p of X_i^4 therefore contributes $3p$ to T_{ii}; a coefficient q of $X_i^2 X_k^2$ contributes q to T_{ik}. Expanding the three displayed quartic polynomials gives the formulas, including the off-diagonal factors from the symmetric quadratic form.

Let T_0 be the full reference cubic-corrected twist, $N_0 = \Omega^{-1} T_0 \Omega^{-1}$ and $N_j = \Omega^{-1} T_j \Omega^{-1}$. For the fixed rational inverse proposal J, let $d < 1$ satisfy $\|\mathrm{Id} - JN_0\|_\infty \leq d$. Take nonnegative entrywise bounds

$$(G_4)_{ik} \geq \sum_{j=1}^{22} |(JN_j)_{ik}|, \qquad b_4 = \max_i \sum_k (G_4)_{ik}. \tag{S114}$$

Theorem S30. A fixed-H_2, H_3 full-coordinate sufficient criterion. *Consider a real analytic conservative perturbation whose terms through degree three vanish and whose quartic part is $\Delta H_4 = \sum_{j=1}^{22} \delta_j P_j$, with $|\delta_j| \leq \eta_4$. If*

$$d + b_4 \eta_4 < 1, \tag{S115}$$

then its full 64×64 quartic twist is nonsingular. The fixed positive Hessian, finite nonresonance and real analyticity then imply the qualitative local elliptic KAM conclusion for the perturbed 64-coordinate Hamiltonian. The higher-order analytic remainder may change the local neighborhood; no numerical torus radius is specified.

Proof. As H_2 and H_3 are unchanged, both the cubic homological solution and its quartic correction remain unchanged. Thus the normalized perturbed twist is exactly $N_0 + \sum_j \delta_j N_j$. For every matrix entry, $|\sum_j \delta_j (JN_j)_{ik}| \leq \eta_4 (G_4)_{ik}$. Taking row sums gives

$$\left\| \mathrm{Id} - J\left(N_0 + \sum_j \delta_j N_j \right) \right\|_\infty \leq d + b_4 \eta_4 < 1.$$

The Neumann criterion makes this product invertible, hence also the perturbed twist. The remaining local KAM hypotheses are inherited from the fixed quadratic part and the stated analyticity. Apply Supplementary Theorem S43.

Exact-source enclosure and numerical instance. The modal projections are interval enclosures from the same exact source graph, rather than rounded eigenvectors treated as exact modes. To retain cancellation in the normal quadratic form, let $Z = PL$, $v = b^T L$ and $f = 1 + c_E + N_e(0)$. The source Hessian identity gives

$$C_\perp = \frac{\Omega - E_{\mathrm{net}} Z^T N_{ll}(0) Z - \kappa v^T v}{f} - \ell^2 U_X^T K_3 U_X, \qquad U_X = \mathrm{diag}(Q/\ell) Z. \tag{S116}$$

Here N_{ll} and N_e are derivatives in the saved neural inputs (l, e_{net}) at the origin, with $e_{\mathrm{net}} = E_0/E_{\mathrm{net}}$. The positive f, the binary feature scales, K_3, modal intervals, baseline twist and fixed preconditioner all refer to this same graph. Outward evaluation of all 22 responses gives

$$\begin{aligned} d &< 2.352888 \times 10^{-6}, & b_4 &< 4.2914981274 \times 10^{10}, \\ \eta_4 &= 2^{-37}, & d + b_4\eta_4 &< 0.312250 < 1. \end{aligned} \tag{S117}$$

The sum of the 22 separately bounded induced norms is less than $5.0443713467 \times 10^{10}$; summing entrywise before taking the maximum row sum yields the smaller b_4 above. Independent exact interval arithmetic verifies all 4096 entrywise bounds and the same strict criterion. Compared with 2^{-69} in (S108), this quartic coefficient radius is 2^{32} times larger, under the essential additional condition $\Delta H_3 = 0$. It is not an optimal radius.

For a single direction let $g_j \geq \|JN_j\|_\infty$. The half-margin allowance $(1-d)/(2g_j)$ ensures $d + g_j|\delta_j| \leq (1+d)/2 < 1$. For E_sA^2,

$$g_{A^2} < 3.108542704 \times 10^9, \qquad \frac{1-d}{2g_{A^2}} \simeq 1.6084669610 \times 10^{-10}. \tag{S118}$$

The ratio is displayed approximately; the exact rational allowance is used in the numerical certificate. It bounds a perturbation about the reference coefficient, rather than the absolute target coefficient. The old-trajectory intervals in (S86)–(S87) are many orders of magnitude too wide to lie in this neighborhood. The seven quadratic directions, changes to H_3, unknown feature factorization, target remainders and physical observation errors are not supplied by the fixed-H_2, H_3 calculation.

S18 A joint full-coordinate cubic and quartic error criterion

Keep the same analytic conservative graph, equilibrium Hessian, canonical chart and order-four nonresonance as in the full-coordinate certificate. Let T denote its cubic-corrected twist, $\Omega = \mathrm{diag}(\omega)$, $N_T = \Omega^{-1}T\Omega^{-1}$, and let J satisfy $\|JN_T - \mathrm{Id}\|_\infty \leq d < 1$. This section permits simultaneous errors in the cubic and quartic jets; only the quadratic jet is fixed exactly.

For the 22 dimensionless quartic coefficients in $\Delta U_4 = E_s[\delta R_4 + A\delta S_2 + \delta c_A A^2]$, write T_j for the action-Hessian response to the jth unit coefficient. These responses are linear in ΔU_4 and do not depend on whether the cubic jet is perturbed. Define the entrywise nonnegative matrix

$$G_{ik} = \sum_{j=1}^{22} |(J\Omega^{-1}T_j\Omega^{-1})_{ik}|, \qquad b_4 = \max_i \sum_k G_{ik}.$$

In numerical use each entry and b_4 is replaced by its outward upper bound. Use the same coefficient one-norm convention in complex canonical variables as in Supplementary Section S16. Suppose $\|H_3\|_1 \leq h_3$, $\|\Delta H_3\|_1 \leq C_3\eta_3$, and every quartic coefficient error has absolute value at most η_4. Let d_3 be the positive cubic divisor lower bound and $\omega_* > 0$ a lower bound on every ω_j.

Proposition S31. Simultaneous cubic and quartic perturbations. *The perturbed full* 64×64 *twist is nonsingular if*

$$d + \frac{\|J\|_\infty}{\omega_*^2 d_3}\left[18h_3C_3\eta_3 + 9(C_3\eta_3)^2\right] + b_4\eta_4 < 1. \tag{S119}$$

Under the stated analytic and fixed-quadratic hypotheses, the local positive-measure 64*-torus conclusion therefore persists.*

Proof. Separate the twist change as $\Delta T = T_4 + T_c$, with $T_4 = \sum_j \delta c_j T_j$ and T_c caused by cubic elimination. The definition of G gives $\|J\Omega^{-1}T_4\Omega^{-1}\|_\infty \leq b_4\eta_4$. For any balanced quartic polynomial $Q = \sum_i q_{ii}I_i^2 + \sum_{i<k} q_{ik}I_iI_k$, its action Hessian has diagonal entries $2q_{ii}$ and off-diagonal entries q_{ik}. Each absolute row sum is at most $2\sum_{i\leq k}|q_{ik}|$. Thus its matrix infinity norm, as well as its Frobenius norm, is bounded by twice its coefficient one-norm. The cubic homological equation and the canonical bracket estimate give

$$\|T_c\|_\infty \leq \{18h_3C_3\eta_3 + 9(C_3\eta_3)^2\}/d_3.$$

The two diagonal factors Ω^{-1} give $\|J\Omega^{-1}T_c\Omega^{-1}\|_\infty \leq \|J\|_\infty\omega_*^{-2}\|T_c\|_\infty$. Consequently the left side of (S119) bounds $\|J\Omega^{-1}(T+\Delta T)\Omega^{-1} - \mathrm{Id}\|_\infty$. The Neumann criterion proves invertibility. The quadratic spectrum and finite-order nonresonance are unchanged, so the analytic local elliptic KAM argument applies. Terms of degree at least five may change the size of the neighborhood, but not this twist test.

For the same 918-parameter exact-source graph with checkpoint prefix `f51bcea12c2f`, outward rational arithmetic gives $\eta_3 = 2^{-57}$ and $\eta_4 = 2^{-37}$ with the left side of (S119) below 0.487747. This combines the nonzero-cubic estimate and the directional quartic estimate; it does not assume $\Delta H_3 = 0$. It remains a sufficient conditional neighborhood. Neither the seven quadratic errors nor the stated cubic and quartic tolerances have been established for an unknown target by the present observations. Exceeding this sufficient neighborhood does not imply that target tori are absent.

The separate contributions to the displayed margin are a base defect below 2.352888×10^{-6}, a cubic budget below 0.175497, and a quartic budget below 0.312248. Independent exact rational reconstruction verifies their sum below

$0.487747 < 1/2$. The cubic interaction with the nonzero reference cubic is included; it dominates the term quadratic in the cubic perturbation at this coefficient radius. This sufficient neighborhood retains all thirteen cubic and twenty-two quartic feature directions, with no terms of degree at most two. It is a qualitative local statement, not a numerical torus-radius or unknown-physical-target certificate.

S19 A quantitative full-coordinate bridge with a moving quadratic jet

Retain the same saved 918-real-parameter candidate and exact source operators as the full-coordinate certificate. Let $H^0 = p^T M^{-1} p/2 + U^0(x)$ be its certified analytic conservative Hamiltonian on 64 configuration coordinates. Let $S^T M S = I$, $S^T U^0_{xx}(0) S = \Omega^2$, $L = S\Omega^{-1/2}$, and retain the sign-selected real eigenbasis from the received modal certificate. Write H^0_3, H^0_4 for its homogeneous potentials in $x = LX$ and $z = (X + iY)/\sqrt{2}$. All polynomial norms below are coefficient one-norms in the independent formal variables $(z, \bar{z})$.

Assume that an analytic real target has the same M, the same equilibrium at zero, and homogeneous differences through degree four of the form

$$\begin{aligned}\Delta U_2 &= E_s[\delta R_2(u) + \delta a A],\\ \Delta U_3 &= E_s[\delta R_3(u) + A\delta S_1(u)],\\ \Delta U_4 &= E_s[\delta R_4(u) + A\delta S_2(u) + \delta c_A A^2],\end{aligned}$$

where $u = Bx/\ell$, $A = x^T K_0 x/(2E_s) - \ell^2 u^T K_3 u/(2E_s)$, $\ell = 1/1000$, and $E_s = 3/500000000000$ exactly. Each ordinary coefficient of the respective 7, 13 and 22 terms has absolute value at most η_2, η_3, η_4. Higher homogeneous terms can be arbitrary analytic terms of degree at least five. These hypotheses concern the target's full-coordinate jet; they do not follow solely from projected observations.

Use received outward constants

$$\|u_i(LX)\|_1 \le U_i, \quad \|A(LX)\|_1 \le a_A, \quad \|H^0_3\|_1 \le h_3, \quad \|H^0_4\|_1 \le h_4, \quad \|\Delta U_r(LX)\|_1 \le C_r\eta_r \quad (r = 3, 4).$$

Set $C_2 = E_s(\sum_{i\le j} U_i U_j + a_A)$. Let w_-, w_+ bound all reference frequencies below and above, $g > 0$ bound every distinct reference squared-frequency separation below, and $d_3, d_4 > 0$ bound all reference nonzero divisors of orders three and at most four, respectively. In particular d_3 also includes the single-mode cubic divisors. Put

$$\begin{aligned}\epsilon &= w_+ C_2 \eta_2, & v &= \epsilon/w_-,\\ t &= 16\sqrt{\frac{w_+}{w_- - v}}\frac{\epsilon}{g-\epsilon} + \frac{v}{2(w_- - v)}, & s &= 1 + t,\\ e_3 &= h_3(s^3 - 1) + C_3\eta_3 s^3, & d_3' &= d_3 - 3v.\end{aligned}$$

Any outward upper bound may replace t in these expressions.

Let T^0 be the reference cubic-corrected Birkhoff twist and let J be the retained exact-binary preconditioner, with $\|J\Omega^{-1}T^0\Omega^{-1} - I\|_\infty \le d < 1$. The received directional quartic response gives b_4 such that $\|J\Omega^{-1}T_4(\Delta U_4)\Omega^{-1}\|_\infty \le b_4\eta_4$, when ΔU_4 is written in the reference chart.

Proposition S32. Moving-Hessian quantitative criterion. *Suppose*

$$0 \le \epsilon < \min\{w_-^2, g/2\}, \qquad v < w_-, \qquad 4v < d_4, \qquad 3v < d_3,$$

and

$$d + b_4\eta_4 + \frac{\|J\|_\infty}{w_-^2}\left[2(h_4 + C_4\eta_4)(s^4 - 1) + \frac{9(2h_3e_3 + e_3^2)}{d_3'} + \frac{27h_3^2 v}{d_3 d_3'}\right] < 1. \tag{S120}$$

Then the target Hessian is strictly elliptic with simple frequencies and nonresonance through order four, and its full 64-dimensional cubic-corrected twist is nonsingular. The same analytic local elliptic KAM theorem gives a positive-measure family of invariant Lagrangian 64-tori in every neighborhood of the equilibrium.

Proof. For a homogeneous polynomial in X alone, the substitution $X_j = (z_j + \bar{z}_j)/\sqrt{2}$ multiplies its coefficient norm by $2^{r/2}$ in degree r: distinct X multiindices have disjoint supports in $(z, \bar{z})$. Thus $\|\nabla_X^2 \Delta U_2(LX)\|_2 \le 2\|\Delta U_2(LX)\|_{1,X} \le C_2\eta_2$. For $E = S^T \Delta U_{2,xx} S$ this gives $\|E\|_2 \le w_+ C_2\eta_2 = \epsilon$. The min–max characterization of the eigenvalues of the real symmetric matrices Ω^2 and $\Omega^2 + E$ gives $|\lambda_i' - \omega_i^2| \le \epsilon$. Consequently all eigenvalues remain positive and separated, and $|\omega_i' - \omega_i| \le \epsilon/w_- = v$. Every divisor with $0 < |k|_1 \le 4$ changes by at most $4v$.

Let R be the real orthogonal matrix whose ith column is the normalized eigenvector of $\Omega^2 + E$ for λ_i'. Projecting its eigenvalue equation onto the complement of the ith reference coordinate gives

$$\|(R_{ji})_{j\ne i}\|_2 \le \frac{\epsilon}{g - \epsilon} < 1.$$

There is therefore a unique sign choice $R_{ii} > 0$. The column distance to the ith coordinate vector is at most $\sqrt{2}\epsilon/(g - \epsilon) \le 2\epsilon/(g - \epsilon)$. Orthogonality makes the squared distance of row i to the corresponding coordinate row equal to the squared distance of column i, namely $2(1 - R_{ii})$. Cauchy–Schwarz in 64 coordinates therefore yields

$$\|R - I\|_\infty \le \frac{16\epsilon}{g - \epsilon}.$$

This explicitly fixes the real phase and does not assume a differentiable eigenvector routine or arbitrarily chosen complex phases.

In the target chart $S' = SR$, $L' = SR(\Omega')^{-1/2}$, define $C = \Omega^{1/2} R (\Omega')^{-1/2}$. The canonical relation between the two charts is $X^0 = CX'$, $Y^0 = C^{-T} Y'$. Since all nonlinear terms are configuration potentials, their change is controlled just by C. Splitting $C - I$ into its $R - I$ part and its diagonal frequency-scaling part proves $\|C - I\|_\infty \le t$. For the latter part use $|\sqrt{\omega_i/\omega_i'} - 1| \le v/[2(w_- - v)]$.

For any homogeneous configuration polynomial P_r, multilinear expansion of $P_r((I + (C - I))X)$ gives

$$\|P_r(CX) - P_r(X)\|_1 \le (s^r - 1)\|P_r\|_1, \qquad \|P_r(CX)\|_1 \le s^r \|P_r\|_1.$$

The same bounds hold in the complex coefficient norm by the degree-dependent factor just proved. Hence the target cubic in its own chart differs from H_3^0 by at most e_3. The direct quartic twist change is the sum of the reference-chart directional response to ΔU_4 and the response to $(H_4^0 + \Delta U_4)(CX) - (H_4^0 + \Delta U_4)(X)$. The latter twist infinity norm is bounded by $2(h_4 + C_4\eta_4)(s^4 - 1)$.

On degree-three polynomials, the homological inverse for ω' has coefficient operator norm at most $1/d_3'$, and its difference from the reference inverse has norm at most $3v/(d_3 d_3')$, coefficient by coefficient. For homogeneous polynomials of degrees r, s, $\|\{f_r, g_s\}\|_1 \le rs\|f_r\|_1\|g_s\|_1$. The action projection is contractive in this norm. The action Hessian of a balanced quartic polynomial has infinity norm at most twice its coefficient norm. Applying these bounds to $\Pi_{\mathrm{action}}\{H_3, \chi_3\}/2$ gives

$$\|T_c' - T_c^0\|_\infty \le \frac{9(2h_3e_3 + e_3^2)}{d_3'} + \frac{27h_3^2 v}{d_3 d_3'}.$$

The last summand controls the moving homological denominators, even when the cubic polynomial itself is unchanged in the common formal chart.

Combining the three twist changes proves that the left side of (S120) bounds $\|J\Omega^{-1}T'\Omega^{-1} - I\|_\infty$. The normalizing Ω here is deliberately the fixed reference matrix; any positive diagonal congruence can test nonsingularity of T'. The Neumann criterion proves T' nonsingular. Analyticity, the verified finite-order nonresonance and the invertible cubic-corrected twist permit the same general fifth-order local KAM reduction as for the reference Hamiltonian. No analyticity norm for an unknown target, finite torus radius, dissipative-flow conclusion or target observation reception is claimed by this finite-jet test.

Same-candidate numerical instance. Outward constants from the same modal, feature-jet and directional-response enclosures give a coefficient neighborhood

$$(\eta_2, \eta_3, \eta_4) = (2^{-86}, 2^{-57}, 2^{-37}).$$

All spectral and divisor inequalities above hold strictly, and the left side of (S120) is below $0.495840 < 1/2$. Here $\epsilon < 5.061 \times 10^{-15}$ and $t < 1.751 \times 10^{-15}$. The quartic norm includes all 22 weighted degree-four network terms and the magnetic cosine quartic. Exact rational recomputation checks both the perturbed criterion and its zero-quadratic-error limit. With the original unrounded constants, the latter agrees exactly with Proposition S31; outwardly shortened constants need only give a larger valid bound.

These are sufficient coefficient tolerances, not observed target errors. For the seven quadratic directions, comparison of complete target and candidate intervals gives a maximum distance of approximately 4.617×10^{-7} among the six restricted quadratic coefficients and 2.636×10^{-4} in the A direction. None fits $\eta_2 = 2^{-86}$. The strict theorem therefore supplies a positive theoretical neighborhood while the unknown-target transfer remains unproved. Its analytic feature and equilibrium hypotheses still require independent support.

S20 From the recorded 31-coefficient loss to a full-coordinate KAM test

Fix the same 918-real-parameter saved candidate used by the full-coordinate reference certificate. Use the exact source scales $\ell = 1/1000$, $E_s = 3/500000000000$, and set

$$u = Bx/\ell, \qquad e = x^T K_0 x/(2E_s), \qquad A = e - \ell^2 u^T K_3 u/(2E_s).$$

Here $BV = I_3$ and $K_3 = V^T K_0 V$, so $A = 0$ on the trial space. For $\nu \in \{\theta, *\}$ assume the material correction is $W_\nu(x) = E_s \Phi_\nu(u, e)$, with Φ_ν real analytic near zero, $\Phi_\nu(0,0) = 0$ and $\partial_u \Phi_\nu(0,0) = 0$. Give u weight one and e weight two. Its weighted Taylor jet in physical degrees two through four, denoted P_ν, has the unique decomposition

$$\begin{aligned} P_\nu &= R_\nu(u) + A S_\nu(u) + c_{\nu,A^2} A^2, \\ R_\nu &= \sum_{2 \le |\alpha| \le 4} R_{\nu,\alpha} u^\alpha, \qquad S_\nu = \sum_{|\gamma| \le 2} S_{\nu,\gamma} u^\gamma. \end{aligned} \tag{S121}$$

There are $31 + 10 + 1 = 42$ ordinary coefficients. Since $A = 0$ on $x = V\ell u$, R_ν is precisely the restricted material jet received by the existing eleven-node design. The terms AS_0, AS_1 and AS_2, with S_j homogeneous of degree j, have physical degrees two, three and four, respectively; A^2 has degree four.

The objective actually used. The recorded coefficient objective is the 31-term saved-center loss

$$\widetilde{L}_{\mathrm{jet}} = \frac{1}{31} \sum_{2 \le |\alpha| \le 4} \left(\frac{k_{\theta,\alpha} - \widetilde{c}_\alpha}{\widetilde{s}_\alpha} \right)^2, \qquad \widetilde{s}_\alpha > 0. \tag{S122}$$

Here $k_{\theta,\alpha}$ denotes its recorded coefficient convention; this is the existing objective, not a 42-term replacement. Require a rigorous upper bound $\overline{L}_{31} \ge \widetilde{L}_{\mathrm{jet}}$ and received bounds

$$|R_{*,\alpha} - \widetilde{c}_\alpha| \le \rho_\alpha, \qquad |R_{\theta,\alpha} - k_{\theta,\alpha}| \le \tau_\alpha.$$

The radius ρ_α includes observation/force-reconstruction, target-tail and original-to-saved-center allowances; τ_α includes any runtime-centering or source-arithmetic transfer. With the main text's exact-source interpretation $k_{\theta,\alpha} = R_{\theta,\alpha}$, one takes $\tau_\alpha = 0$. A float64 loss alone supplies neither a rigorous $\overline{L}_{31}$ nor a transfer bound. The already proved component argument gives

$$|R_{\theta,\alpha} - R_{*,\alpha}| \le E_{R,\alpha} := \widetilde{s}_\alpha \sqrt{31 \overline{L}_{31}} + \rho_\alpha + \tau_\alpha. \tag{S123}$$

For the actual saved-center coefficient contract, its stored radius already includes the original observation radius and center-rounding allowance. Taking that radius as ρ_α and $\tau_\alpha = 0$ in the exact-source convention counts each allowance once. The outward same-weight coefficient evaluation gives $\overline{L}_{31} < 0.010821$. This upper loss bound does not imply that the resulting coefficient errors meet the much smaller KAM tolerances.

A received direct component residual may replace the aggregate term by its tighter bound, retaining the same allowances exactly once.

The additional eleven validation constraints. Independently receive target intervals $I_{*,\gamma} = [l_{*,\gamma}, u_{*,\gamma}]$ for the ten coefficients of S_* and I_{*,A^2} for c_{*,A^2} from additional force or trajectory functionals. Receive candidate intervals $I_{\theta,\gamma}$ and I_{θ,A^2} for the same exact source and saved weights. All force, trajectory, damping-nuisance, arithmetic and analytic-tail errors needed for those target intervals must be included. For any two intervals define

$$d_\times([l,u],[l',u']) = \max\{|l-u'|, |u-l'|\}, \qquad E_{S,\gamma} = d_\times(I_{\theta,\gamma}, I_{*,\gamma}), \quad E_{A^2} = d_\times(I_{\theta,A^2}, I_{*,A^2}).$$

This compares the complete intervals, including their center discrepancy. Here "independently receive" means independently certify the missing constraints; it asserts no statistical independence of samples. Complete forces on the old trial space can receive S but cannot receive A^2; that last coordinate needs a valid off-trial or trajectory sensitivity.

Corollary S33. Conditional 31-loss plus 11-validation composition. *Define*

$$\begin{aligned}
\eta_2 &= \max\{\max_{|\alpha|=2} E_{R,\alpha},\ E_{S,0}\},\\
\eta_3 &= \max\{\max_{|\alpha|=3} E_{R,\alpha},\ \max_{|\gamma|=1} E_{S,\gamma}\},\\
\eta_4 &= \max\{\max_{|\alpha|=4} E_{R,\alpha},\ \max_{|\gamma|=2} E_{S,\gamma},\ E_{A^2}\}.
\end{aligned} \tag{S124}$$

Assume the target autonomous conservative Hamiltonian has the same positive mass matrix, equilibrium and known elastic/magnetic potentials as the reference, and satisfies the analytic and feature-factorization hypotheses above. Retain the source-bound reference constants and hypotheses of Proposition S32. If the triple (S124) satisfies all its strict inequalities, then the target is strictly elliptic with simple linear frequencies, nonresonance through order four, and nonsingular full 64-dimensional cubic-corrected Birkhoff twist. Its local elliptic KAM conclusion follows: invariant Lagrangian 64-tori of positive measure occur in every neighborhood of zero.

Proof. Each nonnegative summand of $31\widetilde{L}_{\mathrm{jet}}$ bounds its individual squared residual; the triangle inequality proves (S123). The maximum distance between two closed real intervals is $d_\times$, proving the eleven other bounds. Subtract (S121) for the two systems. Sorting its terms by physical degree gives exactly the 7, 13 and 22 coefficient perturbations admitted by the moving-Hessian input, bounded by (S124). That input and its analytic local KAM theorem now apply. No IID sampling, sample independence, optimization convergence or zero-loss premise enters this argument.

A finite positive loss is permitted, but must meet the strict KAM budget. For example, for any triple b_2, b_3, b_4 passing that test, monotonicity of its bounds in nonnegative coefficient errors shows that it suffices that the eleven extra errors meet their corresponding b_j and, for every α, $\rho_\alpha + \tau_\alpha < b_{|\alpha|}$ and

$$\overline{L}_{31} \le \frac{1}{31} \min_{2\le|\alpha|\le 4} \left(\frac{b_{|\alpha|} - \rho_\alpha - \tau_\alpha}{\widetilde{s}_\alpha} \right)^2 .$$

This is a conditional budget for the recorded loss, not a newly executed training objective. The present 31-coefficient receipt and incomplete extra validation do not instantiate the corollary. A finite jet is not the full analytic function: higher terms remain analytic and need their own bounds when finite observations are used to receive the target jet. Target force/tail assumptions are not established by fitting the candidate. The conclusion concerns the autonomous conservative system; it gives no KAM statement for the dissipative or driven training trajectories and no specified finite torus radius.

S21 Quadratic positivity and a quartic obstruction in the coefficient intervals

Use the full-coordinate feature factorization and ordinary coefficient convention of Supplementary Sections S8 and S20. In particular,

$$u = Bx/\ell, \qquad e = \frac{x^T K_0 x}{2E_s}, \qquad A = e - \frac{\ell^2 u^T K_3 u}{2E_s} = \frac{x^T C x}{2E_s}, \qquad C = K_0 - B^T K_3 B,$$

where $\ell = 1/1000$, $E_s = 3/500000000000$, $BV = I_3$, and $K_3 = V^T K_0 V$. The weighted material jet is $P = R(u) + AS(u) + c_A A^2$, with 31, 10, and one coefficients, respectively. The results below concern the Cartesian product of the received coefficient intervals. Such an outer set can be larger than the set compatible with all trajectory observations simultaneously. The force and trajectory allowances used to construct these intervals retain their stated reference-equation and target-tail hypotheses.

S21.1 Positivity throughout the seven-dimensional quadratic outer set

Take the source material coefficient $\beta = 3/50$ as a comparison point. Its material correction is βE_0, so that $R_2^\beta(u) = \beta \ell^2 u^T K_3 u/(2E_s)$ and $S_0^\beta = \beta$. Write the seven quadratic coefficient differences as

$$\delta R_2(u) = \sum_{i \le j} \delta r_{ij} u_i u_j, \qquad \delta a = S_0 - \beta.$$

The corresponding full-coordinate equilibrium Hessians satisfy

$$\begin{aligned} \mathsf{H}_\beta &= (1+\beta) K_0 + \kappa b b^T, \\ \mathsf{H}_* &= \mathsf{H}_\beta + \Delta \mathsf{H}, \\ \Delta \mathsf{H} &= \delta a C + B^T D_R B, \qquad D_R = \frac{E_s}{\ell^2} \nabla_u^2 \delta R_2. \end{aligned} \tag{S125}$$

The known magnetic Hessian is retained in both systems. Cubic and quartic material coefficients do not enter this equilibrium Hessian.

Define the positive definite 3-by-3 matrix $\mathcal{G} = B K_0^{-1} B^T$ and set

$$\begin{aligned} c_C &= \max\{1, \operatorname{tr}(K_3 \mathcal{G}) - 1\}, \\ g_{ii} &= \frac{2E_s}{\ell^2} \mathcal{G}_{ii}, \\ g_{ij} &= \frac{E_s}{\ell^2} \left(|\mathcal{G}_{ij}| + \sqrt{\mathcal{G}_{ii} \mathcal{G}_{jj}} \right), \quad i < j. \end{aligned} \tag{S126}$$

For intervals $|\delta a| \le d_a$, $|\delta r_{ij}| \le d_{ij}$, put

$$\rho_K = d_a c_C + \sum_{i \le j} d_{ij} g_{ij}, \qquad \varepsilon_H = \frac{\rho_K}{1+\beta}. \tag{S127}$$

Proposition S34. Structured quadratic interval bound. *If $\varepsilon_H < 1$, every equilibrium Hessian in the specified seven-dimensional coefficient outer set is positive definite and*

$$(1 - \varepsilon_H) \mathsf{H}_\beta \preceq \mathsf{H}_* \preceq (1 + \varepsilon_H) \mathsf{H}_\beta. \tag{S128}$$

For the ordered positive generalized frequencies with the same mass matrix M, this implies

$$\sqrt{1 - \varepsilon_H}\, \omega_i^\beta \le \omega_i^* \le \sqrt{1 + \varepsilon_H}\, \omega_i^\beta, \qquad i = 1, \dots, 64. \tag{S129}$$

This conclusion does not assert frequency simplicity, finite-order nonresonance, or nonsingularity of the quartic twist.

Proof. The matrix $K_0^{-1/2}B^TK_3BK_0^{-1/2}$ is positive semidefinite of rank three. Its nonzero eigenvalues have sum $\operatorname{tr}(K_3\mathcal{G})$, so their maximum does not exceed that trace. It follows that

$$\|K_0^{-1/2}CK_0^{-1/2}\|_2 \le c_C.$$

For a diagonal coefficient, the normalized Hessian direction is a rank-one matrix of norm $2E_s\mathcal{G}_{ii}/\ell^2$. For $i<j$, its rank-two symmetric matrix has its two possible nonzero eigenvalues equal to $(E_s/\ell^2)(\mathcal{G}_{ij}\pm\sqrt{\mathcal{G}_{ii}\mathcal{G}_{jj}})$. Their maximum absolute value is g_{ij}. The triangle inequality therefore gives $-\rho_K K_0 \preceq \Delta\mathsf{H} \preceq \rho_K K_0$. Since $\mathsf{H}_\beta \succeq (1+\beta)K_0$, this proves (S128). Apply the generalized min–max principle with the fixed positive mass M, and then take positive square roots, to obtain (S129).

The received quadratic intervals. For each of the six restricted quadratic coefficients, d_{ij} is the maximum endpoint distance of its received interval from the exact source coefficient of R_2^β. The same construction for the constant normal coefficient uses the interval intersections in Supplementary Section S9. Thus each error includes both the interval radius and its centre discrepancy. Outward arithmetic gives

$$\begin{aligned} \operatorname{tr}(K_3\mathcal{G}) &< 4.288757891, & c_C &< 3.288757891,\\ d_a &< 2.635837185\times10^{-4}, & \rho_K &< 8.736290715\times10^{-4},\\ \varepsilon_H &< 8.241783693\times10^{-4} < 1. \end{aligned} \tag{S130}$$

In particular, the frequency factors in (S129) are bounded below and above by 0.9995878258 and 1.0004120044, respectively. All seven coefficient intervals are admitted simultaneously; no correlation is required for this sufficient positivity bound. This improves the quadratic positivity component of the target analysis without implying that the full moving-Hessian KAM criterion in Proposition S32 is satisfied.

S21.2 A singular quartic twist inside the outer intervals

Consider the following one-parameter slice, with the same mass and known magnetic potential:

$$H_c(x,p)=\frac12 p^TM^{-1}p+\frac{1+\beta}{2}x^TK_0x+\kappa(1-\cos b^Tx)+E_scA(x)^2,\quad -10^{-15}\le c\le 0,\quad \beta=\frac{3}{50}. \tag{S131}$$

Every Hamiltonian on this slice has exactly the same quadratic part and a zero cubic part. Its material feature coefficients are $R=\beta a(u)$, $S=\beta$, and $c_A=c$. Exact endpoint comparisons place the whole slice inside all 31 restricted coefficient intervals, the ten normal coefficient intervals, and both displayed A^2 trajectory outer intervals in (S86) and (S87). Moreover $A(Vq)=0$ implies $\nabla(A^2)(Vq)=0$, so the slice preserves the simultaneous initial-node complete-force values, rather than only their projected components. The initial-node identity alone does not establish subsequent trajectory compatibility. The joint short-window statement is proved separately in Proposition S48.

Use an actual mass-normalized eigenbasis S for H_β, write $\Omega=\operatorname{diag}(\omega_i^\beta)$ and $L=S\Omega^{-1/2}$. In the canonical coordinates $x=LX$, let

$$v_i=b^TL_i,\qquad Q=L^TCL,\qquad I_i=(X_i^2+Y_i^2)/2.$$

The source spectrum is positive, simple, and nonresonant through order four. Since the cubic part vanishes, the quartic Birkhoff twist, defined by the term $I^TT(c)I/2$, is

$$\begin{aligned} T(c)&=T(0)+cT_A,\\ T(0)_{ii}&=-\kappa v_i^4/8, & T(0)_{ij}&=-\kappa v_i^2v_j^2/4\quad (i\ne j),\\ (T_A)_{ii}&=\frac{3Q_{ii}^2}{4E_s}, & (T_A)_{ij}&=\frac{Q_{ii}Q_{jj}+2Q_{ij}^2}{2E_s}\quad (i\ne j). \end{aligned} \tag{S132}$$

The A^2 response agrees with Lemma S29, here in the source-slice canonical chart rather than the learned candidate's chart.

Proposition S35. Obstruction to a uniform quartic twist test. *For the exact source matrices and the slice* (S131)*, there exists* $c_* \in (-10^{-15}, 0)$ *for which* $T(c_*)$ *is singular. Consequently the received coefficient outer set cannot satisfy quartic Kolmogorov-twist nonsingularity uniformly over all its members.*

Proof. All source tip components v_i are nonzero. With $D_v = \mathrm{diag}(v_i^2)$ and $\mathbf{1} \in \mathbb{R}^{64}$,

$$T(0) = -\frac{\kappa}{8} D_v (2\mathbf{1}\mathbf{1}^T - I) D_v.$$

The inner matrix has one positive eigenvalue and 63 negative eigenvalues. Sylvester's law of inertia therefore gives 63 positive and one negative eigenvalue for $T(0)$.

Validated generalized eigenvalue and eigenvector enclosures for all 64 source modes, followed by (S132), give the following strict bounds for the principal block on modes $63, 64$ at $c = -10^{-15}$:

$$\begin{aligned} T_{63,63}(-10^{-15}) &< -1935474, \\ T_{64,64}(-10^{-15}) &< -335169834, \\ \det T_{\{63,64\}}(-10^{-15}) &> 625635089113000. \end{aligned} \tag{S133}$$

This block is negative definite, hence the full real symmetric twist has at least two negative eigenvalues there. The affine symmetric matrix path $T(c)$ is continuous. Its negative index cannot change from one to at least two without an eigenvalue passing through zero. If the first zero occurred at the left endpoint, its two strictly negative eigenvalues would persist on a neighborhood of that endpoint; the same index argument then forces a zero at an interior point. This proves the stated open-interval conclusion.

The obstruction concerns the ordinary fourth-order sufficient KAM test over an outer coefficient set. Its singularity remains valid, but Proposition S47 proves local KAM tori for every finite real member of this source slice using a bordered criterion and an auxiliary energy function. This does not identify an unknown physical material with that slice.

Membership in the trajectory-derived A^2 intervals alone remains a necessary interval test. The additional joint result in Proposition S48 establishes simultaneous compatibility of the whole segment $[-10^{-15}, 0]$ with the unchanged 22-record, 88-cell short-window state balls. Thus those observations do not exclude the singular ordinary-twist member. The result does not assert an exact noise-free fit or a fit to all 257 raw samples. The all-member quadratic positivity, source-slice KAM result, joint short-window compatibility, and separate certificate for the fixed learned Hamiltonian concern their respective stated sets; none certifies the full 42-coefficient target family.

S22 A local KAM criterion for the admitted quadratic target family

Let $I = [\beta_-, \beta_+] \subset (-1, \infty)$ be a compact interval, $A, K_0 \in \mathbb{R}^{3\times 3}$ symmetric positive definite, $d \in \mathbb{R}^3$, and $\kappa > 0$. Consider the conservative family

$$H_\beta(q,p) = \tfrac{1}{2} p^T A^{-1} p + \tfrac{1+\beta}{2} q^T K_0 q + \kappa(1 - \cos(d^T q)). \tag{S134}$$

Here the physical-family hypothesis $W_* = \beta E_0$ is stipulated. Compatibility of observations with I does not establish this constitutive hypothesis for an unknown device.

Lemma S36. Uniform finite-order conditions for the quadratic family. *Write* $K_\beta = (1+\beta)K_0 + \kappa dd^T$*. Suppose that throughout* I *the generalized eigenvalues of* (K_β, A) *are simple, positive and ordered as* $\lambda_1 < \lambda_2 < \lambda_3$*, with* $\omega_j = \sqrt{\lambda_j}$*. Suppose further that*

$$\inf_{\beta\in I} |k \cdot \omega(\beta)| > 0 \quad (0 < |k|_1 \le 4), \qquad \inf_{\beta\in I} (d^T u_j(\beta))^2 > 0 \quad (j = 1,2,3),$$

where the eigenvectors are A-orthonormal. Then the quartic Birkhoff twist, in physical canonical actions, has inertia $(2,1,0)$ *uniformly on* I*. In particular its determinant is bounded away from zero. Every member of* (S134) *satisfies the qualitative local elliptic KAM criterion: sufficiently small neighborhoods of the origin contain a positive-measure family of*

invariant Lagrangian three-tori. This assertion assigns no numerical torus radius or all-integer Diophantine property to the equilibrium frequency vector.

Proof. The Hessian K_β is positive definite because $1+\beta>0$. Choose the signs of the eigenvectors so that $d^T u_j>0$, and let $S=(u_1,u_2,u_3)$. The canonical transformation

$$q=S\,\mathrm{diag}(\omega_j^{-1/2})X,\qquad p=AS\,\mathrm{diag}(\omega_j^{1/2})Y$$

gives $H_2=\sum_j \omega_j(X_j^2+Y_j^2)/2$. There is no cubic term. Put $a_j=(d^Tu_j)^2/\omega_j>0$ and $I_j=(X_j^2+Y_j^2)/2$. The angle averages are $\langle X_j^4\rangle=3I_j^2/2$ and $\langle X_i^2X_j^2\rangle=I_iI_j$ for $i\neq j$. The order-four nonresonance hypothesis makes the resonant projection of $-\kappa(d^Tq)^4/24$ equal to

$$Z_4=-\frac{\kappa}{16}\sum_j a_j^2I_j^2-\frac{\kappa}{4}\sum_{i<j}a_ia_jI_iI_j=\tfrac12 I^TTI,\qquad T=-\frac{\kappa}{8}D(2\mathbf{1}\mathbf{1}^T-I_3)D,\quad D=\mathrm{diag}(a_1,a_2,a_3).$$

The middle matrix has eigenvalues $5,-1,-1$ and inverse $-I_3+(2/5)\mathbf{1}\mathbf{1}^T$. Consequently

$$\det T=-\frac{5\kappa^3}{512}(a_1a_2a_3)^2<0,\qquad T^{-1}=-\frac{8}{\kappa}D^{-1}[-I_3+(2/5)\mathbf{1}\mathbf{1}^T]D^{-1}. \tag{S135}$$

Congruence and the negative scalar give two positive and one negative twist eigenvalues. The potential is entire. Simplicity of the spectrum and nonzero overlaps give analytic canonical coordinates locally in β; the displayed sign convention patches them across I. The compact parameter interval and the strict uniform bounds control the finite normal-form transformations, twist inverse and analytic remainders uniformly. The usual small-action rescaling therefore places each member within the classical nondegenerate KAM theorem. The resulting Diophantine frequencies are selected from the nonlinear frequency image; they need not equal $\omega(\beta)$.

An enclosure of the entire parameter interval. Choose a rational $\beta_c\in I$, solely from the received interval. For $K_c=K_{\beta_c}$, set

$$m=\frac{1+\beta_-}{1+\beta_c},\qquad M=\frac{1+\beta_+}{1+\beta_c}.$$

Since $K_0\preceq K_c/(1+\beta_c)$, $mK_c\preceq K_\beta\preceq MK_c$ for every $\beta\in I$. The generalized min–max principle then gives $m\lambda_j(\beta_c)\le\lambda_j(\beta)\le M\lambda_j(\beta_c)$. Disjoint outward intervals obtained this way label the eigenvalues throughout I.

To avoid substituting center eigenvectors for an interval family, use

$$P_\beta(t)=\det(K_\beta-tA),\qquad N_\beta(t)=d^T\,\mathrm{adj}(K_\beta-tA)d,\qquad (d^Tu_j)^2=-\frac{N_\beta(\lambda_j)}{P'_\beta(\lambda_j)}.$$

This residue identity follows from the symmetric definite spectral decomposition of $(K_\beta-tA)^{-1}$. Exact spectral projectors of $A^{-1}K_\beta$ give the signed eigenvectors

$$u_j=\left[\prod_{k\neq j}\frac{A^{-1}K_\beta-\lambda_kI_3}{\lambda_j-\lambda_k}\right]A^{-1}d\,/\sqrt{(d^Tu_j)^2}.$$

Outward evaluation with both β and all λ_j interval-valued encloses the complete family of canonical charts. Exact identities follow from the spectral theorem; residual intervals containing zero serve as implementation checks rather than proofs of those identities. The executable certificate separately verifies all order-at-most-four divisors, the positive overlap lower bounds and (S135).

The interval I in the numerical instance is the exact scalar compatibility intersection of the received 33 force and 31 coefficient constraints. No material value is assigned to β independently of that interval. This is a target-family certificate under a strong constitutive prior. It supplies no identification of unknown physics, necessity of a neural correction, complex-network advantage, invariant full-field trial subspace, or forced/damped KAM statement.

S22.1 A received numerical instance for the entire target family

The exact compatibility interval received from the 33 force and 31 coefficient inequalities is used unchanged. Its endpoints are approximately 0.0599999981693 and 0.0600000004402; these displays do not replace the exact rational endpoints in the calculation. The matrices are the same native $A = M_3$, $K_0 = K_{03}$, d and exact κ used in the current three-coordinate mechanical restriction. No network weights enter this family calculation.

Corollary S37. Qualitative local KAM for every admitted family member. *Under the explicit constitutive hypothesis $W_* = \beta E_0$, every β in the entire received compatibility interval satisfies Lemma S36. The uniform frequency enclosures in physical radians per second are*

Mode	*Strict enclosing interval*
1	$[11.73745098, 11.73745102]$
2	$[72.26615477, 72.26615488]$
3	$[326.09696048, 326.09696086]$

All three squared modal port overlaps are strictly positive throughout the interval. Every integer combination with $0 < |k|_1 \le 4$ satisfies $|k \cdot \omega| > 11.73745099\ \mathrm{rad\,s^{-1}}$. In the physical-action convention of (S135),

$$-6.217103 \times 10^{17} < \det T(\beta) < -6.217101 \times 10^{17}, \qquad \sigma_{\min}(T(\beta)) > 98709.9.$$

The twist inertia is $(2, 1, 0)$. The target family therefore has the qualitative local invariant-three-torus conclusion of Lemma S36. No numerical action or torus radius is asserted.

Proof. The frozen 512-bit Arb calculation receives the exact beta endpoints from the successful scalar feasibility certificate. Exact rational Sylvester tests verify the mass and the lower-endpoint Hessian. Exact Sturm isolation at the interval-selected dyadic center, followed by the Loewner/min–max enclosure, gives disjoint spectral intervals for every beta. The characteristic-resolvent residues and polynomial spectral projectors are evaluated with both beta and the eigenvalues interval-valued. They receive positive port overlaps and enclose the changing eigenbasis. All 64 plus/minus representatives through order four exclude zero. The analytic twist determinant and inverse formulas then give the stated strict bounds; a separate generic quartic $a, \bar{a}$ expansion checks their factors. The real spectral theorem and the angle-average calculation supply the exact identities. The local elliptic reduction and the classical KAM theorem (Pöschel 2001) give the qualitative conclusion.

The result concerns the admitted conservative family directly. It does not establish perturbative transfer from the learned candidate, unknown physical identification, an invariant full-field trial subspace, or necessity of a neural correction.

S23 Exact finite conditions for the declared full-coordinate family

The matrices in this section are the exact real constants represented by the stored binary64 source entries. Conversion to rational numbers precedes the sum $M = M_0 + M_r$. All 64 free transverse displacement and rotation coordinates are retained, in the declared physical order. Set $b = e_{64}$ and consider only the conservative Hamiltonian

$$H_\beta(x, p) = \tfrac{1}{2} p^T M^{-1} p + \tfrac{1+\beta}{2} x^T K_0 x + \kappa(1 - \cos(b^T x)), \qquad \kappa = \frac{638029437976759647574751488 1383}{106338239662793269832304564822427566 08}.$$

Let $[\beta_-, \beta_+]$ denote the exact rational intersection recorded in the frozen scalar-family receiver. Its decimal display is approximately $[0.059999998169293786, 0.0600000004401771 5]$; the decimal values do not define the interval. The reference parameter is $\beta_c = 3/50$.

Proposition S38. Received full-coordinate finite conditions. *For every $\beta \in [\beta_-, \beta_+]$, the origin of H_β is elliptic. Its 64 positive frequencies are simple, all modal tip overlaps are nonzero, and*

$$|k \cdot \omega(\beta)| \ge \frac{2472231}{1073741824} > 0, \qquad k \in \mathbb{Z}^{64}, \quad 0 < |k|_1 \le 4.$$

Consequently its quartic Birkhoff normal form has a nonsingular action Hessian. In the convention $H_{\mathrm{BNF}} = \omega \cdot I + \frac{1}{2} I^T T I + O_6$*, it is*

$$T = -\frac{\kappa}{8} D(2\mathbf{1}\mathbf{1}^T - I_{64})D, \qquad D = \operatorname{diag}\left(\frac{(b^T u_j)^2}{\omega_j}\right),$$

where $u_i^T M u_j = \delta_{ij}$*. Its inertia is* $(63, 1, 0)$ *and*

$$\det T = -127 \left(\frac{\kappa}{8}\right)^{64} \prod_{j=1}^{64} \left(\frac{(b^T u_j)^2}{\omega_j}\right)^2 \neq 0.$$

Proof. Exact banded rational LDL^T factorizations establish $M \succ 0$ and $K_0 \succ 0$. Thus $(1+\beta)K_0 + \kappa b b^T \succ 0$ for $\beta > -1$. The reduced characteristic polynomial $P_0(t) = \det(K_0 - tM)$ and tip cofactor $N_0(t) = \det[(K_0 - tM)_{1:63,1:63}]$ have degree-preserving reductions of degrees 64 and 63 modulo the verified prime 2147483647. Every source denominator is invertible modulo this prime. Their leading coefficients are respectively 1029774066 and 1236496576. Stored coefficient lists of degrees 62 and 63 give an explicit identity $AP_0 + BN_0 = 1$ over the finite field. Hence P_0, N_0 are coprime over $\mathbb{Q}$ and have no common complex root.

For $s = 1 + \beta$, $t = \lambda/s$, rank-one expansion gives

$$P_\beta(\lambda) = s^{64} P_0(t) + \kappa s^{63} N_0(t), \qquad N_\beta(\lambda) = s^{63} N_0(t).$$

Therefore these polynomials have no common root for any $\beta > -1$. At a repeated eigenvalue of a symmetric definite pencil the adjugate vanishes, whereas at a simple eigenvalue it is a nonzero multiple of uu^T. The absence of a common root excludes repeated eigenvalues and zero tip overlap, uniformly for all such parameters.

At β_c, 128 exact rational endpoint inertia computations isolate all 64 positive eigenvalues in disjoint intervals. The floating eigensolver supplies candidate endpoints only. All 64 brackets passed their first declared padding, without an endpoint retry. For $m = (1+\beta_-)/(1+\beta_c)$ and $M_* = (1+\beta_+)/(1+\beta_c)$,

$$m K_{\beta_c} \preceq K_\beta \preceq M_* K_{\beta_c}.$$

The generalized min–max principle and exact dyadic bounds on $\sqrt{m}, \sqrt{M_*}$ transport the brackets to the whole parameter interval. On a grid with denominator 2^{32} every transported frequency is within $\varepsilon = 337625/2^{32}$ of its stored center.

An exact integer enumeration includes repetitions and covers all signed sums with at most four terms: positive-only sums, one against one, two against one, three against one, and two against two. The identical two-pair multisets represent $k = 0$ and are excluded. Sorting all 2080 pair multisets covers the two-against-two minimum by adjacent distinct sums. The 45760 triple multisets cover all three-against-one cases. The smallest center gap is $11239424/2^{32}$, attained by the signed modes $(2, 23, 30) - (37)$. Subtracting 4ε gives $9888924/2^{32} = 2472231/1073741824$. Canonical oscillator normalization and quartic angle averaging then give the displayed twist formula. Its nonsingularity follows from the exact nonzero overlaps, not from a floating determinant.

The conclusion is relative to the 64-coordinate transverse Hamiltonian, whose canonical phase space has dimension 128. It does not assert a 96-dimensional Lagrangian torus in a model retaining axial coordinates. The source matrices define exact constants; equality with compiled simulator operations or with unknown material laws is a separate premise. The local KAM implication is supplied by the following analytic argument.

S24 Local KAM tori of the current fixed FE conservative model

We retain the received 64 transverse coordinates and the current mass matrix, rather than identifying its momentum quadratic form with the reference one. Let M_c and K_a denote the exact real matrices represented by the current saved binary64

assembly. Under the fixed linear-material, cosine-tip model declaration, set

$$H_c(q,p) = \tfrac{1}{2}p^T M_c^{-1} p + \tfrac{1}{2}q^T K_{e,c} q + \kappa(1-\cos(b^T q)), \qquad b = e_{64}, \tag{S136}$$

$$K_{e,c} = K_a - [\kappa\cos\theta_a + \mu u_a \sin\theta_a] bb^T, \qquad \theta_a = b^T q_a. \tag{S137}$$

Here q_a is the raw initial assembly state and u_a its evaluated nonzero magnetic input. The second term in the bracket is the input contribution to the tangent stiffness; retaining it would change the conservative target. The conservative origin Hessian is consequently

$$K_c = K_{e,c} + \kappa bb^T = K_a + \delta bb^T, \qquad \delta = \kappa(1-\cos\theta_a) - \mu u_a \sin\theta_a. \tag{S138}$$

The declaration specifies the constant matrices and the complete nonlinear potential in (S136). An isolated tangent measurement would not establish that potential identity.

Lemma S39. Cyclicity under a tip stiffness correction. *Let $M \succ 0$ and $K = K^T$, and put $A = M^{-1}K$ and $z = M^{-1}b$. If $\mathcal{C} = [z, Az, \ldots, A^{n-1}z]$ is nonsingular, then, for every real t, the pencil $(K + tbb^T, M)$ has simple eigenvalues and each of its mass-normalized eigenvectors s satisfies $b^T s \neq 0$.*

Proof. Writing $A_t = A + tzb^T$, induction gives $A_t^j z = A^j z + \sum_{i<j} c_{ij}(t) A^i z$. Thus its Krylov matrix is $\mathcal{C}$ times a unit upper triangular matrix and has the same nonzero determinant. The whitened matrix $M^{-1/2}(K + tbb^T)M^{-1/2}$ is symmetric and cyclic for $w = M^{-1/2}b$. A repeated eigenspace, or an eigenvector orthogonal to w, would provide a nonzero vector annihilating every Krylov column. Both are impossible. The whitened eigenvector corresponding to s is $M^{1/2}s$, whose overlap with w equals $b^T s$.

Theorem S40. Current-mass conservative local KAM result. *For the fixed FE Hamiltonian (S136) determined by the received current assembly and the stated potential declaration, every sufficiently small neighborhood of $(q,p) = (0,0)$ contains a positive-measure union of invariant Lagrangian 64-tori with Diophantine frequencies. The conclusion is qualitative and assigns no numerical torus radius.*

Proof. First, a fresh exact rational LDL factorization of M_c has 64 positive pivots. To verify cyclicity independently of the small real correction δ, reduce the rational entries of M_c, K_a modulo the prime $p_* = 2147483647$. Their binary64 denominators are powers of two and are invertible in this field. Exact elimination and an explicit inverse residual give

$$\det M_c \equiv 1157806367, \qquad \det \mathcal{C}(M_c^{-1}K_a, M_c^{-1}b) \equiv 182384660 \pmod{p_*}. \tag{S139}$$

The nonzero latter value proves nonsingularity over the rationals. Lemma S39 therefore applies to the exact real δ in (S138).

For positivity and quantitative frequency separation, let $M_* = M_0 + M_r$ be the exactly added reference mass and $K_* = sK_0 + \kappa bb^T$, where $s = 53/50$. Reference M_*, K_0 are positive definite by exact rational LDL. The received current form bounds are

$$(1-f)M_* \preceq M_c \preceq (1+f)M_*, \qquad f = \frac{21}{10^{11}}, \tag{S140}$$

$$|q^T(K_{e,c} - sK_0)q| \le \eta_K q^T K_0 q, \qquad \eta_K = \frac{11169}{10^{12}}. \tag{S141}$$

Set $r = \eta_K/s$. Since $0 \le r, f < 1$ and $K_* \succeq sK_0$, we obtain $(1-r)K_* \preceq K_c \preceq (1+r)K_*$ and hence $K_c \succ 0$. The generalized min–max principle yields, in increasing order,

$$\sqrt{\frac{1-r}{1+f}}\,\omega_{*,i} \le \omega_{c,i} \le \sqrt{\frac{1+r}{1-f}}\,\omega_{*,i}. \tag{S142}$$

Thus the actual changed mass is included in the frequency calculation.

The 64 reference center frequencies are isolated by exact inertia at rational squared-frequency endpoints. Applying (S142) with outward rational square roots and a common outer endpoint grid 2^{-32} gives the following strict bounds. Repeated indices are allowed in pairs and triples; identical multisets in the last row are excluded.

Signed-sum class	Lower bound multiplied by 2^{32}
Positive-only	50411949809
One versus one, distinct indices	259968759917
Two versus one	7286610902
Three versus one	9172774
Two versus two, distinct multisets	167405463

All 2080 unordered pairs and 45760 unordered triples are included. Each pair or triple interval is compared with every single interval; all distinct pair intervals are separated after ordering their lower endpoints. These cases cover every $0 < |k|_1 \le 4$, including expressions with a shared index on both sides. Consequently

$$|k \cdot \omega_c| \ge \frac{4586387}{2147483648} > 0.00213570, \qquad 0 < |k|_1 \le 4. \tag{S143}$$

Choose real generalized eigenvectors S_c satisfying $S_c^T M_c S_c = I$ and $S_c^T K_c S_c = \Omega_c^2$, and set $L_c = S_c \Omega_c^{-1/2}$. The transformation

$$q = L_c X, \qquad p = L_c^{-T} Y = M_c S_c \Omega_c^{1/2} Y \tag{S144}$$

is canonical and gives $H_{c,2} = \sum_i \omega_{c,i} I_i$, $I_i = (X_i^2 + Y_i^2)/2$. In particular the momentum quadratic form is transformed together with the configuration form. There is no cubic term, and with $t_i = b^T L_{c,i}$ the quartic term is $-\kappa(\sum_i t_i X_i)^4/24$. Its nonresonant Birkhoff normal form is $\frac{1}{2} I^T \mathsf{T}_c I$, where

$$\mathsf{T}_c = -\frac{\kappa}{8} D(2\mathbf{1}\mathbf{1}^T - I)D, \quad D = \operatorname{diag}(t_i^2), \qquad \det \mathsf{T}_c = -127 \left(\frac{\kappa}{8}\right)^{64} \prod_{i=1}^{64} t_i^4 \neq 0. \tag{S145}$$

The last inequality follows from Lemma S39; its inertia is $(63, 1, 0)$ in positive, negative, zero order.

Finally, (S136) is real analytic and even. The fourth-order homological transformation preserves evenness, so the remainder after its quartic normal form begins at degree six. The positivity, finite nonresonance, and nonsingular twist just proved satisfy Theorem S43, which gives the stated local invariant tori. No Diophantine condition is assigned to the equilibrium frequency vector itself.

This result concerns the declared current conservative FE Hamiltonian. It does not identify an unrestricted constitutive target from the training loss, supply the missing coefficient directions of a general 42-term jet, or bound an unknown analytic tail. The learning confidence set must still be related to an admissible target class before a learning-to-target KAM statement follows. Likewise, the forced and dissipative numerical histories and physical execution require their own error and power-balance arguments.

S25 A nonzero-loss interface for the current FE material family

The current-mass result of Theorem S40 concerns a specified conservative Hamiltonian. To connect the saved force loss with that model, consider the explicit material class

$$H_s(q,p) = \tfrac{1}{2} p^T M_c^{-1} p + \tfrac{s}{2} q^T K_{e,c} q + \kappa(1 - \cos(b^T q)), \qquad s \in [99/100, 101/100]. \tag{S146}$$

Changing s scales the current elastic bilinear form while retaining the current mass, mesh, geometry and magnetic potential. This is the declared linear-material scaling, rather than an assertion of bitwise software reassembly for every modulus. Its constitutive restriction is not inferred from the finite observations.

Let V be the fixed 64×3 configuration injection and z_j the ideal reduced observation position for row j. The old normalized force labels subtract the nominal elastic force with stiffness K_0. The correct current-family multipliers are therefore

$$a_j = \frac{\ell}{E_s} [V^T K_0 V z_j]_{r_j}, \qquad c_j = \frac{\ell}{E_s} [V^T K_{e,c} V z_j]_{r_j}, \qquad f_j(s) = s c_j - a_j, \tag{S147}$$

where $r_j \in \{1,2,3\}$ is the observed component. The known magnetic potential is common to both models and cancels in this correction. The c_j are recomputed from the current assembly, including the complete anchor tangent removal in (S137); the old source material multipliers are not substituted for them.

The unchanged network supplies $g_j \in [\ell_j, u_j]$, and y_j and $\eta_j > 0$ are the saved labels and source allowances. Its received loss satisfies

$$L_F = \frac{1}{33}\sum_j \left|\frac{g_j - y_j}{\eta_j}\right|^2, \qquad \frac{3}{100} < L_F \le \overline{L}_F \le \frac{12}{275}. \tag{S148}$$

The normalization remains η_j, even when a model discrepancy is added below.

Proposition S41. Current material parameter and potential enclosure. *Suppose a target in* (S146) *obeys* $|f_j(s) - y_j| \le \eta_j + \Delta_j$. *Then the unchanged nonzero loss implies the necessary outer strips*

$$sc_j \in [\ell_j - \epsilon_j + a_j,\ u_j + \epsilon_j + a_j], \qquad \epsilon_j = \frac{11}{5}\eta_j + \Delta_j. \tag{S149}$$

Let B_θ be their intersection with the stated prior, using outward interval division by the newly received c_j. No direct-data parameter set is intersected into B_θ.

For a force-only transfer from the earlier source class, put

$$\beta(s) = \frac{53}{50}s - 1, \qquad d_j = c_j - \frac{53}{50}a_j, \qquad \Delta_j = \frac{101}{100}|d_j|_{\mathrm{up}}. \tag{S150}$$

Every source-compatible $\beta(s)$ satisfying $|\beta(s)a_j - y_j| \le \eta_j$ then meets the target-label premise with this Δ_j. For the saved checkpoint and all 33 recomputed current multipliers, the resulting exact interval obeys the outward display bounds

$$B_\theta \subset [0.9999999932365705,\ 1.0000000030932600]. \tag{S151}$$

For every s in the exact set B_θ, the unchanged learned potential has

$$\sup_{E_0 \le 4E} \frac{|U_N - U_s|}{E} < 0.003839064485. \tag{S152}$$

The current model $s = 1$ is also compatible with the original unexpanded label allowances: direct outward evaluation gives $\max_j |f_j(1) - y_j|/\eta_j < 0.983855$.

Proof. Equation (S148) gives $|g_j - y_j| \le (6/5)\eta_j$. The triangle inequality and $g_j \in [\ell_j, u_j]$ prove (S149). To transfer source force allowances, use the exact identity $f_j(s) = \beta(s)a_j + sd_j$ and $0 < s \le 101/100$. All 33 c_j receive strict-sign outward intervals; dividing all endpoint combinations and intersecting the resulting strips gives the exact rational set underlying (S151). The direct evaluation at $s = 1$ is performed separately and is not used to tighten this loss-generated set.

Let B_β^{old} denote the entire earlier nonzero-loss-generated outer set. Its exact affine image $S = (1 + B_\beta^{\mathrm{old}})/(53/50)$ lies within the stated positive prior. For $s \in S$, the quantity $\beta(s)a_j$ lies in every old candidate-centred strip. Adding sd_j increases its half-width by at most $(101/100)|d_j|_{\mathrm{up}}$. Thus S lies in every new strip and $S \subseteq B_\theta$. Exact endpoint comparisons independently verify this inclusion; the direct-data set is not used in this step.

For the potential error, write $U_s = U_{\beta(s)} + sq^T\Delta Kq/2$, where $\Delta K = K_{e,c} - (53/50)K_0$. The received form bound implies $|sq^T\Delta Kq|/(2E) \le 4s\eta_K$ on $E_0 \le 4E$. Using the same saved energy skip coefficient c_E, origin derivative interval Γ_4, full-domain feature radius r_4 and energy-head Hessian majorant $\mathcal{H}$ as in the source learning bound gives

$$\frac{|U_N - U_s|}{E} \le 4 \max_{\gamma \in \Gamma_4,\, s \in B_\theta} |c_E + \gamma - \beta(s)| + \tfrac{1}{2}r_4^T\mathcal{H}r_4 + 4(\max B_\theta)\eta_K. \tag{S153}$$

Exact rational arithmetic gives an upper bound 0.003839064484188617, below the stated rational relaxation. The deterministic matrix discrepancy is retained in both the force and potential comparisons.

Corollary S42. Uniform local KAM conditions on the loss-generated set. *Every H_s with s in the exact loss-generated set of Proposition S41 has local positive-measure families of invariant Lagrangian 64-tori. Its frequencies satisfy the parameter-uniform bound*

$$|k\cdot\omega_s| \ge \frac{2064711}{1073741824} > 0, \qquad 0 < |k|_1 \le 4. \tag{S154}$$

Proof. The received form bound gives $K_{e,c} \succeq (53/50 - \eta_K)K_0 \succ 0$. With $c_t = \kappa\cos\theta_a + \mu u_a \sin\theta_a$, its origin stiffness is $K_s = sK_a + (\kappa - sc_t)bb^T$. Scaling the uncorrected pencil by $s > 0$ multiplies its Krylov determinant by s^{2016}. Lemma S39 removes the remaining rank-one correction. Thus all tip projections remain nonzero for every $s > 0$.

For $s \in [s_-, s_+]$ positivity also yields $\min(1, s_-)K_1 \preceq K_s \preceq \max(1, s_+)K_1$. Generalized min–max transfers each of the 64 already received current frequency intervals using outward square roots of these factors. Fresh exact integer endpoint enumeration of all signed sums through order four gives (S154). The cosine quartic twist is nonsingular by (S145), and the Hamiltonians are analytic and even. Theorem S43 applies separately at each s in the exact set.

The old continuous label certificates concern source recorded-state mathematical IVPs using the reference mass. Equation (S150) transfers force functions by a known matrix discrepancy; it does not certify the current saved raw trajectory or transfer a mass-dependent observation residual. The parameter result retains its explicit structural prior and target-label premise. Neither it nor the uniform potential bound identifies the missing directions of a general 42-term jet or an unknown analytic tail.

S26 From fourth-order finite conditions to local KAM tori

Theorem S43. Local elliptic KAM criterion in arbitrary finite dimension. *Let H be real analytic and even near an elliptic equilibrium in $\mathbb{R}^{2n}$, with $n \ge 2$. Suppose its positive linear frequencies satisfy $k\cdot\omega \ne 0$ for $0 < |k|_1 \le 4$, and its quartic Birkhoff normal form is*

$$H\circ\Phi = \omega\cdot I + \tfrac{1}{2}I^T T I + R, \qquad \det T \ne 0, \quad R = O(\|Z\|^6).$$

Then every sufficiently small neighborhood of the equilibrium contains a positive-Lebesgue-measure union of invariant Lagrangian n-tori with Diophantine frequencies. No all-order Diophantine hypothesis on the equilibrium frequency vector is required. This is a qualitative local conclusion, with no specified numerical amplitude threshold.

Proof. Finite fourth-order nonresonance allows the quartic homological equation to be solved except for action monomials. There are no cubic terms. The degree-four generating function has an odd Hamiltonian vector field, so its local analytic time-one map is odd. It preserves the evenness of the Hamiltonian; consequently the remainder begins at degree six. This constructs the stated finite normal form without dividing by any higher-order small divisor.

We apply Theorems A and B of Pöschel (2001). Fix $\tau > n - 1$ and $\nu = \tau + 1$. The action-offset radius in those theorems will be denoted by r, their frequency parameter radius by h_p, and their Diophantine constant by α. Their smallness requirements are

$$|P|_{r,s,h_p} \le \gamma\alpha r s^\nu, \qquad \alpha s^\nu \le h_p, \qquad r, s, h_p \le 1,$$

where $s > 0$ is an angular analytic strip and $\gamma > 0$ depends only on n, τ. These hypotheses must be verified after accounting for the shrinking nonlinear frequency domain.

Let $0 < \rho \ll 1$ and choose real actions in $\rho(1, 2)^n$; use a fixed smaller cube for the frequency parameters. On a complex neighborhood of this cube of action width $c\rho$, and a fixed strip $|\operatorname{Im}\theta| < s$, the oscillator action–angle map is analytic with size $O(\sqrt{\rho})$. The square roots have a fixed analytic branch since the neighborhood stays away from every $I_j = 0$. Consequently $|R| \le C\rho^3$ there. All constants in this proof are fixed before $\rho \to 0$; none is asserted to be numerically small.

Write $h_0(I) = \omega\cdot I + \frac{1}{2}I^T T I$. Since T is invertible, $\varpi = \nabla h_0(I_0) = \omega + TI_0$ is a valid frequency parameter and $I_0 = T^{-1}(\varpi - \omega)$. Its real frequency domain Ω_ρ is a translated parallelepiped of diameter $O(\rho)$ and volume comparable to ρ^n. Take $h_p = c_p\rho$, with $c_p > 0$ sufficiently small, so the complex parameter neighborhood is mapped into the analytic action neighborhood. Translate $I = I_0(\varpi) + J$ for each fixed parameter. The same physical Hamiltonian becomes

$$H = e(\varpi) + \varpi\cdot J + P(J, \theta; \varpi), \qquad P = \tfrac{1}{2}J^T T J + R(I_0 + J, \theta),$$

and therefore $|P|_{r,s,h_p} \le C(r^2+\rho^3)$.

Choose $r=\rho^{3/2}$ and $\alpha=\rho^{5/4}$, keeping $s\le 1$ fixed. For sufficiently small ρ all domains are valid and

$$\frac{|P|_{r,s,h_p}}{\alpha r s^\nu} \le C_s \rho^{1/4} \longrightarrow 0, \qquad \frac{\alpha s^\nu}{h_p} = \frac{s^\nu}{c_p}\rho^{1/4} \longrightarrow 0.$$

Thus the actual quantitative hypotheses of Theorems A and B hold. In particular, the remainder is compared with the shrinking Diophantine constant and shrinking action radius; nonsingularity of T alone is not used as a substitute for this check.

For completeness, the Diophantine parameter set has positive measure in this shrinking frequency domain. The intersection with a resonance slab $|k\cdot\varpi|<\alpha/|k|_1^\tau$ has measure at most $C\rho^{n-1}\alpha/|k|_1^{\tau+1}$. Summing over $k\neq 0$ converges for $\tau>n-1$. A boundary strip of width $O(\alpha)$ has the same order, and the excluded relative measure is therefore $O(\alpha/\rho)=O(\rho^{1/4})$. Theorem B supplies the bi-Lipschitz parameter correction and the Lagrangian torus embeddings. Its estimates give a parameter correction of size $O(|P|/r)=O(\rho^{3/2})=o(\rho)$ and a Lipschitz change tending to zero, since $|P|/(\alpha r s^\nu)=O(\rho^{1/4})$. Together with the invertible frequency-to-action map this preserves positive measure for the union of invariant tori of this single Hamiltonian. Their oscillator amplitude is $O(\sqrt{\rho})$ and the analytic canonical map Φ maps them into arbitrarily small neighborhoods of the original equilibrium.

Corollary S44. Declared full64 family. *For each exact $\beta\in[\beta_-,\beta_+]$ in Proposition S38, the conservative Hamiltonian H_β has positive-measure families of invariant Lagrangian 64-tori arbitrarily near its elliptic equilibrium.*

Proof. The cosine Hamiltonian is even and analytic, the exact finite divisor bound solves the fourth-order normal form, and the received twist is nonsingular. Apply Theorem S43 with, for example, $\tau=64>63$. The Diophantine frequencies are selected from the nonlinear frequency image, rather than assigned to the equilibrium vector. The conclusion concerns this declared family and supplies neither a numerical torus radius nor a certificate for unknown material perturbations.

S27 Local KAM tori across the singular quartic source slice

The failure of the ordinary twist test in Proposition S35 does not preclude a local KAM conclusion for that one-parameter family. An auxiliary scalar function of the energy supplies the missing twist while preserving the original invariant tori.

Lemma S45. Energy reparametrization. *Let H be real analytic and even near an elliptic equilibrium at zero in $\mathbb{R}^{2n}$, $n\ge 2$, with $H(0)=0$, positive linear frequencies ω, and $k\cdot\omega\neq 0$ for $0<|k|_1\le 4$. Write its fourth-order Birkhoff normal form as*

$$H\circ\Phi=\omega\cdot I+\tfrac12 I^T T I+R_6, \qquad R_6=O(\|Z\|^6).$$

If either T or the bordered matrix

$$\mathcal{B}(T,\omega)=\begin{pmatrix} T & \omega \\ \omega^T & 0\end{pmatrix} \tag{S155}$$

is nonsingular, every sufficiently small neighborhood of zero contains a positive-Lebesgue-measure union of invariant Lagrangian tori of H with Diophantine frequencies. This conclusion specifies neither a numerical amplitude nor a prescribed energy surface.

Proof. If T is nonsingular, apply Theorem S43. Otherwise suppose $\det T=0$ and $\det\mathcal{B}(T,\omega)\neq 0$. Fix $E_s>0$ and define

$$G=F(H), \qquad F(h)=h+\frac{h^2}{2E_s}, \qquad \lambda=E_s^{-1}. \tag{S156}$$

The constant λ has units of inverse energy. This is an auxiliary Hamiltonian for the proof; the original equations remain the object of the conclusion. The equilibrium energy has been set to zero, so $F'(H(0))=1$. Consequently G has the same quadratic part, linear frequencies and finite-order nonresonance, and remains even and analytic. Composition with the same canonical map gives

$$G\circ\Phi=\omega\cdot I+\tfrac12 I^T(T+\lambda\omega\omega^T)I+\widetilde{R}_6, \qquad \widetilde{R}_6=O(\|Z\|^6).$$

The only fourth-degree term in $\lambda H^2/2$ is $\lambda(\omega \cdot I)^2/2$. Cofactor expansion and determinant linearity under a rank-one update yield, including when T is singular,

$$\begin{aligned} \det(T + \lambda\omega\omega^T) &= \det T + \lambda\omega^T \operatorname{adj}(T)\omega \\ &= \det T - \lambda \det \mathcal{B}(T, \omega) \neq 0. \end{aligned} \tag{S157}$$

Theorem S43, whose shrinking-domain estimates implement the parameter KAM theorem of Pöschel (2001), now applies to G. No higher-order nonresonance is introduced.

Choose the real neighborhood so that $|H| < E_s/2$. Then $1/2 < F'(H) < 3/2$ and F is injective on the attained energy interval. For each resulting torus embedding K, write $X_G \circ K = DK\,\nu$ with ν Diophantine. Conservation of G and density of the linear flow imply that $G \circ K$ is constant on the whole torus. Injectivity of F implies $H \circ K = E_K$ is constant as well. Thus

$$X_G = (1 + \lambda H)X_H, \qquad X_H \circ K = DK\,\frac{\nu}{1 + \lambda E_K}. \tag{S158}$$

The time multiplier is a positive constant on each torus. If $|k \cdot \nu| \geq \gamma |k|_1^{-\tau}$, the original frequency satisfies the same inequality with constant $\gamma/(1 + \lambda E_K) > 0$. The torus embeddings, Lagrangian property and union of phase-space points are unchanged, so positive phase volume is preserved.

The local construction uses strictly positive action cubes $I \in \rho(1, 2)^n$ and complex neighborhoods avoiding $I_j = 0$. This supplies the analytic branches of the oscillator action–angle map. The transfer in (S158) changes none of those points or embeddings. The parameter ρ may tend to zero after the Hamiltonian and neighborhood have been fixed.

Lemma S46. A constant-pencil criterion. *Let S, U be real symmetric matrices, with S invertible, and let w be real. Suppose every eigenvalue of $S^{-1}U$ is simple over $\mathbb{C}$ and $w^T v \neq 0$ for each right eigenvector v. For every finite real z, either $S + zU$ is nonsingular or its border by w is nonsingular.*

Proof. At $z = 0$, use invertibility of S. At any other singular real parameter, $S + zU = S(I + zS^{-1}U)$ has a one-dimensional real kernel, corresponding to the simple real eigenvalue $-1/z$ of $S^{-1}U$. Let v span this kernel. If $(S + zU)x + tw = 0$ and $w^T x = 0$, multiplication by v^T gives $t = 0$ by symmetry and $v^T w \neq 0$. Then x is proportional to v, and the second equation gives $x = 0$. The bordered matrix is therefore invertible. Complex eigenvalues need not be real; a zero eigenvalue does not correspond to a finite singular parameter. The pairing here is the bilinear transpose $w^T v$.

Proposition S47. Every finite parameter of the source A^2 slice. *Extend the expression for H_c in (S131) to every finite $c \in \mathbb{R}$, retaining its exact source quadratic part and zero cubic part. Each such conservative Hamiltonian has a positive-measure family of invariant Lagrangian 64-tori with Diophantine motion arbitrarily near zero. The neighborhood may depend on c.*

Proof. Put $a = \kappa/8$, $D = \operatorname{diag}(v_i^2)$, $S = I - 2\mathbf{1}\mathbf{1}^T$ and $c_s = 10^{-18}$. Formula (S132) gives

$$T(c) = aD(S + zU)D, \qquad z = c/c_s, \qquad U = (c_s/a)D^{-1}T_A D^{-1}. \tag{S159}$$

All diagonal entries of D are positive. For $n = 64$, $\det S = -127$ and $S^{-1} = I - (2/127)\mathbf{1}\mathbf{1}^T$. Take $w = D^{-1}\omega/2^k$, where $k = -1$ is a fixed normalization. With $P = \operatorname{diag}(a^{-1/2}D^{-1}, a^{1/2}/2^k)$, direct congruence gives

$$P^T \mathcal{B}(T(c), \omega)P = \begin{pmatrix} S + zU & w \\ w^T & 0 \end{pmatrix}.$$

Validated arithmetic at 320 bits encloses all 64 eigenpairs of $S^{-1}U$, separates every pair of eigenvalues, and bounds all 64 quantities $|w^T v|$ strictly away from zero. A separate 384-bit calculation using the inverse generalized source spectrum and tip-normalized modes gives the same one-to-one enclosed spectrum and nonzero pairings. In that normalization, the smallest certified pairing lower bound exceeds 6.2452×10^{-6}. These are complete spectrum and pairing tests, not generic-parameter or almost-everywhere assertions. Lemma S46 therefore proves ordinary or bordered nondegeneracy for every finite real c.

The family remains even and analytic with unchanged positive quadratic part. The finite nonresonance at $\beta = 3/50$ is covered by Proposition S38 and does not depend on c. Apply Lemma S45 to each member. Only singular ordinary-twist members require the auxiliary function (S156); a common transformed twist for all parameters is not assumed.

The proposition applies to this fixed-quadratic, zero-cubic, one-parameter source family. It does not certify the full 42-coefficient outer set, identify an unknown material, select a numerical torus radius, or assert tori on every prescribed energy surface. Forced and dissipative observation dynamics require the separate analysis below.

S28 The source slice in the unchanged joint observation balls

The coefficient intervals alone do not establish a simultaneous trajectory fit. This section verifies a joint statement for the A^2 slice against the original short-window state balls. Throughout, put $K_e = (53/50)K_0$ and

$$D_* = \alpha_* M_0, \qquad \alpha_* = \frac{5754906970707699}{9007199254740992}\ \mathrm{s}^{-1}.$$

This is the stored source damping factor, separate from the candidate's nominal damping factor. Set $f(t) = \mu B_{\max}\cos(\nu t)$, with the same forcing and mass as the original observation certificates. The recorded-state initial value problems are

$$\begin{aligned} M\ddot{x} + D_*\dot{x} + K_e x + \kappa b\sin(b^T x) + g_c(x) &= f(t)b\cos(b^T x), \\ g_c(x) &= \frac{c}{E_s}(x^T C x)Cx, \qquad c \in [-10^{-15}, 0]. \end{aligned} \tag{S160}$$

These forced, damped equations are used only for observation compatibility; Proposition S47 concerns their conservative Hamiltonian counterpart.

S28.1 The fixed observation set

For each of the 22 signed records r, let $Q_{r,j}$ be the exact cubic Hermite interpolant on its original cell $j = 0, 1, 2, 3$, constructed from saved position and velocity at both endpoints. Let $R_{r,j}^{\mathrm{old}}$ be the all-prefix radius from the original source certificate. Define

$$\|e\|_E^2 = e_x^T K_e e_x + e_v^T M e_v, \qquad \sup_{t\in[t_{r,j},t_{r,j+1}]} \|(x,\dot{x}) - (Q_{r,j}, \dot{Q}_{r,j})\|_E \le R_{r,j}^{\mathrm{old}}. \tag{S161}$$

Every record uses its own exact saved initial state. The same coefficient c must satisfy all 22 records, rather than being fitted separately to each record. This is a deterministic admissible error set; no independence of errors or calibrated physical noise level is assumed.

Proposition S48. Joint compatibility in the original short window. *For every $c \in [-10^{-15}, 0]$, the 22 initial value problems (S160) simultaneously satisfy (S161) on all 88 original cells. No observation radius is enlarged. Their solutions also exist on each record's full 257-sample time interval, and their distance from the corresponding declared-source solution is less than 4.269×10^{-19} in the same energy norm. The latter conclusion does not assert a fit to all 257 raw samples.*

Proof. The added potential has the exact gradient g_c displayed in (S160). Set

$$c_k^2 = b^T K_e^{-1} b, \qquad c_m^2 = b^T M^{-1} b,$$
$$q = \frac{\max\{1, \mathrm{tr}(K_3 B K_0^{-1} B^T) - 1\}}{53/50}, \qquad L^2 = \mathrm{tr}(K_e^{-1} C M^{-1} C).$$

The rank-three structure proves $\|K_e^{-1/2} C K_e^{-1/2}\|_2 \le q$. The value L bounds $\|M^{-1/2} C K_e^{-1/2}\|_2$ by the Frobenius norm. Hence, if $\|x\|_{K_e} \le Z$,

$$\|M^{-1/2} g_c(x)\|_2 \le |c| q L Z^3 / E_s. \tag{S162}$$

First establish target existence and a full-interval comparison without assuming that a source tube is valid for the target. The source energy

$$H_0 = \tfrac{1}{2}\|(x,\dot{x})\|_E^2 + \kappa(1 - \cos(b^T x))$$

satisfies $\dot{H}_0 = -\dot{x}^T D_* \dot{x} + f(t)\, d\sin(b^T x)/dt$. Integration by parts, $D_* \succeq 0$, and $\mathrm{TV}(f) \le f_{\max}\nu T$ give

$$H_{0,\max} \le H_0(0) + f_{\max}|\sin(b^T x(0))| + \sqrt{2}c_k f_{\max}(1+\nu T)\sqrt{H_{0,\max}}.$$

Its positive quadratic root supplies a source state cap Z_0. Fix $\eta = 10^{-12}$ in the chosen norm. Until a first exit of the source–target distance E through η, the target norm is bounded by $Z_0 + \eta$, and both angles by $R = c_k(Z_0 + \eta) < 1$. The magnetic force difference and the position–velocity product inequality give

$$E' \le \lambda E + F_c, \qquad \lambda = \tfrac{1}{2}c_m c_k(\kappa + f_{\max}R), \quad F_c = 10^{-15} qL(Z_0+\eta)^3/E_s. \tag{S163}$$

The same initial states give $E(0) = 0$. Outward arithmetic verifies $TF_c e^{\lambda T} < \eta$ for every record, so a first exit is impossible. Boundedness and smooth finite-dimensional ODE continuation give existence on the full saved interval. The maximum received bound is less than 4.269×10^{-19}.

For the sharper short-window result, compare the target directly with the fixed Hermite center. The old certified source residual is

$$\mathcal{R} = M\ddot{Q} + D_*\dot{Q} + K_e Q + \kappa b\sin(b^T Q) - f(t) b\cos(b^T Q), \qquad \|M^{-1/2}\mathcal{R}\|_2 \le \rho_j.$$

Its certificate is retained, including its original source identities and residual bounds; the additional target force is bounded separately. Fix $G_j = 2R_j^{\mathrm{old}}$. Bernstein coefficients provide

$$\begin{aligned} A_j &\ge \sup|Q^T C Q|, \quad V_j \ge \sup\|K_e^{-1/2} C Q\|_2, \\ W_j &\ge \sup\|M^{-1/2} C Q\|_2, \quad \Theta_j \ge \sup|b^T Q|. \end{aligned}$$

Here A_j uses the complete degree-six polynomial $Q^T C Q$; the vector norms use the degree-three Bernstein control-point hull. While the target remains within G_j of Q, expansion of $x = Q + e_x$ gives

$$\begin{aligned} F_j &= \frac{10^{-15}}{E_s}(A_j + 2V_j G_j + qG_j^2)(W_j + LG_j), \\ R_j^\theta &= \Theta_j + c_k G_j < 1, \\ \lambda_j &= \tfrac{1}{2}c_m c_k(\kappa + f_{\max}R_j^\theta). \end{aligned} \tag{S164}$$

The error satisfies $E' \le \lambda_j E + \rho_j + F_j$. Starting with $R_{-1}^{\mathrm{new}} = 0$ and passing each endpoint bound to the next cell, define

$$R_j^{\mathrm{new}} = e^{\lambda_j h_j} R_{j-1}^{\mathrm{new}} + (\rho_j + F_j)\frac{e^{\lambda_j h_j} - 1}{\lambda_j}. \tag{S165}$$

All comparison terms are nonnegative, so this endpoint expression also bounds every earlier point in the cell. The strict inequalities $R_j^{\mathrm{new}} < G_j$ close the first-exit arguments. In addition, the verified $R_j^{\mathrm{new}} \le R_j^{\mathrm{old}}$ proves the unchanged observation-ball conclusion.

The 256-bit computation covers all 22 records and 88 cells with the common bound $|c| \le 10^{-15}$. A separate 384-bit evaluation rebuilds every exact Hermite polynomial from its saved positions and velocities and recomputes the Gram/Bernstein force bounds and radius recursion. The original source residual certificates are inherited with their identities; their entire upstream derivation is not replaced by this recomputation. Both evaluations give

$$\max_{r,j}\frac{R_{r,j}^{\mathrm{new}}}{R_{r,j}^{\mathrm{old}}} < 0.999999938, \qquad \min_{r,j}(R_{r,j}^{\mathrm{old}} - R_{r,j}^{\mathrm{new}}) > 4.6086 \times 10^{-22}. \tag{S166}$$

The full Cartesian product of records and cells is retained, which establishes the simultaneous quantifier for each common c.

Consequences and limitations. The singular ordinary-twist coefficient in Proposition S35 belongs to the same joint short-window admissible set. Those observations therefore do not exclude it or uniquely identify c. Proposition S47 supplies a separate local conservative KAM conclusion even for that member. The joint statement means that each parameter has

trajectories whose error functions lie in the specified balls; it does not mean every arbitrary function in those balls is an ODE solution.

The Hermite centers equal raw position and velocity at the first five saved nodes. Over the remaining 257-sample interval only the source–slice comparison has been established, without an accepted raw-observation tube for that whole interval. The result does not calibrate physical observation noise, establish equivalence with a compiled simulation, identify the unknown physical material, or cover the full 42-coefficient family.

S29 A source-conditioned learning-to-full64 KAM bridge

This section keeps the saved 918-real-parameter network fixed. Its three-configuration-coordinate force evaluations constrain the parameter of the declared *64-coordinate transverse conservative* material family $W_\star = \beta E_0$. This global scalar constitutive assumption is an explicit source-model prior; it does not follow from finite force data. In particular, a three-coordinate neural Hamiltonian is not identified with a 64-coordinate neural Hamiltonian.

Let y_j denote the 33 saved normalized force labels, with deterministic error allowances $\eta_j > 0$, indexed by $j = 1, \dots, 33$. In the declared family their exact targets are βm_j, where m_j is reconstructed from the exact mechanical stiffness and the ideal observation configuration. These 33 observation multipliers are distinct from the 64 modal overlap coefficients used in the twist formula. Thus $|\beta m_j - y_j| \le \eta_j$. The unchanged network graph supplies exact-real values g_j enclosed by received intervals $[\ell_j, u_j]$. Define

$$\delta_j = \max\{|\ell_j - y_j|, |u_j - y_j|\}, \qquad L_F = \frac{1}{33}\sum_{j=1}^{33}\left|\frac{g_j - y_j}{\eta_j}\right|^2, \qquad \overline{L}_F = \frac{1}{33}\sum_{j=1}^{33}\left(\frac{\delta_j}{\eta_j}\right)^2.$$

Then $L_F \le \overline{L}_F$ by interval containment. The normalization in this force-loss interface is η_j; it is not the separate 31-coefficient jet training loss or a fresh validation loss.

Proposition S49. A nonzero force-loss interface. *For every target parameter consistent with the declared family and label allowances, the componentwise and aggregate bounds are*

$$|g_j - \beta m_j| \le \delta_j + \eta_j, \qquad |g_j - \beta m_j| \le \eta_j\big(1 + \sqrt{33\overline{L}_F}\big).$$

More generally, $\overline{L}_F \le \gamma^2/33$ implies the sufficient allowance $\epsilon_j = (1+\gamma)\eta_j$. Each such allowance gives the necessary candidate-centered outer strip

$$\beta m_j \in [\ell_j - \epsilon_j, u_j + \epsilon_j].$$

Intersect these strips over the received rows, dividing with the correct sign whenever $m_j \neq 0$ and retaining the zero-multiplier feasibility tests. Here every intersection uses $j = 1, \dots, 33$. Denote the resulting force-only parameter set by $B_\theta = B_{\theta,33}$.

For any of these choices, the direct-data set

$$B_D = B_{D,33} = \bigcap_{j=1}^{33}\{\beta : |\beta m_j - y_j| \le \eta_j\}$$

is contained in the corresponding B_θ. The same componentwise construction applies separately to the 31 received original-data jet coefficients. Write $B_{D,64}$ for the direct-data set after adding these 31 constraints to the 33 force constraints, and $B_{\mathrm{comp},64}$ for the corresponding componentwise candidate outer set. Write $B_{\mathrm{agg},33}$ for the aggregate-loss outer set using only the 33 force rows. Then

$$B_{D,64} \subseteq B_{\mathrm{comp},64} \subseteq B_{\mathrm{agg},33}.$$

The first inclusion applies the componentwise triangle inequality to all 64 rows. The second uses $\delta_j \le \eta_j\sqrt{33\overline{L}_F}$ on the 33 force rows; adding jet constraints can only shrink the componentwise outer set. These 64 rows are not independent observations.

Proof. The first bound is the triangle inequality. For the second, $|g_j - y_j|/\eta_j \le \sqrt{33 L_F} \le \sqrt{33 \overline{L}_F}$. The rational-budget case is immediate. Since $g_j \in [\ell_j, u_j]$, each error allowance implies the displayed outer strip. Moreover, for any $v \in [\ell_j, u_j]$ and $|\beta m_j - y_j| \le \eta_j$, $|v - \beta m_j| \le \delta_j + \eta_j \le \epsilon_j$. In particular $[y_j - \eta_j, y_j + \eta_j]$ is contained in $[\ell_j - \epsilon_j, u_j + \epsilon_j]$. Dividing by a signed m_j and intersecting proves $B_D \subseteq B_\theta$. No cancellation of a same-sign fit residual is used to tighten the set.

Proposition S50. Received nonzero-loss full64 instance. *For the unchanged saved checkpoint, exact interval evaluation gives*

$$\frac{3}{100} < L_F \le \overline{L}_F \le \frac{12}{275}, \qquad \overline{L}_F \simeq 0.03783786916.$$

The lower inequality is checked from the interval distances of the actual network outputs to the labels; positivity of an upper bound alone is not used as proof of nonzero loss. Take the rational sufficient budget $\gamma = 6/5$, so that $\gamma^2/33 = 12/275$ and $\epsilon_j = 11\eta_j/5$. Without intersecting with the earlier direct-data parameter interval, the candidate-centered construction receives a nonempty exact rational interval B_θ^{rat}. Its endpoints are displayed outward as

$$B_\theta^{\mathrm{rat}} \subset [0.05999999283191808,\ 0.06000000327884982].$$

The exact endpoints, rather than these decimals, define the interval. For every β in that exact interval, the full 64-coordinate linear frequencies obey

$$|k \cdot \omega(\beta)| \ge \frac{2646043}{2147483648} > 0, \qquad 0 < |k|_1 \le 4.$$

Together with the parameter-uniform certificate of nonzero modal tip overlaps, this proves the nondegenerate quartic twist and the qualitative local elliptic KAM conclusion for the declared full-coordinate family.

Proof. The 33 output intervals give strictly positive lower bounds on all 33 absolute fit residuals. Summing their squared normalized lower bounds and comparing exactly gives $L_F > 3/100$. Summing their corresponding upper bounds gives the exact rational $\overline{L}_F \le 12/275$. The parameter strips are intersected using only these evaluations, the saved labels and their deterministic allowances; no material-truth parameter is loaded for the inference.

The new interval is wider than the earlier direct-data interval, so its spectral coverage is recomputed. All 128 exact center-root endpoints from the full64 certificate are retained. Generalized min–max bounds transport every root to the new interval, with exact dyadic square-root enclosures. Exact integer enumeration of all signed frequency sums up to order four, including repeated indices, then gives the displayed strict lower bound. No new eigenvalue proposal or truncated trial space is used. The same computations give strict divisor bounds $2472231/1073741824$ for the componentwise force-and-jet route and $2756265/2147483648$ for the aggregate route using an exact outward enclosure of $\sqrt{33\overline{L}_F}$.

The positive mass and elastic stiffness, nonzero tip overlaps and twist formula hold for the same exact source matrices and all $\beta > -1$. Apply the local elliptic KAM argument of Theorem S43 separately at each accepted parameter. This uses the global scalar constitutive prior to pass from reduced observations to the full-coordinate target family. It does not extend the dimension of the neural graph itself.

Remark S1. Interpretation of the sufficient budget. The threshold $12/275$ is a posteriori sufficient for this checkpoint, these label allowances, and this source-defined scalar material family. It is neither a necessary loss threshold nor a training stopping rule specified before training. Since $B_D \subseteq B_\theta$, this construction does not add identification power to the direct deterministic data. Its contribution is an explicit nonzero-loss interface compatible with the actual saved graph and the full-coordinate local KAM proof. It does not establish generic nonlinear identification, unknown-plant transfer, a numerical torus amplitude, or the necessity or superiority of a neural or complex-valued representation.

S30 A quantitative material-jet neighborhood with fixed quadratic part

This section concerns only the three-coordinate target-family restriction; its constants are not full-coordinate neural bounds. The following result permits general analytic cubic and quartic material corrections to the admitted quadratic target family. It supplies an explicit nonzero sufficient neighborhood; it does not assert that the finite observation interface has identified

the unknown material jet inside it. Let I be the received interval for β, and let

$$H^0_\beta(q,p) = \tfrac{1}{2}p^T A^{-1} p + \tfrac{1+\beta}{2} q^T K_3 q + \kappa(1-\cos(d^T q)).$$

Assume the uniform positive, simple spectrum, order-at-most-four nonresonance and nonsingular twist conditions of the admitted-family certificate. Use its canonical coordinates $q = L_\beta X$ and physical actions $I_j = (X_j^2 + Y_j^2)/2$, and write its twist as T^0_β. For a real analytic perturbation consider

$$R(q) = E_s \left[\sum_{|\alpha|=3} c_\alpha (q/\ell)^\alpha + \sum_{|\alpha|=4} e_\alpha (q/\ell)^\alpha \right] + O(|q|^5), \qquad |c_\alpha| \le \eta_3, \quad |e_\alpha| \le \eta_4. \tag{S167}$$

These are ordinary monomial coefficients, not Taylor derivatives. In particular R has no constant, linear or quadratic terms, and the quadratic part and frequencies remain exactly those of H^0_β. No numerical bound on the higher-order analytic remainder is required for the qualitative local statement below; such a bound would be necessary to assign a numerical neighborhood or torus radius.

Put $a_j = (X_j + iY_j)/\sqrt{2}$ and $\bar a_j = (X_j - iY_j)/\sqrt{2}$, so that $\{a_j, \bar a_k\} = -i\delta_{jk}$ and $I_j = a_j \bar a_j$. Let U_α denote the complex-coordinate polynomial obtained from $E_s(q/\ell)^\alpha$. For $|\alpha| = 3$, let V_α be the unique cubic solution of

$$\{H_2, V_\alpha\} = -U_\alpha.$$

For a homogeneous quartic polynomial P, define $\mathcal{T}(P)$ by

$$\Pi_{\mathrm{res}} P = \tfrac{1}{2} I^T \mathcal{T}(P) I,$$

where the projection retains the balanced monomials $a^\nu \bar a^\nu$. All quartic resonances are balanced by the order-four nonresonance condition. Enumerate the ten cubic indices in a fixed order. Define the matrices

$$A_\gamma = \mathcal{T}(U_\gamma), \quad |\gamma| = 4, \qquad B_{\alpha\alpha} = \tfrac{1}{2}\mathcal{T}(\{U_\alpha, V_\alpha\}),$$

$$B_{\alpha\rho} = \tfrac{1}{2}\mathcal{T}(\{U_\alpha, V_\rho\} + \{U_\rho, V_\alpha\}), \qquad \alpha < \rho.$$

All matrices depend on β through the frequencies and chart. Let G_4, G_3 be nonnegative entrywise upper-bound matrices satisfying, uniformly throughout the whole interval I,

$$(G_4)_{ij} \ge \sum_{|\gamma|=4} |((T^0_\beta)^{-1} A_\gamma)_{ij}|, \qquad (G_3)_{ij} \ge \sum_{\alpha \le \rho} |((T^0_\beta)^{-1} B_{\alpha\rho})_{ij}|.$$

Set $b_k = \max_i \sum_j (G_k)_{ij}$ for $k = 3, 4$.

Lemma S51. A fixed-quadratic robust elliptic KAM criterion. *If*

$$b_4 \eta_4 + b_3 \eta_3^2 < 1, \tag{S168}$$

then, for every $\beta \in I$ and every real analytic perturbation (S167)*, the Hamiltonian $H^0_\beta + R$ has nonsingular quartic Birkhoff twist and satisfies the qualitative local elliptic KAM criterion. In particular sufficiently small neighborhoods of its elliptic equilibrium contain a positive-measure family of invariant Lagrangian three-tori. The size of a suitable neighborhood may depend on the higher-order analytic remainder.*

Proof. The base Hamiltonian has no cubic term. The cubic homological equation and the degree-four Lie-series formula therefore give

$$H_4^{\mathrm{eff}} = H_4^0 + R_4 + \tfrac{1}{2}\{R_3, \chi_3\}, \qquad \chi_3 = \sum_{|\alpha|=3} c_\alpha V_\alpha.$$

The sign follows from $\{H_2,\chi_3\}=-R_3$: in the pullback Lie series $\{R_3,\chi_3\}+\frac{1}{2}\{\{H_2,\chi_3\},\chi_3\}$ the two cubic-generated contributions combine to $\frac{1}{2}\{R_3,\chi_3\}$. Consequently the twist perturbation is exactly

$$\Delta T=\sum_{|\gamma|=4} e_\gamma A_\gamma+\sum_{\alpha\le\rho} c_\alpha c_\rho B_{\alpha\rho}.$$

The triangle inequality and induced infinity norm imply

$$\|(T_\beta^0)^{-1}\Delta T\|_\infty \le b_4\eta_4+b_3\eta_3^2<1.$$

The Neumann-series criterion makes $T_\beta^0+\Delta T=T_\beta^0[\mathrm{Id}+(T_\beta^0)^{-1}\Delta T]$ invertible. The equilibrium Hessian and finite-order divisors are unchanged. Real analyticity, the finite normal-form construction and the nonsingular twist give the same qualitative local elliptic KAM conclusion as for the admitted family. Terms of degree at least five do not alter this twist.

An independent coefficient-norm majorant. For a polynomial in $(a,\bar a)$ write $\|P\|_1$ for its coefficient one-norm. Let

$$t_i=\frac{\sqrt{2}}{\ell}\sum_j|(L_\beta)_{ij}|,\qquad C_k=E_s\sum_{|\alpha|=k}\prod_i t_i^{\alpha_i}.$$

Then $\|R_3\|_1\le C_3\eta_3$ and $\|R_4\|_1\le C_4\eta_4$. If δ_3 bounds all relevant cubic divisors away from zero, then $\|\chi_3\|_1\le C_3\eta_3/\delta_3$. For homogeneous polynomials of degrees m,n, $\|\{P,Q\}\|_1\le mn\|P\|_1\|Q\|_1$; this follows term by term because the sum of the absolute Poisson multipliers is bounded by mn. Resonant projection does not increase the coefficient one-norm, and extraction of the symmetric action Hessian gives $\|\mathcal{T}(P)\|_F\le 2\|\Pi_{\mathrm{res}}P\|_1$. Thus, uniformly on I,

$$\|\Delta T\|_2\le 2C_4\eta_4+\frac{9C_3^2}{\delta_3}\eta_3^2.$$

This estimate independently verifies that the cubic correction is quadratic in the cubic jet and is not omitted. The executable result uses the sharper 15 linear and 55 quadratic matrix responses, with outward Arb arithmetic for the full interval of canonical charts.

Scope of a received coefficient tolerance. The numerical tolerance applies to material coefficients in the exact normalization $\ell=10^{-3}$ and $E_s=6\times10^{-12}$, with the quadratic family jet fixed. The current finite-observation coefficient radii are compared with this sufficient bound, without declaring the unknown target received. An upper error bound exceeding the tolerance does not prove absence of KAM tori. It shows that this particular uniform robustness certificate has not been instantiated by that bound. Uncertainty in the quadratic jet requires a separate spectral and coordinate perturbation analysis.

Received nonzero target-jet tolerance. For the entire received scalar interval, outward evaluation of all 15 quartic responses and 55 symmetric cubic products gives the safe rounded constants

$$b_4<1.123431\times10^{11},\qquad b_3<2.663083\times10^{12}.$$

The normalization is $\ell=10^{-3}$ and $E_s=6\times10^{-12}$. An independent computation in real (X,Y) coordinates inverts the 56-dimensional cubic homological operator and extracts the action Hessian by circle moments, without using the complex-polynomial implementation. Both implementations receive the coefficient rectangle

$$\eta_3=2^{-22}\simeq2.384186\times10^{-7},\qquad \eta_4=2^{-38}\simeq3.637979\times10^{-12},\qquad b_4\eta_4+b_3\eta_3^2<0.561<1.$$

The common radius $\eta_3=\eta_4=2^{-38}$ gives the stronger inverse margin below 0.409. These are sufficient radii, with no assertion that they are optimal.

The current candidate-to-target quartic interface radius of approximately 2.066906×10^{-6} is about 5.68×10^5 times the received common radius. The largest prescribed magnetic quartic coefficient, approximately 5.363314×10^{-9}, is about 1.47×10^3 times that radius. These comparisons use the ordinary normalized coefficient convention; they do not identify the unknown material perturbation relative to the quadratic target family. In particular, the known magnetic term is already

included exactly in the base Hamiltonian and is not an uncertain material coefficient. The present certificate fixes the quadratic jet exactly and therefore does not receive the current unknown-target learning-to-KAM transfer. Failure of a sufficient bound is not a proof that target KAM tori are absent.

Comparison with the received material coefficient boxes. The comparison reference is the admitted material potential βE_0, whose cubic and quartic material coefficients vanish for every admitted β. The prescribed magnetic cosine remains in the base Hamiltonian. Using the actual eleven-configuration coefficient centers m_α and radii r_α gives the exact box majorants

$$\widehat{\eta}_k = \max_{|\alpha|=k}(|m_\alpha| + r_\alpha), \qquad k = 3, 4.$$

Their numerical values are

$$\widehat{\eta}_3 \simeq 8.168713334 \times 10^{-7}, \qquad \widehat{\eta}_4 \simeq 2.067036334 \times 10^{-6}.$$

Thus the respective box-to-safe-radius ratios are approximately 3.426 and 5.682×10^5 for the rectangle $(\eta_3, \eta_4) = (2^{-22}, 2^{-38})$. The sufficient inverse-margin majorant evaluates to approximately 232219, which exceeds one. This is a direct material-to-family comparison, independently of the candidate network's coefficients. It does not establish that a specific compatible target lacks KAM tori.

These boxes were themselves received under an explicit zero fifth-order tail and equilibrium prior for the declared material family. They cannot be promoted to coefficient bounds for an arbitrary unknown analytic material without a separately justified tail bound. Moreover, the robustness theorem fixes the quadratic jet exactly; the finite-data quadratic uncertainty has not been received by that theorem. Consequently this comparison supplies no completed unknown-target learning-to-KAM transfer.

S31 From the executed dissipative loss to a full-state work budget

The candidate storage is $H_N(x, v) = v^T M v/2 + U_N(x)$, with $D_N(x) = a_N(x) D_0$ and $D_0 = \alpha_{\mathrm{nom}} M_0$, where $\alpha_{\mathrm{nom}} = 5004266931050173/9007199254740992\ \mathrm{s}^{-1}$ is the candidate's nominal coefficient. The actual dissipative objective uses eleven signed pairs and four successive windows of 64 panels each, hence $J = 44$ paired windows. Its constant weak test functions lie in the three-dimensional trial space. The integrals in that objective use the composite trapezoidal rule. Adjacent windows share an endpoint but no panel. The conservative gradient is detached in the damping update; final readout calibration nevertheless changes its numerical value, so the loss is evaluated again after calibration. This objective is distinguished below from a velocity-weighted full-coordinate work residual.

S31.1 An exact nodal work identity

For any full-coordinate nodal sequence, without assuming it is an exact solution, define

$$h_i = t_{i+1} - t_i > 0, \qquad \bar{v}_i = (v_i + v_{i+1})/2, \tag{S169}$$

$$n_i = D_N(x_i) v_i - g(x_i) u_i, \qquad f_i = \nabla U_N(x_i) + n_i, \tag{S170}$$

$$m_i = M(v_{i+1} - v_i) + h_i (f_i + f_{i+1})/2, \qquad P_i = v_i^T n_i. \tag{S171}$$

Introduce the conservative and covariance corrections

$$C_i = U_N(x_{i+1}) - U_N(x_i) - \frac{h_i}{2} \bar{v}_i^T [\nabla U_N(x_i) + \nabla U_N(x_{i+1})], \tag{S172}$$

$$Q_i = \frac{h_i}{4} (v_{i+1} - v_i)^T (n_{i+1} - n_i). \tag{S173}$$

Then the following is an identity of real arithmetic:

$$\Delta H_{N,i} + \frac{h_i}{2}(P_i + P_{i+1}) = \bar{v}_i^T m_i + C_i + Q_i. \tag{S174}$$

Indeed, symmetry of the constant mass gives $\Delta(v^T M v/2) = \bar{v}_i^T M \Delta v_i$. The definition of C_i supplies the potential increment, while

$$\frac{v_i^T n_i + v_{i+1}^T n_{i+1}}{2} - \bar{v}_i^T \frac{n_i + n_{i+1}}{2} = \frac{1}{4} \Delta v_i^T \Delta n_i$$

supplies Q_i. Summation gives the identity for each nodal prefix. Neither C_i nor Q_i may be discarded merely because the weak momentum loss is small. In particular, C_i can also contain a kinematic inconsistency in the observed position–velocity sequence.

S31.2 The weak component and its executed-loss interface

The identities below use a consistent mathematical operator set. In particular $B = (V^T M V)^{-1} V^T M$ is an exact definition, rather than an assertion about a stored rounded projection buffer. The executed program instead uses its saved matrices and cached projected-gradient contractions. Its finite diagnostics are related to the mathematical residuals through a separate allowance below. Let $B = (V^T M V)^{-1} V^T M$, so $BV = I_3$. On a paired window of length H, the corresponding mathematical weak residuals are

$$R_\pm = \frac{k}{H} V^T \sum_i m_i^\pm, \quad O = (R_+ - R_-)/2, \quad E = (R_+ + R_-)/2, \quad k = Q_1/E_{\mathrm{net}}.$$

Here Q_1 and E_{net} are the exact saved binary feature scales of the executed model, distinct from the exact decimal scale E_s. The normalization $d_s > 0$ is the saved damping-loss scale. Write $\beta_i^\pm = B\bar{v}_i^\pm$ and $z_i^\pm = \bar{v}_i^\pm - V\beta_i^\pm$, so $V^T M z_i^\pm = 0$. Let $\overline{\beta}_\pm$ be the arithmetic mean of the panel values, and set $a = (\overline{\beta}_+ - \overline{\beta}_-)/2$ and $e = (\overline{\beta}_+ + \overline{\beta}_-)/2$. Direct substitution gives

$$\begin{aligned} \sum_{\sigma=\pm,i} (\bar{v}_i^\sigma)^T m_i^\sigma = {} & \frac{2H}{k} a^T O + \frac{2H}{k} e^T E \\ & + \sum_{\sigma,i} (\beta_i^\sigma - \overline{\beta}_\sigma)^T V^T m_i^\sigma + \sum_{\sigma,i} (z_i^\sigma)^T m_i^\sigma. \end{aligned} \tag{S175}$$

These are respectively the odd component, the even component, within-window modal variation, and the component outside the trial space. Opposite initial velocities do not imply that the latter three terms cancel along forced nonlinear trajectories.

The mathematical normalized loss is $L_D = (3J)^{-1} \sum_j \|O_j/d_s\|_2^2$. Cauchy–Schwarz yields

$$\left| \sum_j \frac{2H_j}{k} a_j^T O_j \right| \le \frac{2d_s}{k} \sqrt{3JL_D} \left(\sum_j H_j^2 \|a_j\|_2^2 \right)^{1/2}. \tag{S176}$$

This is a quantitative loss-to-work connection for the first term, rather than a bound for the complete right-hand side of (S174). The same inequality may be applied to a single paired window. Summing different trajectories does not provide a safety budget for one trajectory.

For an interface to actual execution, let $\widehat{O}_j$ denote the stored weak odd vectors, interpreted as exact binary values. Define $L_D^{\mathrm{rec}} = (3J)^{-1} \sum_j \|\widehat{O}_j/d_s\|_2^2$ in real arithmetic. If a received allowance satisfies $\left(\sum_j \|O_j - \widehat{O}_j\|_2^2\right)^{1/2} \le \eta_O$, then

$$\left| \sum_j \frac{2H_j}{k} a_j^T O_j \right| \le \frac{2}{k} \left(d_s \sqrt{3JL_D^{\mathrm{rec}}} + \eta_O \right) \left(\sum_j H_j^2 \|a_j\|_2^2 \right)^{1/2}. \tag{S177}$$

The triangle inequality proves this form. The allowance must include operator-buffer differences, cached-gradient consistency and floating evaluation and summation errors. A printed floating-point loss also needs its norm-evaluation allowance before it can replace L_D^{rec} in a strict bound.

For example, the stored projection B_b and basis V_b need not satisfy $B_b V_b = I_3$ exactly. If the correction gradient is $B_b^T F_q + F_E K_b x$, its exact projection onto V_b contains $(B_b V_b)^T F_q$. Replacing this by F_q introduces $[I_3 - (B_b V_b)^T] F_q$. Similarly, replacing $V_b^T b$ by a saved port vector d_b introduces a port-consistency error. These terms are included in η_O, not set to zero by an approximate numerical left-inverse test. Likewise the stored B_b is not assigned exact M-orthogonality.

For the frozen checkpoint, exact arithmetic gives $\|B_b V_b - I_3\|_{\max} < 4.53781 \times 10^{-17}$; the saved port vector happens to equal $V_b^T b$ exactly. A separate global saved-weight derivative bound receives the operator part on all 88 signed windows. If O^{cache} denotes the cached weak formula interpreted in real arithmetic, its stacked odd-row allowance is

$$\left(\sum_{j=1}^{44} \|O_j - O_j^{\mathrm{cache}}\|_2^2\right)^{1/2} < 1.318083006 \times 10^{-11}. \tag{S178}$$

The bound retains native-to-buffer changes in mass, trial basis, projection, damping, magnetic coefficient and network features. It uses the maxima of the original nodal coordinates and endpoint velocity rates, the exact trapezoidal weights, and global first- and second-derivative bounds obtained from absolute realified weights, $|\tanh'| \le 1$ and $|\tanh''| \le 2$. The direct operator-to-cache bound already includes the cache shortcut; its separate full-buffer shortcut estimate is not added again. The remaining cache-to-execution gap is received separately below.

S31.2.1 Direct reception of the original executed rows The original vectorized PyTorch evaluation order is reconstructed from the frozen calibrated checkpoint and all 22 original records. Its saved scalar MSE is reproduced bit for bit. The 132 odd scalar rows, intermediate caches, exact binary normalization and original times are archived separately from the older NumPy diagnostic. Interpreting every saved buffer as an exact binary rational, direct 256-bit outward evaluation reconstructs the mathematical cache on all 5654 nodes. The energy-feature derivative, origin subtraction in the first three features only, and both dissipative readout components are retained. The mathematical windows use exact time differences, exact endpoint momentum and exact composite-trapezoidal weights. Thus comparison with the saved executed rows includes rounding in nonlinear evaluation, matrix contractions, normalization, quadrature weights and summation. No correctly rounded implementation of the executed transcendental functions is assumed.

A separate 384-bit implementation splits complex affine maps into real and imaginary maps and propagates four forward derivatives using sech^2; it also integrates unnormalized panels before dividing by the window length. It independently recomputes every node and odd row. These receptions give

$$\left\|O^{\mathrm{cache}} - \widehat{O}\right\|_{\mathrm{stack}} < 2.672835191 \times 10^{-12}, \tag{S179}$$

$$\eta_O = \eta_{\mathrm{op}} + \eta_{\mathrm{arith}} < 1.585366525 \times 10^{-11}, \tag{S180}$$

$$\|O\|_{\mathrm{stack}} < 4.959697033 \times 10^{-7}, \tag{S181}$$

$$\frac{\|O\|_{\mathrm{stack}}^2}{132 d_s^2} < 9.473763848 \times 10^{-11}. \tag{S182}$$

The weak-row units include k and have not been divided by d_s. The real-arithmetic L_D^{rec} differs from the printed $\widetilde{L}_D = 9.473158199312406 \times 10^{-11}$ by less than 5.465034×10^{-27}. Equivalently, replacing $d_s\sqrt{132 L_D^{\mathrm{rec}}}$ by $d_s\sqrt{132\widetilde{L}_D}$ requires an additional norm allowance below 1.430571×10^{-23}. Using L_D^{rec} avoids that replacement. Equation (S177) therefore has a received numerical allowance for its finite odd-work term. It remains distinct from the continuous-ledger arithmetic allowance $\eta_{A,j}$ and does not control the other three work components, the between-node trajectory or physical actuation.

Proposition S52. A finite functional obstruction. *For a finite linear observation operator L and a covector w, an inequality $|w^T r| \le C\|Lr\|$ for all residual vectors r, with a finite constant C, exists if and only if $w \in \mathrm{range}(L^T)$.*

Proof. Necessity follows because the inequality forces w to annihilate $\ker L$, whose orthogonal complement is $\mathrm{range}(L^T)$. For sufficiency write $w = L^T c$ and use Cauchy–Schwarz.

For each of the saved-buffer and exact-source trial bases, and an actual saved panel-average velocity, exact rational elimination constructs a single-panel full-coordinate impulse $m^\dagger$ with $V^T m^\dagger = 0$ and $\bar{v}^T m^\dagger = 1$. Placing this impulse in that panel annihilates every weak odd row while preserving nonzero work. Thus the present weak loss alone has no uniform full-work bound for arbitrary residuals. This witness is a statement about the finite observation operator, not a claim that the impulse is generated by the learned model or an unknown physical rod. Additional target-class assumptions may constrain such omitted residuals.

S31.3 A trajectory-wise continuous safety interface

Along an actual absolutely continuous trajectory with $\dot{x} = v$, the same constant $M \succ 0$ and input map g, define

$$r = M\dot{v} + \nabla U_N(x) + D_N(x)v - g(x)u, \qquad W(t) = \int_0^t v^T r \, ds. \tag{S183}$$

Local integrability of these quantities and the chain rule give

$$H_N(t) = H_N(0) - \int_0^t v^T D_N v \, ds + \int_0^t uy \, ds + W(t), \qquad y = g(x)^T v. \tag{S184}$$

The residual may incorporate conservative mismatch, damping mismatch and other mechanical work. No separate identification of a physical pointwise damping matrix is used in this identity.

Theorem S53. Safety from a one-sided prefix-work budget. *Let Ω be bounded and open, U_N be C^1 on a neighborhood of its closure, and $D_N \succeq 0$. Assume local existence and continuation of the actual closed-loop solutions whenever their position and velocity remain bounded in the stated domain. For each solution under consideration, suppose before any exit and for every $0 \le t \le T$ that*

$$W(t) \le \varepsilon_W, \quad uy \le 0, \quad x(0) \in \Omega, \quad H_N(0) + \varepsilon_W < \inf_{\partial\Omega} U_N, \qquad \varepsilon_W \ge 0.$$

Then that solution exists on $[0, T]$, remains in Ω and satisfies $H_N(t) \le H_N(0) + \varepsilon_W$ there. A conclusion uniform over an initial-state or disturbance family requires all these bounds uniformly over that family.

Proof. Equation (S184) bounds the storage before exit. At a first boundary contact the nonnegative kinetic energy gives $H_N \ge U_N \ge \inf_{\partial\Omega} U_N$, a contradiction. The potential is bounded below on the closure, so the storage bound and $M \succ 0$ bound the velocity. The assumed continuation property excludes finite-time loss of the solution.

For a sampled or imperfect magnetic port, a certified upper bound on $\int_0^t uy \, ds$ can instead be added to the same energy budget. The sign condition is not inherited from a command based on stale or noisy output measurements.

S31.4 Reception from nodes to all prefixes

The continuous work also equals

$$W(t) = H_N(z(t)) - H_N(z(0)) + \int_0^t (v^T D_N v - uy) \, ds.$$

Let $\widehat{W}_j$ be the mathematical nodal ledger, and let a computed value $\widetilde{W}_j$ have a certified arithmetic allowance $|\widetilde{W}_j - \widehat{W}_j| \le \eta_{A,j}$. Suppose, for the same actual path, endpoint storage errors are at most $\eta_{H,i}$ and the cumulative power-quadrature error is at most $\eta_{P,j}$. If additionally $\sup_{t_j \le t \le t_{j+1}} [W(t) - W(t_j)] \le \eta_{I,j}$, then

$$\sup_{0 \le t \le T} W(t) \le \max_j \{\widetilde{W}_j + \eta_{A,j} + \eta_{H,j} + \eta_{H,0} + \eta_{P,j} + \eta_{I,j}\}. \tag{S185}$$

Include the terminal node with $\eta_{I,j} = 0$. This follows by the triangle inequality at nodes and the one-sided increment bound between them. State, input, function-evaluation and quadrature allowances must refer to the same path and input. Agreement of two time steps alone supplies none of these certified allowances.

S31.5 Finite saved-data diagnostic

The post-calibration floating-point loss recomputed in the original vectorized order is $\widetilde{L}_D = 9.473158199312406 \times 10^{-11}$. On the 22 saved records, the odd-work sum is approximately $4.0660 \times 10^{-10} E_{\mathrm{net}}$ and its Cauchy bound is $9.9192 \times 10^{-10} E_{\mathrm{net}}$. The saved-projection remainder contribution is approximately $1.12705 \times 10^{-6} E_{\mathrm{net}}$; the largest positive nodal prefix among the individual records is approximately $8.2198 \times 10^{-8} E_{\mathrm{net}}$. These floating-point diagnostics use the saved rounded operator buffers and projection; the projection remainder need not be exactly mass-orthogonal for those buffers.

The archived odd-row diagnostic uses a different numerical evaluation order from the reproduced vectorized scalar loss. Its real-arithmetic norm is therefore not asserted to equal that printed scalar bit for bit. They use the original 20 μs records, with no new training or integration. They verify the implemented decomposition numerically; they do not receive the continuous allowances in (S185) or establish physical safety.

S31.6 Continuous reception on two existing mathematical IVPs

A separate reception uses the existing full-state tubes for the positive and negative direction-three records, each on its original first four cells, $0 \le t \le 3.125 \times 10^{-7}$ s. These are the declared source IVPs initialized by their own saved states. Their source damping is $D_* = \alpha_* M_0$, with $\alpha_* = 5754906970707699/9007199254740992\ \mathrm{s}^{-1}$; it differs from the candidate's α_{nom}. The candidate uses the same exact source matrices, exact projected-coordinate map and saved binary weights and feature scales as the conservative certificate. Rounded runtime buffers are not identified with these exact operators.

Let Q be the exact cubic Hermite center and R_i its certified whole-cell error radius in $\mathrm{diag}(K_e, M)$, where $K_e = (53/50)K_0$. Exact rational Bernstein coefficients enclose $Q^T K_0 Q/2$ and $\dot{Q}^T M \dot{Q}/2$ by upper bounds E_i^+ and T_i^+. With $r_x = R_i/\sqrt{53/50}$ and $S_i = \sqrt{2T_i^+}$, quadratic expansion gives

$$|E_0(x) - E_0(Q)| \le \sqrt{2E_i^+}\, r_x + r_x^2/2, \tag{S186}$$

$$v^T M v/2 \le T_i^+ + S_i R_i + R_i^2/2. \tag{S187}$$

The dual bounds for Bx, $b^T x$ and $b^T v$ use, respectively, the diagonal of $BK_e^{-1}B^T$, $b^T K_e^{-1} b$ and $b^T M^{-1} b$. These enclose all candidate features throughout each cell. Outward evaluation of the realified split-tanh heads then gives $H_N(t) \le \mathcal{H}_i^+$ and $a_N(x(t)) \le a_i^+$. The initial storage is evaluated separately with a lower bound $\mathcal{H}_0^-$. The energy-feature chain term and both dissipative readout components are retained.

Exact factorization receives $M_r \succeq 0$, hence $D_0 \preceq \alpha_{\mathrm{nom}} M$. If $V_{b,i}^+$ bounds $|b^T v|$ on the same cell and $|u| \le B_{\max}$, then

$$v^T D_N v - uy \le \mathcal{P}_i^+ := a_i^+ \alpha_{\mathrm{nom}} (S_i + R_i)^2 + \mu B_{\max} V_{b,i}^+.$$

For every t in cell i, including its endpoints,

$$W(t) \le \mathcal{H}_i^+ - \mathcal{H}_0^- + \sum_{j \le i} h_j \max(0, \mathcal{P}_j^+). \tag{S188}$$

Using the positive part covers both completed cells and every partial current cell. Thus this is a direct continuous-ledger enclosure, without replacing an integral by an uncertified nodal quadrature. All function evaluations use 256-bit outward arithmetic. The resulting separate all-prefix upper bounds are

$$\begin{aligned} \sup_t W_+(t)/E_{\mathrm{net}} &< 1.027949038 \times 10^{-5}, \\ \sup_t W_-(t)/E_{\mathrm{net}} &< 2.844323240 \times 10^{-5}. \end{aligned} \tag{S189}$$

These are upper bounds, not estimates of the actual errors. They cover two specified mathematical solutions over this short interval, not the 22 training trajectories, a long-time experiment or a physical plant. The open-loop cosine input does not establish $uy \le 0$, so this reception alone does not instantiate Theorem S53 for magnetic feedback.

S32 A source-conditioned uniform residual-work bound

This section receives a work bound along a declared source family while retaining the nonzero force-loss allowance of the selected network. Let $E = E_{\mathrm{net}}$, $E_0 = x^T K_0 x/2$ and $\Omega_4 = \{x : E_0(x) < 4E\}$. Consider

$$\dot{x} = v, \qquad M\dot{v} = -\nabla U_\beta(x) - D_*(x)v + g(x)u, \tag{S190}$$

where

$$\begin{aligned} U_\beta(x) &= (1+\beta)E_0(x) + \kappa[1-\cos(b^T x)], \\ D_*(x) &= \alpha(x)M_0, \qquad \alpha(x) \geq \alpha_{\min} > 0. \end{aligned} \tag{S191}$$

The matrices M, K_0, M_0 are positive definite, $\kappa \geq 0$, and α is locally Lipschitz. The candidate uses the same exact native mass and magnetic map $g(x) = \mu\cos(b^T x)b$, with

$$H_N = v^T M v/2 + U_N(x), \qquad D_N(x) = a_N(x)\alpha_{\mathrm{nom}} M_0, \quad a_N \geq 0.$$

The actual port is $y = g(x)^T v$ and there is no additional work term. Inputs are bounded and satisfy the energy balance for (S190); the following hypotheses concern their actual values, rather than a proposed or reconstructed feedback signal.

The force-loss reception uses all 33 original force rows and gives $3/100 < L_F \leq \overline{L}_F \leq 12/275$. The sufficient budget $\gamma = 6/5$ therefore gives componentwise target allowances $11\eta_j/5$ and the candidate-centered parameter set

$$B_\theta = \bigcap_{j=1}^{33} \{\beta : \beta m_j \in [\ell_j - 11\eta_j/5, u_j + 11\eta_j/5]\}. \tag{S192}$$

The exact rational endpoints of this nonempty interval are retained; outward decimal bounds are

$$B_\theta \subset [0.05999999283191808,\ 0.06000000327884982].$$

We assume $\beta \in B_\theta$ in the declared global source family. The inference does not intersect this set with the smaller direct-data interval. Replacing the scalar interval in the same-checkpoint global Taylor bound gives the strict rational relaxations

$$\sup_{\overline{\Omega}_4} |U_N - U_\beta| < \varepsilon E, \qquad \inf_{\partial\Omega_4} U_N > BE, \tag{S193}$$

where

$$\varepsilon = 0.003839019804, \qquad B = 4.236161085119.$$

Thus the value allowance includes the nonzero force fit and its deterministic label allowances for this constitutive family.

Let $p = (4.595063567868, 1.819558891632, 0.927250710462)$ be the same-source compliance bounds, and let L_R, L_I be the global componentwise Lipschitz rows of the actual split-tanh damping head. For $h > 0$, choose any rational $q_h > \sqrt{h}$ and define

$$\begin{aligned} r_h &= (q_h p_1, q_h p_2, q_h p_3, h), \\ a_{\max}(h) &= (|1 + c_C| + L_R r_h)^2 + (L_I r_h)^2. \end{aligned} \tag{S194}$$

Weighted Cauchy–Schwarz and the exact weight products imply $a_N(x) \leq a_{\max}(h)$ whenever $E_0(x) \leq hE$. This estimate uses the full native configuration, without restricting it to a three-coordinate subspace.

Proposition S54. A loss-conditioned bound for every work prefix. *Suppose $x_0 \in \Omega_4$, $H_N(x_0, v_0) \leq CE$, and a nonnegative number P bounds every actual input-work prefix:*

$$S(t) := \int_0^t u(s)y(s)\,ds \leq PE \qquad (t \geq 0). \tag{S195}$$

Put $\beta_{\min} = \min B_\theta$ and

$$h = \frac{C + \varepsilon + P}{1 + \beta_{\min}} < 4, \quad A = \frac{\alpha_{\mathrm{nom}}}{\alpha_{\min}} a_{\max}(h), \quad \sigma = (A-1)_+. \tag{S196}$$

Define the candidate residual on the source trajectory and its signed accumulated work by

$$r = M\dot{v} + \nabla U_N + D_N v - gu, \qquad W(t) = \int_0^t v^T r\,ds.$$

Then the solution exists for all positive time and, simultaneously for all prefixes,

$$\frac{W(t)}{E} \le \mathcal{W} := 2\varepsilon + \sigma(C + \varepsilon + P). \tag{S197}$$

If $C + P + \mathcal{W} < B$, this work budget also receives the candidate first-exit guard

$$x(t) \in \Omega_4, \qquad H_N(t)/E \le C + P + \mathcal{W} < B \quad (t \ge 0). \tag{S198}$$

Exact continuous saturated feedback $u = -\operatorname{sat}_U(ky)$, $k, U > 0$, satisfies (S195) with $P = 0$.

Proof. Let $H_* = v^T M v/2 + U_\beta$. The source power balance is

$$H_*(t) + \int_0^t v^T D_* v \, ds = H_*(0) + S(t).$$

The initial value comparison gives $H_*(0) \le (C + \varepsilon)E$. Since $H_* \ge (1 + \beta_{\min})E_0 \ge 0$, both $E_0(x(t)) \le hE$ and $\int_0^t v^T D_* v \, ds \le (C + \varepsilon + P)E$ follow independently of candidate dissipation. In particular $h < 4$ keeps the source trajectory strictly inside the configuration domain. The positive source energy also bounds velocity; local regularity and bounded input give continuation.

Writing $\delta U = U_N - U_\beta$, the common mass and complete port give

$$r = \nabla \delta U + (D_N - D_*)v, \quad W(t) = \delta U(x(t)) - \delta U(x_0) + \int_0^t v^T (D_N - D_*) v \, ds.$$

On the source domain, $D_N \preceq A D_*$, hence $D_N - D_* \preceq \sigma D_*$. Combining this inequality with (S193) and the source dissipation budget proves (S197). The positive part is essential when $A < 1$. Finally, the candidate storage identity on this source trajectory is

$$H_N(t) = H_N(0) + S(t) - \int_0^t v^T D_N v \, ds + W(t).$$

Dropping its nonnegative candidate-dissipation integral gives the cap in (S198); a first boundary crossing would require $H_N \ge U_N > BE$, contradicting that cap.

For the exact constant-viscosity source, retain separately

$$\alpha_* = \frac{5754906970707699}{9007199254740992}\ \mathrm{s}^{-1}, \qquad \alpha_{\mathrm{nom}} = \frac{5004266931050173}{9007199254740992}\ \mathrm{s}^{-1}.$$

Their ratio is not replaced by $23/20$. Set $C_{22} = 0.972247611362$. Each of the 22 independently received saved numerical initial states satisfies $x_0 \in \Omega_4$ and $H_N(x_0, v_0) < C_{22}E$, using its own full position and velocity. Exact rational reception gives the following strict outward bounds:

C	$\alpha_{\min}$ (s^{-1})	P	$\mathcal{W} <$	$C + P + \mathcal{W} <$
C_{22}	α_*	0	1.079983443038	2.052231054400
$3/2$	α_*	0	2.293563097602	3.793563097602
1	$1/3$	0	3.085450496352	4.085450496352
C_{22}	α_*	$1/4$	1.602240623572	2.824488234934

All four caps are below B. The first two rows cover the declared source viscosity and, more generally, $\alpha(x) \ge \alpha_*$. The third allows the larger viscosity class $\alpha(x) \ge 1/3\ \mathrm{s}^{-1}$. The last row provides an interface for positive input-work supply; it does not establish that a particular sampled input satisfies its prefix budget. The common C_{22} reception concerns saved initial conditions, not the subsequent saved cosine-driven records as feedback trajectories.

This result adds a uniform residual-work reception involving both actual trained heads. It does not add physical identification: the global scalar potential family and positive viscosity lower bound remain assumptions, and the latter has not been inferred from the weak damping loss. Source energy already gives the sharper storage comparison $H_N(t)/E \le C + 2\varepsilon + P$; the work-based guard above is a sufficient verification of a separate interface. No physical sensing, actuator, timing or initial-state-error bound is received here.

Compatibility with conservative learning and KAM. The parameter interval in (S192) is exactly the loss-generated interval in Proposition S50. Its conservative source Hamiltonian is $p^T M^{-1} p/2 + U_\beta(x)$. Thus the local conservative KAM theorem and the dissipative work theorem apply to the same admitted material parameters and native operators, with the same nonzero force-loss allowance. KAM is applied after removing damping and time-dependent input. The work theorem is applied to the driven dissipative source equation. No invariant conservative torus is used as a safety certificate for the dissipative closed loop.

Positive source viscosity cannot be removed from the all-time work claim. For an undamped source harmonic oscillator with $u = 0$, $U_N = U_*$ and strictly positive candidate damping, a nontrivial periodic orbit has $W(t) = \int_0^t v^T D_N v \, ds \to \infty$ despite bounded storage. Thus the earlier safety hypothesis $D_* \succeq 0$ alone does not imply a uniform residual-work budget.

S32.1 All original source-IVP horizons from a bounded input

The prefix-supply condition can be received without solving another IVP. For a source in (S191), assume $x(0) \in \Omega_4$ and $H_N(x(0), v(0)) \le CE$, so that $H_*(0) \le (C + \varepsilon)E$. Let $c_M \ge \sqrt{b^T M^{-1} b}$ and suppose the actual prescribed mathematical input satisfies $|u| \le U$ on $[0, T]$. Positivity of the source potential and the common port imply

$$|uy| \le \mu U c_M \sqrt{2H_*}, \qquad \sqrt{H_*(t)/E} \le \sqrt{C + \varepsilon} + \frac{\mu U c_M t}{\sqrt{2E}}.$$

The second inequality follows from source dissipation and the first, using $\sqrt{H_* + \delta}$ and then $\delta \downarrow 0$ if $H_* = 0$. Integrating the first bound therefore gives, simultaneously at every prefix $0 \le t \le T$,

$$\frac{|S(t)|}{E} \le \frac{1}{E} \int_0^t |uy| \, ds \le 2 q_T \sqrt{C + \varepsilon} + q_T^2, \qquad q_T = \frac{\mu U c_M T}{\sqrt{2E}}. \tag{S199}$$

This is an absolute supply estimate; it assumes neither feedback nor nonpositive instantaneous power. It uses the input law over the whole time interval, not only its values at stored nodes.

The original mathematical source law is the exact-decimal bounded cosine with amplitude $U = 10^{-4}$ T. The 22 saved clocks each start at zero and end at their own original binary64 time near $20\,\mu$s. Use their maximum endpoint for T, the common exact native mass compliance and $C = C_{22} = 0.972247611362$ from the independently received initial-state calculation. Strict rational square-root enclosures give an absolute supply bound below $0.096606E$, hence the exact relaxation $P = 1/10$ is valid simultaneously on all 22 source-IVP horizons. The original source parameter $\beta = 3/50$ is in the retained loss-generated interval, and its viscosity is α_*. Proposition S54, restricted to these finite horizons, therefore gives a continuous all-prefix residual-work budget and candidate boundary guard for every one of these mathematical solutions. Strict outward rational bounds are

$$\sup_{j,t} W_j(t)/E < 1.277474991512, \qquad H_{N,j}(t)/E < 2.349722602874 < B, \tag{S200}$$

where each t belongs to that IVP's own saved horizon. No time step, trajectory interpolation or new solver output enters this argument.

These are the declared mathematical IVPs from the 22 saved initial conditions and prescribed source input. Their entire-time work bounds do not identify the compiled simulator trajectories with those IVPs or bound physical state, sensing or actuation errors.

S32.2 A finite-horizon supply reception for the recorded reference history

The input-work hypothesis can be received on the existing three-hold reference history without assuming a pointwise negative port throughout that history. Keep its own saved-seed initial state, native exact mass $M = M_0 + M_r$, source coefficients $\beta = 3/50$ and $\alpha = \alpha_*$, and the same three recorded constant total fields. The interval is $[t_0, t_3]$, where

$$t_0 = \frac{5902958103587057}{295147905179352825856}\ \text{s}, \qquad t_3 = \frac{7378697629483821}{14757395258967641 2928}\ \text{s}.$$

These are the saved binary64 clocks, approximately 20 and 50 μs. They do not define a future-time guarantee.

Define the actual reference-input supply by

$$S_0(t) = \int_{t_0}^{t} u(s)y(s)\,ds.$$

On hold k, its exact increment is

$$S_0(t) - S_0(t_k) = \mu u_k[\sin\theta(t) - \sin\theta(t_k)].$$

This identity contains no dissipation or storage term. The existing full64 comparison supplies unrestarted error radii on 192 retained Hermite cells. Exact reconstruction of the tip Bezier controls and their derivative controls, followed by rational degree-nine sine enclosures with eleventh-order remainder, bounds the signed increment from each hold anchor. On cells whose received port satisfies $uy \le 0$, the propagated left-endpoint bound for S_0 is an additional valid prefix bound. Taking the smaller valid bound and propagating endpoints gives directly

$$S_0(t) \le 0 \qquad (t_0 \le t \le t_3). \tag{S201}$$

In normalized units, the first hold endpoint has $S_0(t_1)/E < -0.002523710912$ and the next two endpoints have $S_0(t_2)/E, S_0(t_3)/E < -0.004805955433$. All 15 failed pointwise nonpositive-port tests in the first hold are retained; their prefixes are received using the signed hold-anchor sine differences. Therefore (S201) does not assert that S_0, or the source energy, is nonincreasing between every pair of times. No energy cap is subtracted to infer this supply bound.

The same saved seed has independently received $H_*(t_0)/E < 0.832211815$. Source positivity first gives $x(t_0) \in \Omega_4$, and the loss-conditioned potential allowance then gives $H_N(t_0)/E < 1$. Applying the proof of Proposition S54 on this finite interval with $C = 1$, $P = 0$ and $\alpha_{\min} = \alpha_*$ yields

$$\begin{aligned} &\frac{1}{E}\int_{t_0}^{t} v^T r\,ds < 1.133261675220, \\ &\quad H_N(t)/E < 2.133261675220 < B \qquad (t_0 \le t \le t_3). \end{aligned} \tag{S202}$$

Thus the new residual-work interface has an input-supply instance for this particular declared reference history. This reception preserves the original failed third sensor contract and receives no subsequent input, compiled-equation identity, physical initial-state error, sensor or actuator guarantee. It uses neither a replacement training initial state nor a new feedback integration.

S33 Physical power, learning errors, and bounded magnetic safety

All derivatives in this section use physical time. The full observation space has $n = 64$ coordinates; its three-coordinate restriction is a different dynamical system. Let

$$M = M_0 + M_r \succ 0, \qquad K_0 \succ 0, \qquad D_0 = \alpha M_0 \succeq 0, \qquad \alpha > 0,$$

with constant symmetric matrices. In particular, D_0 is not αM. For a full-column-rank $V \in \mathbb{R}^{64\times 3}$, put

$$A = V^T M V, \qquad B = A^{-1}V^T M, \qquad q = Bx, \qquad E_0(x) = \tfrac{1}{2}x^T K_0 x, \qquad \theta = b^T x.$$

Thus $BV = I_3$. Fix $\mu > 0$ and a constant bias coefficient $\kappa > 0$. The transverse magnetic field $u = B_y$ is measured in tesla. Define

$$\begin{aligned} U_N(x) &= E_0(x) + \kappa(1 - \cos\theta) + F_N(Bx, E_0(x)), \\ D_N(x) &= a_N(Bx, E_0(x))D_0, \\ H_N(x,v) &= \tfrac{1}{2}v^T M v + U_N(x), \\ g(x) &= \mu\cos\theta\, b, \qquad y(x,v) = g(x)^T v. \end{aligned} \tag{S203}$$

Here F_N is real-valued on real inputs. The complex network is decoded in real coordinates before differentiation.

The actual network parameterization. With positive saved feature scales Q_j and E_*, distinct from the exact decimal normalization E_s, write $s = (q_1/Q_1, q_2/Q_2, q_3/Q_3, E/E_*)$. Let $N(s)$ be the real energy readout and $(R(s), I(s))$ the two real readouts of the separate complex dissipation network. The candidate is

$$\begin{aligned} F_N(q,E) &= c_E E + E_* \left[N(s) - N(0) - \sum_{j=1}^{3} \gamma_j s_j \right], \qquad \gamma_j = \partial_j N(0), \\ a_N(q,E) &= [1 + c_C + R(s) - R(0)]^2 + [I(s) - I(0)]^2. \end{aligned} \tag{S204}$$

The fourth linear feature term is retained. Removing $\partial_4 N(0) E/E_*$ would define another potential.

Proposition S55. Full-field structure and real-domain bounds. *For differentiable F_N, the full candidate equation*

$$\dot{x} = v, \qquad M\dot{v} = -\nabla U_N(x) - D_N(x)v + g(x)u$$

has the identities

$$\nabla_x F_N(Bx, E_0(x)) = B^T F_q + F_E K_0 x, \tag{S205}$$

$$\dot{H}_N = -a_N v^T D_0 v + uy. \tag{S206}$$

In particular, $D_N \succeq 0$. Suppose the split-tanh network is written as two real hidden layers with matrices W_1, W_2 and final real output rows C_R, C_I. For a real feature box $|s_j| \le r_j$, define

$$\delta_R = |C_R|\,|W_2|\,|W_1| r, \qquad \delta_I = |C_I|\,|W_2|\,|W_1| r,$$

where the products use entrywise absolute values and the displayed outputs are scalars. Then throughout that real box

$$a_- \le a_N \le a_+, \qquad a_- = (\max\{0, |1 + c_C| - \delta_R\})^2, \quad a_+ = (|1 + c_C| + \delta_R)^2 + \delta_I^2. \tag{S207}$$

If, in addition, $D_0 \succeq \lambda_0 M$ for $\lambda_0 > 0$ and $a_- > 0$, then $D_N \succeq a_- \lambda_0 M$ on the box.

Proof. Since $dq = B\,dx$ and $dE_0 = x^T K_0\,dx$, the chain rule gives (S205). Multiplication of the candidate acceleration equation by v^T proves (S206). For the Rayleigh function $\mathcal{R}_N(x, v) = a_N(Bx, E_0(x)) v^T D_0 v/2$, the derivative with respect to v, with x fixed, is $a_N D_0 v$. There is no spatial derivative of a_N in its dissipated power. Real tanh is globally 1-Lipschitz. Applying this fact to the two real-coordinate hidden layers bounds their output differences from the origin by the two stated matrix products. The reverse triangle inequality bounds the first squared factor from below; the ordinary triangle inequality bounds both factors from above. The Loewner bound follows on multiplication by $D_0 \succeq 0$.

The lower bound in (S207) must be recomputed for the selected trained weights and the relevant full-state feature range. A sum of two squares alone does not exclude a common zero. For a fixed field value u_c, the alternative storage $H_N - \mu u_c \sin\theta$ dissipates at rate $-v^T D_N v$. For a differentiable varying field, including $-\mu u \sin\theta$ in the storage gives supply $-\mu \dot{u} \sin\theta$. With the storage in (S203), the supply is uy and has no $\dot{u}$ term. These are two storage conventions for the same force.

A sufficient coercivity test. For finite split-tanh weights, the real readout satisfies $|N(s)| \le C_N$ on all real features, for an explicit finite constant C_N obtained from its final output row and bias. Set

$$\xi_c = E_* B^T \operatorname{diag}(Q_1^{-1}, Q_2^{-1}, Q_3^{-1})(\gamma_1, \gamma_2, \gamma_3)^T.$$

If $1 + c_E > 0$, then for every $0 < \eta < 1 + c_E$,

$$U_N(x) \ge (1 + c_E - \eta) E_0(x) - E_*(C_N + |N(0)|) - \frac{\xi_c^T K_0^{-1} \xi_c}{2\eta}. \tag{S208}$$

Indeed, the centered position term is $-\xi_c^T x$ and $|\xi_c^T x| \le \sqrt{2E_0 \xi_c^T K_0^{-1} \xi_c} \le \eta E_0 + \xi_c^T K_0^{-1} \xi_c/(2\eta)$. Thus H_N has compact sublevels under this sufficient test. The architecture by itself does not constrain $c_E > -1$.

The physical plant and its errors. On an open configuration region Ω, suppose the physical plant is

$$\dot{x} = v, \qquad M\dot{v} = -\nabla U_*(x) - D_*(x)v + g(x)u + w(x,v,t), \qquad D_*(x) = D_*(x)^T \succeq 0. \tag{S209}$$

It has the same declared mass and magnetic port as the candidate. The generalized force w covers additional physical work, rather than conservative or damping errors already retained in U_*, D_*. Define

$$H_* = \tfrac{1}{2} v^T M v + U_*(x), \quad \Delta U = U_N - U_*, \quad G = \nabla \Delta U, \quad d_* = v^T D_* v, \quad d_N = v^T D_N v.$$

Unlike a full-phase network with learned momentum dependence, this candidate uses exactly $\dot{x} = M^{-1}p$. At the same observed full state (x, v), its port output is therefore exactly y; no potential gradient error changes this equality. Replacing (x, v) by a reduced state or a stale observation requires a separate output-error bound. Along the physical plant the two storage derivatives are

$$\dot{H}_* = -d_* + uy + v^T w, \qquad \dot{H}_N = -d_* + uy + v^T(G + w). \tag{S210}$$

Theorem S56. Physical safety and potential-value transfer. *Let Ω be bounded with compact closure. Assume U_N, U_* are C^2 and D_* is locally Lipschitz on a neighborhood of that closure, and*

$$|U_N - U_*| \le \varepsilon_U \quad \text{on } \overline{\Omega}. \tag{S211}$$

Let $U > 0$, $k > 0$ and use the actual continuous feedback

$$u = -\operatorname{sat}_U(ky), \qquad \operatorname{sat}_U(r) = \max(-U, \min(U, r)), \qquad \Phi(y) = |y| \min(k|y|, U).$$

Suppose $w = 0$, $x(0) \in \Omega$, and for some energy cap c,

$$H_N(x(0), v(0)) + \varepsilon_U \le c, \qquad \inf_{x \in \partial\Omega} U_N(x) - \varepsilon_U > c. \tag{S212}$$

Then the physical solution exists for all $t \ge 0$, remains in Ω, and satisfies

$$H_*(t) \le H_*(0) \le c, \qquad H_N(t) \le c + \varepsilon_U. \tag{S213}$$

Consequently any physical admissible set containing $\{(x, v) : x \in \overline{\Omega},\ H_N(x, v) \le c + \varepsilon_U\}$ contains this trajectory. Neither a gradient-error bound nor an approximation of D_* by D_N is needed for this conclusion.

Proof. The actual field has magnitude at most U and is locally Lipschitz in the state. The first identity in (S210) gives $\dot{H}_* = -d_* - \Phi(y) \le 0$ as long as the solution remains in the declared region. The initial inequality and (S211) give $H_*(0) \le c$. At a first configuration exit, continuity and the strict boundary inequality would imply $H_* \ge U_* \ge U_N - \varepsilon_U > c$, a contradiction. On the compact configuration closure, U_* is bounded below. The energy cap therefore bounds v because $M \succ 0$. The solution stays in a compact subset of its smooth domain, so the standard continuation theorem excludes finite-time escape. Finally $H_N = H_* + \Delta U \le c + \varepsilon_U$ proves the set inclusion.

The theorem concerns the true storage and the exact measured physical port. The learned energy supplies an initial-energy estimate and a guard conversion. Its role here is not a claim that a neural controller outperforms a physical passivity controller. If a positive-work force, an incorrect mass, or an incorrect input map is present, the corresponding extra power cannot be omitted.

Proposition S57. Quantitative error and observation budgets. *Set $r = (v^T M v)^{1/2}$. Suppose, on a common declared state region,*

$$\|M^{-1/2}(D_N - D_*)M^{-1/2}\|_2 \le \delta_D, \quad \|M^{-1/2}G\|_2 \le \varepsilon_g, \quad \|M^{-1/2}w\|_2 \le \varepsilon_w.$$

Then

$$d_* \geq \underline{d} := \max\{0, d_N - \delta_D r^2\}. \tag{S214}$$

For exact port feedback the two powers obey

$$\dot{H}_* \leq -\underline{d} - \Phi(y) + \varepsilon_w r, \tag{S215}$$

$$\dot{H}_N \leq -d_N + \delta_D r^2 - \Phi(y) + (\varepsilon_g + \varepsilon_w) r. \tag{S216}$$

If the applied feedback instead uses $\widehat{y}$ with $|y - \widehat{y}| \leq \varepsilon_y$, let $a = \min(k|\widehat{y}|, U)$. The term $-\Phi(y)$ in either bound may be replaced by

$$-a(|\widehat{y}| - \varepsilon_y). \tag{S217}$$

Proof. The matrix error gives $|d_N - d_*| \leq \delta_D r^2$; retain also $d_* \geq 0$. Weighted Cauchy–Schwarz gives $|v^T G| \leq \varepsilon_g r$ and $v^T w \leq \varepsilon_w r$. Substitution into (S210) gives the assertions. For the observed output, $uy = u\widehat{y} + u(y - \widehat{y}) \leq -a|\widehat{y}| + a\varepsilon_y$.

For example, a certified $D_N \succeq \lambda_N M$ and $\delta_D < \lambda_N$ give true damping at least $(\lambda_N - \delta_D) r^2$. A velocity-proportional disturbance bound $\|M^{-1/2} w\| \leq \nu r$ with $\nu \leq \lambda_N - \delta_D$ is absorbed by this damping. A constant $\varepsilon_w > 0$ gives a linear term in r and is not absorbed uniformly by quadratic damping near $r = 0$. The same issue affects a nonzero constant gradient bound in (S216). Empirical weak-loss values alone provide none of the uniform constants in this proposition.

Theorem S58. Boundary invariance and bounded-port feasibility. *Let H denote either H_* or H_N, with H of class C^2, and suppose the sublevel $S_c = \{(x, v) : H(x, v) \leq c\}$ is compact, lies in the declared smooth domain, and has a regular boundary: $\nabla_{x,v} H \neq 0$ on ∂S_c. For a specified locally Lipschitz physical force $w(x, v, t)$, uniformly locally Lipschitz in the state on finite time intervals, define*

$$P_* = -d_* + v^T w \quad (H = H_*), \qquad P_N = -d_* + v^T (G + w) \quad (H = H_N).$$

Assume the corresponding functions are continuous in time. The saturated feedback above renders S_c positively invariant if

$$P(x, v, t) \leq \Phi(y(x, v)) \quad \textit{for all } (x, v) \in \partial S_c, \quad t \geq 0. \tag{S218}$$

At a fixed boundary state and time, the smallest available energy derivative among all fields $|u| \leq U$ equals

$$\min_{|u| \leq U} (P + uy) = P - U|y|. \tag{S219}$$

Thus $P - U|y| \leq 0$ at every boundary point is necessary for an invariant cap under a bounded continuous feedback. This pointwise feasibility condition alone asserts neither a continuous feedback selection nor an implemented controller.

Proof. Along the saturated closed loop, $\dot{H} = P - \Phi(y)$. The boundary condition is the inward tangent condition for the regular sublevel. To make the invariance argument explicit, a compact regular level has a tubular neighborhood in which distance to the level is bounded by a constant times $|H - c|$. The locally Lipschitz closed-loop field and the locally Lipschitz level normal therefore bound the positive part $h = (H - c)_+$ by $\dot{h} \leq L_T h$ in that neighborhood on each finite interval $[0, T]$. Gronwall's inequality with $h(0) = 0$ precludes exit. Initial states strictly inside the sublevel are handled by their first boundary contact. Compactness gives continuation. The scalar minimization is obtained with $u = -U \operatorname{sign} y$ when $y \neq 0$, and with any input when $y = 0$. If its minimum were positive at a boundary state, every available continuous closed loop would initially point strictly out of the sublevel there, contradicting invariance from that initial state.

In particular, at $y = 0$ no magnetic field changes the instantaneous energy derivative. This occurs when $\cos\theta = 0$ or $b^T v = 0$. Restricting $|\theta| < \pi/2$ removes only the former possibility. On a verified part Γ of a boundary where $|y| \geq y_0 > 0$, choosing $k \geq U/y_0$ gives $\Phi(y) = U|y|$. A budget $P \leq U y_0 - m$ with $m > 0$ then provides a strict inward margin m on Γ. The remaining boundary still requires (S218); a local nonzero-port calculation does not cover the turning points or the velocity directions orthogonal to b.

A centered, realizable obstruction for a learned-energy cap. Take two coordinates with $M = K_0 = I_2$, $D_* = dI_2$ for $d > 0$, $b = (1, 0)^T$, $\mu > 0$, and

$$U_* = E_0 + \kappa(1 - \cos x_1), \qquad U_N = (1 + \epsilon)E_0 + \kappa(1 - \cos x_1), \qquad \epsilon > 0.$$

This discrepancy is precisely the affine energy skip $F_N = \epsilon E_0$ with zero neural branch, not an uncentered linear force. At $x = (0, 1)^T$, $v = (0, t)^T$ and $0 < t < \epsilon/d$,

$$y = 0, \qquad \dot{H}_N = \epsilon t - dt^2 > 0, \qquad \dot{H}_* = -dt^2 < 0. \tag{S220}$$

Every magnetic input has the same derivative at this state. The H_N sublevel through it has a regular boundary and cannot be invariant under any bounded continuous magnetic feedback, although the true passive energy decreases. The parameters ϵ, t can be arbitrarily small. Hence a small conservative-gradient error and positive damping do not, by themselves, certify invariance of learned energy sublevels on the physical plant.

Restriction and execution scope. On $x = Vq$, $v = VA^{-1}p$, the restricted storage is

$$H_N^{\mathrm{r}}(q, p) = \tfrac{1}{2}p^T A^{-1}p + U_N(Vq), \qquad y^{\mathrm{r}} = \mu \cos(b^T Vq) b^T VA^{-1}p.$$

It has its own reduced power identity. It does not identify the physical output with y^{r} on an arbitrary full trajectory, prove invariance of $\mathrm{range}(V)$, or transfer a restricted KAM torus to the full forced dissipative system. A sampled, delayed, or native numerical realization additionally needs full-prefix work, observation, actuation, and numerical-defect bounds for this same physical model. The continuous boundary theorem does not silently inherit a previous held-command record or its safety certificate.

S33.1 Dissipation and coercivity for the calibrated weights

Fresh calculation with the calibrated weights gives $c_C \simeq 0.07237264643$ and $c_E \simeq 0.06002256805$. In particular, $1 + c_E > 0$, so the sufficient global real coercivity condition in (S208) holds for this same network. Its nonnegative dissipation-factor parameterization and full-field power identity hold algebraically. Exact native LDL factorizations again give $D_0 > (6/25)M\,\mathrm{s}^{-1}$, with the exact sum $M = M_0 + M_r$. Exact saved-weight absolute products give the following strict downward relaxations on the full-state feature boxes $|s|_\infty \le r$:

r	a_N lower bound	D_N/M lower bound in s^{-1}
$1/4$	1.0346097	0.2483063
$1/2$	0.9253337	0.2220801
1	0.7250739	0.1740177
$3/2$	0.5492037	0.1318088
2	0.3977230	0.0954535

The last column is a strict Loewner matrix lower bound for every full velocity in the stated configuration domain. The same-checkpoint calculation proves candidate dissipation and coercivity; it does not by itself identify physical damping or certify sampled magnetic feedback. A conditional physical potential guard requires its own uniform potential allowance and implementation assumptions.

S33.2 A conditional physical guard for the calibrated potential

The same saved weights also admit a full-configuration potential-value guard under an explicit physical family. Suppose

$$U_*(x) = (1 + \beta)E_0(x) + \kappa(1 - \cos(b^T x)),$$

with constant β in the exact scalar compatibility interval in Section S14.1. This is a constitutive family and parameter-interval assumption, not an identification of an arbitrary physical potential from the simulation records. Let E_{net} be the exact saved

binary energy scale and set

$$\Omega_4 = \{x : E_0(x) < 4E_{\mathrm{net}}\}, \qquad c = 2.05E_{\mathrm{net}}.$$

Fresh exact-weight bounds on the energy-head gradient and Hessian, an outward origin derivative calculation, and the exact native compliance matrix give the strict relaxations

$$\sup_{\overline{\Omega}_4} |U_N - U_*| < 0.003838998456E_{\mathrm{net}}, \qquad \inf_{\partial\Omega_4} U_N > 4.236161085119E_{\mathrm{net}}.$$

These constants are recomputed for the calibrated weights and the new compatibility interval. They do not reuse the earlier single-configuration checkpoint's margins. With $\varepsilon_U = 0.003838998456E_{\mathrm{net}}$, the certified boundary margin is

$$4.236161085119 - 0.003838998456 - 2.05 = 2.182322086663 > 0.$$

The initial set

$$\mathcal{I} = \{(x, v) : x \in \Omega_4, \quad H_N(x, v) \le 2.046161001544E_{\mathrm{net}}\}$$

contains the origin and an open neighborhood, by continuity and the algebraic centering $H_N(0, 0) = 0$. Corollary S60 separately receives all 22 numerical initial records into this fixed set.

Assume the exact same native mass and physical magnetic port, locally Lipschitz physical damping $D_*(x) \succeq 0$, and no unmodeled work. Under actual continuous feedback $u = -\mathrm{sat}_U(ky)$ with the exact full-state port $y = \mu\cos(b^T x)b^T v$, any $U, k > 0$, Theorem S56 applies to every initial state in $\mathcal{I}$. It gives existence for all $t \ge 0$ and

$$x(t) \in \Omega_4, \qquad H_*(t) \le H_*(0) \le c, \qquad H_N(t) \le c + \varepsilon_U.$$

The physical damping assumption in this theorem is distinct from the candidate damping bounds. The candidate bound on the full Ω_4 domain is proved separately in Proposition S59; it does not identify physical damping. This is a conditional continuous-feedback safety conclusion; no sampled controller, sensor error, actuation delay, controller rollout or physical-target KAM transfer is certified here.

S33.3 Strict candidate dissipation on the full safety configuration domain

All constants in this subsection belong to the unchanged 918-real-parameter checkpoint used for the calibrated energy guard. Saved binary64 weights and feature scales are exact constants of the mathematical source graph. The native matrices are the exact source operators, with $M = M_0 + M_r \succ 0$, $K_0 \succ 0$ and $D_0 \succ (6/25)M\,\mathrm{s}^{-1}$. This is a statement about the candidate equations, not an identification of the physical dissipative force.

Let $T = BK_0^{-1}B^T$, $E_0 = x^T K_0 x/2$ and

$$\Omega_4 = \{x : E_0(x) < 4E_{\mathrm{net}}\}, \qquad s = (q_1/Q_1, q_2/Q_2, q_3/Q_3, E_0/E_{\mathrm{net}}), \quad q = Bx.$$

The positive Q_j carry the units of q_j in the declared native-coordinate normalization, and E_{net} is an energy in joules; the four features are dimensionless. The same-source compliance calculation supplies strict rational bounds $p_j^2 Q_j^2 > 2E_{\mathrm{net}}T_{jj}$. Here

$$(p_1, p_2, p_3) = (4.595063567868,\ 1.819558891632,\ 0.927250710462).$$

The displayed terminating decimals are exact rational upper bounds, not rounded evaluations of square roots. Put

$$r = (2p_1, 2p_2, 2p_3, 4) = (9.190127135736, 3.639117783264, 1.854501420924, 4).$$

Proposition S59. Anisotropic-domain damping bound. *For every real $x \in \overline{\Omega}_4$, the selected candidate satisfies*

$$0.005110793448 < a_N(x) < 5.344160513025, \tag{S221}$$

$$D_N(x) \succ 0.00122659042752M\,\mathrm{s}^{-1}. \tag{S222}$$

The conclusion holds on the full 64-coordinate ellipsoid, without assuming $x \in \mathrm{range}(V)$. For the exact candidate equation, define $H_N = v^T M v/2 + U_N(x)$ and $y = \mu \cos(b^T x) b^T v$. While $x(t) \in \overline{\Omega}_4$,

$$\dot{H}_N \le -0.00122659042752\,\mathrm{s}^{-1} v^T M v + uy. \tag{S223}$$

The inequality is strict when $v \neq 0$. In particular, zero total field or exact continuous feedback $u = -\,\mathrm{sat}_U(ky)$ makes this candidate storage nonincreasing on that domain.

Proof. Weighted Cauchy–Schwarz with $K_0 \succ 0$ gives

$$|B_j x|^2 \le (x^T K_0 x)\, T_{jj}.$$

Thus, for $E_0(x) \le 4E_{\mathrm{net}}$, $|q_j/Q_j| < 2p_j = r_j$, and $0 \le E_0/E_{\mathrm{net}} \le 4$. Consequently the entire configuration ellipsoid maps into the stated anisotropic feature box. No reduction of a full state is used in this argument.

For completeness, write each complex affine weight as $A_\ell + iB_\ell$. The four real features enter the complex network as $s + i0$. In real coordinates its first weight block is

$$W_1 = \begin{pmatrix} A_1 \\ B_1 \end{pmatrix} \in \mathbb{R}^{24\times 4}, \qquad W_2 = \begin{pmatrix} A_2 & -B_2 \\ B_2 & A_2 \end{pmatrix} \in \mathbb{R}^{24\times 24},$$

and its two output rows are $C_R = (A_3, -B_3)$ and $C_I = (B_3, A_3)$. Both hidden activations apply ordinary real tanh separately to these 24 real coordinates. Since $|\tanh'(t)| \le 1$ for every real t, the global componentwise Lipschitz estimate, including arbitrary fixed biases, is

$$\begin{pmatrix} |R(s) - R(0)| \\ |I(s) - I(0)| \end{pmatrix} \le L|s|, \qquad L = \left| \begin{pmatrix} C_R \\ C_I \end{pmatrix} \right| |W_2|\,|W_1|.$$

Absolute values and inequalities are componentwise. The estimate is valid for all real features, not merely near zero. It uses no holomorphic-complex-tanh derivative assertion: the implemented activation is the realification of split tanh.

Reconstructing these blocks from the exact saved complex weights gives, on $|s_j| \le r_j$,

$$\delta_R = L_R r \approx 1.0008828328328028, \qquad \delta_I = L_I r \approx 1.0226300556580044.$$

These two displays are explanatory floating-point approximations; the certificate uses their exact rational values. The saved scalar is $c_C = 2607499387914309/36028797018963968$. Hence

$$m = |1 + c_C| - \delta_R > 0, \qquad m^2 \le a_N(x) \le (|1 + c_C| + \delta_R)^2 + \delta_I^2.$$

Exact rational arithmetic proves

$$m^2 > 0.005110793448, \quad (|1 + c_C| + \delta_R)^2 + \delta_I^2 < 5.344160513025, \quad (6/25)m^2 > 0.00122659042752.$$

Combining the lower factor bound with the received strict native mass-metric inequality for D_0 proves (S221)–(S222).

Finally, differentiate H_N along $M\dot{v} = -\nabla U_N - D_N v + \mu \cos(b^T x) b u$. The conservative terms cancel, with the complete energy-feature chain rule retained, yielding $\dot{H}_N = -v^T D_N v + uy$. Substitution of (S222) proves (S223). Its damping term has units $(\mathrm{s}^{-1})(\mathrm{J}) = \mathrm{J\,s}^{-1}$, as does uy. For saturated continuous feedback $uy = -|y| \min(k|y|, U) \le 0$.

Candidate trajectory scope. The already received guard gives $\inf_{\partial\Omega_4} U_N > 4.236161085119E_{\mathrm{net}}$ and the initial set has $H_N(0) \le 2.046161001544E_{\mathrm{net}}$ with $x(0) \in \Omega_4$. For the exact candidate under zero total field or the specified continuous saturated feedback, global nonnegative dissipation implies $\dot{H}_N \le 0$. A first configuration exit would require $H_N \ge U_N > 4.236161085119E_{\mathrm{net}}$, contradicting the initial bound. The same selected weights satisfy $1 + c_E > 0$ and the received real coercivity test, so bounded-energy trajectories are in a compact phase-space set and smooth-vector-field continuation gives existence for all positive time. Thus this candidate retains the strict mass-metric damping estimate throughout those trajectories. No exponential decay of the full storage, asymptotic convergence to a specified equilibrium, sampled-control

passivity, or physical-target damping error is asserted. On a physical plant, the derivative of the learned storage has its separate potential/damping/work error terms; the candidate inequality cannot replace that analysis.

S33.4 Current-network reception of the 22 recorded numerical initial states

Corollary S60. Fixed-guard reception of all 22 recorded numerical initial states. *Keep the current calibrated complex split-tanh potential U_N, the same exact native full-coordinate matrices $M = M_0 + M_r, K_0$, and the conditional physical-family assumptions of Section S33.2. Let E_{net} be its exact saved binary64 energy scale and set $E_0(x) = x^T K_0 x/2$. Retain the already certified domain, cap and error allowance without adjustment:*

$$\Omega_4 = \{x : E_0(x) < 4E_{\mathrm{net}}\}, \qquad c = \frac{41}{20}E_{\mathrm{net}}, \qquad \varepsilon_U = \frac{479874807}{125000000000}E_{\mathrm{net}}, \qquad c - \varepsilon_U = \frac{255770125193}{125000000000}E_{\mathrm{net}}.$$

Consider the finest signed records selected in the frozen 11-configuration training protocol: the original shifted configuration, directions 2 and 3, and design nodes 3 through 10, with each of their two velocity-sign labels. For each record use its own unique saved sample with time exactly zero and form

$$x_0 = (w_1, \theta_1, \ldots, w_{32}, \theta_{32})^T, \qquad v_0 = (w_{t,1}, \theta_{t,1}, \ldots, w_{t,32}, \theta_{t,32})^T.$$

Every saved binary64 component is interpreted exactly. Neither vector is replaced by Vq, by an intended preparation, or by the opposite sign. Write $H_N(x, v) = v^T M v/2 + U_N(x)$ and $m_0 = c - \varepsilon_U - H_N(x_0, v_0)$. Both signs in each row satisfy the following strict rational bounds; all terminating decimals are exact outward relaxations:

configuration	$E_0(x_0)/E_{\mathrm{net}} <$	$H_N(x_0, v_0)/E_{\mathrm{net}} <$	$m_0/E_{\mathrm{net}} >$
original shifted	0.011840151418	0.972247611362	1.073913390182
direction 2	0.075607037915	0.237377388001	1.808783613543
direction 3	0.665892667009	0.785898582383	1.260262419161
jet node 3	0.699184876801	0.818661343066	1.227499658478
jet node 4	0.672114306437	0.795381639556	1.250779361988
jet node 5	0.248565856915	0.290319882622	1.755841118922
jet node 6	0.688816911554	0.745602603503	1.300558398041
jet node 7	0.686085823522	0.817887305482	1.228273696062
jet node 8	0.757581829134	0.887146476049	1.159014525495
jet node 9	0.198955734425	0.276686835522	1.769474166022
jet node 10	0.705414332297	0.819033952757	1.227127048787

Thus all 22 saved numerical initial states satisfy the same frozen initial guard. Across these states,

$$0.237377387997 < \frac{H_N(x_0, v_0)}{E_{\mathrm{net}}} < 0.972247611362, \qquad 1.073913390182 < \frac{m_0}{E_{\mathrm{net}}} < 1.808783613547.$$

The unchanged boundary guard has exact margin $2182322086663E_{\mathrm{net}}/10^{12} > 0$.

Proof. The receiver verifies the current guard's actual zero exit code and hashed result before receiving its fixed constants. Each of the 22 observation files is bound to its own completed producer receipt and a launcher receipt with actual MATLAB exit code zero, including the different receipt structures of the original and later producers. The 257 saved times, 33 node identities, and all four 257×33 position/velocity field arrays are checked separately for every record. Exact rational coordinate gaps at most 10^{-15} m associate each saved node with a unique native finite-coordinate field row; this is a node-identity check, not an interpolation-error estimate. The saved clamped-root values at the initial sample are exactly zero.

The receiver extracts all 64 free coordinates and evaluates E_0 and $v^T M v/2$ by exact rational arithmetic, adding $M_0 + M_r$ before use. The same-source successful native positive-pivot proof is received; no new LDL factorization is performed. For the energy features only, it computes the exact mass projection $q = Bx_0$, where $B = (V^T M V)^{-1} V^T M$; the full state remains x_0. The centered current energy-head expression is evaluated with outward Arb arithmetic at 512 bits, including its

actual complex realification, the exact skip coefficient, the origin readout and the first three origin-gradient subtractions. The magnetic term uses the actual saved tip angle θ_{32}. This yields strict rational lower and upper endpoints for every H_N/E_{net} and m_0/E_{net}. Independent exact rational rounding proves the displayed relaxations. All 22 tests resolve strictly and pass; rejection would have been reported as a valid negative decision rather than as arithmetic failure. Since $E_0 < 4E_{\mathrm{net}}$ and $H_N + \varepsilon_U < c$, each saved vector pair satisfies the theorem's initial inequalities when used as an exact mathematical initial state. The 11 positive/negative pairs have nonzero recorded position and velocity-sum differences, which are retained and evaluated separately.

This corollary receives numerical recorded initial states, not physical measurements. No certified error bound relates a saved binary64 state to a true physical initial state. The continuous safety conclusion still requires the declared physical potential family, the same exact full-state port, $w = 0$, and actual continuous saturated feedback. These forced observation records do not show that such feedback was executed, or certify any sampled, delayed or sensor-error realization. No new native solve, training, gradient update, weight selection, guard adjustment, control run, or physical KAM transfer is established.

S34 Input-variation storage for the declared current finite-element model

The source-IVP bounds in Sections S32 and S32.1 use the exact reference mass and the initial states of the original short records. Here a different constant-matrix model is defined from the current raw assembly and its declared finite-element structure. Its mass, initial state and original clock are retained separately. A nonautonomous storage exploits the variation of its prescribed field and gives a bound over the complete current 0.6-second horizon.

S34.1 The declared model and the magnetic tangent

Use the same 64 transverse coordinates x, tip covector $b = e_{64}$, reference stiffness $K_0 \succ 0$ and positive magnetic constants κ, μ as in the article. The model considered in this section is

$$\begin{aligned} \dot{x} &= v, \\ M_c\dot{v} &= -K_{e,c}x - \kappa\sin(b^T x)b - D_c v + \mu u(t)\cos(b^T x)b, \\ U_c(x) &= \tfrac{1}{2}x^T K_{e,c}x + \kappa[1-\cos(b^T x)], \\ H_c(x,v) &= T_c(v) + U_c(x), \qquad T_c(v) = \tfrac{1}{2}v^T M_c v, \\ \psi(x) &= \mu\sin(b^T x), \qquad g(x) = \nabla\psi(x). \end{aligned} \tag{S224}$$

Here $M_c = M_c^T \succ 0$, $D_c = D_c^T \succeq 0$ and $K_{e,c} = K_{e,c}^T$ are constant. The stored declarations specify Hermitian beam interpolation, a material reference frame, linear elasticity, constant density and rotary inertia, consistent mass, mass-proportional damping and a fixed mesh with geometric nonlinearity disabled. Interpreting these declarations as fixed Galerkin bilinear forms gives the constant matrices in (S224); the stated magnetic end moment is retained nonlinearly. This structural interpretation is an explicit modeling condition of the result.

The stored stiffness is a tangent at the assembly state (x_a, u_a), with $\theta_a = b^T x_a$. Differentiating the restoring force at fixed input gives

$$\begin{aligned} K_a &= K_{e,c} + [\kappa\cos\theta_a + \mu u_a \sin\theta_a]bb^T, \\ K_{e,c} &= K_a - [\kappa\cos\theta_a + \mu u_a\sin\theta_a]bb^T. \end{aligned} \tag{S225}$$

The input-dependent sine term has a positive sign in the tangent. Equation (S225) assumes the same coordinate ordering, constraint elimination, moment covector and restoring-force sign convention as (S224). An additional nonlinear potential would contribute its own Hessian and would require a corresponding error allowance. The saved floating-point matrix and initial-state entries are interpreted as exact dyadic real numbers; the scalar trigonometric functions in this declared model are enclosed outwardly. A single tangent is used to bound a constant elastic matrix under this structural condition, rather than to infer the structure from a finite sample of states.

For completeness, the zero-input conservative companion has

$$\begin{aligned} U_{c,xx}(0) &= K_a + [\kappa(1-\cos\theta_a) - \mu u_a \sin\theta_a] bb^T, \\ (H_c)_{pp} &= M_c^{-1}, \qquad p = M_c v. \end{aligned} \tag{S226}$$

Thus the driven assembly tangent and the origin conservative Hessian are different matrices. Section S24 separately verifies the finite-order nonresonance and twist conditions for this companion.

S34.2 Generalized matrix bounds and domain coercivity

Retain the energy normalization and position domain

$$E = E_{\mathrm{net}}, \qquad E_0(x) = \tfrac{1}{2} x^T K_0 x, \qquad \Omega_4 = \{x : E_0(x) < 4E\}. \tag{S227}$$

The candidate retains its original mass $M = M_0 + M_r$ and energy $H_N(x,v) = v^T M v/2 + U_N(x)$. Comparisons below evaluate both energies at the same physical state (x,v).

Lemma S61. Constant-matrix relative bounds. *Let $R = R^T \succ 0$ and $A = A^T$. If $\|R^{-1}A\|_\infty \le \eta$, then*

$$-\eta R \preceq A \preceq \eta R. \tag{S228}$$

In particular, define

$$\beta_c = 3/50, \qquad \Delta K_c = K_{e,c} - (1+\beta_c) K_0.$$

An outward bound $\|K_0^{-1}\Delta K_c\|_\infty \le \eta_K$ implies, for all x,

$$|x^T \Delta K_c x| \le \eta_K x^T K_0 x, \qquad |U_c - U_{\beta_c}| \le \eta_K E_0. \tag{S229}$$

Similarly, an outward bound $\|M_c^{-1}(M - M_c)\|_\infty \le m_+$ gives $M \preceq (1+m_+) M_c$.

Proof. The matrix $R^{-1}A$ is similar to the real symmetric matrix $R^{-1/2} A R^{-1/2}$. Its real eigenvalues have absolute value at most $\|R^{-1}A\|_\infty$. Applying the spectral theorem to the symmetric matrix and then congruence by $R^{1/2}$ proves (S228). The elastic conclusion follows because U_c and U_{β_c} have the same magnetic potential; their difference is $x^T \Delta K_c x/2$. The mass conclusion is the upper half of (S228).

The similarity to a symmetric matrix is essential to the quadratic-form conclusion. Exact matrix products and outward scalar intervals in (S225) give a bound on every row sum of $K_0^{-1}\Delta K_c$. Once the constant-matrix model is specified, (S229) consequently holds on the entire position domain. In particular,

$$\sup_{\overline{\Omega}_4} |U_c - U_{\beta_c}| \le 4\eta_K E. \tag{S230}$$

If the magnetic coefficient also changes, its potential difference must be added to this bound.

Put

$$\begin{aligned} c_0 &= b^T K_0^{-1} b, \qquad \Theta^2 = 8Ec_0, \\ \lambda &= 1 + \beta_c - \eta_K > 0, \qquad r_\theta = 1 - \Theta^2/12 \ge 0, \\ \overline{K} &= \lambda K_0 + \kappa r_\theta b b^T. \end{aligned} \tag{S231}$$

On $\overline{\Omega}_4$, quadratic-form Cauchy–Schwarz gives $|b^T x|^2 \le \Theta^2$. The global inequality $1 - \cos\theta \ge \theta^2/2 - \theta^4/24$ then yields

$$U_c(x) \ge \tfrac{1}{2} x^T \overline{K} x \ge \lambda E_0(x) \ge 0. \tag{S232}$$

One proof of the cosine inequality differentiates $1-\theta^2/2+\theta^4/24-\cos\theta$ four times: its fourth derivative is $1-\cos\theta\geq 0$, its first four initial data vanish at zero, and the function is even. Thus no equality based on a small-angle approximation is used.

Since $\overline{K}\succ 0$, the rank-one inverse formula gives

$$c_K^2 := b^T\overline{K}^{-1}b = \frac{c_0}{\lambda+\kappa r_\theta c_0}. \tag{S233}$$

The fixed sine gauge for ψ and $|\sin\theta|\leq|\theta|$ therefore imply

$$|\psi(x)|\leq \mu c_K\sqrt{x^T\overline{K}x}\leq \mu c_K\sqrt{2U_c(x)}. \tag{S234}$$

The positive bias stiffness in $\overline{K}$ improves this bound relative to using only λK_0.

S34.3 A storage theorem for bounded-variation input

Let u be right-continuous with left limits and of bounded variation on $[0,T]$. Assume that, throughout this interval,

$$|u(t)|\leq U_{\max},\qquad \mathrm{TV}_{[0,T]}(u)\leq \mathcal{V}_T. \tag{S235}$$

The variation counts the atoms in $(0,T]$, and $u(0)$ is the post-initial value used in the storage. Ordinary input jumps are bounded forces in (S224); the mechanical state remains continuous and has no impulse reset. Define

$$\begin{aligned} a^2 &= \frac{\mu^2U_{\max}^2c_K^2}{2E},\qquad \mathcal{B}_\rho=\frac{a^2E}{\rho},\qquad 0<\rho<1,\\ Z(t) &= H_c(x(t),v(t))-u(t)\psi(x(t))+\mathcal{B}_\rho. \end{aligned} \tag{S236}$$

The scalar a is nonnegative and is distinct from the learned damping factor a_N. The constant $\mathcal{B}_\rho$ is a storage shift; B below continues to denote the candidate boundary value in (S193).

Theorem S62. Finite-horizon input-variation storage bound. *Suppose* (S224), (S232), (S234) *and* (S235) *hold, with* $x(0)\in\Omega_4$. *Let* $z_{\mathrm{in}}\geq Z(0)/E$ *be a nonnegative upper bound, and set*

$$\begin{aligned} q_T &= \frac{\mu c_K\mathcal{V}_T}{\sqrt{2(1-\rho)E}},\\ \mathcal{Z}_T &= (\sqrt{z_{\mathrm{in}}}+q_T)^2,\\ h_T &= \left[a+\sqrt{a^2+\mathcal{Z}_T-a^2/\rho}\right]^2. \end{aligned} \tag{S237}$$

If

$$\frac{\mathcal{Z}_T}{(1-\rho)\lambda}<4, \tag{S238}$$

the solution exists on $[0,T]$ *and, simultaneously for every prefix,*

$$\begin{aligned} Z(t)/E &\leq \mathcal{Z}_T,\\ E_0(x(t))/E &\leq \frac{\mathcal{Z}_T}{(1-\rho)\lambda}<4,\\ H_c(x(t),v(t))/E &\leq h_T. \end{aligned} \tag{S239}$$

The radicand defining h_T *is nonnegative under these feasible initial-data hypotheses. If, in addition,*

$$M\preceq(1+m_+)M_c,\quad m_+\geq 0,\qquad U_N-U_c\leq\varepsilon_+E\quad \text{on } \overline{\Omega}_4, \tag{S240}$$

then

$$H_N(x(t),v(t))/E\leq(1+m_+)h_T+\varepsilon_+. \tag{S241}$$

For the same-checkpoint allowance in (S193) and the potential comparison (S230), one may take $\varepsilon_+ = \varepsilon + 4\eta_K$. If $(1+m_+)h_T + \varepsilon_+ < B$, this also receives the candidate boundary-storage guard for this declared mathematical model and horizon.

Proof. By (S234) and Young's inequality,

$$|u\psi| \le \mu U_{\max} c_K \sqrt{2U_c} \le \rho U_c + \mathcal{B}_\rho. \tag{S242}$$

Consequently, on every pre-exit prefix,

$$Z \ge T_c + (1-\rho)U_c \ge (1-\rho)\lambda E_0, \qquad Z \ge T_c \ge 0. \tag{S243}$$

Multiplying (S224) by v^T gives $dH_c = -v^T D_c v\, dt + u\, d\psi$. The function $\psi(x(t))$ is absolutely continuous. BV integration by parts therefore yields the signed measure identity

$$dZ = -v^T D_c v\, dt - \psi\, du. \tag{S244}$$

In particular the field-variation term has a negative sign.

First consider the diffuse part of these measures, including both the absolutely continuous and the singular-continuous part of du. For $\delta > 0$, the BV chain rule and (S243) give

$$\begin{aligned} \left(d\sqrt{Z+\delta}\right)_{\text{diff}} &= \frac{-v^T D_c v\, dt - \psi\,(du)_{\text{diff}}}{2\sqrt{Z+\delta}} \\ &\le \frac{\mu c_K}{\sqrt{2(1-\rho)}} |du|_{\text{diff}}. \end{aligned} \tag{S245}$$

Here $|du|$ denotes the total-variation measure of du; its diffuse part includes absolutely continuous and singular-continuous variation.

At an input jump, x, v, H_c and ψ are continuous, whereas

$$\Delta Z = -\psi\, \Delta u. \tag{S246}$$

If $U_c > 0$, coercivity on both sides of the jump implies

$$\begin{aligned} |\sqrt{Z^+} - \sqrt{Z^-}| &= \frac{|\psi \Delta u|}{\sqrt{Z^+} + \sqrt{Z^-}} \\ &\le \frac{\mu c_K |\Delta u|}{\sqrt{2(1-\rho)}}. \end{aligned} \tag{S247}$$

The same upper bound holds with $Z^\pm + \delta$ in the two roots. If $U_c = 0$, (S234) gives $\psi = 0$, so that the jump vanishes. Integrating (S245), summing the countably many jump increments and letting $\delta \downarrow 0$ proves

$$\sqrt{Z(t)/E} \le \sqrt{Z(0)/E} + \frac{\mu c_K}{\sqrt{2(1-\rho)E}}\, \mathrm{TV}_{[0,t]}(u). \tag{S248}$$

This establishes the first bound in (S239) while the solution remains in the domain. For $U_{\max} > 0$, the second term at T equals $a(\mathcal{V}_T/U_{\max})/\sqrt{1-\rho}$, which also checks its normalization.

Let τ be the first position exit. Up to $\min\{\tau, T\}$, (S243) and (S248) imply the strict interior bound (S238). By continuity of x, a first boundary point would have $E_0 = 4E$, contradicting this strict bound. A field jump at the putative exit does not change the mechanical state; the same argument follows from its left limit. The inequalities $E_0 < 4E$ and $T_c \le Z \le E\mathcal{Z}_T$ bound both x and v. The smooth finite-dimensional coefficients and bounded input in (S224) then give continuation through T. Thus the pre-exit argument closes without assuming future retention in the domain.

To recover the mechanical energy, put $h = H_c/E$. Since $U_c \le H_c$, (S234) gives

$$h - 2a\sqrt{h} \le Z/E - a^2/\rho \le \mathcal{Z}_T - a^2/\rho. \tag{S249}$$

Completing the square gives $(\sqrt{h} - a)^2 \le a^2 + \mathcal{Z}_T - a^2/\rho$ and hence $h \le h_T$. In particular, a feasible initial state implies $Z(0)/E \ge a^2(1/\rho - 1)$, so that the radicand is nonnegative. The constants a^2, $\mathcal{B}_\rho$ and the initial storage must use the same normalization; the negative shift $-a^2/\rho$ in (S237) is retained.

Finally, $T_N \le (1 + m_+)T_c$ and $U_N \le U_c + \varepsilon_+ E$, with $U_c \ge 0$, imply

$$H_N \le (1 + m_+)H_c + \varepsilon_+ E.$$

This proves (S241). The asserted value of ε_+ follows by adding the two uniform potential allowances.

The result controls the candidate storage directly. It does not assert a bound of the same size for the residual-work integral of Section S32, or identify the compiled numerical history with a solution of (S224). A finite $\mathcal{V}_T$ gives the stated finite horizon. Extending T requires a new variation budget or an additional argument.

S34.4 The current raw initial state and complete clock

The current instance uses the first raw solver state (x_c^0, v_c^0) in the received 64-coordinate map, interpreted as exact real initial data for (S224). It uses the raw first-state velocity and the raw mass M_c in $H_c(0)$. The observed/interpolated first state and the 22 original short-record initial states are distinct data. In particular, the earlier C_{22} bound is not substituted for this initial energy. The current first state satisfies

$$x_c^0 \in \Omega_4, \qquad H_c(0)/E < 0.364015095, \qquad H_N(x_c^0, v_c^0)/E < 3/8. \tag{S250}$$

The original current clock contains 96001 saved times, starts at zero and ends at

$$T_c^{\mathrm{end}} = \frac{5404319552844595}{2^{53}}\ \mathrm{s}, \tag{S251}$$

the saved binary64 value of 0.6 s. The count records the clock's provenance; the storage proof uses its complete analytic input law.

The prescribed mathematical input family is

$$\begin{aligned} u(t) &= A_0 \cos(\omega t), \qquad 0 \le t \le T_c^{\mathrm{end}}, \\ A_0 &\in \left[10^{-4}, \frac{7378697629483821}{2^{66}}\right]\ \mathrm{T}, \\ \omega &\in \left[\frac{1141692154656479}{10^{14}}, \frac{3213577726426711}{2^{48}}\right]\ \mathrm{s}^{-1}. \end{aligned} \tag{S252}$$

Each small interval includes the exact source decimal and its saved binary64 interpretation. The field u_a in (S225) is the field evaluated at the stored assembly anchor; $u(0) = A_0$ in the initial nonautonomous storage is varied over (S252). These two uses are kept distinct when bounding the common elastic matrix and the input family's initial storage.

The complete frequency box satisfies $2\pi < \omega T_c^{\mathrm{end}} < 3\pi$. Consequently

$$\mathrm{TV}_{[0,T_c^{\mathrm{end}}]}(u) = A_0[5 - \cos(\omega T_c^{\mathrm{end}})]. \tag{S253}$$

The two completed half-cycles each contribute $2A_0$ and the remaining decreasing segment contributes $A_0[1 - \cos(\omega T_c^{\mathrm{end}})]$. This calculation uses the known input between stored nodes.

For this instance, the constant-matrix calculation gives the exact rational relaxation

$$\begin{aligned} &\eta_K = \frac{11169}{10^{12}}, \qquad \lambda = 1.059999988831, \\ &M \preceq (1+m_+)M_c, \qquad 0 \le m_+ < 2.031 \times 10^{-10}, \\ &\Theta^2 < 7.545124 \times 10^{-6}. \end{aligned} \tag{S254}$$

The declared $\beta_c = 3/50$ belongs to the exact loss-generated B_θ in (S192). The potential allowance retained for this nonzero-force-loss family is

$$\begin{aligned} \beta_c = 3/50 \in B_\theta, \\ \varepsilon = \frac{959754951}{250000000000} = 0.003839019804, \\ B = 4.236161085119. \end{aligned} \tag{S255}$$

This is the allowance in (S193). It is slightly larger than 0.003838998456 in the earlier conditional comparison (89); the present calculation uses (S255) in $\varepsilon_+ = \varepsilon + 4\eta_K$. Choosing $\rho = 1/10$ before the calculation, outward evaluation over the full box (S252) gives

$$\begin{aligned} \mathcal{V}_{T_c^{\mathrm{end}}} < 0.000415646653\ \mathrm{T}, \\ a^2 < 0.045392061, \qquad Z(0)/E < 0.886094565. \end{aligned} \tag{S256}$$

The trigonometric enclosures use 384-bit outward interval arithmetic; the final square-root ceilings and comparisons are rational. The displayed constants are outward relaxations of the retained exact calculation. In particular, the shift and the subtracted term in (S237) use the same underlying a^2.

For every member of (S252), from this same initial state, the strict printed bounds are

$$\begin{aligned} Z(t)/E &< 3.514800373, \\ E_0(x(t))/E &< 3.684277159 < 4, \\ H_c(x(t),v(t))/E &< 3.902663433, \\ H_N(x(t),v(t))/E &< 3.906502498 < B \qquad (0 \le t \le T_c^{\mathrm{end}}). \end{aligned} \tag{S257}$$

The candidate boundary margin exceeds $0.329658587E$. The energy-domain comparison is strict before the boundary-storage comparison is used. These estimates cover the original current clock through its own endpoint, rather than the earlier approximately $20\,\mu$s horizons or a restarted feedback seed.

The conclusion is a conditional bound for the declared constant-matrix FE model. Section S35 separately receives the signed residual-work ledger of an explicit raw-data Hermite curve. Section S36 supplies a separate current-IVP reference tube and a strict saved-knot state separation. Individual modal phases and identity with COMSOL internal dense output remain unreceived. Applying these results to a physical device additionally requires model, initial-state, sensing and actuation allowances. The analytic input box in (S252) specifies a mathematical field family and supplies no bound for arbitrary numerical cosine or actuator errors. The forced, dissipative storage result is separate from the conservative local KAM theorem in Section S24, which uses its own full-coordinate jet, spectral and twist conditions. Neither result identifies an unknown physical target.

S35 Signed residual work for the current saved-data reconstruction

This section receives a continuous curve defined explicitly by the current saved finite-element data. It does not identify that curve with the internal dense output of COMSOL or with an exact solution of the declared continuous initial-value problem. The distinction is essential: the residual below is evaluated on the reconstructed curve, rather than assumed to vanish.

S35.1 The raw finite-element path and its local injection

The original record contains 96001 raw solver states and their saved time derivatives at strictly increasing raw times

$$0 = t_0 < t_1 < \cdots < t_N = T, \qquad N = 96000, \qquad T = \frac{5404319552844595}{2^{53}}\ \mathrm{s}. \tag{S258}$$

Thus T is the saved binary64 value of 0.6 s. The full saved vectors have 101 solver entries. The individually received field–node–coordinate map selects the 64 transverse mechanical coefficients: displacement `comp1.v` and rotation `comp1.thZ`, alternating over the 32 non-root nodes. Write S for this fixed coordinate-selection injection, so that

$$x_i = S^T U_i, \qquad v_i = S^T \dot{U}_i, \qquad S^T S = I_{64}. \tag{S259}$$

Here Sx places the declared transverse coefficients in their received solver slots. It is a local coordinate injection for this declared transverse FE model, not an assertion that a 64-vector reconstructs every auxiliary solver entry or every physical field. The transverse matrices and tip covector use exactly this field and node ordering, with $b = e_{64}$. No modal truncation, modal transformation or arbitrary variable rescaling is performed. The metadata name `canonical_row` denotes a common FE row order; the stored velocity is not a canonical momentum. In the declared constant-mass Hamiltonian chart that momentum would be $p = M_c v$.

The x_i, v_i, t_i in (S259) are the original raw arrays, not the separately stored interpolated observations. In particular their own first state and complete raw clock are retained. All saved binary64 entries are interpreted as exact real dyadic numbers. For $h_i = t_{i+1} - t_i$ and $s = (t - t_i)/h_i$, define the unique cubic Hermite interpolant

$$\begin{aligned} &q(t_i) = x_i, \quad q(t_{i+1}) = x_{i+1}, \\ &\dot{q}(t_i) = v_i, \quad \dot{q}(t_{i+1}) = v_{i+1}, \qquad v(t) = \dot{q}(t). \end{aligned} \tag{S260}$$

Consequently q is C^1, v is continuous and absolutely continuous, and the acceleration is piecewise continuous. Velocity is defined by differentiating q; it is not interpolated by a separate unrelated rule. Acceleration may jump at a saved time, but continuous velocity and constant mass produce no momentum impulse or additional node-work term.

For $B_{j,n}(s) = \binom{n}{j} s^j (1-s)^{n-j}$, the cubic position, quadratic velocity and linear acceleration control points are

$$\begin{aligned} Q_{i,0} &= x_i, & Q_{i,1} &= x_i + \tfrac{1}{3} h_i v_i, \\ Q_{i,2} &= x_{i+1} - \tfrac{1}{3} h_i v_{i+1}, & Q_{i,3} &= x_{i+1}, \\ V_{i,0} &= v_i, & V_{i,1} &= 3(x_{i+1} - x_i)/h_i - v_i - v_{i+1}, \\ V_{i,2} &= v_{i+1}, & & \\ A_{i,0} &= 2(V_{i,1} - V_{i,0})/h_i, & A_{i,1} &= 2(V_{i,2} - V_{i,1})/h_i. \end{aligned} \tag{S261}$$

These temporal control points are distinct from the spatial Hermitian beam basis used to define the FE coefficients.

S35.2 The declared model and its residual power

Retain the declared current constant matrices $M_c = M_c^T \succ 0$, $D_c = D_c^T \succeq 0$ and $K_{e,c} = K_{e,c}^T$, with

$$\begin{aligned} U_c(q) &= \tfrac{1}{2} q^T K_{e,c} q + \kappa[1 - \cos\theta], \\ H_c(q,v) &= \tfrac{1}{2} v^T M_c v + U_c(q), \\ \psi(q) &= \mu \sin\theta, \qquad \theta = b^T q, \\ K_{e,c} &= K_{\mathrm{raw}} - c_a b b^T, \\ c_a &= \kappa \cos\theta_a + \mu u_a \sin\theta_a. \end{aligned} \tag{S262}$$

The tangent anchor (θ_a, u_a) is the received raw assembly anchor. Its evaluated field u_a is kept distinct from the variable initial amplitude of the analytic input family. The original, unsymmetrized assembled entries and their received coordinate

map are retained; c_a is enclosed outwardly. The constant-matrix interpretation of the stored linear, fixed-mesh FE declarations remains a modeling condition. A single stored tangent does not establish arbitrary-state commercial-software equivalence.

For a bounded input u, define the force residual and signed power on the explicit curve by

$$\begin{aligned} r(t) &= M_c\dot{v} + D_c v + K_{e,c} q + \kappa \sin\theta\, b - \mu u(t) \cos\theta\, b, \\ P(t) &= v(t)^T r(t), \qquad R(t) = \int_0^t P(\tau)\, d\tau. \end{aligned} \tag{S263}$$

Then r is integrable and R is absolutely continuous, with $R(0) = 0$. The defining residual identity gives

$$dH_c = -v^T D_c v\, dt + u\, d\psi + dR. \tag{S264}$$

It holds on each open segment and hence on the whole interval in the absolutely continuous sense. No zero-residual assertion is made for the saved data.

S35.3 Bernstein envelopes and all-time signed prefixes

The certificate constructs fixed polynomial lower and upper envelopes on every segment:

$$\sum_{k=0}^{5} \ell_{i,k} B_{k,5}(s) \;\le P(t_i + h_i s) \le \sum_{k=0}^{5} u_{i,k} B_{k,5}(s), \qquad 0 \le s \le 1. \tag{S265}$$

The true P need not be a polynomial: the envelopes include the trigonometric terms and the full analytic drive-parameter box. To construct them, the linear acceleration and quadratic velocity in (S261) are elevated to degree three in $M_c\dot{v} + D_c v$. The linear term $\kappa\theta b$ is included in the stiffness, and the remaining magnetic terms satisfy

$$\begin{aligned} |\kappa(\sin\theta - \theta)| &\le \kappa|\theta|^3/6, \\ |\mu u(\cos\theta - 1)| &\le \mu U_{\max}|\theta|^2/2. \end{aligned} \tag{S266}$$

The scalar drive cosine on each whole time segment is enclosed outwardly using an Arb midpoint value and the global inequality $|\cos a - \cos b| \le |a - b|$. Multiplying the resulting cubic residual envelopes by the quadratic velocity and summing coordinates gives (S265). In particular the scalar product retains inertial, elastic and coordinate cancellations before an all-time bound is taken.

The Bernstein derivative formula implies that the degree-six control points for an antiderivative with initial value zero are

$$C^-_{i,0} = C^+_{i,0} = 0, \quad C^-_{i,j} = \frac{h_i}{6} \sum_{k=0}^{j-1} \ell_{i,k}, \quad C^+_{i,j} = \frac{h_i}{6} \sum_{k=0}^{j-1} u_{i,k}, \quad 1 \le j \le 6. \tag{S267}$$

Indeed, differentiating with respect to s multiplies successive control-point differences by six. Integrating the two pointwise envelopes in (S265) therefore bounds every partial segment integral. Define full-segment prefix bounds by

$$\underline{S}_i = \sum_{m<i} C^-_{m,6}, \qquad \overline{S}_i = \sum_{m<i} C^+_{m,6}. \tag{S268}$$

For every $s \in [0,1]$ this gives

$$\begin{aligned} \underline{S}_i + \sum_{j=0}^{6} C^-_{i,j} B_{j,6}(s) &\le R(t_i + h_i s) \\ &\le \overline{S}_i + \sum_{j=0}^{6} C^+_{i,j} B_{j,6}(s). \end{aligned} \tag{S269}$$

Since the Bernstein basis is nonnegative and sums to one,

$$\mathcal{A} = \max\left\{0,\ \max_{0\le i<N}\left[\overline{S}_i + \max_{0\le j\le 6} C^{+}_{i,j}\right]\right\} \quad\Longrightarrow\quad R(t) \le \mathcal{A} \quad (0 \le t \le T). \tag{S270}$$

Outward enclosures of the displayed quantities preserve this implication. The initial zero is included explicitly. No monotonicity of R is used.

An upper bound only for $R(T)$ would be insufficient. Cancellation after an earlier positive excursion can make the terminal integral small while its earlier supremum is large; for example, $R(t) = C\sin^2(\pi t/T)$ has $R(T) = 0$ and supremum C for any $C > 0$. Formula (S270) receives the whole time domain, including times strictly between saved nodes.

S35.4 A signed-prefix storage theorem, including BV input jumps

Let $E > 0$, $E_0(q) = q^T K_0 q/2$ and $\Omega_4 = \{q : E_0(q) < 4E\}$. Assume the received fixed-model comparisons give, on $\overline{\Omega}_4$,

$$U_c \ge \lambda E_0 \ge 0, \qquad |\psi| \le \mu c_K\sqrt{2U_c}, \qquad \lambda > 0. \tag{S271}$$

For a right-continuous BV input with left limits, assume $|u| \le U_{\max}$ and $\mathrm{TV}_{[0,T]}(u) \le \mathcal{V}_T$. Its variation includes input jumps in $(0, T]$, and $u(0)$ denotes the post-initial value. Ordinary input jumps do not reset q or v. Fix $0 < \rho < 1$ and define

$$\begin{aligned} a^2 &= \frac{\mu^2 U_{\max}^2 c_K^2}{2E}, \qquad \mathcal{B}_\rho = \frac{a^2 E}{\rho}, \\ Z &= H_c - u\psi + \mathcal{B}_\rho, \qquad q_T = \frac{\mu c_K \mathcal{V}_T}{\sqrt{2(1-\rho)E}}. \end{aligned} \tag{S272}$$

Theorem S63. Storage from a signed residual-work prefix. *Let the curve (S260) start in Ω_4, and suppose the residual and input satisfy (S263)–(S272). Let $z_{\mathrm{in}} \ge Z(0)/E$ and let $\mathcal{A} \ge 0$ satisfy $R(t) \le \mathcal{A}$ on $[0, T]$. Set*

$$\begin{aligned} \mathcal{Z}_{\mathcal{A}} &= \left[\sqrt{z_{\mathrm{in}} + \mathcal{A}/E} + q_T\right]^2, \\ h_{\mathcal{A}} &= \left[a + \sqrt{a^2 + \mathcal{Z}_{\mathcal{A}} - a^2/\rho}\right]^2. \end{aligned} \tag{S273}$$

If

$$\frac{\mathcal{Z}_{\mathcal{A}}}{(1-\rho)\lambda} < 4, \tag{S274}$$

then, for every $t \in [0, T]$,

$$\begin{aligned} Z(t)/E &\le \mathcal{Z}_{\mathcal{A}}, \\ E_0(q(t))/E &\le \frac{\mathcal{Z}_{\mathcal{A}}}{(1-\rho)\lambda} < 4, \\ H_c(q(t), v(t))/E &\le h_{\mathcal{A}}. \end{aligned} \tag{S275}$$

If the same candidate, evaluated at the same (q, v), satisfies

$$M_N \preceq (1 + m_+)M_c, \quad m_+ \ge 0, \qquad U_N - U_c \le \varepsilon_+ E \quad \text{on } \overline{\Omega}_4, \tag{S276}$$

then

$$H_N(q(t), v(t))/E \le (1 + m_+)h_{\mathcal{A}} + \varepsilon_+. \tag{S277}$$

Thus a strict upper bound below the same candidate boundary value B receives its storage guard for this curve.

Proof. On every pre-exit prefix, Young's inequality and (S271) imply

$$|u\psi| \le \rho U_c + \mathcal{B}_\rho, \qquad Z \ge \tfrac{1}{2} v^T M_c v + (1-\rho)U_c \ge (1-\rho)\lambda E_0 \ge 0. \tag{S278}$$

The function $\psi(q(t))$ is absolutely continuous. BV integration by parts in (S264) gives the measure identity

$$dZ = -Q\,dt - \psi\,du + dR, \qquad Q = v^T D_c v \geq 0. \tag{S279}$$

Define

$$Y(t) = Z(t) + \mathcal{A} - R(t). \tag{S280}$$

The all-time prefix hypothesis gives $Y \geq Z \geq 0$, while $R(0) = 0$ and (S279) give

$$Y(0) = Z(0) + \mathcal{A}, \qquad dY = -Q\,dt - \psi\,du, \qquad |\psi| \leq \mu c_K \sqrt{\frac{2Y}{1-\rho}}. \tag{S281}$$

The residual cancels without requiring R to be monotone.

For $\delta > 0$, the BV chain rule on the diffuse parts, including singular-continuous input variation, yields

$$\big(d\sqrt{Y+\delta}\big)_{\mathrm{diff}} = \frac{-Q\,dt - \psi(du)_{\mathrm{diff}}}{2\sqrt{Y+\delta}} \leq \frac{\mu c_K}{\sqrt{2(1-\rho)}}|du|_{\mathrm{diff}}. \tag{S282}$$

At an input jump, q, v, H_c, ψ and R are continuous, so $\Delta Y = -\psi\Delta u$. The last inequality in (S281) holds on both sides. If the sum of the two square roots is positive, it follows that

$$\begin{aligned} |\sqrt{Y^+} - \sqrt{Y^-}| &= \frac{|\psi\Delta u|}{\sqrt{Y^+} + \sqrt{Y^-}} \\ &\leq \frac{\mu c_K}{\sqrt{2(1-\rho)}}|\Delta u|. \end{aligned} \tag{S283}$$

To see the factor $1/2$ in the last step, bound $|\psi|$ by $\mu c_K\sqrt{2/(1-\rho)}$ times the smaller of the two roots. If a root vanishes then this bound forces $\psi = 0$, so the jump vanishes; the same conclusion also follows by using $Y^{\pm} + \delta$ and passing to the limit.

Integrating (S282), summing the jump increments and letting $\delta \downarrow 0$ proves

$$\sqrt{Y(t)/E} \leq \sqrt{[Z(0)+\mathcal{A}]/E} + \frac{\mu c_K\,\mathrm{TV}_{[0,t]}(u)}{\sqrt{2(1-\rho)E}}. \tag{S284}$$

Since $Z \leq Y$, this gives the first two caps in (S275) on every pre-exit prefix. At a first configuration exit, continuity of q would give $E_0(q) = 4E$, contradicting (S274). An input jump does not alter that position. This proves retention throughout the already defined curve's interval; it is not an existence or accuracy assertion for another numerical or exact IVP. The bound on $\mathcal{A}$ used here is constructed on the entire explicit curve from a globally defined residual, without assuming future retention in Ω_4, so this first-exit step is not circular.

Put $h = H_c/E$. Since $U_c \leq H_c$, the port bound gives

$$h - 2a\sqrt{h} \leq Z/E - a^2/\rho \leq \mathcal{Z}_{\mathcal{A}} - a^2/\rho.$$

Completing the square proves $h \leq h_{\mathcal{A}}$, with the same a^2 retained in the positive and negative terms. Feasible initial storage makes the radicand nonnegative. Finally, (S276), $U_c \geq 0$ and $T_N \leq (1+m_+)T_c$ imply $H_N \leq (1+m_+)H_c + \varepsilon_+ E$, proving (S277).

S35.5 The complete current-record instance

This instance uses the exact saved binary64 network scale $E = E_{\mathrm{net}} \approx 6 \times 10^{-12}$ J, the current raw initial state, $\rho = 1/10$ and the same analytic drive family

$$\begin{aligned} u(t) &= A_0 \cos(\omega t), \\ A_0 &\in \left[10^{-4}, \frac{7378697629483821}{2^{66}}\right] \mathrm{T}, \\ \omega &\in \left[\frac{1141692154656479}{10^{14}}, \frac{3213577726426711}{2^{48}}\right] \mathrm{s}^{-1}. \end{aligned} \tag{S285}$$

The intervals cover the source decimals and their saved binary64 interpretations. They do not supply an arbitrary software-cosine or physical-actuator error allowance. The same received constants are retained:

$$\begin{aligned} \lambda &= 1.059999988831, \qquad \eta_K = 1.1169 \times 10^{-8}, \\ 0 \le m_+ &< 2.031 \times 10^{-10}, \qquad \varepsilon_+ = 0.003839019804 + 4\eta_K, \\ B &= 4.236161085119. \end{aligned} \tag{S286}$$

The actual computation uses retained exact rational quantities, including the common a^2, initial storage and input-variation ceiling, rather than substituting separately rounded display values.

All 96000 segments are included. Outward evaluation gives

$$\begin{aligned} \sup_{0 \le t \le T} R(t)/E &< 0.006085625, \\ E_0(q(t))/E &< 3.696971113 < 4, \\ H_c(q(t), v(t))/E &< 3.916235955, \\ H_N(q(t), v(t))/E &< 3.920075020 < B. \end{aligned} \tag{S287}$$

The same-candidate boundary margin exceeds $0.316086065E$. These caps result from Theorem S63; they receive the continuous signed-work ledger of the explicit saved-data reconstruction over its complete original clock.

For comparison, define cumulative positive and absolute residual work by

$$W_+ = \int_0^T \max\{P(t), 0\}\, dt, \qquad W_{\mathrm{abs}} = \int_0^T |P(t)|\, dt.$$

The retained earlier calculation gives

$$\begin{aligned} 296.323170414 &< W_+/E < 416.298782743, \\ 592.646139811 &< W_{\mathrm{abs}}/E < 832.597362913, \\ -0.003537547 &< R(T)/E < 0.003939583. \end{aligned} \tag{S288}$$

Thus terminal signed cancellation alone is not the certificate; the all-time bound in (S287) is. The earlier sufficient relaxation that replaces $\mathcal{A}$ by a valid upper bound on W_+ cannot pass here. Indeed its domain cap is at least

$$\frac{W_+/E}{(1-\rho)\lambda} > 310.611293015 > 4. \tag{S289}$$

This excludes that particular cumulative-positive-work sufficient budget, rather than proving an unsafe curve, an inaccurate IVP solution, or failure of the signed-prefix criterion. The difference is the retained cancellation at every prefix, rather than only at the terminal time.

As a separate check, convexity of the received positive-definite quadratic energies on the position and velocity Bernstein control points gives the stronger direct curve caps $E_0/E < 0.505908505$ and $H_N/E < 0.828326835$. They are not substituted for the signed-prefix work construction or used to assume future domain retention in its first-exit proof.

S35.6 Arithmetic reception and the remaining scope

The residual-power enclosures use the exact saved matrix and state entries, outward binary64 interval operations enlarged with `nextafter` after each basic operation, and 256-bit Arb enclosures for the scalar transcendental parameters and drive. Matrix and scalar products are accumulated with the interval operations rather than an assumed BLAS rounding model. Final segment bounds are summed as exact rational numbers. Source and code hashes are checked before and after the calculation.

A separate exact-rational calculation re-integrates the cached power-control intervals over every one of the 96000 segments, using exact differences of the original raw times. It reconstructs the node prefixes, the degree-six antiderivative controls and the storage arithmetic, and gives a slightly tighter prefix upper bound than the displayed one. This is an independent aggregation check of the same cached residual-power envelopes; it does not replace independent validation of those upstream residual-control-point enclosures. An additional exact-rational check on eight original segments verifies the Hermite derivative identities and the linear mass–damping–stiffness energy-integral identity.

The result concerns the stated coordinate injection, constant-FE interpretation, raw initial data, analytic input box and explicit Hermite curve. It does not recover the internal COMSOL dense output or bound the curve's phase or state error relative to the exact current IVP. Signed residual work only measures the residual's projection onto this curve's velocity and cannot alone furnish a state-error tube. No unknown physical constitutive law, sensing or actuation error is received, and no KAM transfer for a changed Hamiltonian follows. No new COMSOL solve, ODE integration or training is used, and no earlier trajectory certificate is substituted for this raw record.

S36 A full-time state tube for the current raw-state IVP

S36.1 Current operators, initial state and exact input box

Retain the 64 non-root transverse displacement/rotation coordinates and their physical velocities in the received source row order. The current mass and damping are the exact dyadic entries of the original symmetric raw E and D submatrices. Symmetry is checked before evaluation; no symmetrization is applied. The separated elastic tangent in article (100) gives

$$\begin{aligned} K_{l,c} &= K_{e,c} + \kappa b b^T \\ &= K_{\mathrm{raw}} + [\kappa(1-\cos\theta_a) - \mu u_a \sin\theta_a] b b^T, \qquad b = e_{64}. \end{aligned} \tag{S290}$$

Here $\theta_a = b^T x_0$ uses the actual raw first state and u_a is the received binary64 field value at that assembly anchor. The trigonometric correction is an exact real quantity enclosed outward, rather than a rounded exact matrix entry. This is the origin pencil used by the current conservative companion in Section S24; K_{raw} alone is not used as the origin stiffness. Fresh interval LDL factorizations receive $M_c, D_c, K_{l,c} \succ 0$, retaining all 64 pivots of each matrix.

The source constants are interpreted as the exact rationals

$$\mu = \frac{5902958103587057}{295147905179352825856}, \qquad \kappa = \frac{6380294379767596475747514881383}{1063382396627932698323045648224275660 8}. \tag{S291}$$

Let hull denote the closed interval with the specified endpoints, and fl_{64} the saved binary64 interpretation of the displayed decimal. For $T = 3/5$ s, the input is

$$\begin{aligned} &u(t) = A\cos(\omega t), \\ &A \in \mathrm{hull}\{1/10000, \mathrm{fl}_{64}(0.0001)\}, \\ &\omega \in \mathrm{hull}\{1141692154656479/10^{14}, \mathrm{fl}_{64}(11.41692154656479)\}. \end{aligned} \tag{S292}$$

All conclusions hold for every pair in this box, using its own solution and the same fixed operators. The vectors x_0, v_0 are read from the original `source64_x0/source64_v0` fields and checked bitwise against the first raw position and velocity columns. The interpolated observation state is not substituted. The raw clock ends at $5404319552844595/9007199254740992$ s, which is below T, so the certified interval covers the entire original history.

Set $P_c = \mathrm{diag}(K_{l,c}, M_c)$ and retain

$$N_0 = \sqrt{x_0^T K_0 x_0 + v_0^T M_c v_0} \approx 2.029128417959113 \times 10^{-6},$$
$$N_c = \|z_0\|_{P_c} \approx 2.0900194096895563 \times 10^{-6}. \tag{S293}$$

The two constants normalize the same current-energy distance. The reference matrix K_0 occurs only in N_0; the dissipative error dynamics below use $K_{l,c}$ and M_c throughout.

S36.2 The fixed function and the current semigroup defect

The already stored full-coordinate function is defined by

$$\widehat{x}(t) = \Re\left\{\sum_{j=1}^{128} Q_j e^{\lambda_j t} + Q_p e^{i\widehat{\omega}t}\right\},$$
$$\widehat{v}(t) = \Re\left\{\sum_{j=1}^{128} V_j e^{\lambda_j t} + V_p e^{i\widehat{\omega}t}\right\}. \tag{S294}$$

Every stored real and imaginary component of the absorbed amplitudes, poles and $\widehat{\omega}$ is interpreted as its exact binary64 dyadic. The parameter file has SHA-256 prefix `bf1d1725ac58` and is the same function used in Section S40.3. This definition does not require multiplying a separately rounded basis and coefficient vector. In particular, $\widehat{x}'$ need not equal $\widehat{v}$; both defects are retained below. No new ODE integration, eigensolution, simulation or fit defines this reference.

Write $f(t) = \mu A \cos(\omega t)$ and $\widehat{f}(t) = \mu(1/10000)\cos(\widehat{\omega}t)$. For $\theta = b^T x$, the exact current equation takes the form

$$\dot{x} = v,$$
$$M_c\dot{v} + D_c v + K_{l,c}x = b\{f(t) + r(t,\theta)\}, \tag{S295}$$
$$r(t,\theta) = \kappa(\theta - \sin\theta) + f(t)(\cos\theta - 1).$$

Define

$$d_x = \widehat{x}' - \widehat{v}, \qquad d_f = M_c\widehat{v}' + D_c\widehat{v} + K_{l,c}\widehat{x} - b\widehat{f}. \tag{S296}$$

For the linear matrix

$$\mathcal{A}_c = \begin{pmatrix} 0 & I \\ -M_c^{-1}K_{l,c} & -M_c^{-1}D_c \end{pmatrix},$$

direct multiplication yields

$$\mathcal{A}_c^T P_c + P_c\mathcal{A}_c = \mathrm{diag}(0, -2D_c) \preceq 0. \tag{S297}$$

Hence $e^{t\mathcal{A}_c}$ contracts in $\|\cdot\|_{P_c}$ for $t \geq 0$. This estimate does not assume simultaneous diagonalization and introduces no highest-frequency Gronwall factor.

Lemma S64. Current-model defect bound. *The solution from z_0 is unique and exists on $[0,T]$. For $e = z_c - \widehat{z}$ and $\beta_c = (b^T M_c^{-1} b)^{1/2}$,*

$$\|e(t)\|_{P_c} \leq \|e(0)\|_{P_c} + \int_0^t \sqrt{d_x^T K_{l,c} d_x + d_f^T M_c^{-1} d_f}\, ds$$
$$+ \beta_c \int_0^t |f - \widehat{f}|\, ds + \beta_c \int_0^t |r(s, b^T x_c(s))|\, ds. \tag{S298}$$

Proof. The smooth vector field has at most linear growth in x, v, because the sine and cosine terms are bounded and M_c is invertible. Finite-dimensional continuation therefore gives a unique solution on every finite interval. Subtraction of (S296) from (S295) gives the linear error dynamics with forcing $(-d_x, -M_c^{-1}d_f)$ and the scalar port terms $f - \widehat{f} + r$. Variation of constants and (S297) give (S298). The saved path is not assumed to solve this equation.

For a complex residual pair (a, w), the norm of its real rotated value is bounded by

$$\left\{\Re(a)^T K_{l,c}\Re(a) + \Im(a)^T K_{l,c}\Im(a) + \Re(w)^T M_c^{-1}\Re(w) + \Im(w)^T M_c^{-1}\Im(w)\right\}^{1/2}.$$

Each of the 128 homogeneous columns is integrated using $(1 - e^{T\Re\lambda_j})/(-\Re\lambda_j)$, with the strict negative real parts checked individually. The harmonic particular column uses the factor T. The additional input-box allowance is

$$\beta_c\mu\left(\Delta A\, T + A_{\max}\Delta\omega\, T^2/2\right), \tag{S299}$$

where the differences are measured from the reference amplitude and $\widehat{\omega}$. It follows from the real cosine Lipschitz bound. The absolute budgets are approximately $2.300229316517212 \times 10^{-18}$ for the initial mismatch, $5.579950460923778 \times 10^{-13}$ for the homogeneous columns, $6.419741347274808 \times 10^{-13}$ for the particular column and $5.217324506437489 \times 10^{-19}$ for the input box. Their outward sum, denoted L, is below $1.199972003 \times 10^{-12}$. All values are recomputed with the current matrices; no earlier source-model tube is imported.

S36.3 First exit, acyclic refinement and the saved-knot witness

Let $\chi_c = (b^T K_{l,c}^{-1} b)^{1/2}$ and $\widehat{\theta} = b^T\widehat{x}$. Cauchy–Schwarz and the exact trigonometric remainder bounds give

$$\begin{aligned} |\theta - \widehat{\theta}| &\le \chi_c\|e\|_{P_c}, \\ |r(t,\theta)| &\le \kappa|\theta|^3/6 + \mu A_{\max}|\theta|^2/2. \end{aligned} \tag{S300}$$

The received bound for this same explicit function is $|\widehat{\theta}| < 0.000755947808515$, and $\chi_c \approx 369.020557634$. Before a hypothetical first exit from $\|e\|_{P_c} < 10^{-7}$, the true angle is below 0.000792849864278. Substitution into (S298) bounds its right-hand side by $1.684108870 \times 10^{-9} < 10^{-7}$. Since the initial error is also strictly below 10^{-7}, continuity excludes the exit and proves this preliminary bound on the whole interval.

For sharper integration, divide $[0, T]$ into 3000 exact slabs of width $\Delta t = 1/5000$ s. Let m_k be the slab midpoint, and let S and F contain the 36 slow and 92 remaining poles. Fresh interval evaluation takes an outward upper bound of

$$\begin{aligned} h_k = &\left|\Re\left\{Q_{p,64}e^{i\widehat{\omega}m_k} + \sum_{j\in S} Q_{j,64}e^{\lambda_j m_k}\right\}\right| \\ &+ \frac{\Delta t}{2}\left\{|\widehat{\omega}Q_{p,64}| + \sum_{j\in S}|\lambda_j Q_{j,64}|\right\} + \sum_{j\in F}|Q_{j,64}|. \end{aligned} \tag{S301}$$

The midpoint and derivative terms include the harmonic particular response as well as the slow poles. Since every homogeneous pole decays for $t \ge 0$, these derivative and tail terms bound the full slab, giving $|\widehat{\theta}| \le h_k$. This scalar partition retains every coordinate and pole; it is not a reduced dynamical model. For an already proved absolute tube $\varepsilon_{\mathrm{prev}}$, integration of (S300) gives the new bound

$$L + \beta_c \sum_{k=1}^{3000} \frac{1}{5000}\left\{\frac{\kappa}{6}(h_k + \chi_c\varepsilon_{\mathrm{prev}})^3 + \frac{\mu A_{\max}}{2}(h_k + \chi_c\varepsilon_{\mathrm{prev}})^2\right\}. \tag{S302}$$

The complete exact-endpoint calculation successively receives approximately $1.684108869141321 \times 10^{-9}$, $4.532057571744762 \times 10^{-10}$ and $4.522835936235115 \times 10^{-10}$. Each refinement takes the outward endpoint established by the previous stage, rather than assuming its own conclusion. The displayed decimal relaxations are not fed back into that calculation.

Theorem S65. Uniform current-IVP tube and strict saved-state separation. *For every input in* (S292),

$$\begin{aligned}\sup_{t\in[0,T]}\|z_c(t)-\widehat{z}(t)\|_{P_c} &< 4.522835937\times10^{-10},\\ \sup_{t\in[0,T]}\|z_c(t)-\widehat{z}(t)\|_{P_c}/N_0 &< 0.000222895501.\end{aligned} \tag{S303}$$

At the exact raw time $t_* = 5584463537939415/18014398509481984$ *s, with zero-based index* 49600*, the Hermite path satisfies*

$$\|z_H(t_*)-z_c(t_*)\|_{P_c}/N_0 > 0.653273417462. \tag{S304}$$

With denominator N_c*, the same two bounds are respectively* 0.000216401624 *and* 0.634240835240.

Proof. The first-exit argument and two acyclic refinements give the first bound; division by the strict lower enclosure of N_0 gives the second. At a raw knot, z_H equals the exact saved raw position and velocity columns. Evaluating all 64 coordinates of (S294) at the exact binary64 clock value and using the same P_c yields

$$0.653496312962 < \|z_H(t_*)-\widehat{z}(t_*)\|_{P_c}/N_0 < 0.653496312964.$$

Reverse triangle with the exact outward tube and normalization endpoints proves (S304); its absolute lower bound exceeds $1.325575656070055\times10^{-6}$. Division by the enclosure of N_c gives the intrinsic bounds. Only the exact endpoints, rather than subtraction of rounded display values, establish the strict printed constants.

The primary interval calculation uses Arb at 320 bits; an independent 512-bit implementation rereads the original raw operators and state and recomputes every residual column, slab and witness. Both successful actual exits and the interface audit are indexed in the reproducibility manifest. The compact current assembly, raw history and parameter file identities agree with the existing Hermite and conservative-companion components.

For current mass-normalized modes satisfying $K_{l,c}S_c = M_cS_c\Omega_c^2$ and $S_c^TM_cS_c = I$, the canonical change $x = S_c\Omega_c^{-1/2}X$, $p_c = M_cS_c\Omega_c^{1/2}Y$, with $p_c = M_cv$, gives

$$\|z_1-z_2\|_{P_c}^2 = \sum_j \omega_{c,j}\{(\Delta X_j)^2+(\Delta Y_j)^2\}. \tag{S305}$$

It is therefore the current canonical energy metric. If a particular reference modal radius $\widehat{\rho}_j$ has a certified positive lower bound and $\varepsilon/\sqrt{\omega_{c,j}} < \widehat{\rho}_j$, the circular phase distance is at most $\arcsin\{\varepsilon/(\sqrt{\omega_{c,j}}\widehat{\rho}_j)\}$. No numerical modal-radius lower bound or individual phase bound is received here, and the saved-state gap is not labelled a phase gap.

The theorem concerns the declared current fixed-matrix equation. It proves neither its arbitrary-state identity with COMSOL's compiled equations nor an identity with internal dense output or a physical device. The saved-knot separation therefore does not diagnose a solver error. Small signed residual work and the storage bound in Section S35 remain valid for the Hermite curve and do not imply a small state discrepancy. Conservative KAM, forced-IVP state approximation, learned-model trajectory accuracy and physical model correspondence retain their separate premises.

S37 Dissipative work for the current fixed-FE model

Here the candidate denotes the exact native realization of the saved weights and source constants; floating-point evaluation errors remain covered only by their separate arithmetic interfaces.

We use the fixed-FE declaration and the current initial state specified above. This section concerns its exact continuous mathematical model; it does not treat the saved solver arrays as an exact solution. The same trained candidate has mass $M_N = M_0 + M_r$ and damping $D_N(x) = \alpha_{\mathrm{nom}}a_N(x)M_0$; the current model has constant symmetric mass M_c, damping D_c, and potential U_c. Both equations use the declared common port $g = \nabla\psi$, $\psi = \mu\sin(b^Tx)$. The already received scalar structural family contains the current member and gives $|U_N - U_c| \le \varepsilon E$ on $\Omega_4 = \{E_0 = x^TK_0x/2 < 4E\}$, with $\varepsilon < 0.003839064485$.

Proposition S66. Signed work in the common velocity chart. *Set* $H_c = v^T M_c v/2 + U_c$, $H_N = v^T M_N v/2 + U_N$, $\Delta T = v^T(M_N - M_c)v/2$, *and* $\Delta U = U_N - U_c$. *Along a current-model trajectory, define*

$$r_N = M_N \dot{v} + \nabla U_N + D_N v - gu, \quad W(t) = \int_0^t v^T r_N \, ds, \quad Q_c(t) = \int_0^t v^T D_c v \, ds.$$

Then every pre-exit prefix satisfies

$$W(t) = \Delta T(t) - \Delta T(0) + \Delta U(t) - \Delta U(0) + \int_0^t v^T (D_N - D_c) v \, ds. \tag{S306}$$

If $M_N - M_c \preceq m_+ M_c$, $H_c/E \le H_*$, $D_N \preceq A_* D_c$, *and* $Q_c/E \le Q_*$, *then*

$$\frac{W(t)}{E} \le m_+ H_* - \frac{\Delta T(0)}{E} + 2\varepsilon + (A_* - 1)_+ Q_*. \tag{S307}$$

Proof. Subtract the current equation $M_c \dot{v} + \nabla U_c + D_c v = gu$ from the definition of r_N, multiply by v^T, and integrate. Symmetry and the chain rule give (S306). The mass comparison gives $\Delta T \le m_+ v^T M_c v/2 \le m_+ H_c$, since $U_c \ge 0$. The potential and damping bounds prove (S307). The two canonical momenta are $p_c = M_c v$ and $p_N = M_N v$. In the common p_c chart the candidate kinetic term is $p_c^T M_c^{-1} M_N M_c^{-1} p_c/2$; using $p_c^T M_N^{-1} p_c/2$ would define a different storage.

For $E_0/E \le h$, the global real componentwise derivative matrix L of the saved split-tanh damping head gives radii $r_h = (p_1\sqrt{h}, p_2\sqrt{h}, p_3\sqrt{h}, h)$ and $\delta_R = L_R r_h$, $\delta_I = L_I r_h$. Consequently

$$a_N \le a_*(h) := (|1 + c_C| + \delta_R)^2 + \delta_I^2, \qquad A_*(h) = \frac{\alpha_{\mathrm{nom}} a_*(h)}{\alpha_c - \eta_D},$$

where the received current matrices satisfy $(\alpha_c - \eta_D)M_0 \preceq D_c \preceq (\alpha_c + \eta_D)M_0$ and $\alpha_c > \eta_D$. These estimates use the actual saved weights and fixed matrices. They are additional structure, rather than consequences of the odd loss value alone.

For the original 0.6-second analytic cosine family, let $Z = H_c - u\psi + \mathrm{shift}\, E$ be the received nonautonomous storage. Applying its square-root BV argument to the separate auxiliary storage $Z_+ = Z + Q_c$, for which $dZ_+ = -\psi \, du$ and $Z_+ \ge Z \ge 0$, gives $Z_+/E \le Z_*$. Completing the square gives $Z/E \ge \mathrm{shift} - a^2$, and hence

$$\frac{Q_c}{E} \le Z_* - \mathrm{shift} + a^2.$$

Also $M_0 \preceq M_N \preceq M_c/(1-f)$ yields the independent bound $Q_c/E \le 2(\alpha_c + \eta_D)T H_*/(1-f)$. The smaller bound is used in (S307). Its resulting independent work allowance is approximately $10.13E$ and does not close the boundary test. This failed coarse estimate is retained. The signed energy ledger instead cancels the common integral of $v^T D_N v$ and gives

$$H_N = H_c + \Delta T + \Delta U \le (1 + m_+)H_* E + \varepsilon E < 4.236161085119E.$$

Thus the full-horizon conditional storage guard does not require the independent work allowance to be small.

Corollary S67. Nonpositive-power inputs from the current initial state. *For the same fixed-FE model and its current initial state, zero total field or exact continuous feedback* $u = -\mathrm{sat}_U(ky)$, $y = g^T v$, $k > 0$, $U \ge 0$, *admits an all-time, all-prefix residual-work allowance less than* $0.211330E$. *The corresponding guard satisfies*

$$H_N(0) + W_* < (3/8 + 0.211330)E < 4.236161085119E.$$

Proof. Here $uy \le 0$ and $H_c(t) + Q_c(t) \le H_c(0) \le C_c E$, where $C_c < 0.364016$. The fixed quadratic declaration implies globally $U_c \ge \lambda E_0$, $\lambda = 53/50 - \eta_K > 0$, so $h = C_c/\lambda < 0.343412 < 4$. The smaller feature domain gives $A_*(h) < 1.559456$. Use $H_* = Q_* = C_c$ in (S307) and evaluate its signed initial mass term from the actual current initial velocity. Exact rational arithmetic gives $W_* < 0.211330E$ and the already received initial candidate bound gives $H_N(0) < 3E/8$. Coercivity bounds position and velocity, and locally Lipschitz fixed-model dynamics give continuation for all positive times. This establishes the asserted all-prefix bound and strict guard.

The last corollary is an analytic input branch; it is not a feedback experiment and does not reinterpret the archived cosine records. For the original 132 scalar weak odd rows, the executed loss and its arithmetic/operator reception give only $\|O_{\mathrm{native}}\|_2 \le d_s\sqrt{132L_{\mathrm{rec}}} + \eta_O$, with $L_{\mathrm{rec}} \approx 9.473158199312406\,10^{-11}$ and $\eta_O < 1.585366525\,10^{-11}$ in weak-row units. The exact weak-loss kernel, the even and normal components, and node-to-continuous errors remain separate. No current raw-history observation certificate, unknown physical damping identification, sampled-controller accuracy, or complex network superiority follows from the present composition.

S38 A conditional sampled magnetic power guard

Let $x, v \in \mathbb{R}^{64}$ have the native transverse finite-element ordering, let $b = e_{64}$, and set

$$E_0(x) = \tfrac{1}{2}x^T K_0 x, \qquad U_\beta(x) = (1+\beta)E_0(x) + \kappa(1 - \cos b^T x). \tag{S308}$$

Here M, K_0 are the received positive definite native matrices and $\beta \in [\beta_-, \beta_+] \subset (-1, \infty)$ is constant. Consider the explicitly assumed physical family

$$M\dot v = -\nabla U_\beta(x) - a(x)D_0 v + \mu u b\cos(b^T x), \qquad \dot x = v, \tag{S309}$$

where $D_0 \succeq 0$, a is locally Lipschitz, and $0 \le a(x) \le \bar a$. The input u denotes the actual total transverse magnetic field, after quantization, with $|u| \le U$. The preparation cosine field is replaced, not superposed. Define

$$H_\beta = \tfrac{1}{2}v^T M v + U_\beta(x), \qquad y = \mu\cos(b^T x) b^T v. \tag{S310}$$

Then $\dot H_\beta = -a(x)v^T D_0 v + uy \le uy$. Discarding the nonnegative dissipation requires no bound on its derivative.

A full-state output-rate majorant. Fix an energy cap $c > 0$, and write $z = M^{-1}b$. On $H_\beta \le c$, set

$$V_c = \sqrt{2c\, b^T M^{-1} b}, \qquad \Theta_c = \sqrt{\frac{2c\, b^T K_0^{-1} b}{1+\beta_-}}, \qquad S_c = \min(1, \Theta_c), \tag{S311}$$

$$A_K = (1+\beta_+)\sqrt{\frac{2c\, z^T K_0 z}{1+\beta_-}}, \qquad A_D = \bar a\sqrt{2c\,(D_0 z)^T M^{-1}(D_0 z)}, \tag{S312}$$

$$A_B = (b^T M^{-1} b)(\kappa S_c + \mu U). \tag{S313}$$

For positive μ, κ, a valid majorant is

$$L_y = \mu\{S_c V_c^2 + A_K + A_D + A_B\}. \tag{S314}$$

Indeed, the energy cap bounds $|b^T v|$ and $|b^T x|$ by V_c and Θ_c. Cauchy–Schwarz in the K_0 and M metrics bounds the elastic and dissipative terms in $b^T\dot v$ by A_K and A_D, respectively. The remaining magnetic term is bounded by A_B. Differentiating $y = \mu\cos(b^T x)b^T v$ gives $|\dot y| \le L_y$. The assumption $\bar a < \infty$ is needed for this particular majorant; positive semidefinite dissipation alone does not provide it.

Sampled work and a strict energy ledger. At the actual hold start t_k, suppose a valid observation supplies $|y(t_k) - \widehat{y}_k| \le \varepsilon_k$. Apply a constant actual input u_k for a prescribed duration $h_k > 0$. For every prefix $0 \le s \le h_k$ remaining below the energy cap,

$$H_\beta(t_k + s) - H_\beta(t_k) \le F_k(s) := s(u_k\widehat{y}_k + |u_k|\varepsilon_k) + \tfrac{1}{2}s^2|u_k|L_y. \tag{S315}$$

Since F_k is convex and $F_k(0) = 0$, $F_k(s) \le \max\{0, F_k(h_k)\}$. Let $\Delta_k = F_k(h_k)$ and let $\lceil\cdot\rceil_g$ denote upward rounding to a fixed positive energy grid g. Assume a trusted initial ledger satisfies $H_\beta(t_0) \le L_0 < c$ and $L_0 \in g\mathbb{Z}_{\ge 0}$. Accept a nonzero proposal only when it satisfies the actual input limit, the current observation is valid, and

$$\Delta_k \le \alpha(c - L_k), \qquad L_{k+1} = \left\lceil \max\{0, L_k + \Delta_k\}\right\rceil_g < c, \qquad 0 < \alpha < 1. \tag{S316}$$

Otherwise apply exactly zero and retain the ledger. The accepted hold obeys $H_\beta(t_{k+1}) \le L_{k+1}$ and remains strictly below c throughout. To prove the latter without circularity, suppose there is a first time at which $H_\beta = c$. Before that time (S314) applies, and (S315) implies

$$H_\beta(t_k + s) \le L_k + \max\{0, \Delta_k\} < c,$$

contradicting the first exit. A rejected or zero proposal uses $\dot{H}_\beta \le 0$ directly. Induction proves the strict energy cap for all holds. A fixed $h_k = h > 0$, or another schedule without finite-time accumulation, excludes Zeno accumulation. The coercive energy then bounds the state for all finite times.

For an opposite-sign proposal, the stronger condition $|\widehat{y}_k| \ge \varepsilon_k + L_y h_k/2$ makes $\Delta_k \le 0$ and makes every prefix work bound nonpositive. This integral statement does not assert instantaneous $u_k y(t) \le 0$; the sufficient instantaneous condition instead uses $|\widehat{y}_k| \ge \varepsilon_k + L_y h_k$.

Connection to the learned potential guard. Suppose the current checkpoint satisfies $|U_N - U_\beta| \le \varepsilon_U$ on $\Omega_4 = \{E_0 < 4E_{\mathrm{net}}\}$. Each separately received numerical initial state can initialize the ledger by rounding an outward upper enclosure of $H_N(t_0) + \varepsilon_U$ upward to the grid, provided the rounded value is strictly below c. For the present $c = 41E_{\mathrm{net}}/20$ and received β_-, $c/(1+\beta_-) < 4E_{\mathrm{net}}$, so the physical energy cap also stays inside Ω_4. The theorem concerns the declared physical family, not a derivative match between the learned and physical potentials.

Observation and execution scope. A rigorous midpoint computation at a saved numerical state supplies an arithmetic observation enclosure. It does not certify a physical sensor. For a tip observation with actual hold-start errors $|\theta - \widehat{\theta}| \le r_\theta$ and $|\omega - \widehat{\omega}| \le r_\omega$, a sufficient additional port error allowance is

$$\mu\{r_\omega + |\widehat{\omega}| \min(2, r_\theta)\},$$

plus the outward error of computing $\mu\cos(\widehat{\theta})\widehat{\omega}$. Timing, sensing and total-field actuation must satisfy their declared contracts to apply the result to an experiment. This initial-proposal reception alone does not execute a controller. The separate numerical execution is treated in Section S38.3; neither reception claims a physical sampled controller run, physical parameter identification, convergence, or control performance.

S38.1 A received sampled-power instance for the calibrated guard

Use the same current checkpoint, $c = 41E_{\mathrm{net}}/20$, $\varepsilon_U = 479874807E_{\mathrm{net}}/125000000000$, the unchanged scalar compatibility interval, and the full 64-coordinate native matrices. For this instance assume additionally $D_*(x) = a(x)D_0$ with $0 \le a(x) \le 6/5$. This finite operator bound is an extra physical hypothesis; it is not inferred from positive semidefiniteness or from the candidate's feature-box dissipation bound.

Exact rational matrix operations and outward square roots give

$$|b^T x| < 0.001909972751, \qquad |b^T v| < 20.504956934$$

on $H_* \le c$. With the actual total-field limit $U = 10^{-4}$ T, (S314) has the strict bound

$$L_y < 49.953396682 \ \mathrm{J\,T^{-1}s^{-2}}.$$

Here M is the exact sum of the saved M_0 and M_r entries, and $D_0 = \alpha M_0$ with the exact nominal coefficient from the native operator record. The shifted trajectory damping coefficient and translational-plus-rotary mass are not substituted into D_0.

For each of the 22 separately received numerical initial states, compute an outward enclosure of $y = \mu\cos\theta_{32}\, v_{\theta,32}$ and take its rational midpoint $\widehat{y}_0$. Its arithmetic enclosure radius is strictly below the prescribed allowance $\varepsilon_y = 10^{-8}\,\mathrm{J\,T^{-1}s^{-1}}$. Set the energy grid to $g = 10^{-9}E_{\mathrm{net}}$, the reserve fraction to $\alpha = 1/2$, and initialize

$$L_0 = \left\lceil \overline{H}_N(x_0, v_0) + \varepsilon_U \right\rceil_g.$$

The strict common bounds are $0.241216386 < L_0/E_{\mathrm{net}} < 0.976086611 < c/E_{\mathrm{net}}$. For the nonzero opposite-sign proposal $u_0 = -U\,\mathrm{sign}(\widehat{y}_0)$, exact evaluation of (S316) gives

Hold duration	Accepted nonzero proposals	Upper next ledger/E_{net}
$1\,\mu$s	$22/22$	0.976499156
$10\,\mu$s	$22/22$	1.017677105
$100\,\mu$s	$0/22$	0.976086611

For the $10\,\mu$s holds the computed work majorants obey $0.041462830554 < \Delta_0/E_{\mathrm{net}} < 0.041617370734$. The rejected $100\,\mu$s proposals use the exact zero fallback and retain their initial ledger. These comparisons keep the same field limit, error allowance, cap and rounding grid throughout.

The arbitrary-horizon conclusion follows from the work-ledger theorem only if every subsequent hold also receives a valid current observation and applies the accepted actual total command with the stated timing. This initial-proposal calculation evaluates saved numerical states without executing a trajectory. A separate numerical feedback execution and its reference-IVP certificate appear in Section S38.3. The error allowance is not a physical sensor certificate. Acceptance establishes a conditional energy guard, not monotone decrease on accepted holds or improved control performance.

S38.2 Continuous reception of the three held-field segments

Fix the declared 64-coordinate mathematical equation

$$M\ddot{x} + D\dot{x} + Kx + b\{\kappa \sin(b^{\top}x) - \mu u(t)\cos(b^{\top}x)\} = 0, \quad M = M_0 + M_r, \quad K = \tfrac{53}{50}K_0, \quad D = \alpha_{\mathrm{shift}}M_0,$$

where the saved matrix entries and $\alpha_{\mathrm{shift}} = 5754906970707699/9007199254740992$ are interpreted as exact rationals. Here $b = e_{64}$, time is in seconds, u is in tesla, and $y = \mu\cos(b^{\top}x)b^{\top}\dot{x}$ has units $\mathrm{J}/(\mathrm{T\,s})$. The field u is the actual, constant binary64 total transverse field on each segment; it replaces the preparation cosine. This declared IVP starts at the fresh seed's own saved position and velocity. It asserts neither an exact physical initial state nor identity with a compiled native equation.

Let Q be the piecewise cubic Hermite curve through every retained native position and velocity endpoint. Each cell uses its own exact dyadic duration. All 101 predecessor handoffs, including the physical-time coordinate, are received bitwise. Thus Q and $\dot{Q}$ are continuous across both interior cells and field switches. The exact equation has continuous position and velocity at a field switch, although acceleration may jump. Define

$$e = x - Q, \quad r^2 = \dot{e}^{\top}M\dot{e} + e^{\top}Ke, \quad c_M = \sqrt{b^{\top}M^{-1}b}, \quad c_K = \sqrt{b^{\top}K^{-1}b}.$$

The initial radius is zero once, at the declared saved seed, and is not reset at a continuation boundary. For $|u| \le U = 10^{-4}\,\mathrm{T}$, the scalar magnetic force is globally Lipschitz in angle with constant $\kappa + \mu U$. If ρ bounds the Hermite residual in the M^{-1} norm, then

$$D^{+}r \le ar + \rho, \qquad a = \tfrac{1}{2}(\kappa + \mu U)c_M c_K.$$

Indeed, the damping term is nonpositive, the residual contributes at most $r\rho$, and $|\dot{e}^{\top}b|\,|b^{\top}e| \le c_M c_K r^2/2$. For a cell of duration h with initial radius r_ℓ,

$$\bar{r}(s) = e^{as}r_\ell + \rho\,\frac{e^{as} - 1}{a}, \qquad 0 \le s \le h$$

is a nondecreasing prefix majorant. An outward grid ceiling of $\bar{r}(h)$ supplies the next initial radius. The global force bound removes a tube-closure assumption.

On one cell put $\theta_Q = b^{\top}Q$ and $\Theta \ge \sup|\theta_Q|$. Form the degree-nine polynomial residual

$$R_{\mathrm{pol}} = M\ddot{Q} + D\dot{Q} + KQ + b\{\kappa(\theta_Q - \theta_Q^3/6) - \mu u(1 - \theta_Q^2/2)\}.$$

The omitted scalar force is bounded on the whole cell by

$$\epsilon_{\mathrm{nl}} = \kappa\Theta^5/120 + \mu|u|\Theta^4/24.$$

These real Taylor bounds do not require a small-angle assumption. If $B_0, \dots, B_9$ are the degree-nine Bernstein coefficient vectors of R_{pol}, convexity of the norm gives

$$\rho = \max_j \sqrt{B_j^\top M^{-1} B_j} + c_M \epsilon_{\mathrm{nl}}.$$

The exact rational matrices and inverse are received once, then enclosed in 512-bit Arb balls. All cell arithmetic, including the quadratic forms and stable $\mathrm{expm1}(ah)$ evaluation, uses outward intervals; the reported radius is rounded upward. No per-cell matrix inverse or sampled-residual substitution is used.

At a saved endpoint with tip velocity $v_{\theta,Q}$,

$$|y(x,\dot{x}) - y(Q,\dot{Q})| \le \mu\{c_M r + |v_{\theta,Q}| \min(2, c_K r)\}.$$

For a whole-cell comparison, replace r by its prefix endpoint majorant and $|v_{\theta,Q}|$ by the Bernstein bound on the Hermite velocity. For the controller's actual logged exact rational value $\widehat{y}$, the separate outward readout allowance $|\,\mu\cos(\theta_Q)v_{\theta,Q} - \widehat{y}\,|$ must be added. In particular an exact midpoint of an Arb enclosure need not be converted to binary64. Only a receipt binding the actual logged value to the actual hold-start observation can receive the fixed prospective $\varepsilon_y = 10^{-8}\,\mathrm{J/(T\,s)}$ contract. A saved finite state alone is insufficient. Neither a successful IVP tube nor that sensor inequality alone receives the other causal ledger, cap, timing, actuation, or initial guard hypotheses of the sampled safety theorem.

S38.3 Continuous reception of the recorded calibrated feedback

Proposition S68. Recorded three-hold energy ledger. *Consider the 64-coordinate mathematical initial-value problem*

$$M\ddot{x} + D\dot{x} + Kx + b\{\kappa \sin\theta - \mu u(t)\cos\theta\} = 0, \qquad \theta = b^\top x, \quad b = e_{64},$$

with the exact saved matrices $M = M_0 + M_r$, $K = (53/50)K_0$, *and* $D = \alpha_{\mathrm{shift}} M_0$, *where* $\alpha_{\mathrm{shift}} = 5754906970707699/9007199254740992$. *Its initial position and velocity are the actual saved numerical seed state at* $t_0 \simeq 20\,\mu\mathrm{s}$. *Define*

$$H_\star = \tfrac{1}{2}\dot{x}^\top M\dot{x} + \tfrac{1}{2}x^\top Kx + \kappa(1-\cos\theta), \qquad y = \mu\cos\theta\, b^\top\dot{x}.$$

The three recorded holds have their own exact saved binary64 endpoints $t_0 < t_1 < t_2 < t_3 \simeq 50\,\mu\mathrm{s}$. *On these holds the actual total input is, respectively,* $-u_q, +u_q, 0$, *with*

$$u_q = \frac{1844674407370955}{18446744073709551616}\,\mathrm{T} < 10^{-4}\,\mathrm{T}.$$

The held field replaces the preparation cosine. For this declared IVP, the received bounds imply

$$H_\star(t_j) \le L_j, \qquad H_\star(t) \le 0.910436431\,E_{\mathrm{net}} < c = \tfrac{41}{20}E_{\mathrm{net}} \quad (t_0 \le t \le t_3),$$

where

$$(L_0, L_1, L_2, L_3)/E_{\mathrm{net}} = (0.836050803,\ 0.870553256,\ 0.910436431,\ 0.910436431).$$

This is a statement about the recorded three-hold declared IVP. It does not assert identity with the compiled native equation, a physical initial-state or sensor certificate, native dense-output accuracy, or any future rollout.

Proof. The actual native execution contains one fresh preparation solve and three continuation solves. Each continuation retains all 101 saved solution and time-derivative entries bitwise from its predecessor. The mathematical IVP starts only at the saved seed; the preceding preparation trajectory is not certified by this proposition. The three holds contain 514 retained raw states each, including an extremely short terminal cell of approximately $4.066\,10^{-19}\,\mathrm{s}$. The original all-cell reception is preserved and does not receive the fixed sensor budget.

The separate selected-partition protocol predeclares strides two, four and eight. For each hold, the stride-eight curve retains the original raw indices $0, 8, \dots, 504, 513$, with zero-based indexing. It has 64 Hermite cells per hold and keeps every actual

controller endpoint and its original position and velocity. The final short cell is merged into a wider cell. No source state, input, controller time or error budget is changed. For this new piecewise C^1 curve Q, every selected cell is certified on its entire interval using the residual of the declared 64-coordinate equation. The reconstruction need not interpolate omitted raw samples; no enclosure of those samples or native dense output is inferred.

The comparison radius satisfies

$$r^2 = (\dot{x} - \dot{Q})^\top M(\dot{x} - \dot{Q}) + (x - Q)^\top K(x - Q), \qquad D^+ r \le ar + \rho,$$

where $a = (\kappa + \mu 10^{-4})\sqrt{(b^\top M^{-1} b)(b^\top K^{-1} b)}/2$. It is initialized once at $r(t_0) = 0$ and propagated across all three holds. The polynomial residual uses cubic sine and quadratic cosine terms. Its whole-cell remainder is bounded by $\kappa\Theta^5/120 + \mu|u|\Theta^4/24$. Degree-nine Bernstein coefficient vectors, the M^{-1} norm and outward 512-bit Arb arithmetic give the cell residual bound. Thus neither residual samples nor resetting the radius at field switches is used.

At a hold start, with $c_M = \sqrt{b^\top M^{-1} b}$ and $c_K = \sqrt{b^\top K^{-1} b}$, the received port allowance is

$$\mu\{c_M r + |v_{\theta,Q}| \min(2, c_K r)\} + \left|\mu \cos(\theta_Q) v_{\theta,Q} - \widehat{y}\right|.$$

The second term independently receives the exact rational midpoint $\widehat{y}$ actually used by the bridge; it is not converted to binary64. The resulting upward bounds at the three hold starts are

$$\delta_{y,0} \le 10^{-50}, \qquad \delta_{y,1} < 6.873047\,10^{-9}, \qquad \delta_{y,2} < 1.639569\,10^{-8} \quad \mathrm{J/(T\,s)}.$$

Only the first two holds have nonzero input. Their allowances fit the unchanged prospective $\varepsilon_y = 10^{-8}$. The computed third bound exceeds that budget, and the all-three-sensor contract remains unreceived.

Direct exact-model energy evaluation gives $H_\star(t_0) < 0.832211815\,E_{\mathrm{net}} \le L_0 < c$. An independent energy-dual calculation on $H_\star \le c$ gives an output-rate bound below $49.953391663\,\mathrm{J/(T\,s^2)}$; it fits the already frozen uniform bound $L_y < 49.953396682\,\mathrm{J/(T\,s^2)}$. For each nonzero hold, its actual exact duration $h_j = t_{j+1} - t_j$ is used in

$$\Delta_j = h_j\{u_j \widehat{y}_j + |u_j|\varepsilon_y\} + \tfrac{1}{2} h_j^2 |u_j| L_y.$$

Independent rational arithmetic receives the reserve gate, upward $10^{-9}E_{\mathrm{net}}$ grid ceiling, strict cap gate and matching single transactional commit. The quadratic prefix majorant and first-exit argument then give $H_\star(t) < c$ throughout the first two holds and $H_\star(t_2) \le L_2$. These holds spend a positive energy reserve; they are not certified to have nonincreasing energy.

On the final recorded hold the actual total field is identically zero. Since $D = \alpha_{\mathrm{shift}} M_0 \succeq 0$,

$$\dot{H}_\star = -\dot{x}^\top D \dot{x} \le 0, \qquad H_\star(t) \le L_2 = L_3 < c \quad (t_2 \le t \le t_3).$$

This branch requires neither a port observation nor L_y. It completes the recorded-interval ledger statement without receiving the third sensor contract.

Comparison of reconstructions. The original all-cell reconstruction and the predeclared stride-two and stride-four reconstructions did not receive the required port budget at the second nonzero hold. Their computed upper bounds exceed 10^{-8}; upward rounded displays are $4.972088\,10^{-4}$, $2.132616\,10^{-8}$ and $1.094943\,10^{-8}$, respectively. These are upper bounds, not lower bounds on the actual error. The stride-eight reconstruction receives both required nonzero-hold contracts; its third, unused sensor contract remains unreceived. All original negative outcomes, raw data and actual commands are retained.

S39 Signed work for the received feedback history

S39.1 Signed work on a received constant-field history

Fix the previously declared, saved-seed, full64 reference IVP

$$M\ddot{x} + D\dot{x} + Kx + b\{\kappa \sin\theta - \mu u(t)\cos\theta\} = 0, \quad \theta = b^\top x, \quad b = e_{64},$$

where $M = M_0 + M_r \succ 0$, $K = (53/50)K_0 \succ 0$, $D = \alpha_{\mathrm{shift}} M_0 \succeq 0$, and $\mu > 0$. The matrices, scalar parameters, initial data and executed held total fields have the exact-source conventions of the received reference IVP. There are no impulses at a change of field. The physical-storage function used in this reference equation is

$$H_\star(x, v) = \tfrac{1}{2} v^\top M v + \tfrac{1}{2} x^\top K x + \kappa(1 - \cos\theta).$$

It does not contain u, so it is continuous at a field switch. Multiplying the equation by $v^\top$ gives

$$\dot{H}_\star = -v^\top D v + \mu u \cos\theta\, \dot{\theta}.$$

Consequently, on one constant-field hold $[t_k, t_{k+1}]$,

$$H_\star(t) - H_\star(t_k) = -\int_{t_k}^{t} v^\top D v \, ds + \mu u_k \{\sin\theta(t) - \sin\theta(t_k)\}. \tag{S317}$$

Dropping the nonnegative dissipation yields an upper bound, without requiring a numerical velocity quadrature or a bound on a sensor readout.

Suppose the already received continuous comparison with the piecewise cubic Hermite curve Q supplies on every retained cell I_i a bound

$$(\dot{x} - \dot{Q})^\top M (\dot{x} - \dot{Q}) + (x - Q)^\top K (x - Q) \le r_i^2.$$

The propagated radius is initialized to zero only at the saved seed and is never reset at a cell boundary or field switch. Write $c_K = \sqrt{b^\top K^{-1} b}$ and $c_M = \sqrt{b^\top M^{-1} b}$. If $[q_i^-, q_i^+]$ and $[v_i^-, v_i^+]$ enclose the ranges of $b^\top Q$ and $b^\top \dot{Q}$ on I_i, respectively, then

$$\theta(I_i) \subseteq [q_i^- - c_K r_i, q_i^+ + c_K r_i],$$
$$\dot{\theta}(I_i) \subseteq [v_i^- - c_M r_i, v_i^+ + c_M r_i].$$

This follows from Cauchy–Schwarz in the K and M inner products. Bernstein coefficients of the exact cubic and its quadratic derivative provide the required whole-cell ranges, rather than sampled extrema. At a retained endpoint use its saved angle and the corresponding propagated endpoint radius instead of the whole-cell range.

For a real interval A, let $S(A)$ be any outward interval enclosing $\sin A$. For intervals A, B define

$$\mathcal{W}_u(A, B) = \mu u\{S(A) - S(B)\}, \qquad W_u^+(A, B) = \sup \mathcal{W}_u(A, B).$$

The subtraction is interval subtraction and is allowed to overestimate correlation between its two arguments. In particular, the same uncertain state is not treated as two independent physical states. If A_i encloses $\theta(I_i)$, A_k encloses the hold-start angle, and $E_k \ge H_\star(t_k)$, then every $t \in I_i$ satisfies

$$H_\star(t) \le E_k + W_{u_k}^+(A_i, A_k). \tag{S318}$$

For the hold endpoint interval A_{k+1},

$$E_{k+1} = E_k + W_{u_k}^+(A_{k+1}, A_k) \quad \Longrightarrow \quad E_{k+1} \ge H_\star(t_{k+1}).$$

The same estimates may be based at the left endpoint of an individual cell. The minimum of several valid upper bounds is again a valid upper bound. This permits interval dependence to be reduced by comparing both the hold anchor and the previous cell endpoint.

There is a separate sign test. If $\cos\theta \geq 0$ and $u_k\dot{\theta} \leq 0$ throughout a cell, then $\dot{H}_\star \leq 0$ throughout that cell, so its energy is nonincreasing. For $u_k = 0$ this conclusion needs no angle or velocity sign test. A proved sign test permits replacement of a cell's prefix and endpoint bounds by the already proved left-endpoint energy bound. A failed sign test is retained as a failure of that sufficient condition; it is not a proof that the energy increases. If this sign test fails anywhere on a hold, a bound $H_\star(t) \leq H_\star(t_k)$ throughout that hold must not be described as pairwise monotonicity on the hold.

All bounds used below can be rational. On $[-1, 1]$, $\cos z \geq 1 - z^2/2 > 0$, so sine is increasing. Taylor's theorem gives

$$P_7(z) = z - z^3/6 + z^5/120 - z^7/5040, \qquad |\sin z - P_7(z)| \leq |z|^9/9!.$$

For $[a, b] \subset [-1, 1]$ use

$$S([a, b]) = [P_7(a) - |a|^9/9!,\ P_7(b) + |b|^9/9!].$$

An exact rational ceiling for c_K, c_M and exact rational Bernstein coefficients therefore suffice for signed magnetic-work bounds. For comparison, the weaker absolute-work estimate is

$$H_\star(t) \leq H_\star(t_0) + \sum_{I_i \text{ before } t} \mu|u_i|h_i\{\sup_{I_i}|b^\top\dot{Q}| + c_M r_i\},$$

where the final partial cell may be enlarged to its entire duration.

Taking the maximum of the received prefix bounds and the initial bound gives a uniform energy cap on precisely the recorded finite time interval. This is an offline theorem about the same declared continuous reference IVP and the same executed fields. It changes neither the controller nor its original prospective guard, ledger, sensor budget or negative sensor receptions. It does not enclose a compiled native equation, COMSOL dense output, omitted native samples, a physical plant, an actuator or a sensor, and implies no future-time guarantee, tracking performance or complex-network advantage.

Offline signed-work reception of the executed reference history. For the same saved-seed reference IVP, the same three executed total fields, and the independently received stride-eight comparison, exact rational reconstruction of all 192 retained tip-angle cubics gives

$$H_\star(t) < 0.832211815\,E_{\mathrm{net}}, \qquad t_0 \leq t \leq t_3, \qquad (t_0, t_3) \simeq (20, 50)\,\mu\mathrm{s}.$$

The exact endpoints are the saved binary64 clocks $t_0 = 5902958103587057/29514790517935282 5856\,\mathrm{s}$ and $t_3 = 7378697629483821/147573952589676412928\,\mathrm{s}$. The original prospective ledger bound $0.910436431\,E_{\mathrm{net}}$ remains valid and unchanged; the smaller offline bound uses the signed identity $\mu u_k[\sin\theta(t) - \sin\theta(t_k)]$ on each constant-field hold. No new native solve, training update, ODE integration, or inferred damping work enters this calculation. The nonnegative dissipation is discarded.

The received upper bounds at the three hold endpoints are, with outward decimal ceilings,

$$H_\star(t_1) < 0.829688104\,E_{\mathrm{net}}, \qquad H_\star(t_2), H_\star(t_3) < 0.827405859\,E_{\mathrm{net}}.$$

On every hold, the received energy is bounded by that hold's actual starting energy. A stronger pairwise monotonicity conclusion is received throughout the second hold, where the continuous angle tube has positive cosine and the tip-velocity tube has nonpositive sign, and throughout the third hold, whose total field is exactly zero. Fifteen cells of the first hold do not satisfy the nonpositive-port sufficient condition; therefore no monotonicity claim is made for the entire three-hold history.

The calculation receives existing continuous reference-IVP radii and rebuilds the exact tip Hermite ranges and signed work, rather than rederiving all full64 residuals. It preserves the original failed third sensor-budget reception. Its scope remains this particular reference IVP and recorded time interval; it supplies no compiled-equation, native dense-output, physical sensing or actuation, future-rollout, tracking, or complex-network superiority certificate.

Table S2. Post hoc nominal-mode decomposition for the fixed original 918-parameter checkpoint and one positive coarse-step case. All shares refer to squared mechanical-energy coordinates; the peak share uses the maximum of the unchanged full-state metric.

$\omega_j h_c$	Count	Initial energy	Peak error	Integrated error
$[0, 0.1]$	16	0.89088316	0.11128504	0.14585627
$(0.1, 1]$	33	0.08326342	0.62204564	0.65501561
$(1, 2]$	15	0.02585342	0.26666932	0.19912812
$(2, \infty)$	0	0	0	0

S40 Long-time numerical prediction diagnostics

S40.1 Post hoc decomposition in all nominal mechanical modes

We examine the fixed original 918-parameter checkpoint on the single positive-initial-velocity coarse-step case over $0 \le t \le 0.6\,\mathrm{s}$. This analysis describes the already observed prediction error. It neither selects a model nor replaces the full-state evaluation by a selected mode. Let L_K and L_M denote the saved Cholesky factors, and define the rounded nominal matrices $K_0 = L_K L_K^{\mathsf{T}}$ and $M = L_M L_M^{\mathsf{T}}$. The mass-normalized nominal modes satisfy

$$K_0 S = M S \Omega^2, \qquad S^{\mathsf{T}} M S = I_{64}, \qquad \Omega = \mathrm{diag}(\omega_1, \ldots, \omega_{64}). \tag{S319}$$

These are the modes of the saved nominal quadratic model, rather than instantaneous nonlinear modes or an exact eigensystem of the physical device.

For the saved positive coordinate scale s, the full-state observation is $z^{\mathrm{obs}} = (L_K^{\mathsf{T}} x/s, L_M^{\mathsf{T}} \dot{x}/s)$. Write the prediction error as $\delta z = (\delta z_q, \delta z_p)$ and retain the same initial-state denominator as the primary metric, $D = \|z^{\mathrm{obs}}(0)\|_2^2$. The nominal modal transforms and normalized squared errors are

$$T_q = \Omega^{-1} S^{\mathsf{T}} L_K, \qquad T_p = S^{\mathsf{T}} L_M, \tag{S320}$$

$$e_j^2(t) = \frac{(T_q \delta z_q(t))_j^2 + (T_p \delta z_p(t))_j^2}{D}, \qquad E^2(t) = \frac{\|\delta z(t)\|_2^2}{D}. \tag{S321}$$

Both transforms are orthogonal in exact arithmetic, hence $\sum_{j=1}^{64} e_j^2(t) = E^2(t)$. Equivalently, the position modal coordinates are $\Omega S^{\mathsf{T}} M x/s$ and the velocity modal coordinates are $S^{\mathsf{T}} M \dot{x}/s$. Their squared magnitudes partition the nominal quadratic mechanical energy. The independently recomputed full-error curve agreed with the saved primary curve within 3.4×10^{-16}; the maximum absolute discrepancy in the pointwise squared-error decomposition was 2.8×10^{-14}. These are floating-point consistency checks.

Table S2 reports all 64 modes, grouped after observation by $\omega_j h_c$, where $h_c = 12.5\,\mu\mathrm{s}$ is the native coarse-step setting. For a band B, the initial energy share is $\sum_{j \in B}[(T_q z_q^{\mathrm{obs}}(0))_j^2 + (T_p z_p^{\mathrm{obs}}(0))_j^2]/D$. The peak-error share is $\sum_{j \in B} e_j^2(t_*)/E^2(t_*)$, evaluated at the common full-state maximum $t_* = 0.3131125\,\mathrm{s}$. It does not sum the separate maxima of individual modes. The integrated share is $\sum_{j \in B} \int_0^{0.6} e_j^2(t)\,dt / \int_0^{0.6} E^2(t)\,dt$, evaluated by the composite trapezoidal rule on all 96,001 common output times, with spacing $6.25\,\mu\mathrm{s}$.

The modes with $\omega_j h_c > 0.1$ carry about 10.91% of the initial nominal mechanical energy but 88.87% of the squared error at the full-state maximum and 85.41% of its time integral. This is a descriptive association with frequency. The band boundaries are not certified time-integration error thresholds, and this decomposition does not identify COMSOL step-size error as the cause of the mismatch. The positive coarse/fine comparison is reported separately in Section S40.2; it does not certify a converged reference. The full-state maximum remains approximately 0.654780; a much smaller first-mode error cannot replace this primary result. A single fixed checkpoint and one coarse-step case also provide no paired-case conclusion or evidence of a general advantage of complex-valued networks.

S40.2 Fine-step observation scores from retained numerical predictions

The positive coarse- and fine-step COMSOL records have bitwise-identical output times, prescribed magnetic inputs, and observed initial positions and velocities in all 64 free coordinates. Inspection of the frozen prediction code establishes that the predictor right-hand side depends only on time, its current state, the prescribed input, and the saved model parameters.

Table S3. All five positive fine-step maximum-error contrasts. Entries are multiplied by 10^{-5}; each row retains all three fixed training seeds. SD uses divisor two and describes training-seed variability on this one record. The original 918-parameter checkpoint is outside these contrasts.

Contrast	Seed 43	Seed 44	Seed 45	Mean	Sample SD
R−GR	11.79209	3.54427	20.44787	11.92808	8.45262
C−GC	10.31492	11.67295	3.65521	8.54769	4.29108
R−C	-8.54026	-8.92145	16.76004	-0.23389	14.71841
GR−GC	-10.01743	-0.79277	-0.03262	-3.61427	5.55831
(GR−GC)−(R−C)	-1.47717	8.12869	-16.79266	-3.38038	12.56921

Subsequent observation rows enter the error evaluation rather than the predictor dynamics. The archived predictions retain their saved numerical initial vectors. Thus the 26 existing IVP integrations, comprising 13 frozen predictors under both numerical settings, were reused to score the fine-step observations. No new integration, parameter fitting, or checkpoint selection was performed. Recomputing the original coarse error curves reproduced the saved curves within 3.4×10^{-16}. The direct observation-coordinate reconstruction and the archived momentum-first initialization differ by at most 1.84×10^{-15} per component in the scaled initial chart. The two formulas coincide in exact arithmetic for an exact factorization $M = L_M L_M^T$; this is not a bitwise floating-point claim. Reuse here scores the fixed archived numerical trajectories; it does not supply a rigorous roundoff-to-exact-IVP or physical initial-state transfer certificate.

All fine-step scores retain the full 128-dimensional energy coordinates and the actual initial-state denominator. Over the 96,001 common output times, the primary maximum errors of all 13 predictors range from 0.58083122 to 0.58113085. The original 918-parameter checkpoint has maximum, RMS, and endpoint errors of 0.58092591, 0.50106948, and 0.48303912, respectively. Here RMS is the square root of the unweighted mean of the squared normalized errors over all output times. The maximum primary/reference predictor discrepancies range from 4.2470×10^{-5} to 4.2541×10^{-5}, and all 13 fine-step numerical-sensitivity flags are false under the original criterion. These discrepancies remain numerical diagnostics rather than certified integration-error bounds or a certificate of resolved model rankings. The original checkpoint's fixed first-mode maximum error is 5.6290×10^{-6} with the same full-state initial denominator; the full-state result remains the primary metric.

The five fixed comparisons retain all three training seeds, with the original checkpoint reported separately. For the primary maximum error, the $R - C$ differences change sign across seeds: their mean is -2.3389×10^{-6} and sample standard deviation is 1.4718×10^{-4}. The fine-step record therefore supplies no general advantage of complex-valued predictors. These are scores against a numerically generated fine-step record. The coarse/fine native discrepancy itself reaches 0.64373173 under the common initial-state normalization; neither record is established as converged physical truth by this two-setting comparison. The negative-initial-velocity cases remain outside this result, and the fine-step tip and weak-momentum diagnostics have not been recomputed.

Native numerical step sensitivity. The fine record uses an internal step $6.25\,\mu$s and the same 0.6 s horizon, $96\,001$ output times and source model as the coarse record. The initial velocity discrepancy from the prescribed field is 9.012×10^{-9} in the native component maximum, inside the original guard; that guard is not a rigorous solver-error certificate. Under the common full-state initial normalization, the coarse/fine native difference has maximum 0.643731731, RMS 0.544956606 and endpoint 0.539518613. The maximum occurs at $t = 0.3400375$ s. The same nominal-mode partition assigns about 92.13% of the squared peak discrepancy and 90.03% of its time integral to modes with $\omega_j h_c > 0.1$. Both native records therefore require further resolution evidence before either serves as a converged reference for long-time accuracy. This comparison does not establish a convergence order, a physical error bound, or a causal partition of model and native-solver errors.

S40.3 Common explicit-reference scores and native phase replay

The continuous comparison uses the physical function

$$\widehat{x}(t) = \mathrm{Re}\left\{ \sum_{j=1}^{128} X_j e^{\lambda_j t} + X_p e^{i\omega t} \right\}, \tag{S322}$$

$$\widehat{v}(t) = \mathrm{Re}\left\{ \sum_{j=1}^{128} V_j e^{\lambda_j t} + V_p e^{i\omega t} \right\}. \tag{S323}$$

		Primary predictor setting			Reference predictor setting				
Model	Seed	Max	RMS	End	Max	RMS	End	$\Delta_{\max}$	Flag
Original 918	43	0.404262	0.241485	0.362148	0.626699	0.299439	0.578607	0.424965	Yes
R	43	3.229611	1.680458	3.228633	3.129018	1.613348	3.128217	0.425014	Yes
C	43	13.865406	7.016893	13.863760	13.833395	6.995524	13.831998	0.425199	No
GR	43	13.055120	6.549756	13.055120	13.018918	6.525994	13.018918	0.424700	No
GC	43	3.514403	2.129596	3.505876	3.171466	1.935207	3.162771	0.424981	Yes
R	44	3.740712	1.963873	3.740432	3.705764	1.940107	3.705584	0.425025	Yes
C	44	15.493636	7.822983	15.493516	15.490206	7.820223	15.490206	0.425208	No
GR	44	4.040732	2.289744	4.035185	3.734727	2.109618	3.729397	0.424921	Yes
GC	44	0.967077	0.550724	0.965894	0.607852	0.342907	0.606656	0.424961	Yes
R	45	26.207681	12.990085	26.207371	26.200723	12.984536	26.200723	0.425401	No
C	45	5.559406	2.874230	5.558599	5.505569	2.838342	5.504516	0.425064	Yes
GR	45	2.214167	1.317406	2.209361	1.866691	1.119072	1.861651	0.424964	Yes
GC	45	3.441853	2.044959	3.433812	3.100270	1.850031	3.092051	0.424956	Yes

Table S4. All 13 frozen numerical predictors scored against the same explicit continuous function at all $96\,001$ saved times. Every numerical entry except the seed is multiplied by 10^{-4}. $\Delta_{\max}$ is the maximum primary/reference predictor discrepancy with the unchanged full-state initial denominator. The flag applies the original 10^{-4} or 5% rule to the primary explicit-reference maximum. The original 918-parameter checkpoint is outside the five contrasts. These are sampled numerical scores; source-IVP enclosure and evaluation-roundoff reception are separate.

Every binary64 pole and physical amplitude is treated as a fixed coefficient of this function. The homogeneous coefficients are already absorbed into X_j, V_j; no new multiplication of modal bases by initial coefficients defines the comparison target. All 128 poles and all 64 physical positions and velocities are retained. The harmonic drive is based on the original prescribed input; the stored binary frequency and the exact decimal source frequency are distinguished by the frequency-transfer allowance in Section S6. The construction uses the observed initial state, $M = M_0 + M_r$, $D = \alpha M_0$ and the source linearized stiffness $(53/50)K_0 + \kappa bb^T$. In particular, D is not replaced by αM. The residual reception and nonlinear source-IVP enclosure are stated in Section S6.

Scoring conventions and scope. The 26 existing numerical predictions comprise all 13 frozen models under the primary and reference settings of the main text. No weights, initial conditions or solver settings are changed. At each of the $96\,001$ archived output times, the explicit physical function is mapped with the original saved factors L_K, L_M and scale s:

$$\widehat{z}(t) = s^{-1} \begin{pmatrix} L_K^T \widehat{x}(t) \\ L_M^T \widehat{v}(t) \end{pmatrix}, \qquad e_a(t_i) = \frac{\|z_a(t_i) - \widehat{z}(t_i)\|_2}{\|z_a(0)\|_2}. \tag{S324}$$

Here a is the primary or reference predictor setting. The norm and denominator agree with the archived full-state mechanical score; the denominator is not rescaled by mode, time or reference amplitude. All models retain identical saved clocks, factors, scale and actual initial vectors. The small difference between the physical reconstruction and archived initial chart is retained rather than removed by alignment. The exponential evaluation uses column phasors in blocks of 512 times. RMS is the square root of the unweighted mean of $e_a(t_i)^2$, and the endpoint is $e_a(0.6\,\mathrm{s})$.

Table S4 gives maximum, RMS and endpoint scores for both settings of every model. The predictor discrepancy is $\max_i \|z_{\mathrm{primary}}(t_i) - z_{\mathrm{reference}}(t_i)\|_2 / \|z(0)\|_2$. The unchanged flag is true when this discrepancy exceeds 10^{-4} or $0.05 \max_i e_{\mathrm{primary}}(t_i)$. Nine models trigger it for the explicit-function target. The false flags obtained against the archived coarse and fine native records apply only to those earlier, much larger target discrepancies. Neither the two-setting check nor the sampled score proves a rigorous integration-error bound for the learned IVP. Section S41 receives sampling and scoring arithmetic in the exact saved-factor metric, then combines the source-IVP tube by the triangle inequality within its stated scope. No between-output learned-trajectory or physical-device claim follows.

Complete comparisons. Tables S5, S6 and S7 retain all five fixed comparisons for both predictor settings and all three training seeds. Mean and sample SD use these three seed values, with divisor two for the sample variance; the output times are not independent replications. The original 918-parameter checkpoint is reported separately and is excluded from these contrasts. C$-$GC is positive in every seed for the primary maximum score, but R$-$C and GR$-$GC change sign. There is no seed-uniform ordering of real and complex predictors in these data.

Native records against the identical explicit function. The raw coarse and fine MAT files are independently streamed and decompressed, and the SHA-256 of every decompressed field is checked against the observation caches. The physical

Contrast	Seed 43	Seed 44	Seed 45	Mean	Sample SD
Primary predictor setting					
R−GR	-9.825510	-0.300020	23.993514	4.622662	17.438641
C−GC	10.351004	14.526559	2.117554	8.998372	6.314116
R−C	-10.635796	-11.752925	20.648275	-0.580149	18.392837
GR−GC	9.540717	3.073655	-1.227686	3.795562	5.420378
(GR−GC)−(R−C)	20.176513	14.826579	-21.875961	4.375711	22.891442
Reference predictor setting					
R−GR	-9.889900	-0.028963	24.334032	4.805057	17.616617
C−GC	10.661929	14.882354	2.405299	9.316527	6.346401
R−C	-10.704377	-11.784442	20.695155	-0.597888	18.448222
GR−GC	9.847452	3.126875	-1.233579	3.913583	5.582248
(GR−GC)−(R−C)	20.551828	14.911317	-21.928733	4.511471	23.070915

Table S5. All five maximum-score contrasts against the common explicit function. Entries are multiplied by 10^{-4}. Both predictor settings and all three fixed training seeds are retained. A positive A−B value means B has the smaller score. Means and sample SD use the three seeds, with variance divisor two. No statistical significance or physical variability is inferred.

Contrast	Seed 43	Seed 44	Seed 45	Mean	Sample SD
Primary predictor setting					
R−GR	-4.869297	-0.325871	11.672679	2.159171	8.546392
C−GC	4.887297	7.272259	0.829271	4.329609	3.257497
R−C	-5.336435	-5.859110	10.115855	-0.359897	9.076030
GR−GC	4.420160	1.739020	-0.727553	1.810542	2.574602
(GR−GC)−(R−C)	9.756595	7.598129	-10.843408	2.170439	11.321877
Reference predictor setting					
R−GR	-4.912646	-0.169511	11.865464	2.261102	8.649113
C−GC	5.060317	7.477316	0.988311	4.508648	3.279490
R−C	-5.382176	-5.880116	10.146195	-0.372033	9.112454
GR−GC	4.590787	1.766711	-0.730959	1.875513	2.662541
(GR−GC)−(R−C)	9.972964	7.646827	-10.877154	2.247546	11.425674

Table S6. All five RMS-score contrasts against the common explicit function. Entries are multiplied by 10^{-4}. Both predictor settings and all three fixed training seeds are retained. A positive A−B value means B has the smaller score. Means and sample SD use the three seeds, with variance divisor two. No statistical significance or physical variability is inferred.

Contrast	Seed 43	Seed 44	Seed 45	Mean	Sample SD
Primary predictor setting					
R−GR	-9.826488	-0.294754	23.998010	4.625590	17.440800
C−GC	10.357884	14.527622	2.124787	9.003431	6.311378
R−C	-10.635127	-11.753084	20.648772	-0.579813	18.392990
GR−GC	9.549244	3.069292	-1.224452	3.798028	5.423691
(GR−GC)−(R−C)	20.184371	14.822376	-21.873223	4.377841	22.891626
Reference predictor setting					
R−GR	-9.890700	-0.023813	24.339073	4.808186	17.619038
C−GC	10.669227	14.883550	2.412465	9.321747	6.343798
R−C	-10.703781	-11.784623	20.696208	-0.597399	18.448721
GR−GC	9.856147	3.122741	-1.230401	3.916162	5.585698
(GR−GC)−(R−C)	20.559928	14.907363	-21.926608	4.513561	23.071623

Table S7. All five endpoint-score contrasts against the common explicit function. Entries are multiplied by 10^{-4}. Both predictor settings and all three fixed training seeds are retained. A positive A−B value means B has the smaller score. Means and sample SD use the three seeds, with variance divisor two. No statistical significance or physical variability is inferred.

observations are then encoded directly by L_K, L_M, s, without a modal projection, and compared with the identical absorbed-amplitude function in (S323). All clocks, input values and 64-coordinate observed initial positions and velocities are bitwise identical across the native pair. The normalization uses the same archived primary initial vector as the model scores. Over all $96\,001$ times, the coarse native maximum, RMS and endpoint discrepancies from the function are 0.654781417, 0.557657813 and 0.535618167; the fine native values are 0.580928598, 0.501070317 and 0.483040258. The directly reconstructed native pair has maximum 0.643731730, RMS 0.544956606 and endpoint 0.539518613. These are sampled discrepancies from a common specified function; the independently received source-IVP tube and evaluation-roundoff bounds, rather than the coarse/fine agreement alone, determine any source-trajectory interpretation.

Cayley phase replay on the common native-step grid. The recorded generalized-α settings use $\rho_\infty = 1$ and internal steps $h_c = 12.5\,\mu\mathrm{s}$ and $h_f = 6.25\,\mu\mathrm{s}$. The corresponding linear average-acceleration signature maps a source pole to

$$R_h(\lambda) = \frac{1 + h\lambda/2}{1 - h\lambda/2}. \tag{S325}$$

The known harmonic particular response is retained using the discrete frequency $2\tan(\omega h/2)/h$. This is a numerical replay of the recorded configuration; the native solver implementation is not certified by assigning it this map. Only the $48\,001$ common native-step times are used for these comparisons. In particular, a coarse midpoint output is not treated as a Cayley step.

The coarse native–Cayley maximum, RMS and endpoint discrepancies are 3.137621×10^{-7}, 1.791643×10^{-7} and 3.134612×10^{-7}. The fine values are 4.524499×10^{-7}, 2.572934×10^{-7} and 4.425433×10^{-7}. The Cayley coarse/fine maximum, RMS and endpoint are 0.643731730, 0.552877237 and 0.539518584 on this grid. The RMS of the native pair on all output times is different, as given above; the two grids must not be interchanged. All 64 positive-imaginary poles are retained and satisfy the fixed phase-reporting reliability conditions. For the largest imaginary pole part $138\,759.402560\,\mathrm{rad\,s^{-1}}$, the observed coarse phase rate is $114\,307.454906\,\mathrm{rad\,s^{-1}}$ and the Cayley rate is $114\,307.454903\,\mathrm{rad\,s^{-1}}$; the fine rates are $130\,928.572134$ and $130\,928.572129\,\mathrm{rad\,s^{-1}}$.

Finally, if c and f denote paired coarse and fine complex pole coordinates after subtracting the known particular response, the algebraic identity

$$f - c = \underbrace{|f|c/|c| - c}_{\text{amplitude component}} + \underbrace{f - |f|c/|c|}_{\text{phase component}} \tag{S326}$$

defines the order used in the decomposition. Conjugate pole pairs are mapped back to the full nominal mechanical norm. The maximum amplitude and phase component norms are 0.005293283 and 0.641781934. The full norm reconstruction residual is below 3.34×10^{-16}; the squared-norm reconstruction, including the cross term, differs by less than 2.58×10^{-16}. The components are not orthogonal and cannot be assigned additive squared-error percentages. This identity is descriptive; the quantitative native–Cayley replay supplies the separate evidence linking the record-pair discrepancy to phase dispersion. No new COMSOL solve, ODE integration, model fit or weight loading is used in these calculations.

S41 Arithmetic reception of the sampled continuous-reference scores

The continuous bound in Supplementary Section S6 concerns an exact finite-exponential function with fixed binary-rational coefficients. Its connection to the reported numerical scores requires reception of the function evaluation, coordinate factors, denominator, and metric arithmetic. We give this interface for all $N = 96001$ output times. Throughout this section, a saved numerical prediction vector is treated as a fixed vector; it is not identified with the exact solution of the corresponding learned-model IVP.

S41.1 Metric and arithmetic premises

Let $s > 0$ be the saved coordinate scale, and interpret every entry of the saved factors L_K, L_M as an exact binary rational. Define

$$\mathcal{C}(x, v) = s^{-1}(L_K^T x, L_M^T v), \qquad d = \|p_0\|_2, \tag{S327}$$

where p_0 is the actual saved initial chart vector and d is its exact Euclidean norm. The common factors, scale, initial vector, and full clock are equal across all thirteen prediction archives. We distinguish d from its floating evaluation d_{fp}, receiving their ratio from an exact sum of the 128 binary-rational squared entries.

The arithmetic bounds assume round-to-nearest IEEE binary64 basic real arithmetic and square root, gradual underflow, and the standard operation-count error model for the real and complex dot products. Set

$$u = 2^{-53}, \qquad \eta = 2^{-1074}, \qquad \gamma_n = \frac{nu}{1 - nu}.$$

The conservative operation counts allow fused multiply-add and reassociation within the stated envelopes. These are explicit platform premises, rather than a formal verification of an arbitrary BLAS implementation or a flush-to-zero arithmetic mode. Absolute tiny-result terms are retained. Positive magnitude envelopes exclude overflow in the received matrix, chart, norm, division, and aggregation operations. Correct rounding of the NumPy complex exponential is not assumed.

S41.2 Reception of every returned phasor

Let $h = 1/160000$, $\tau_i = ih$, and denote the stored output time by t_i. Exact binary-rational comparisons at every index give

$$\Delta_t = \max_i |t_i - \tau_i| < 1.006\,10^{-16}. \tag{S328}$$

For each of the 128 homogeneous poles and the additional drive pole $i\widehat{\nu}$, form 256-bit Arb enclosures of $a = \exp(\lambda k h)$ and $q = \exp(\lambda h)$ at each block start k. Every actual binary64 anchor and step conversion is checked to have complex Euclidean error below $\delta = 2^{-51}$. Each block contains at most $B = 512$ values; its comparison sequence is the binary64 cumulative product of the converted anchor and repeated converted step. Since $\operatorname{Re}\lambda \le 0$, the exact factors have modulus at most one. A uniform recurrence error bound is

$$R_B = (1+\delta)^B - 1 + \gamma_{8B}(1+\delta)^B + 8B\eta(1+\delta)^B < 6.822\,10^{-13}. \tag{S329}$$

The terms respectively receive input conversion, complex-product roundoff, and absolute underflow. The operation count is conservative for the basic real operations of each complex multiplication.

At every index and pole, the real and imaginary differences between the actual NumPy phasor and the comparison sequence are rounded outward using successive `nextafter` operations. Let D_j be the largest resulting complex ℓ^1 difference for pole j. Then

$$\left|e^{\mathrm{fp}}_{j,i} - e^{\lambda_j t_i}\right|_1 \le E_j := D_j + 2R_B + 2|\lambda_j|\Delta_t. \tag{S330}$$

Here $|e^{\lambda t} - e^{\lambda\tau}| \le |\lambda||t-\tau|$ for nonnegative t, τ; the factors two also cover conversion from complex Euclidean norm to ℓ^1 norm. The largest D_j is below $2.585\,10^{-11}$. This receives all $129N = 12384129$ returned phasors, including their floating argument multiplication, without an exponential-library correct-rounding premise.

S41.3 Physical evaluation and chart arithmetic

For either physical amplitude matrix X and particular vector x_p, write

$$A_{ij} = |\operatorname{Re} X_{ij}| + |\operatorname{Im} X_{ij}|, \qquad a_{p,i} = |\operatorname{Re} x_{p,i}| + |\operatorname{Im} x_{p,i}|.$$

All these positive quantities are formed exactly. The actual returned phasor components have absolute value below two, so $S = 4(A\mathbf{1} + a_p)$ is a common positive magnitude envelope. With $E = (E_1, \ldots, E_{128})^T$ and drive error E_p, a componentwise physical evaluation bound is

$$\epsilon_{\mathrm{phys}} \le AE + a_p E_p + \gamma_{1040} S + 8192\eta\mathbf{1}. \tag{S331}$$

The $1040 = 8(128) + 16$ count covers the complex dot, particular product, and addition. A corresponding envelope for the actual physical floating result is $S_{\mathrm{fp}} = (1+\gamma_{1040})S + 8192\eta\mathbf{1}$. For a real factor L, the chart error is bounded by

$$\epsilon_{\mathrm{chart}} \le \frac{|L|^T\epsilon_{\mathrm{phys}} + \gamma_{130}|L|^T S_{\mathrm{fp}} + 1024\eta\mathbf{1}}{s} + \eta\mathbf{1}. \tag{S332}$$

This includes the real 64-term dot and scale division. Combining the two physical components and dividing by d gives the uniform sampled reference-evaluation bound

$$\epsilon_{\mathrm{eval}} < 3.818896813\,10^{-8}. \tag{S333}$$

The positive component bounds and their Euclidean combination are evaluated with outward Arb arithmetic. Relevant physical and chart magnitude envelopes are below 2^{200}; the saved prediction entries obey the same guard. Together with the received nonzero denominator, these guards also exclude overflow in the squared norms and RMS.

S41.4 Transfer of the exact-source tube to the saved metric

The saved factors need not be exact factors of the source operators. Form their exact products $\widetilde{K} = L_K L_K^T$ and $\widetilde{M} = L_M L_M^T$, and compute

$$\begin{aligned} \rho_K &= \|K_0^{-1}(\widetilde{K} - K_0)\|_\infty, \qquad \rho_M = \|M^{-1}(\widetilde{M} - M)\|_\infty, \\ \rho &= \max(\rho_K, \rho_M) < 2.843855207\,10^{-9}. \end{aligned} \tag{S334}$$

These are exact maximum absolute row sums. The first inverse defect is similar to the symmetric matrix $K_0^{-1/2}(\widetilde{K} - K_0)K_0^{-1/2}$, whose spectral radius is therefore at most ρ_K. The same argument applies to mass. Thus

$$(1-\rho)K_0 \preceq \widetilde{K} \preceq (1+\rho)K_0, \qquad (1-\rho)M \preceq \widetilde{M} \preceq (1+\rho)M.$$

Let $\varepsilon_{\mathrm{abs}}$ be the received full-time absolute source tube in the $\mathrm{diag}(K_l, M)$ norm for the actual observed initial state. Since $K_l \succeq K_0$, its bound in the saved chart is

$$\mu_{\mathrm{src}} = \frac{\sqrt{1+\rho}\,\varepsilon_{\mathrm{abs}}}{sd} < 0.000222887809874. \tag{S335}$$

This uses the absolute tube and actual archive denominator; it does not add relative errors normalized by different initial states or metrics.

S41.5 Source score intervals and their scope

For a saved prediction curve p_i, define the exact sampled source error

$$e_i = \frac{\|p_i - \mathcal{C}\zeta(t_i)\|_2}{d}. \tag{S336}$$

Let c_i be its archived numerical score against the explicit reference, and let $c_{\max} = \max_i c_i$. With $a = d_{\mathrm{fp}}/d$, $g = \gamma_{1040}$ and $U_n = \sqrt{4096\eta} + 4096\eta$, the subtraction, norm, and final division admit the uniform allowance

$$r_{\mathrm{met}} = \left(|a-1| + \frac{ag}{1-g}\right) c_{\max} + \frac{aU_n}{d_{\mathrm{fp}}(1-g)} + \eta. \tag{S337}$$

Combining the reference and source bounds gives

$$|e_i - c_i| \le \mu_{\mathrm{src}} + \epsilon_{\mathrm{eval}} + r_{\mathrm{met}} \quad \text{for every saved output index } i. \tag{S338}$$

Maximum and endpoint are one-Lipschitz under this uniform bound. The equal-weight sampled RMS is also one-Lipschitz; its numerical aggregation receives the further allowance

$$\frac{\gamma_{4N+16}}{1-\gamma_{4N+16}}\,\mathrm{RMS}_{\mathrm{fp}}(c) + \sqrt{8N\eta} + 8N\eta.$$

Each resulting interval is clipped at zero on the lower side. All 26 archived numerical curves were reproduced at every point with zero numerical difference before this reception, yielding all $13 \times 2 \times 3 = 78$ source score intervals.

For the original checkpoint, all six lower endpoints are zero and the outward upper endpoints are

Setting	Maximum	RMS	Endpoint
Primary	0.000263352179	0.000247074486	0.000259140815
Reference	0.000285595879	0.000252869856	0.000280786708

For the fixed native export samples, the raw-to-chart transformation has an additional componentwise allowance

$$\epsilon_{\mathrm{raw},j} \le \frac{\gamma_{130}}{s} \sum_i m_i |L_{ij}| + \frac{1024\eta}{s} + \eta,$$

where m_i is the authenticated exact maximum absolute saved raw coordinate. With the Euclidean norm of the combined position and velocity allowances, divided by the exact archive denominator d, added to (S338), the coarse and fine native exports have maximum source-score intervals, respectively,

$$[0.65455849,\ 0.65500435], \qquad [0.58070567,\ 0.58115153]. \tag{S339}$$

The exact fixed-raw native-pair sampled maximum lies in $[0.64373173030, 0.64373173041]$; here the common reference cancels, so neither the source tube nor its evaluation error is added.

These are bounds on fixed numerical prediction or exported native vectors at the stated sample times relative to the declared mathematical source IVP. They do not certify a saved prediction as an exact learned-model IVP trajectory, behavior between outputs, the COMSOL compiled-equation interface, or physical-system equivalence. They also do not establish a general model ranking or complex-network advantage. The arithmetic reception adds no training update, time integration, or native solve.

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